
OpenSTREAM Documentation

*Open Solvers for Two-phase flow Research,
Engineering Analysis and Modeling*

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CONTENTS

1	Welcome	1
1.1	What's inside?	1
1.2	Why OpenSTREAM?	1
1.3	Links	1
1.4	About this documentation	2
2	Usage	3
2.1	Getting started	3
2.1.1	Prerequisites	3
2.1.2	Install OpenSTREAM	3
2.1.3	MATLAB and Python compatibility	4
2.1.4	Configure MATLAB and Python	4
2.1.5	Install the Python dependencies	6
2.1.6	Verify the installation	6
2.1.7	Next steps	7
2.2	Run sample script	7
2.3	Release numbering and versioning	8
2.3.1	Prerelease identifiers	9
2.3.2	Release documentation	9
2.3.3	Development and released versions	9
3	Guides	11
3.1	Theory overview	11
3.1.1	Solver capability overview	11
3.1.2	Detailed theory pages	12
3.2	Glossary	24
3.2.1	Recommended terminology	26
3.2.2	Style notes	27
3.3	Notation	27
3.3.1	General conventions	27
3.3.2	Primary variables	28
3.3.3	Phase and field subscripts	28
3.3.4	Source-term notation	28
3.3.5	Closure model notation	29
3.3.6	Units	29
3.4	Global reference list	29
3.5	Testing and verification	30
3.5.1	Test organization	30
3.5.2	Running all tests	30
3.5.3	Running individual test categories	31

3.5.4	Continuous integration	31
3.5.5	Reference solutions	32
3.5.6	Adding new tests	32
3.5.7	Unit test guidelines	33
3.5.8	Integration test guidelines	33
3.5.9	Environment validation	34
3.5.10	Generated outputs	34
3.5.11	Troubleshooting	34
3.5.12	Contributor checklist	35
3.6	Tutorials	35
3.6.1	1. Quick Start	35
3.6.2	2. Mixture Solver Capabilities	35
3.6.3	3. Two-Fluid Solver Capabilities	35
3.6.4	4. Three-Field Solver Capabilities	36
3.6.5	5. Four-Field Solver Capabilities	36
3.6.6	6. Input Files	36
3.6.7	7. Numerical Convergence and Solver Diagnostics	36
3.6.8	8. Mesh and Time-Step Sensitivity	36
3.6.9	9. Advanced Visualization and Post-Processing	36
4	Applications	39
4.1	Contributing to the database	39
4.1.1	Types of contributions	39
4.1.2	Dataset organization	40
4.1.3	Dataset template	40
4.1.4	Naming convention	41
4.1.5	Source data	43
4.1.6	Lightweight datasets	44
4.1.7	Dataset fields	44
4.1.8	Data provenance	46
4.1.9	Dataset class	47
4.1.10	OpenSTREAM dependency	48
4.1.11	Application projects	48
4.1.12	Comparison and validation	49
4.1.13	Plotting and post-processing	49
4.1.14	Testing and verification	50
4.1.15	Documentation requirements	51
4.1.16	Commit messages	51
4.1.17	Submitting a contribution	52
4.1.18	Contributor checklist	52
4.2	OpenSTREAM-database	53
4.2.1	Purpose	54
4.2.2	Repository organization	54
4.2.3	Dataset source files	55
4.2.4	Lightweight in-memory datasets	55
4.2.5	Units	56
4.2.6	Data preparation and provenance	56
4.2.7	Application projects	57
4.2.8	Requirements	57
4.2.9	Installation	58
4.2.10	MATLAB setup	58
4.2.11	Typical workflow	58
4.2.12	Generated files	59
4.2.13	Data provenance and citation	59

4.2.14	Validation and interpretation	59
4.2.15	Testing and verification	60
4.2.16	Development status	60
4.2.17	Contributing	60
4.3	Application projects	61
4.3.1	NURETH21	61
4.3.2	NUTHOS15_MRM	62
4.4	Publications	62
4.4.1	Overviews	62
4.4.2	Mixture model	63
4.4.3	Two-fluid model	63
4.4.4	Three-field model	63
4.4.5	Four-field model	63
5	Packages	65
5.1	InputEnums	65
5.2	Inputs	75
5.3	Session	92
5.4	Solver	94
5.4.1	Four-field solver	95
5.4.2	Mixture solver	112
5.4.3	Three-field solver	134
5.4.4	Two-fluid solver	143
6	ExternalPackages	175
6.1	CoolPropWrapper	175
7	Community	177
7.1	Authors	177
7.1.1	Maintainers	177
7.1.2	Contributors and users	177
7.2	Contribute to the code	178
7.2.1	How to contribute	178
7.2.2	Coding standards	179
7.2.3	Documentation	180
7.2.4	Testing	180
7.2.5	Reporting issues	182
7.2.6	Code of conduct	182
7.2.7	Communication	182
7.2.8	Contributor checklist	182
7.3	Contribute to the documentation	183
7.3.1	Setup Sublime Text as text editor	183
7.3.2	Configure Python and Sphinx	183
7.3.3	Start writing	185
7.3.4	Build the documentation	185
7.4	OpenSTREAM FAQ	185
7.4.1	Getting started	186
7.4.2	Running simulations	190
7.4.3	Choosing a solver	194
7.4.4	Numerical convergence	195
7.4.5	Results and post-processing	198
7.4.6	Physical scope and limitations	205
7.4.7	Extending and contributing	212
7.5	Support	214

Bibliography	215
MATLAB Module Index	219
Index	221

WELCOME

OpenSTREAM (**O**pen **S**olvers for **T**wo-phase flow **R**esearch, **E**ngineering **A**nalysis and **M**odeling) is an open-source, object-oriented computational environment designed for simulating one-dimensional, multi-field, two-phase flows in straight geometries (Le Corre *et al.* [2025a]) (Le Corre *et al.* [2025b]). It supports phenomena such as wall boiling, phase interactions and non-equilibrium effects, making it a relevant tool for researchers working in thermal-hydraulics and two-phase flow systems.

1.1 What's inside?

OpenSTREAM offers a suite of solver frameworks tailored to different modeling needs:

- A **mixture solver** – with thermal non-equilibrium capabilities
- A generic **two-fluid solver** – for separate-phase modeling
- A **three-field solver** – for annular two-phase flow
- An advanced **four-field solver** – for annular two-phase flow explicitly capturing disturbance waves

These solvers are designed to simulate single-component, thermally expandable, steady-state and transient boiling two-phase flows in straight multi-wall channels, which may be uniformly or non-uniformly heated. The implementation relies on a set of reasonable simplifying assumptions to ensure computational efficiency and stability. Each solver includes a set of baseline closure models, which can be customized by modifying the corresponding class methods in the source code.

1.2 Why OpenSTREAM?

OpenSTREAM is built to support open, transparent research and to lower the barrier to entry for developing and validating fundamental models in two-phase flow simulations. Its goal is to make advanced thermal-hydraulic modeling more accessible to engineers, scientists, and students alike. Whether you're conducting research or learning the fundamentals, OpenSTREAM provides a collaborative and extensible environment for model development, performance evaluation, and cross-institutional validation, advancing the state of the art in thermal-hydraulic simulation.

1.3 Links

OpenSTREAM is developed openly on GitHub:

- **Source code:** [OpenSTREAM-solvers/openstream](#)
- **Report an issue:** [Issue tracker](#)
- **Contribute:** see the [Community](#) section

1.4 About this documentation

This guide will walk you through:

- Installation and setup instructions
 - The design philosophy behind OpenSTREAM
 - How to run simulations and interpret results
 - Example applications and use cases
 - How to contribute
-

1. Jean-Marie Le Corre, Jason Chan, Elisabeth Walter, Evan T. Hurlburt, and Roman W. Morse. OpenSTREAM: an open-source platform for two-phase flow modeling and simulation. In *Proceedings of the 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*. Busan, Korea, August 31–September 5 2025a. URL: https://openstream-solvers.github.io/openstream/_static/publications/LeCorre2025OpenSTREAM.pdf.
2. Jean-Marie Le Corre, Jason Chan, Elisabeth Walter, Evan T. Hurlburt, and Roman W. Morse. OpenSTREAM: a new open-source platform for two-phase flow model development. In *Proceedings of the 12th International Conference on Multiphase Flow (ICMF 2025)*. Toulouse, France, May 12–16 2025b. URL: https://openstream-solvers.github.io/openstream/_static/publications/LeCorre2025ICMF.pdf.

2.1 Getting started

OpenSTREAM is a MATLAB program for simulating one-dimensional two-phase flows. Fluid properties are calculated using the Python version of [CoolProp](#) through [CoolPropWrapper](#), a MATLAB interface to CoolProp.

This guide explains how to obtain OpenSTREAM, configure MATLAB and Python, install the required dependencies, and verify the installation.

2.1.1 Prerequisites

Before installing and configuring OpenSTREAM, ensure that the following software is available:

- MATLAB.
- A Python version compatible with the installed MATLAB release.
- Git or GitHub Desktop.

CoolProp and its associated Python dependencies are installed later in this guide.

No additional MathWorks toolbox is currently required by the core OpenSTREAM solver frameworks.

2.1.2 Install OpenSTREAM

OpenSTREAM is hosted in the [OpenSTREAM GitHub repository](#).

There are two principal ways to obtain and maintain a local copy of the repository:

- The [Git command-line interface](#).
- [GitHub Desktop](#).

Install using the Git command-line interface

Clone the repository and its submodules using SSH:

```
git clone --recursive git@github.com:OpenSTREAM-solvers/openstream.git
```

Alternatively, clone the repository using HTTPS:

```
git clone --recursive https://github.com/OpenSTREAM-solvers/openstream.git
```

The `--recursive` option also clones the repository submodules, including [CoolPropWrapper](#).

For information about authentication, see:

- [Connecting to GitHub with SSH](#).

- [Caching GitHub credentials](#).
- [Working with Git submodules](#).

Enter the cloned repository:

```
cd openstream
```

Install using GitHub Desktop

Open GitHub Desktop, sign in, and select **Clone repository**. Select the OpenSTREAM repository and choose the local destination folder.

After cloning, verify that the `CoolPropWrapper` submodule contains its expected files. If the submodule is empty, initialize it from a terminal opened in the repository root:

```
git submodule update --init --recursive
```

2.1.3 MATLAB and Python compatibility

OpenSTREAM requires MATLAB. No additional MathWorks toolbox is currently required by the core solver frameworks. Fluid properties are calculated externally using the Python version of CoolProp through CoolPropWrapper.

OpenSTREAM is tested with selected MATLAB and Python version combinations through its continuous-integration workflow.

Table 2.1: CI-tested MATLAB and Python configurations

MATLAB release	Python version
R2024b	3.11
R2025b	3.12
R2026a	3.13

These combinations represent configurations tested by the OpenSTREAM continuous-integration workflow. Other MATLAB and Python combinations may work but are not necessarily tested. Python 3.14 is not currently included in the tested CI configuration.

The tested software configurations apply to the OpenSTREAM version or development revision documented on this site. See [Release numbering and versioning](#) for the project versioning convention.

Compatibility policy

OpenSTREAM distinguishes between tested and untested software configurations. A tested configuration is a combination of MATLAB, Python, CoolProp, and associated dependencies that is exercised by the OpenSTREAM continuous-integration workflow.

The configurations listed above represent the environments tested for the current OpenSTREAM development version or release. Other software combinations may work but are not guaranteed and may not be covered by automated testing.

Compatibility information may change as new MATLAB, Python, and CoolProp versions become available. Consult the compatibility information corresponding to the OpenSTREAM version being used.

2.1.4 Configure MATLAB and Python

Before using OpenSTREAM, verify that a suitable Python version is installed and that MATLAB is configured to use the corresponding Python executable.

The general compatibility requirements are available in the [MathWorks Python compatibility table](#). The **MATLAB Interface** column identifies the Python versions supported by each MATLAB release.

The following configuration instructions are adapted from the [CoolPropWrapper repository](#).

Inspect the configured Python environment

In MATLAB, inspect the current Python environment:

```
pythonEnvironment = pyenv;  
disp(pythonEnvironment)
```

If no Python environment is configured, or if the configured version is not suitable for the installed MATLAB release, install a compatible Python version and configure MATLAB to use it.

Install a compatible Python version

Windows

Download the required version from [Python.org](#) and follow the installation instructions.

Linux

Check whether Python is already available:

```
python3 --version
```

If necessary, install an appropriate version using the operating-system package manager. For example, on a Debian-based distribution:

```
sudo apt-get install python3.x
```

Replace `x` with the required minor version.

macOS

Check the available Python version:

```
python3 --version
```

If an appropriate version is not available, install one using Homebrew:

```
brew install python
```

Alternatively, download the required version from [Python.org](#).

Configure the Python executable in MATLAB

Configure MATLAB to use the required Python executable:

```
pyenv('Version', '<path-to-python-executable>')
```

Replace `<path-to-python-executable>` with the complete path to the Python executable.

Typical locations include:

- **Windows:** `C:\Users\Username\AppData\Local\Programs\Python\Python311\python.exe`
- **Linux:** `/usr/bin/python3.11`
- **macOS:** `/usr/local/bin/python3` or `/opt/homebrew/bin/python3`

On Linux and macOS, commands such as `which python3` can help locate the executable.

2.1.5 Install the Python dependencies

The recommended approach is to install the dependencies from the requirements file supplied with CoolPropWrapper.

Alternatively, install the dependencies used by the current continuous-integration environment directly:

```
python -m pip install CoolProp==8.0.0 nanobind
```

In the commands above, `python` must refer to the same Python executable configured through `pyenv`. If necessary, replace `python` with the complete path to that executable.

The current OpenSTREAM continuous-integration environment installs `CoolProp==8.0.0` together with `nanobind`. Other CoolProp versions may work but are not necessarily tested.

When several Python installations are available, ensure that the dependencies are installed using the same Python executable configured in MATLAB.

2.1.6 Verify the installation

After configuring MATLAB and Python, verify that MATLAB is using the intended Python environment and that CoolProp is accessible through CoolPropWrapper.

Inspect the Python environment

```
pythonEnvironment = pyenv;  
disp(pythonEnvironment)
```

Confirm that the reported executable and Python version correspond to the intended installation.

Verify the Python packages

From a terminal, use the same Python executable configured in MATLAB:

```
python -m pip show CoolProp  
python -m pip show nanobind
```

Create a CoolPropWrapper object

In MATLAB, create a fluid-property object:

```
fluidProperties = ...  
    CoolPropWrapper.CoolPropWrapper('WATER');
```

Successful construction confirms that MATLAB can access Python, import CoolProp, and initialize the requested fluid.

Evaluate a representative fluid property

Evaluate the saturation temperature of water at 6 MPa:

```
saturationTemperature = ...  
    fluidProperties.temperature('P',6.0e6,'Q',0);  
disp(saturationTemperature)
```

A finite numerical result confirms that the MATLAB-Python-CoolProp interface is functioning.

Run the automated tests

From the OpenSTREAM tests folder, run the complete automated test suite:

```
runOpenSTREAMTests
```

The complete suite includes environment, unit, and integration tests. The environment tests verify Python availability, package installation, and basic CoolProp functionality. The same complete test suite is executed by the MATLAB continuous-integration workflow when changes are pushed to the repository.

Run the sample script

Run the *sample script* to perform a complete OpenSTREAM calculation and confirm that the installation is functioning correctly.

The sample calculation uses the mixture solver to simulate boiling two-phase flow in a uniformly heated tube followed by an adiabatic section. It constructs the required input objects, evaluates the fluid properties, solves the case, generates representative axial plots, and saves the solver object.

Run the commands from the OpenSTREAM repository root, as described on the sample-script page.

2.1.7 Next steps

Congratulations! At this point, you should have a local copy of OpenSTREAM and a functioning MATLAB, Python, and CoolProp configuration.

Continue with the *OpenSTREAM tutorials*, beginning with Tutorial 1: Quick Start. Tutorial 1 constructs the input objects, runs the example case with the mixture, two-fluid, three-field, and four-field solver frameworks, generates representative plots, and saves the solver objects.

You can then continue with the solver-specific and workflow-oriented tutorials according to your interests.

2.2 Run sample script

After installation, you're all set to launch your first simulation!

As a first exercise, open MATLAB and run the commands listed below from within the `openstream` folder, which will simulate an example of boiling two-phase flow in a uniformly heated tube followed by an adiabatic region. The **OpenSTREAM** mixture model will be used, which defaults to the Homogeneous Equilibrium Model (HEM) when the physical models are not modified by the user.

This is the quickest and easiest way to start exploring what OpenSTREAM can do, and to confirm everything is working smoothly under the hood.

```
% Clear existing variables from workspace
clearvars

% Import the necessary classes for inputs and solvers
import Inputs.*
import Solvers.*
import Solvers.Mixture.*

% Turn off warnings during input setup
warning off

% Create the input set using paths to model, options, geometry, and boundary condition.
↪ files
inputSet = InputSet( ...
```

(continues on next page)

(continued from previous page)

```

modelFilePath      = './inputs/models.inp', modelID      = 'TUTORIAL1', ...
optionsFilePath    = './inputs/options.inp', optionsID   = 'DEFAULT', ...
geometryFilePath   = './inputs/geom.inp',   geometryID  = 'TUTORIAL1', ...
bcFilePath         = './inputs/tutorial1.inp', ...
sessionParentDir   = fullfile(pwd,'outputs'), ...
overwriteSessionFiles = true, ...
LOGMODE            = 'BOTH');

% Create a mixture solver object
mixSolver = MixtureSolver(inputSet);

% Solve
mixSolver.solve();

% Default axial plots
mixSolver.plotz();

% Save results
mixSolver.save();

```

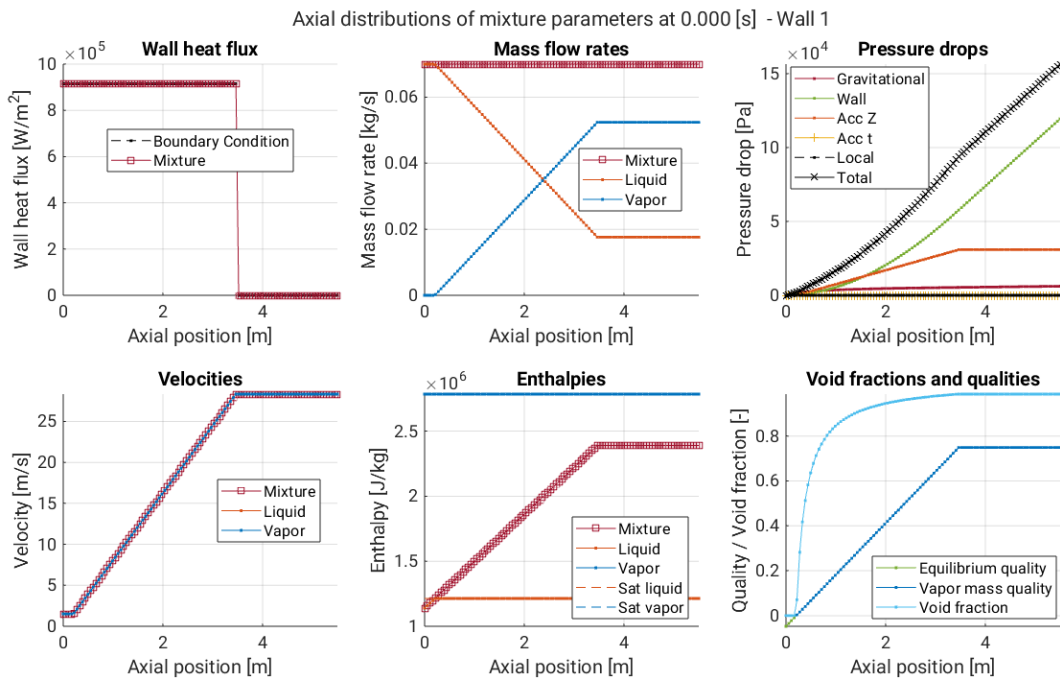


Figure 2.1: Sample figure of axial distributions of mixture parameters.

2.3 Release numbering and versioning

OpenSTREAM uses a hybrid calendar-based and semantic version-numbering scheme. Release identifiers have the following general form:

```
vYYYY.N
```

where YYYY identifies the release year and N identifies the successive release within that year.

Examples include:

- `v2026.0`: the first principal OpenSTREAM release associated with 2026;
- `v2026.1`: a subsequent OpenSTREAM update released in 2026;
- `v2026.1-alpha`: a prerelease associated with the planned `v2026.1` release.

Principal releases are intended to collect significant code, model, interface, testing, and documentation changes. The project aims to publish principal releases when justified by the accumulated changes. Additional releases may be published as needed.

2.3.1 Prerelease identifiers

A suffix may be added when a release is made available before its final publication. For example:

```
v2026.1-alpha
```

identifies an alpha prerelease associated with `v2026.1`.

Prerelease versions may contain incomplete, experimental, or insufficiently validated features. They should not be interpreted as stable releases.

2.3.2 Release documentation

Each released OpenSTREAM version should be accompanied by release notes summarizing, as applicable:

- new solver and physical-model capabilities;
- corrections to existing implementations;
- changes to input parameters or default values;
- numerical-method changes;
- documentation and tutorial updates;
- tested MATLAB, Python, and CoolProp configurations;
- known limitations;
- migration information for existing input files or user-developed models.

2.3.3 Development and released versions

The development version may contain changes that are not available in the latest released version. Users should verify that the documentation, tutorials, and source code correspond to the same OpenSTREAM version or source revision.

When citing, distributing, or reproducing an OpenSTREAM calculation, record the release identifier or source revision used for the calculation.

3.1 Theory overview

This page provides a high-level overview of the theoretical foundations of the simulation frameworks implemented in **OpenSTREAM**, together with the relevant references.

The frameworks are designed to simulate one-dimensional, two-phase flow in straight channels. They provide increasing levels of physical resolution, from a mixture formulation to the explicit representation of liquid films, entrained droplets, and disturbance waves.

The appropriate framework depends on the flow regime, quantities of interest, and assumptions acceptable for the intended application. Detailed equations, derivations, assumptions, and implementation notes are provided on the individual theory pages linked below.

3.1.1 Solver capability overview

Framework	Principal fields	Thermal treatment	Time dependence	Principal applications
Mixture	Mixture, with liquid and vapor properties	Thermal equilibrium or nonequilibrium	Steady-state and transient	Initialization, system-level boiling, and post-CHF calculations
Two-fluid	Separate liquid and vapor fields	Thermal equilibrium or nonequilibrium	Steady-state and transient	Separate-phase transport and interfacial exchange
Three-field	Vapor, liquid film, and entrained droplets	Thermal equilibrium in the current annular-flow formulation	Steady-state and transient	Annular flow, film transport, entrainment, deposition, and dryout
Four-field	Vapor, base film, disturbance waves, and entrained droplets	Thermal equilibrium in the current annular-flow formulation	Steady-state and transient	Wave-resolved annular flow and developing wave behavior

The table summarizes the current solver formulations. Availability of a solver framework does not imply that every physical model, closure relation, flow regime, or application has been comprehensively validated.

OpenSTREAM is primarily intended for fundamental model development, assessment, and validation in one-dimensional straight-channel configurations. The applicability of a calculation depends on the selected closure models, their validity ranges, the numerical settings, and the conditions under consideration.

3.1.2 Detailed theory pages

Mixture model

- **Purpose:** Provide initialization and robust predictions under relevant simplifications.
- **Assumptions:** Represent the flow using a single mixture field, with optional hydrodynamic and thermal nonequilibrium models.
- **Equations:** Conserve mixture mass, momentum, and energy.

Two-fluid model

- **Purpose:** Resolve hydrodynamic and thermal nonequilibrium between the liquid and vapor phases.
- **Assumptions:** Represent the liquid and vapor as separate fields coupled through interfacial exchange models.
- **Equations:** Conserve mass, momentum, and energy separately for the liquid and vapor phases.

Three-field model

- **Purpose:** Model annular two-phase flow using separate vapor, liquid-film, and entrained-droplet fields.
- **Assumptions:** Apply the current annular-flow formulation under thermal-equilibrium conditions up to liquid-film dryout.
- **Equations:** Conserve mass and momentum for the liquid film and entrained droplets together with the mixture-model vapor solution.

Four-field model

- **Purpose:** Resolve the continuous base film and disturbance waves as separate liquid fields in annular flow.
- **Assumptions:** Apply the current annular-flow formulation under thermal-equilibrium conditions up to liquid-film dryout.
- **Equations:** Conserve mass and momentum for the base film, disturbance waves, and entrained droplets together with the mixture-model vapor solution and a disturbance-wave number-density transport equation.
- **Nonequilibrium behavior:** Represent the spatial and temporal development of disturbance-wave properties without assuming immediate local hydrodynamic equilibrium.

Select one of the theory pages above for the detailed formulation, implemented assumptions, closure relations, and references.

Mixture model

The mixture simulation framework in **OpenSTREAM** provides a simplified yet powerful approach to modeling two-phase flows. Based on a three-equation formulation, it treats the phases as a single continuum with averaged properties. This abstraction enables efficient simulation of systems where phase separation is minimal or where a fully resolved multi-field model is not required.

The mixture model serves two key roles within OpenSTREAM:

- **Robust initialization** for more complex solvers, ensuring robust starting conditions.
- **Efficient simulation** of flows with hydrodynamically well-coupled phases, where the mixture approximation remains physically meaningful.
- **Industry relevance** due to its simplicity and robustness, offering a practical approach for simulating averaged flow behavior in complex thermal-hydraulic systems.

In addition, a Mixture Relaxation Model (MRM) is available (Le Corre *et al.* [2026]). This extended formulation introduces vapor mass and energy conservation equations, allowing for thermal non-equilibrium between phases and expanding the model's applicability to more dynamic flow regimes.

By reducing complexity while preserving essential dynamics, the mixture model offers a practical entry point for both model development and exploratory analysis in thermal-hydraulic systems.

An overview of the mixture model implemented in OpenSTREAM is provided below. A more detailed derivation and theoretical background can be found in Le Corre *et al.* [2025a].

Governing equations

The mixture model solves the following conservation equations:

1. Mass conservation

$$\frac{\partial}{\partial t} \left(\frac{W}{u} \right) + \frac{\partial W}{\partial z} = 0$$

where:

- W is the mixture mass flow rate
- u is the mixture (static) velocity

2. Momentum conservation

$$\frac{\partial W}{\partial t} + \frac{\partial}{\partial z} (u^+ \cdot W) = -A \cdot \left(\frac{\partial p}{\partial z} + \frac{\partial p_K}{\partial z} + \cos \theta \cdot g \cdot \rho \right) - \sum \Pi_{wall}^n \cdot \tau_{wall}^n$$

where:

- ρ is the mixture density
- u^+ is mixture (advection) velocity
- A is the cross-sectional area
- p is the pressure
- K relates to obstruction form loss
- θ is the inclination angle
- g is the gravitational acceleration
- n is the wall index
- Π_{wall}^n is the wall perimeter for wall index n
- τ_{wall}^n is the wall shear stress for wall index n

3. Energy conservation

$$\frac{\partial}{\partial t} \left(h \cdot \frac{W}{u} \right) + \frac{\partial}{\partial z} (h^+ \cdot W) = \sum \Pi_{wall}^n \cdot q''_{wall}^n$$

where:

- h is the mixture (static) specific enthalpy
- h^+ is the mixture (advection) specific enthalpy
- q''_{wall}^n is the wall heat flux for wall index n

Mixture relaxation model

When selecting the Mixture Relaxation Model (MRM), two supplementary conservation equations are solved for the vapor phase. These equations provide the information required to determine the vapor mass fraction and vapor enthalpy while accounting for delayed interfacial heat and mass transfer. The resulting five-equation framework remains computationally efficient while allowing departures from thermodynamic equilibrium in subcooled boiling, saturated boiling, and post-boiling transition conditions.

1. Vapor mass conservation

$$\frac{\partial}{\partial t} \left(\frac{W_v}{u_v} \right) + \frac{\partial W_v}{\partial z} = A \cdot a_i \cdot (\Gamma - \Lambda) + \sum \Pi_{wall}^n \cdot \Gamma_{wb}^n$$

where:

- W_v is the vapor mass flow rate
- u_v is the vapor velocity
- a_i , is the volumetric interfacial area
- Γ is the interfacial evaporation mass flux
- Λ is the interfacial condensation mass flux
- Γ_{wb}^n is the wall boiling mass flux for wall index n

2. Vapor energy conservation

$$\frac{W_v}{u_v} \cdot \frac{\partial}{\partial t} (h_v) + W_v \cdot \frac{\partial}{\partial z} (h_v) = A \cdot a_i \cdot (h_l - h_v) \cdot \Gamma + \sum \Pi_{wall}^n \cdot q''_{wall,v}$$

where:

- h_v is the vapor specific enthalpy
- $q''_{wall,v}$ is the wall heat flux vapor for wall index n

3. Time relaxation approximation

The fundamental assumption of the MRM is to consider that the interfacial mass exchanges are governed by the deviation from thermal equilibrium under a time relaxation principle:

$$A \cdot a_i \cdot (\Gamma - \Lambda) = \frac{x_{eq} \cdot W - W_v}{u_v \cdot t_{Relax}}$$

where:

- x_{eq} is the thermodynamic equilibrium quality
- t_{Relax} is the relaxation time for the interfacial mass transfer (evaporation or condensation)

Closure models

To complete the conservation equations, several closure models are required:

- Wall shear stress: τ_w^n
- Form pressure losses: $\frac{\partial p_K}{\partial z}$
- Wall heat transfer models
- Relaxation time for the interfacial mass transfer (required for the MRM model only): t_{Relax}

For each simulation, the selected closure models are defined in the OpenSTREAM model file, chosen from the available options listed in [InputEnums](#). If not explicitly specified by the user, default models are applied as defined in [Inputs.Model](#). All relevant closure models are implemented in [Solvers.Mixture.Mixture](#), which the users can modify to suit specific simulation needs.

In addition, the thermodynamic properties for each phase are computed using [CoolProp](#), an open-source thermophysical property library that provides accurate equations of state and transport properties for a wide range of fluids.

Features and assumptions

- Supports both steady-state and transient simulations in straight channels
- Supports uniform and non-uniform wall heat flux distribution
- Can include thermal non-equilibrium modeling (subcooled boiling or post Critical Heat Flux) via closure models or MRM

- Allows phase velocity slip using drift flux models
- Neglects minor contributions such as frictional heating and temporal pressure gradient contributions

Role in OpenSTREAM

The mixture model provides a robust tool for simulating homogeneous two-phase flow, including cases with thermal non-equilibrium. It also provides an initialization framework for more complex solvers (i.e., two-fluid, three-field, four-field). Currently, its pressure gradient solution is reused across all solver frameworks to enhance robustness and numerical stability.

Implementation notes

Package

- *Solvers*

Module

- *Solvers.Mixture*

Mixture solver class

- *Solvers.Mixture.MixtureSolver*

Field and phase classes:

- *Solvers.Mixture.Mixture*
- *Solvers.Mixture.Liquid*
- *Solvers.Mixture.Vapor*

Key solver methods

- *Solvers.Mixture.MixtureSolver.solve()*

Key field properties:

- *Solvers.Mixture.Mixture.W*
- *Solvers.Mixture.Mixture.P*
- *Solvers.Mixture.Mixture.H*

Two-fluid model

The two-fluid simulation framework in **OpenSTREAM** provides a detailed representation of two-phase flow by solving six conservation equations—mass, momentum, and energy for both liquid and vapor phases. This formulation enables the simulation of key non-equilibrium effects and is foundational in nuclear reactor thermal-hydraulic code systems.

The two-fluid model serves several key roles within OpenSTREAM:

- **Full separate phase treatment:** Liquid and vapor are modeled as distinct continua, each with its own set of governing equations.
- **Non-equilibrium dynamics:** Captures both thermal and hydrodynamic non-equilibrium, essential for accurate transient and boiling flow simulations.
- **Industry relevance:** Commonly used in reactor system codes to simulate phenomena such as evaporation, condensation, and phase separation.

By explicitly modeling both liquid and vapor phases, the two-fluid approach preserves critical non-equilibrium dynamics while offering a flexible foundation for detailed model development and advanced thermal-hydraulic analysis.

An overview of the two-fluid model implemented in OpenSTREAM is provided below. A more detailed derivation and theoretical background can be found in Le Corre *et al.* [2025a] and Walter [2024].

Governing equations

1. Mass conservation

$$\text{Liquid: } \frac{\partial}{\partial t} \left(\frac{W_l}{u_l} \right) + \frac{\partial W_l}{\partial z} = -Aa_i(\Gamma - \Lambda) - \sum \Pi_{wall}^n \Gamma_{wb}^n$$

$$\text{Vapor: } \frac{\partial}{\partial t} \left(\frac{W_v}{u_v} \right) + \frac{\partial W_v}{\partial z} = Aa_i(\Gamma - \Lambda) + \sum \Pi_{wall}^n \Gamma_{wb}^n$$

where:

- W_l, W_v are the liquid and vapor mass flow rates
- u_l, u_v are the liquid and vapor velocities
- a_i is the volumetric interfacial area
- Γ is the interfacial evaporation mass flux
- Λ is the interfacial condensation mass flux
- Γ_{wb}^n is the wall boiling mass flux for wall index n

2. Momentum conservation

$$\text{Liquid: } \rho_l A_l \left(\frac{\partial u_l}{\partial t} + u_l \frac{\partial u_l}{\partial z} \right) = -Aa_i \Lambda (u_l - u_v) - A_l \left(\frac{\partial p}{\partial z} + \cos \theta g \rho_l \right) + Aa_i \tau_{v,l} - \sum \Pi_{wall}^n \tau_{wall,l}^n$$

$$\text{Vapor: } \rho_v A_v \left(\frac{\partial u_v}{\partial t} + u_v \frac{\partial u_v}{\partial z} \right) = (Aa_i \Gamma + \sum \Pi_{wall}^n \Gamma_{wb}^n) (u_l - u_v) - A_v \left(\frac{\partial p}{\partial z} + \cos \theta g \rho_v \right) - Aa_i \tau_{v,l} - \sum \Pi_{wall}^n \tau_{wall,v}^n$$

where

- A_l, A_v are the liquid and vapor cross-sectional areas
- $\tau_{v,l}$ is the interfacial shear stress
- $\tau_{wall,l}, \tau_{wall,v}$ are the liquid and vapor wall shear stresses

3. Energy conservation

$$\text{Liquid: } \rho_l A_l \left(\frac{\partial h_l}{\partial t} + u_l \frac{\partial h_l}{\partial z} \right) = -Aa_i \Lambda (h_l - h_v) + \sum \Pi_{wall}^n (q''_{wall,l} - (h_v - h_l) \Gamma_{wb}^n)$$

$$\text{Vapor: } \rho_v A_v \left(\frac{\partial h_v}{\partial t} + u_v \frac{\partial h_v}{\partial z} \right) = Aa_i \Gamma (h_l - h_v) + \sum \Pi_{wall}^n q''_{wall,v}$$

where

- h_l, h_v are the liquid and vapor specific enthalpies
- $q''_{wall,l}, q''_{wall,v}$ are the wall heat flux to liquid and vapor for wall index n

Closure models

To complete the conservation equations, several closure models are required:

- Volumetric interfacial area: a_i
- Interfacial evaporation and condensation mass fluxes: Γ, Λ
- Wall boiling mass flux: Γ_{wb}^n
- Interfacial shears stress: $\tau_{v,l}$
- Wall shear stress for each phase: $\tau_{wall,l}, \tau_{wall,v}$

- Wall heat flux for each phase: $q''_{wall,l}^n, q''_{wall,v}^n$
- Wall heat transfer models

These relations depend on the local flow regime, which is determined by an additional flow regime identification model.

The selected closure models are defined in the OpenSTREAM model file, chosen from the available options listed in [InputEnums](#). If not explicitly specified by the user, default models are applied as defined in [Inputs.Model](#). All closure models are implemented in [Solvers.TwoFluid.Liquid](#) and [Solvers.TwoFluid.Vapor](#), which the users can modify to suit specific simulation needs.

In addition, the thermodynamic properties for each phase are computed using [CoolProp](#), an open-source thermophysical property library that provides accurate equations of state and transport properties for a wide range of fluids.

Features and assumptions

- Supports both steady-state and transient simulations in straight channels
- Supports uniform and non-uniform wall heat flux distribution
- Captures phase-specific velocities and temperatures
- Includes interfacial mass, momentum, and energy exchange
- Supports wall boiling and condensation
- Neglects minor contributions such as frictional heating and temporal pressure gradient contributions

Role in OpenSTREAM

The two-fluid model is the legacy framework for simulating non-equilibrium two-phase flows. It is validated against system codes like TRACE and supports both steady-state and transient simulations.

Implementation notes

Package

- [Solvers](#)

Module

- [Solvers.TwoFluid](#)

Two-fluid solver class

- [Solvers.TwoFluid.TwoFluidSolver](#)

Phase classes

- [Solvers.TwoFluid.Liquid](#)
- [Solvers.TwoFluid.Vapor](#)

Flow regime enumeration class

- [Solvers.TwoFluid.REGIMES](#)

Key solver methods

- [Solvers.TwoFluid.TwoFluidSolver.solve\(\)](#)

Key field properties:

- [Solvers.TwoFluid.Liquid.W](#)

- `Solvers.TwoFluid.Liquid.U`
- `Solvers.TwoFluid.Liquid.H`
- `Solvers.TwoFluid.Vapor.W`
- `Solvers.TwoFluid.Vapor.U`
- `Solvers.TwoFluid.Vapor.H`

Three-field model

The three-field simulation framework in **OpenSTREAM** provides a structured representation of annular two-phase flow by solving relevant conservation equations for the liquid film and droplet fields, with the vapor field derived from the mixture solver. This formulation is tailored for saturated flow conditions within a theoretical framework used in other codes, such as in subchannel analysis for Boiling Water Reactor (BWR) fuel simulations (Adamsson and Le Corre [2011])(Adamsson and Le Corre [2014]).

The three distinct flow fields: vapor, entrained droplets and liquid film, are illustrated in Figure 3.1.

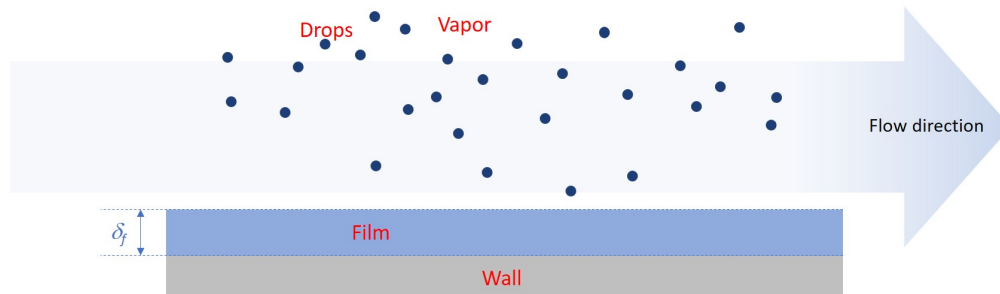


Figure 3.1: Field geometrical characteristics in the three-field simulation framework (vapor, drops and film).

The three-field model serves several key roles within OpenSTREAM:

- **Field separation:** Distinguishes between vapor, entrained droplets, and liquid film, enabling detailed modeling of annular flow dynamics.
- **Film and droplet transport:** Captures deposition and entrainment processes critical to predicting film dryout and droplet behavior.
- **Thermal equilibrium assumption:** Simplifies energy conservation while retaining essential mass and momentum exchanges.

By explicitly modeling the liquid film and droplet fields, the three-field framework enables accurate simulation of saturated annular flow regimes in single channels and supports advanced thermal-hydraulic analysis for various industrial applications.

An overview of the three-field model implemented in OpenSTREAM is provided below. A more detailed derivation and theoretical background can be found in Le Corre *et al.* [2025a].

Governing equations

The conservation equations are formulated at the wall level, indexed by n , to support multi-wall geometries with different heating rates. These equations are solved starting from the onset of annular two-phase flow. Upstream of this onset, the solution from the mixture solver is used to initialize the flow fields.

1. Mass conservation

Film:
$$\frac{\partial}{\partial t} \left(\frac{W_f^n}{u_f^n} \right) + \frac{\partial W_f^n}{\partial z} = \Pi_{all}^n (D - E^n - \Gamma_{wb}^n)$$

where:

- W_f^n is the liquid film mass flow rate for wall index n
- u_f^n is the liquid film velocity for wall index n
- D is the drop deposition mass flux
- E is the film entrainment mass flux
- Γ_{wb}^n is the wall boiling mass flux for wall index n

The drop mass flow rate, W_d , is simply computed by subtracting W_f from the total liquid mass flow rate obtained from the mixture model.

2. Momentum conservation

Liquid Film: $\rho_{ls}\delta_f^n(\frac{\partial u_f^n}{\partial t} + u_f^n \frac{\partial u_f^n}{\partial z}) = (u_d - u_f^n)D - \delta_f^n(\frac{\partial p}{\partial z} + \cos\theta g\rho_{ls}) + \tau_{v,f}^n - \tau_{wall,f}^n$

Droplets: $\rho_{ls}(\frac{\partial u_d}{\partial t} + u_d \frac{\partial u_d}{\partial z}) = (\sum \Pi_{wall}^n(u_f^n - u_d)E^n)\frac{\rho_{ls}u_d}{W_d} - (\frac{\partial p}{\partial z} + \cos\theta g\rho_{ls}) + \frac{A_d}{V_d}\tau_{v,d}$

where:

- ρ_{ls} is the saturated liquid density
- δ_f^n is the liquid film thickness for wall index n
- u_d is the drop velocity
- $\tau_{v,f}^n$ is the vapor/film interfacial shear stress for wall index n
- $\tau_{v,d}^n$ is the vapor/drop interfacial shear stress
- $\tau_{wall,f}^n$ is the wall shear stress on the liquid film for wall index n
- A_d, V_d are the drop interfacial area and volume (based on surface averaging)

3. Energy conservation

Under thermal equilibrium: $\Gamma_{wb}^n = \frac{q''_w^n}{h_{vs} - h_{ls}}$

where:

- h_{vs}, h_{ls} are the saturated vapor and liquid specific enthalpies
- q''_w^n is the wall heat flux for wall index n

Closure models

To complete the conservation equations, several closure models are required:

- Onset of annular two-phase flow
- Film/drop mass flow rate split at onset of annular two-phase flow
- Drop deposition mass flux: D
- Film entrainment mass flux: E
- Vapor/film interfacial shear stress: $\tau_{v,f}^n$
- Vapor/drop interfacial shear stress: $\tau_{v,d}^n$
- Wall shear stress on the liquid film: $\tau_{w,f}^n$
- Drop interfacial area and volume: A_d, V_d

The selected closure models are defined in the OpenSTREAM model file, chosen from the available options listed in *InputEnums*. If not explicitly specified by the user, default models are applied as defined in *Inputs.Model*. All closure models are implemented in *Solvers.ThreeField.Film* and *Solvers.ThreeField.Drop*, which the users can modify to suit specific simulation needs.

In addition, the thermodynamic properties for each phase are computed using *CoolProp*, an open-source thermophysical property library that provides accurate equations of state and transport properties for a wide range of fluids.

Features and assumptions

- Supports both steady-state and transient simulations in straight channels
- Supports uniform and non-uniform wall heat flux distribution
- Applicable up to film dryout
- Models drop deposition and film entrainment
- Assumes thermal equilibrium (no temperature difference between phases)
- Supports multi-wall geometries (e.g., annuli)
- Neglects minor contributions such as frictional heating and temporal pressure gradient contributions

Role in OpenSTREAM

The three-field model is the preferred framework for simulating annular two-phase flow up to the point of liquid film dryout. It offers detailed modeling of liquid film and droplet dynamics, making it particularly well-suited for advanced thermal-hydraulic analyses in Boiling Water Reactor (BWR) applications.

Implementation notes

Package

- *Solvers*

Module

- *Solvers.ThreeField*

Three-field solver class

- *Solvers.ThreeField.ThreeFieldSolver*

Field classes

- *Solvers.ThreeField.Film*
- *Solvers.ThreeField.Drop*

Key solver methods

- *Solvers.ThreeField.ThreeFieldSolver.solve()*

Key field properties:

- *Solvers.ThreeField.Film.W*
- *Solvers.ThreeField.Film.U*
- *Solvers.ThreeField.Drop.W*
- *Solvers.ThreeField.Drop.U*

Four-field model

The four-field simulation framework in **OpenSTREAM** extends the traditional three-field model by explicitly representing disturbance waves, in addition to vapor, droplets, and the base liquid film. This modeling approach was originally developed in Le Corre [2022a] and Le Corre [2022b]. It provides improved resolution of annular two-phase flow dynamics, enabling more accurate and detailed simulations.

The four distinct flow fields: vapor, entrained droplets, base liquid film and disturbance waves, are illustrated in Figure 3.2.

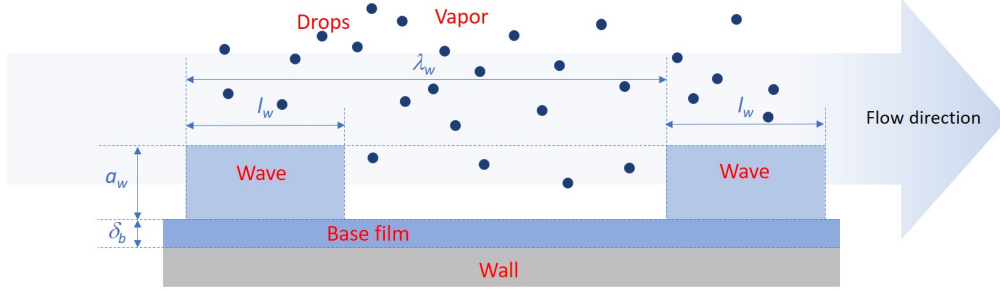


Figure 3.2: Field geometrical characteristics in the four-field simulation framework (vapor, drops, base film and waves).

The four-field model serves several key roles within OpenSTREAM:

- **Wave-resolved modeling:** Separates the liquid film into base film and disturbance waves, capturing their distinct transport behaviors and interactions.
- **Non-equilibrium dynamics:** Includes a Boltzmann-type wave number density transport equation to simulate wave formation, merging, and dissipation.
- **Enhanced predictive capability:** Enables detailed simulation of film dryout, wave-driven mass transport, and hydrodynamic transitions in developing annular flow.

By explicitly modeling disturbance waves and their interactions with other flow fields, the four-field framework offers state-of-the-art capabilities for simulating complex annular flow phenomena, including intermittent film dryout.

An overview of the four-field model implemented in OpenSTREAM is provided below. A more detailed derivation and theoretical background can be found in Le Corre [2022a] and Le Corre *et al.* [2025a].

Governing equations

The conservation equations are formulated at the wall level, indexed by n , to support multi-wall geometries with different heating rates. These equations are solved starting from the onset of annular two-phase flow. Upstream of this onset, the solution from the mixture solver is used to initialize the flow fields.

1. Mass conservation

$$\text{Base Film: } \frac{\partial}{\partial t} \left(\frac{W_b^n}{u_b^n} \right) + \frac{\partial W_b^n}{\partial z} = \Pi_{wall}^n (D_b - \Gamma_{wb,b}^n + \Psi_w^n - \Psi_b^n)$$

$$\text{Disturbance Waves: } \frac{\partial}{\partial t} \left(\frac{W_w^n}{u_w^n} \right) + \frac{\partial W_w^n}{\partial z} = \Pi_{wall}^n (D_w - E^n - \Gamma_{wb,w}^n - \Psi_w^n + \Psi_b^n)$$

where:

- W_b^n, W_w^n are the base film and wave mass flow rates for wall index n
- u_b^n, u_w^n are the base film and wave velocities for wall index n
- D_b, D_w are the drop deposition fluxes on the base film and wave fields
- $\Gamma_{wb,b}^n, \Gamma_{wb,w}^n$ are the base film and wave wall boiling mass fluxes for wall index n

- Ψ_w^n, Ψ_b^n are exchange mass fluxes between film and wave fields for wall index n

2. Momentum conservation

Base Film: $\rho_{ls}\delta_b^n(\frac{\partial u_b^n}{\partial t} + u_b^n \frac{\partial u_b^n}{\partial z}) = D_b(u_d - u_b^n) + \Psi_w^n(u_w^n - u_b^n) - \delta_b^n(\frac{\partial p}{\partial z} + g\rho_{ls}) + \beta_b^n\tau_{v,b}^n + (1 - \beta_b^n)\tau_{w,b}^n - \tau_{wall,b}^n$

Disturbance Waves: $\rho_{ls}\delta_w^n(\frac{\partial u_w^n}{\partial t} + u_w^n \frac{\partial u_w^n}{\partial z}) = D_w(u_d - u_w^n) + \Psi_b^n(u_b^n - u_w^n) - \delta_w^n(\frac{\partial p}{\partial z} + g\rho_{ls}) + (1 - \beta_b^n)(\tau_{v,w}^n - \tau_{w,b}^n)$

where

- ρ_{ls} is the saturated liquid density
- δ_b^n, δ_w^n are the base film and wave (equivalent) thicknesses for wall index n
- u_d is the drop velocity
- β_b^n is the base film interfacial fraction
- $\tau_{v,b}^n, \tau_{v,w}^n$ are the vapor/base film and vapor/wave interfacial shear stresses for wall index n
- $\tau_{w,b}^n$ is the base film/wave interfacial shear stress
- $\tau_{wall,b}^n$ is the wall shear stress on the base film for wall index n

3. Energy conservation

Base Film under thermal equilibrium: $\Gamma_{wb,b}^n = \frac{q''_{wall,b}{}^n}{h_{vs} - h_{ls}}$

Disturbance Waves under thermal equilibrium: $\Gamma_{wb,w}^n = \frac{q''_{wall,w}{}^n}{h_{vs} - h_{ls}}$

where:

- h_{vs}, h_{ls} are the saturated vapor and liquid specific enthalpies
- $q''_{wall,b}{}^n, q''_{wall,w}{}^n$ are the wall heat flux to the base film and wave fields for wall index n

4. Wave number density transport

$\frac{\partial N_w^n}{\partial t} + \frac{\partial}{\partial z}(u_w^n N_w^n) = \frac{N_w^{eq,n} - N_w^n}{t_{Relax}^n}$

where:

- N_w^n is the wave number density (spatial density of waves)
- $N_w^{eq,n}$ is the equilibrium wave number density
- t_{Relax}^n is the relaxation time associated with wave formation, merging, and dissipation.

Closure models

To complete the conservation equations, several closure models are required:

- Same closure models as for the *three-field model*
- Base film/wave mass flow rate split at onset of annular two-phase flow
- Drop deposition split between base film and wave fields
- Exchange mass fluxes between film and waves: Ψ_w^n, Ψ_b^n
- Wall boiling split between base film and wave fields
- Base film interfacial fraction: β_b^n
- Vapor/base film and vapor/wave interfacial shear stresses: $\tau_{v,b}^n, \tau_{v,w}^n$
- Base film/wave interfacial shear stress: $\tau_{w,b}^n$
- Wall shear stress on the base film: $\tau_{wall,b}^n$

- Equilibrium wave number density: $N_w^{eq,n}$
- Relaxation time associated with wave formation, merging, and dissipation: t_w^{Relax}

The selected closure models are defined in the OpenSTREAM model file, chosen from the available options listed in [InputEnums](#). If not explicitly specified by the user, default models are applied as defined in [Inputs.Model](#). All closure models are implemented in [Solvers.FourField.Wave](#) and [Solvers.FourField.Base](#), which the users can modify to suit specific simulation needs.

In addition, the thermodynamic properties for each phase are computed using [CoolProp](#), an open-source thermophysical property library that provides accurate equations of state and transport properties for a wide range of fluids.

Features and assumptions

- Supports both steady-state and transient simulations in straight channels
- Supports uniform and non-uniform wall heat flux distribution
- Models intermittent wave transport and non-equilibrium wave dynamics
- Includes wave-film exchange and wave number density evolution
- Assumes thermal equilibrium (no temperature difference between phases)
- Supports multi-wall geometries (e.g., annuli)
- Neglects minor contributions such as frictional heating and temporal pressure gradient contributions

Role in OpenSTREAM

The four-field model provides state-of-the-art simulation capabilities for annular two-phase flow, including in developing flow regions. It captures wave dynamics and their impact on mass and momentum transfer.

Implementation notes

Package

- [Solvers](#)

Module

- [Solvers.FourField](#)

Four-field solver class

- [Solvers.FourField.FourFieldSolver](#)

Field classes

- [Solvers.FourField.Wave](#)
- [Solvers.FourField.Base](#)
- [Solvers.FourField.Drop](#)

Key solver methods

- [Solvers.FourField.FourFieldSolver.solve\(\)](#)

Key field properties:

- [Solvers.FourField.Wave.W](#)
- [Solvers.FourField.Wave.U](#)

- *Solvers.FourField.Wave.FREQUENCY*
- *Solvers.FourField.Base.W*
- *Solvers.FourField.Base.U*
- *Solvers.ThreeField.Drop.W*, inherited by *Solvers.FourField.Drop*
- *Solvers.ThreeField.Drop.U*, inherited by *Solvers.FourField.Drop*

↗

3.2 Glossary

This glossary defines terminology used throughout the **OpenSTREAM** documentation. The intent is to make the documentation more consistent and to clarify the distinction between closely related concepts used in one-dimensional, multi-field, two-phase flow modeling.

Annular flow

A two-phase flow regime in which a continuous vapor core is surrounded by a liquid film along the wall. Liquid droplets may also be entrained in the vapor core. In OpenSTREAM, annular flow models may represent vapor, droplets, liquid film, and, in advanced formulations, disturbance waves as separate fields.

Boundary condition

A prescribed condition imposed at the boundary of the computational domain. Typical boundary conditions specify inlet flow properties (mass flow, enthalpy), outlet pressure, and wall heat flux.

Closure model

A model used to close the system of conservation equations by providing relationships for physical processes that are not resolved directly. Examples include wall heat transfer, interfacial drag, interfacial heat transfer, phase change, entrainment, deposition, and wall friction.

Conservation equation

A governing equation expressing the conservation of a physical quantity, such as mass, momentum, or energy. In OpenSTREAM, conservation equations are formulated for the mixture, phase, or field variables depending on the selected solver.

Deposition

The transfer of liquid droplets from the vapor core back to the wall film. Deposition is typically represented through a closure model in annular flow formulations.

Disturbance wave

A coherent liquid structure traveling along the liquid film in annular flow. Disturbance waves can influence wall wetting, film mass transport, entrainment, deposition, and interfacial momentum transfer.

Droplet field

A dispersed liquid field representing entrained droplets transported within the vapor core. In annular flow modeling, the droplet field is distinct from the wall liquid film field.

Entrainment

The transfer of liquid from the wall film into the vapor core as droplets. Entrainment is typically represented through a closure model in annular flow formulations.

Field

A computationally resolved constituent with its own set of transported variables or conservation equations. A field is not always identical to a thermodynamic phase. For example, liquid droplets and a wall liquid film are both liquid phase constituents, but they may be represented as separate fields.

Film dryout

The disappearance of the continuous wall liquid film. In annular flow modeling, film dryout marks the loss of wall wetting by the liquid film and is often associated with a change in heat transfer behavior.

Four-field model

A multi-field model for annular two-phase flow in which vapor, droplets, liquid film, and disturbance waves are represented explicitly.

Governing equation

A mathematical equation describing the evolution of a conserved quantity or transported variable. Governing equations in OpenSTREAM are based on one-dimensional conservation laws with source terms.

Hydrodynamic equilibrium

A modeling assumption in which the phases or fields share the same velocity, or in which relative motion is neglected. When hydrodynamic non-equilibrium is considered, different phases or fields may have different velocities.

Hydrodynamic non-equilibrium

A condition in which two or more phases or fields have different velocities or momentum balances. Two-fluid and multi-field formulations commonly account for hydrodynamic non-equilibrium.

Interfacial area

The area of contact between phases or fields per unit volume. Interfacial area appears in closure models for interfacial transfer of mass, momentum, and energy.

Interfacial transfer

Exchange of mass, momentum, or energy between phases or fields. Examples include evaporation, condensation, interfacial drag, and interfacial heat transfer.

Liquid film

A continuous liquid layer flowing along the wall in annular flow. The liquid film is treated separately from entrained liquid droplets in multi-field annular flow models.

Mixture model

A reduced two-phase flow model in which the phases are represented through mixture quantities. Depending on the implementation, the mixture model may include additional relations to account for thermal non-equilibrium.

Multi-field model

A model in which different constituents of the flow are represented as separate computational fields. Multi-field models allow different phases or flow structures, such as vapor, droplets, liquid film, and disturbance waves, to be described with different transported variables.

Multi-wall channel

A channel representation that may include more than one heated or unheated wall surface. This allows wall-specific thermal or hydraulic conditions to be represented in the model formulation.

Non-equilibrium

A condition in which one or more variables are not shared among phases or fields. Non-equilibrium may be thermal, hydrodynamic, mechanical, or chemical depending on the variables considered.

One-dimensional model

A model in which flow variables vary primarily along one spatial coordinate, usually the axial direction. Cross-sectional effects are represented through averaged quantities and closure models rather than resolved explicitly.

Phase

A physically distinct thermodynamic state of matter, such as liquid or vapor. A phase may be represented by one or more fields in a multi-field model.

Phase change

Conversion of mass from one phase to another, such as evaporation or condensation. In two-phase thermal-hydraulic models, phase change is usually represented as an interfacial or wall source term.

Relaxation model

A model that drives a variable toward an equilibrium or target state over a characteristic relaxation time. Relaxation models are often used to represent unresolved non-equilibrium processes in a numerically robust way.

Relaxation time

A characteristic timescale over which a variable approaches an equilibrium or target value in a relaxation model. Smaller relaxation times correspond to faster return toward the target state.

Solver

A computational implementation of a mathematical model and numerical method used to solve a specific set of governing equations. In OpenSTREAM, different solvers correspond to different levels of physical detail and modeling assumptions.

Source term

A term in a conservation or transport equation representing production, destruction, or exchange of a quantity. Source terms may arise from wall interaction, interfacial transfer, external forces, or phase change.

Straight channel

A flow path without geometric bends or complex three-dimensional features. In one-dimensional modeling, a straight channel is represented along a single axial coordinate.

Thermal equilibrium

A modeling assumption in which phases or fields share the same temperature, or in which temperature differences are neglected.

Thermal non-equilibrium

A condition in which two or more phases or fields may have different temperatures or energy balances. Thermal non-equilibrium is important when heat transfer and phase change cannot be represented accurately by a single shared temperature.

Three-field model

A multi-field model for annular two-phase flow in which vapor, droplets, and liquid film are represented as separate fields.

Two-fluid model

A model in which the liquid and vapor phases are described by separate conservation equations. Two-fluid models can represent hydrodynamic and thermal non-equilibrium between phases.

Two-phase flow

Flow involving two thermodynamic phases, commonly liquid and vapor in the context of thermal-hydraulic applications.

Wall boiling

Boiling that occurs at a heated wall. Wall boiling models typically include mechanisms such as nucleation, evaporation, convection, quenching, and associated wall heat partitioning, depending on the level of model detail.

Wall friction

Momentum loss associated with shear stress at the wall. Wall friction is usually represented by a closure model or friction factor correlation.

Wall heat flux

The heat flux imposed at, or transferred through, a wall boundary. In heated channel simulations, wall heat flux is a key input or result depending on the problem formulation.

3.2.1 Recommended terminology

For consistency across the documentation, the following preferred terms are recommended:

Preferred term	Use when referring to	Avoid or use only with care
OpenSTREAM	The project, code, and documentation set	OpenStream, openstream
two-phase flow	Liquid-vapor flow modeled by OpenSTREAM	multiphase flow, unless more than two phases are intended
field	Computationally resolved constituent	phase, when the distinction is important
phase	Thermodynamic state, such as liquid or vapor	field, when referring to physical state
closure model	Implemented model used to close governing equations	closure law, correlation, constitutive law, unless a specific distinction is intended
thermal non-equilibrium	Different temperatures or energy balances among phases or fields	thermal disequilibrium, nonequilibrium thermal effects
hydrodynamic non-equilibrium	Different velocities or momentum balances among phases or fields	velocity slip, unless specifically referring to relative velocity
source term	Additive term in a conservation or transport equation	forcing term, production term, unless more specific
straight channel	One-dimensional channel without bends or complex 3D geometry	straight geometry, if channel-specific wording is clearer

3.2.2 Style notes

- Use `OpenSTREAM` consistently as the project name.
- Use `two-phase flow` when referring specifically to liquid-vapor applications.
- Use `multi-field` with a hyphen when it modifies a noun, for example `multi-field model`.
- Use `one-dimensional` with a hyphen when it modifies a noun, for example `one-dimensional solver`.
- Use `thermal non-equilibrium` and `hydrodynamic non-equilibrium` consistently.
- Prefer `closure model` for implemented models and reserve `constitutive relation` for broader physical relationships.
- Distinguish carefully between `phase` and `field` in theory pages.

3.3 Notation

This page summarizes the notation conventions used in the **OpenSTREAM** documentation. The theory pages, solver descriptions, closure model documentation, and implementation notes are kept consistent across the mixture, two-fluid, three-field, and four-field formulations.

3.3.1 General conventions

OpenSTREAM uses one-dimensional, cross-section averaged conservation equations. Unless otherwise stated, the independent variables are time and the axial coordinate:

$$t \quad \text{and} \quad z$$

where t is time and z is the axial coordinate along the flow channel.

The following conventions are used throughout the documentation:

- z for the axial coordinate.

- t for time.
- k as a generic field or phase index when no specific constituent is intended.
- Subscripts identify phases, fields, interfaces, and walls.
- Superscripts identify time levels, iteration counters, or model-specific states when needed.
- All symbols are defined immediately after the equation in which they first appear.

3.3.2 Primary variables

Symbol	Suggested name	Meaning
t	time	Time coordinate.
z	axial coordinate	One-dimensional spatial coordinate along the flow path.
A	flow area	Cross-sectional flow area.
P_w	wetted perimeter	Wall perimeter in contact with the fluid or liquid film.
D_h	hydraulic diameter	Hydraulic diameter of the channel.
α_k	volume fraction	Volume fraction of field or phase k .
ρ_k	density	Density of field or phase k .
u_k	velocity	Axial velocity of field or phase k .
p	pressure	Pressure (shared among fields).
T_k	temperature	Temperature of field or phase k .
h_k	specific enthalpy	Specific enthalpy of field or phase k .
e_k	specific internal energy	Specific internal energy of field or phase k .
E_k	total specific energy	Total specific energy of field or phase k , if used.
x	vapor quality	Vapor mass fraction in a two-phase mixture.
q_w''	wall heat flux	Heat flux imposed at, or transferred through, the wall.
g	gravitational acceleration	Gravitational acceleration.

3.3.3 Phase and field subscripts

The notation distinguishes between thermodynamic phases and computational fields. A phase is a physical state of matter, such as liquid or vapor. A field is a computationally resolved constituent with its own transported variables.

Subscript	Meaning	Notes
l	liquid phase	Used for liquid phase.
g	vapor phase	Used consistently for vapor or gas.
f	liquid film	Used for the wall liquid-film field in annular flow models.
d	droplets	Used for entrained liquid droplets in the vapor core.
$wall$	wall	Used for wall quantities, such as T_{wall} or q_{wall}'' .
i	interface	Used for interfacial quantities, such as a_i or htc_i .
w	disturbance wave	Use for disturbance wave quantities when a compact subscript is needed.

3.3.4 Source-term notation

Source terms make it clear which conserved quantity is affected and which interaction is represented.

Symbol	Name	Meaning
Γ_k	mass source	Net mass source to field or phase k .
M_k	momentum source	Net momentum source to field or phase k .
Q_k	energy source	Net energy source to field or phase k .
$\Gamma_{a \rightarrow b}$	mass transfer rate	Mass transferred from field a to field b .
$M_{a \rightarrow b}$	momentum transfer	Momentum transferred from field a to field b .
$Q_{a \rightarrow b}$	heat or energy transfer	Energy transferred from field a to field b .

3.3.5 Closure model notation

Closure models use notation that clearly identifies the modeled process. Examples include:

Symbol	Suggested meaning	Typical use
C_D	drag coefficient	Interfacial or droplet drag model.
f_{wall}	wall friction factor	Wall momentum loss.
htc_{wall}	Wall heat-transfer coefficient	Wall heat transfer.
htc_i	Interfacial heat-transfer coefficient	Interfacial heat transfer.
a_i	interfacial area concentration	Interfacial transfer models.
τ_{wall}	wall shear stress	Wall friction and pressure drop.
τ_i	interfacial shear stress	Momentum exchange between fields.
t_{relax}	relaxation time	Relaxation models.

3.3.6 Units

SI units are used consistently unless a specific model, correlation, or input file requires otherwise.

Quantity	Unit	Notes
Length	m	Use for z , hydraulic diameter, and channel dimensions.
Time	s	Use for t , time steps, and relaxation times.
Pressure	Pa	Use for absolute pressure unless otherwise stated.
Temperature	K	Use kelvin in equations. Celsius may be used in user-facing examples if clearly stated.
Density	kg/m ³	Use for phase, field, and mixture density.
Velocity	m/s	Use for axial velocities.
Mass flow rate	kg/s	Use for total or field-specific mass flow rates.
Mass flux	kg/m ² /s	Use for area-normalized mass flow rate.
Heat flux	W/m ²	Use for wall heat flux.
Specific enthalpy	J/kg	Use for phase or field enthalpy.

3.4 Global reference list

This page gathers all the sources cited throughout the **OpenSTREAM** documentation in one convenient place. Whether you're tracing a model back to its theoretical roots or looking for further reading, this is your go-to resource for exploring the academic and technical foundations behind the code.

3.5 Testing and verification

OpenSTREAM includes an automated testing framework to support code verification, numerical consistency, and software environment compatibility.

The test suite is intended to:

- Detect regressions when modifying existing functionality.
- Verify representative solver workflows against reference solutions.
- Confirm that required external dependencies are available and functional.
- Support reproducible development.

The current test suite focuses primarily on core infrastructure and a subset of solver capabilities. Additional tests will be added over time to improve coverage of numerical models, closure laws, input processing, and post-processing functionality.

Note

OpenSTREAM's automated test suite is under active development.

The current tests verify selected functionality, solver workflows, numerical results, and software dependencies, but do not provide comprehensive coverage of the complete OpenSTREAM code base or every implemented physical model and numerical option.

Passing all tests indicates that the covered functionality behaves as expected for the tested cases. It should not be interpreted as complete verification or validation of OpenSTREAM.

3.5.1 Test organization

The OpenSTREAM tests are organized into three categories:

Folder	Purpose
tests/ environment	Verification of the MATLAB-Python environment and required external packages, including basic CoolProp functionality.
tests/unit	Focused tests of individual OpenSTREAM classes, methods, and utilities.
tests/ integration	End-to-end solver workflows, physical consistency checks, numerical regression comparisons, and solver persistence tests.

Unit tests should execute quickly and focus on individual components. Integration tests construct and solve representative OpenSTREAM cases and compare selected calculated quantities with approved reference solutions.

3.5.2 Running all tests

The recommended way to execute the complete test suite is to run `runOpenSTREAMTests.m` from the `tests` folder:

```
runOpenSTREAMTests
```

The test runner:

- Recursively discovers test classes in the `tests` folder and its subfolders.
- Executes the environment, unit, and integration tests, including the numerical regression comparisons.
- Displays detailed progress and failure diagnostics.

- Displays a summary table containing the test class, test method, and result.
- Reports the total execution time of the complete test suite.
- Raises an error if any test fails or does not complete.

After execution, the MATLAB test results and generated summary table are available in the workspace as `testResults` and `testSummary`.

A successful run finishes without errors and reports all tests as passed.

3.5.3 Running individual test categories

Individual test categories can also be run separately from the `tests` folder.

Environment validation tests:

```
suite = matlab.unittest.TestSuite.fromFolder('environment');
runner = matlab.unittest.TestRunner.withTextOutput;
results = runner.run(suite);
```

Unit tests:

```
suite = matlab.unittest.TestSuite.fromFolder('unit');
runner = matlab.unittest.TestRunner.withTextOutput;
results = runner.run(suite);
```

Integration tests:

```
suite = matlab.unittest.TestSuite.fromFolder('integration');
runner = matlab.unittest.TestRunner.withTextOutput;
results = runner.run(suite);
```

The command `results = runner.run(suite)` returns an array of `matlab.unittest.TestResult` objects, with one element for each test method executed. Each element records the test name, pass, failure, and incomplete status, execution duration, and additional framework details.

The results can be inspected directly or converted to a table:

```
disp(results)
resultTable = table(results);
disp(resultTable)
```

Failed or incomplete tests can be identified using:

```
failed = results([results.Failed]);
incomplete = results([results.Incomplete]);
notPassed = results(~[results.Passed]);
```

3.5.4 Continuous integration

The complete available test suite is executed automatically by the MATLAB CI workflow when changes are pushed to the repository.

After pushing a change, review the MATLAB CI result in GitHub Actions. The workflow fails if any automated test fails or does not complete.

The documentation build and MATLAB CI are separate workflows. A successful documentation build or GitHub Pages deployment does not indicate that the MATLAB tests passed. The status of each workflow should therefore be reviewed independently.

When MATLAB CI fails:

1. Open the failed MATLAB CI workflow.
2. Identify the failing test class and test method in the diagnostics.
3. Reproduce the failure locally using `runOpenSTREAMTests`.
4. Determine whether the failure results from a software defect, an unintended numerical change, a changed environment, or an intentional modification that requires an approved update to the test or reference solution.
5. Run the complete test suite locally before pushing the correction.

3.5.5 Reference solutions

Integration tests compare selected generated results with approved reference solutions.

Reference data are stored under:

```
tests/integration/references/<session-directory>/
```

The reference solutions represent the expected solver behavior for defined test cases. They are version-controlled and form part of the approved regression baseline.

Reference solutions should only be updated when:

- A defect has been corrected.
- A model formulation has intentionally changed.
- Numerical behavior has been intentionally modified and reviewed.
- The approved test environment has changed in a way that justifiably changes the calculated reference results.

Reference solutions should **not** be updated merely to make a failing test pass. Before replacing a reference solution, the numerical differences and their cause should be understood and reviewed.

3.5.6 Adding new tests

When introducing new functionality:

1. Add unit tests whenever possible.
2. Add integration tests if the change affects solver results.
3. Verify that all existing tests continue to pass.
4. Update the documentation if user-visible behavior changes.

As a general rule:

- New testable classes and methods should have focused unit tests.
- New numerical and closure models should have unit tests where possible and integration tests demonstrating their solver-level behavior.
- New external dependencies should have environment tests.

3.5.7 Unit test guidelines

Unit tests should:

- Focus on a single class or function.
- Use small and deterministic inputs.
- Avoid unnecessary file I/O.
- Execute quickly.
- Be independent of one another.

A good unit test verifies one piece of behavior and has a clear failure mode.

3.5.8 Integration test guidelines

Integration tests should, where practical:

- Exercise representative solver workflows.
- Use representative and reproducible input files.
- Verify that solvers reach their expected states.
- Verify propagation of prescribed boundary conditions.
- Check relevant conservation or physical-consistency relationships.
- Compare important calculated quantities with approved reference values.
- Verify that saved solver objects can be reloaded consistently.
- Produce deterministic results suitable for automated execution.

The current solver integration tests verify selected behavior including:

- Final solver states.
- Mixture-solver inlet boundary conditions.
- The mixture steady-state energy balance.
- Transfer of the final pseudo-time initialization state to the first stored physical state.
- Field mass flow rate distributions.
- Field velocity distributions.
- Field enthalpy distributions.
- Solver save-and-load behavior.

Calculated field distributions are compared with the approved reference values using a strict relative tolerance. This avoids false failures caused by insignificant floating-point differences while retaining sensitivity to numerical regressions. Reference values equal to zero remain subject to exact comparison because no absolute tolerance is currently applied.

Future integration tests may additionally verify pressures, void fractions, film thicknesses, exchange terms, transition locations, and other solver-specific quantities.

Not all quantities, solver options, model combinations, physical regimes, error conditions, or post-processing capabilities are currently tested.

3.5.9 Environment validation

Environment tests verify external software requirements.

The current environment tests verify:

- Python availability through MATLAB.
- Python-version accessibility.
- Installation and import of required Python packages.
- Basic CoolProp functionality.

Environment tests help identify installation problems before running larger simulations.

3.5.10 Generated outputs

Some tests generate temporary files, session directories, logs, and saved solver objects.

Generated outputs are stored in category-specific outputs directories, including:

```
tests/unit/outputs/  
tests/integration/outputs/
```

Unit-test artifacts are removed automatically during test teardown. Integration-test outputs are retained after execution to support inspection and troubleshooting, and are recreated or replaced by a subsequent test run.

Generated outputs should not normally be committed to the repository. Repository `.gitignore` rules exclude directories named `outputs` from version control.

Approved reference solutions are stored separately under `tests/integration/references` and remain version-controlled as part of the regression baseline.

3.5.11 Troubleshooting

A test failure does not necessarily indicate a software defect.

When a test fails:

1. Read the diagnostic message carefully.
2. Identify the failing test class and method.
3. Determine whether the failure is caused by:
 - A coding error.
 - An unintended numerical change.
 - An intentional model or numerical change.
 - An outdated or inappropriate reference solution.
 - A software-environment or dependency problem.
 - A platform-dependent floating-point difference.
4. Re-run the failing test class or test category locally.
5. Inspect generated outputs and compare the actual and expected values.
6. Verify the expected behavior manually when needed.
7. Update reference solutions only when the change is intentional, understood, justified, and reviewed.
8. Run the complete test suite before pushing the correction.

3.5.12 Contributor checklist

Before submitting a pull request:

- The complete local test suite passes.
- New or modified functionality includes appropriate tests.
- Numerical differences from approved references have been reviewed.
- Reference solutions have been updated only when justified.
- Documentation has been updated when user-visible behavior changes.
- Generated outputs are not committed.
- The MATLAB CI workflow passes after the changes are pushed.

3.6 Tutorials

Ready to get hands-on with **OpenSTREAM**? These tutorials introduce the principal solver frameworks, input formats, numerical workflows, and post-processing capabilities of OpenSTREAM, progressing from a basic example to more advanced simulation and analysis tasks.

Each tutorial is provided as a MATLAB Live Script in the OpenSTREAM `tutorials` folder. The Live Scripts can be executed, modified, and extended directly in MATLAB. Exported HTML versions are also available through the links below.

Note

The tutorials are intended to be executed interactively as MATLAB Live Scripts. Some tutorials include animations and interactive figures that are displayed in separate pop-up windows when run from the MATLAB Live Editor. These interactive features and their controls are generally not available in the exported HTML versions.

3.6.1 1. Quick Start

[Open Tutorial 1: Quick Start](#)

Run a provided OpenSTREAM case using the mixture, two-fluid, three-field, and four-field solver frameworks. This tutorial introduces input loading, solver construction, basic axial plots, inspection of selected results, and saving solver objects. It is the recommended starting point for new users.

3.6.2 2. Mixture Solver Capabilities

[Open Tutorial 2: Mixture Solver Capabilities](#)

Explore homogeneous and nonhomogeneous mixture formulations, including fixed-slip and drift-flux models. The tutorial also introduces empirical and relaxation-based thermal-nonequilibrium models, pressure-drop components, wall-friction and local-loss options, and a physical transient calculation.

3.6.3 3. Two-Fluid Solver Capabilities

[Open Tutorial 3: Two-Fluid Solver Capabilities](#)

Construct and run two-fluid calculations with separate liquid and vapor fields. Examine phase mass flow rates, velocities, enthalpies, temperatures, hydrodynamic and thermal nonequilibrium, interfacial area, flow-regime indicators, phase exchange terms, pseudo-time convergence, and physical transient behavior.

3.6.4 4. Three-Field Solver Capabilities

Open Tutorial 4: Three-Field Solver Capabilities

Model annular flow using separate liquid-film and entrained-droplet fields together with the mixture-model vapor solution. Compare onset-of-annular-flow initialization methods, momentum formulations, entrainment and deposition closures, film thickness, mass and momentum exchange terms, pseudo-time convergence, and transient behavior.

3.6.5 5. Four-Field Solver Capabilities

Open Tutorial 5: Four-Field Solver Capabilities

Extend the three-field formulation by separating the liquid film into base-film and disturbance-wave fields. Examine film decomposition, wave geometry and frequency, field fractions, onset-of-annular-flow splitting, equilibrium and relaxation models, momentum formulations, field exchange terms, pseudo-time convergence, and a physical transient calculation.

3.6.6 6. Input Files

Open Tutorial 6: Input Files

Learn how OpenSTREAM cases are assembled from physical-model, numerical- option, geometry, and boundary-condition inputs. This tutorial demonstrates input-set selection, inspection of imported objects, loading equivalent model definitions from INP and JSON files, and conversion of native input files to JSON.

3.6.7 7. Numerical Convergence and Solver Diagnostics

Open Tutorial 7: Numerical Convergence and Solver Diagnostics

Distinguish local pointwise convergence, pseudo-time convergence, and physical-time advancement. Inspect solver states and initialization histories, examine the influence of relaxation factors and the pseudo-time step, troubleshoot convergence difficulties, and assess whether a converged solution is numerically consistent.

3.6.8 8. Mesh and Time-Step Sensitivity

Open Tutorial 8: Mesh and Time-Step Sensitivity

Evaluate the sensitivity of mixture-solver results to spatial and temporal discretization. Compare steady-state solutions calculated with different axial meshes and transient responses calculated with different physical time steps. The tutorial also discusses local resolution, peak values, transition timing, computational effort, and the distinction between a sensitivity assessment and a systematic convergence study.

3.6.9 9. Advanced Visualization and Post-Processing

Open Tutorial 9: Advanced Visualization and Post-Processing

Apply the principal OpenSTREAM visualization and post-processing capabilities to a transient solution. Generate animated axial distributions, time histories at selected elevations, and time-elevation surfaces. Extract calculated quantities, assemble MATLAB tables, export selected results to CSV, and save and reload solver objects for additional analysis.

Warning

The physical models, closure relations, numerical settings, and solver options selected throughout these tutorials are intended for educational and demonstration purposes.

Their use in a tutorial does not imply that they are appropriate, recommended, calibrated, verified, or validated for a particular application. Before applying OpenSTREAM to a technical study, assess the assumptions, applicability range, calibration basis, numerical sensitivity, and validation evidence associated with every selected model.

APPLICATIONS

4.1 Contributing to the database

Contributions to OpenSTREAM-database are welcome. Contributions may include new publicly available datasets, corrections to existing dataset implementations, additional application cases, improved plotting and comparison methods, uncertainty information, validation metrics, documentation, and project workflows.

OpenSTREAM-database complements the core OpenSTREAM repository. Contributions related to solver frameworks, physical models, closure models, numerical methods, input handling, or other core functionality should normally be made in the OpenSTREAM repository.

Contributions related to experimental datasets and their application, assessment, validation, or publication companion workflows belong in OpenSTREAM-database.

4.1.1 Types of contributions

Contributions may include:

- New publicly available experimental datasets.
- Additional cases from datasets already implemented.
- Corrections to source-data transcription, digitization, conversion, or metadata.
- Improved dataset documentation and bibliographic information.
- Additional OpenSTREAM solver or model configurations.
- New calculated-versus-measured comparisons.
- Experimental uncertainty information.
- Validation metrics and statistical analyses.
- Improved visualization and post-processing methods.
- MATLAB Live Script application and publication companion projects.
- Corrections to existing application workflows.
- Tests and verification capabilities.
- Development of an automated testing and regression framework.

4.1.2 Dataset organization

Dataset implementations should follow the established MATLAB package organization:

```
+DatasetName/  
├── +src/  
│   └── DatasetName.xml  
├── @DatasetName/  
│   ├── DatasetName.m  
│   └── plotResults.m  
└── README.md
```

Additional dataset-specific methods may be included when required. For example:

```
+DatasetName/  
├── +src/  
│   └── DatasetName.xml  
├── @DatasetName/  
│   ├── DatasetName.m  
│   ├── plotResults.m  
│   └── plotTrends.m  
└── README.md
```

The package components have the following roles:

+src

Contains the authoritative XML or JSON source-data representation. All stored numerical values must use SI units, with absolute temperatures expressed in kelvin.

@DatasetName

Contains the dataset-specific class and separately implemented methods.

DatasetName.m

Defines the dataset-specific class, which inherits from the generic `Dataset` class.

plotResults.m

Provides dataset-specific plotting and calculated-versus-measured comparison functionality.

README.md

Documents the original experimental source, implemented data, units, uncertainty, assumptions, processing, usage, verification, attribution, and limitations.

A dataset package should normally maintain only one authoritative machine-readable source file. If XML and JSON representations are both retained, the dataset README must identify the authoritative representation, explain how the alternative representation is generated, and describe how consistency is verified.

The dataset-specific class should inherit from the generic `Dataset` class and use the shared dataset interface where practical.

4.1.3 Dataset template

OpenSTREAM-database includes a `+Template` package that provides the starting structure for a new dataset implementation.

The template contains:

- A `+src` folder containing guidance for adding the authoritative XML or JSON source-data file.
- An `@Template` class folder.

- A template `Template.m` class that inherits from the generic `Dataset` class.
- A template `plotResults.m` method.
- A template `README.md` containing the expected dataset documentation structure.

The template follows this organization:

```
+Template/
├── +src/
│   └── README.md
├── @Template/
│   ├── Template.m
│   └── plotResults.m
└── README.md
```

To begin implementing a new dataset:

1. Copy the complete `+Template` package.
2. Select a valid dataset name following the naming convention described below.
3. Rename the package folder, class folder, and MATLAB class file consistently.
4. Update the class definition, `addPath` method, comments, and documentation.
5. Add the authoritative XML or JSON source-data file under `+src`.
6. Add all fields required by each supported experimental case.
7. Add and document any required dataset-specific fields.
8. Complete the dataset-specific properties and methods.
9. Complete the dataset `README` using the template headings.
10. Implement dataset-specific result plotting where applicable.
11. Verify source-data loading, input generation, case execution, and post-processing.

The template supplies the expected structure and documentation placeholders. Contributors remain responsible for completing and verifying all dataset-specific source data, metadata, methods, assumptions, unit conversions, uncertainties, plotting functionality, and application workflows.

4.1.4 Naming convention

Dataset packages follow a strict naming convention. The package folder, class folder, class file, class definition, dataset name, and source-data filename must use consistent names.

For a dataset named `AuthorYear`, the expected structure is:

```
+AuthorYear/
├── +src/
│   └── AuthorYear.xml
├── @AuthorYear/
│   ├── AuthorYear.m
│   └── plotResults.m
└── README.md
```

The following names must agree:

Element	Required form
MATLAB package folder	<code>+AuthorYear</code>
MATLAB class folder	<code>@AuthorYear</code>
MATLAB class file	<code>AuthorYear.m</code>
MATLAB class definition	<code>classdef AuthorYear < Dataset</code>
Dataset name assigned in <code>addPath</code>	<code>data.name = 'AuthorYear'</code>
Source-data filename	<code>AuthorYear.xml</code>
Source-data location	<code>+AuthorYear/+src/AuthorYear.xml</code>

The leading `+` and `@` characters are part of MATLAB package and class-folder syntax:

- `+AuthorYear` defines the MATLAB package.
- `@AuthorYear` defines the class folder within that package.
- `AuthorYear.m` contains the class definition.
- The class is referenced using the package-qualified name `AuthorYear.AuthorYear`.

For publication-based datasets, the package name should generally combine the surname of the first author and the four-digit publication year, following the convention used by existing dataset packages.

If this combination is already used by another package, add a concise qualifier that unambiguously identifies the publication. Prefer the second author's surname or a short title keyword, for example `AuthorCoauthor2020` or `Author2020CHF`.

The suffixes `a`, `b`, and so forth may be used when closely related publications are conventionally cited as `Author2020a`, `Author2020b`, and so forth. The assignment of these suffixes must be documented and remain stable.

The capitalization of author names and title keywords must be used consistently throughout the package name, class name, source filename, and documentation. For example, do not mix `Author2020`, `author2020`, and `AUTHOR2020`.

The dataset name must be a valid MATLAB identifier. Spaces, hyphens, punctuation, and other invalid identifier characters must not be used.

When copying the template, update every occurrence of `Template` in:

1. The package-folder name.
2. The class-folder name.
3. The class filename.
4. The `classdef` declaration.
5. `data.name` in `addPath`.
6. The source-data filename.
7. Code examples and headings in `README.md`.
8. Project scripts that construct or refer to the dataset class.

For example:

```
classdef AuthorYear < Dataset
    % AUTHORYEAR Dataset implementation for Author et al. (Year).

    methods

        function addPath(data)
```

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```

% ADDPATH Define the dataset name and source-data file.

data.name = 'AuthorYear';
data.path = ...
    data.getSourceFilePath('AuthorYear.xml');
end

end

end

```

Use `getSourceFilePath` rather than a hard-coded relative or absolute path. This allows the dataset package to locate its source file independently of the current MATLAB working directory.

Separately implemented methods in the `@AuthorYear` folder do not need redundant declarations in the main class file.

The resulting object is constructed using the package-qualified class name:

```
data = AuthorYear.AuthorYear(...);
```

A mismatch between package, class, or source-data names may prevent MATLAB from locating the class or may cause the dataset implementation to reference the wrong source-data file.

4.1.5 Source data

Dataset source files are stored under the package `+src` directory.

XML and JSON are supported source formats. A dataset package should normally maintain only one authoritative machine-readable source representation.

If XML and JSON representations are both retained, the dataset README must state:

- Which representation is authoritative.
- How the alternative representation is generated.
- Whether contributors may edit the alternative representation.
- How consistency between the representations is verified.

Equivalent XML and JSON files should not be maintained independently without a documented synchronization and verification procedure.

All numerical data stored in OpenSTREAM-database source files must use SI units. Absolute temperatures must be expressed in kelvin. This requirement applies even when the original experimental publication reports data using another system of units.

The source-data implementation should preserve the original experimental information as faithfully as possible. Any transcription, correction, character recognition, digitization, filtering, interpolation, reconstruction, unit conversion, or derivation must be documented.

Missing numerical information must be represented explicitly and must not be interpreted as zero unless zero is the reported experimental value. The missing-value convention must be documented in the dataset README and handled explicitly by the dataset-specific implementation.

The dataset object reads and interprets the corresponding XML or JSON source file during construction. Reading the source file is therefore part of dataset-object construction and is not a separate step in the application workflow.

4.1.6 Lightweight datasets

OpenSTREAM-database can also be used without an XML or JSON source-data file.

A case can be defined directly using a one-row MATLAB table and passed to a lightweight Dataset object:

```
data = Dataset( ...
    1, ...
    'isLightWeight',true, ...
    'lightWeightEntryData',entryData);
```

This workflow is suitable for:

- Publication demonstration cases.
- Concise examples and tutorials.
- Exploratory calculations.
- User-defined cases that do not require a maintained dataset package.
- Comparisons of several OpenSTREAM solver frameworks using a common case definition.

The table must contain the geometry, boundary-condition, wall-heating, and fluid information required by the selected OpenSTREAM workflow. All numerical values must follow the same SI-unit convention as file-based datasets.

A lightweight case should not replace a maintained dataset package when experimental provenance, measurement uncertainty, repeated case selection, or long-term reuse must be documented.

4.1.7 Dataset fields

Each experimental case in the XML or JSON source file must define the fields required by its intended OpenSTREAM workflow. Additional dataset-specific fields may be added when required to represent the available experimental information.

Mandatory fields

The following fields are normally required for a complete heated-channel case:

Field	Description	SI unit
Fluid	Working-fluid identifier used by OpenSTREAM.	Not applicable
Pressure	System pressure.	Pa
MassFlow	Inlet mass flow rate.	kg/s
InletEnthalpy	Inlet specific enthalpy.	J/kg
Length	Axial length of the modeled channel.	m
Area	Channel flow area.	m ²
Perimeter	Channel perimeter used by the OpenSTREAM geometry.	m
Power	Total applied power.	W
WallMesh	Axial wall-interval lengths associated with the wall-power distribution.	m
WallPower	Relative axial wall-power distribution.	Dimensionless

Field names are case-sensitive and must be written exactly as expected by the input-generation implementation.

The mandatory fields must satisfy the following requirements:

- Every supported case must define the fields required by its workflow.
- All numerical values must use SI units.

- Absolute temperatures must be expressed in kelvin.
- Fluid must use an identifier supported by the OpenSTREAM fluid-property interface.
- WallMesh and WallPower must have compatible lengths and ordering.
- The sum of WallMesh must be consistent with the modeled channel length.
- Geometry fields must represent the channel modeled by OpenSTREAM.
- Boundary-condition fields must represent the intended experimental case.
- Any conversion, reconstruction, or interpretation used to obtain a field must be documented in the dataset README.

Fields that are not required for a particular workflow may be omitted only when the generic input-generation implementation and dataset-specific methods support their absence.

Dataset-specific fields

Additional fields may be included when required to represent information specific to the dataset.

Dataset-specific fields may describe:

- Measured quantities.
- Instrument and measurement locations.
- Experimental uncertainties.
- Facility or test-section characteristics not represented by the mandatory geometry fields.
- Inlet, outlet, or local experimental conditions.
- Flow-regime or transition information.
- Dataset-specific case classifications.
- Information required for calculated-versus-measured comparisons.
- Quantities derived during source-data preparation or post-processing.

Each dataset-specific field must be documented in the dataset README. The documentation should state:

- The exact field name.
- The physical meaning of the field.
- Whether the field is required or optional.
- Confirmation that all numerical data use SI units.
- The unit used for the implemented quantity whenever clarification is useful.
- The unit reported in the original experimental source, when different.
- The source table, figure, page, appendix, or data file.
- Any unit conversion, transcription, character recognition, digitization, interpolation, derivation, or other processing applied.
- The reported measurement uncertainty, where available.
- The behavior when the field is absent or unavailable.
- Whether the quantity is measured, prescribed, transcribed, digitized, reconstructed, derived, or calculated.

A general SI-unit statement is sufficient where the representation is unambiguous. Units must be stated explicitly where clarification is needed. Absolute temperatures must be identified as being expressed in kelvin.

A field name must not imply an incorrect physical unit. For example, a field named `InletSubcooling` may represent an enthalpy difference in J/kg rather than a temperature difference. Such conventions must be stated explicitly.

Dataset-specific fields must not replace, rename, or change the meaning of the mandatory fields.

When the same physical quantity is represented by several dataset implementations, a common field name and representation should be used where practical.

Source-file requirements

The XML or JSON source representation should use a consistent structure across all implemented cases.

Before completing a dataset implementation, confirm that:

- Every supported case contains the fields required by its workflow.
- Mandatory and dataset-specific field names are used consistently.
- Numerical fields contain valid SI values.
- Absolute temperatures are expressed in kelvin.
- Related array dimensions and ordering are mutually consistent.
- `WallMesh` and `WallPower` describe compatible axial distributions.
- Case identifiers are unique.
- Missing experimental information is represented explicitly.
- Missing numerical information is not represented as zero unless zero is the reported value.
- Undocumented replacement values are not introduced.
- Derived, reconstructed, or calculated fields are identified and documented.
- Unsupported or incomplete cases are identified clearly.

The dataset class should validate the fields required by its implementation and report missing, inconsistent, or unsupported data with a clear diagnostic.

4.1.8 Data provenance

Every dataset contribution must identify the authoritative experimental source.

The dataset README should include:

- The complete bibliographic reference.
- A link to the publication or public data source, when available.
- A description of the experimental facility.
- The relevant test section and geometry.
- The measured and prescribed quantities.
- The range of experimental conditions.
- Confirmation that all stored numerical data use SI units, with absolute temperatures expressed in kelvin.
- The reported measurement uncertainties.
- Known limitations or qualifications of the data.

- The relationship between the original data and the implemented source-data file.
- Any licensing, redistribution, attribution, or citation requirements.

The source-data preparation method must be stated explicitly. Applicable methods include:

- Manual transcription from publication tables.
- Character recognition from scanned reports.
- Figure digitization.
- Import from publicly available machine-readable data.
- Unit conversion.
- Interpolation or reconstruction.
- Derivation of additional quantities required for OpenSTREAM input generation or post-processing.

Where applicable, document the checks performed to improve accuracy and internal consistency, including:

- Geometry consistency.
- Mass-flow, mass-flux, and area consistency.
- Power reconstruction.
- Energy-balance checks.
- Unit-conversion checks.
- Array-length and ordering checks.
- Comparison of repeated or related cases.
- Review against the original tables or figures.

The original publication or public data source remains authoritative for the experimental configuration, measurements, uncertainties, and interpretation.

The OpenSTREAM-database implementation must not imply greater accuracy, completeness, or certainty than is supported by the published information.

4.1.9 Dataset class

The dataset-specific MATLAB class should:

- Inherit from the generic `Dataset` class.
- Read and interpret its source-data representation during object construction.
- Define its package name and source-data file through `addPath`.
- Use `getSourceFilePath` to locate the authoritative source file.
- Preserve the generic dataset interface where practical.
- Identify the available experimental cases and quantities.
- Generate valid OpenSTREAM inputs.
- Operate consistently using SI units.
- Handle missing experimental values explicitly.
- Avoid hard-coded local file-system paths.
- Avoid redundant declarations for methods already implemented separately in the `@DatasetName` folder.

- Use clear property and method names.
- Document assumptions and transformations.
- Validate inputs and report unsupported selections clearly.
- Support reproducible execution across the intended environments.

Functionality common to several datasets should be considered for implementation in the generic `Dataset` class rather than duplicated across dataset-specific classes.

4.1.10 OpenSTREAM dependency

OpenSTREAM is maintained as a separate repository and is not included as a Git submodule of OpenSTREAM-database.

Dataset implementations and application projects should rely on the required OpenSTREAM classes being available on the MATLAB path. They should not assume a fixed relative directory arrangement between the two repositories.

Installation and MATLAB path instructions are provided on the [OpenSTREAM-database](#) page.

4.1.11 Application projects

Application projects are stored under the repository `projects` folder. They may use a dataset package or a lightweight in-memory `Dataset` object.

A contributed project should provide a reproducible workflow for:

1. Constructing a dataset-specific object, which reads and interprets the corresponding XML or JSON source file, or constructing a lightweight `Dataset` object from an in-memory table.
2. Selecting one or more experimental or demonstration cases.
3. Selecting an OpenSTREAM solver and model configuration.
4. Generating the required OpenSTREAM input files.
5. Running the selected cases.
6. Reviewing solver convergence.
7. Extracting calculated and measured quantities.
8. Comparing the results.
9. Visualizing and interpreting the comparison.
10. Documenting assumptions, numerical settings, and limitations.

Projects should avoid hard-coded local paths and should state the expected working directory or MATLAB path configuration clearly.

MATLAB Live Scripts are recommended when explanatory text, executable code, figures, calculated results, convergence information, and interpretation form part of the documented application workflow.

Publication companion workflows

A publication companion Live Script should reproduce or illustrate selected calculations, figures, and results presented in the corresponding publication.

The companion workflow must not be described as a complete record of the research programme. Research activities generally include additional test cases, closure-model development, sensitivity studies, uncertainty analysis, numerical investigations, intermediate results, and unsuccessful model variants that are not included in the published figures.

The project documentation should identify:

- The corresponding publication.
- The figures or results reproduced.
- The experimental datasets used.
- Important nondefault physical and numerical settings.
- Calculations or comparisons from the publication that are not reproduced.
- The limitations of the companion workflow.

The MATLAB Live Script is the authoritative executable version. A completed execution may also be exported to HTML for viewing without MATLAB. Interactive Live Editor features may not be preserved in the HTML export.

4.1.12 Comparison and validation

Calculated-versus-measured comparisons should identify:

- The experimental quantity being compared.
- Whether the experimental value is directly measured or derived.
- The corresponding OpenSTREAM quantity.
- The location and time associated with the comparison.
- The SI unit used for the comparison.
- The selected solver and physical models.
- The numerical settings.
- The spatial and temporal discretization.
- The experimental uncertainty, where available.
- The comparison metric, where applicable.
- Any filtering, averaging, smoothing, interpolation, or alignment procedure.

A favorable comparison for one case, quantity, dataset, or model selection must not be described as general validation outside the investigated conditions.

Contributors should distinguish between:

- Verification that an implementation behaves as intended.
- Validation against experimental observations.
- Numerical sensitivity to mesh, time step, convergence settings, and solver options.
- Sensitivity to closure-model and parameter selections.
- Experimental, model, parameter, and numerical uncertainty.

4.1.13 Plotting and post-processing

Dataset-specific plotting methods should:

- Use SI units.
- Use kelvin for absolute temperatures.
- Identify measured, derived, and calculated quantities clearly.
- Include readable axis labels, units, legends, and captions.
- Distinguish solvers and model configurations consistently.

- Represent experimental uncertainty when available and relevant.
- Avoid implying agreement beyond the precision or uncertainty of the available data.
- Support reproducible use from the corresponding application project.
- Ensure that smoothing used for visualization does not modify the stored source data.
- Keep original measurement points visible or otherwise traceable when smoothed curves are shown.

Common plotting or comparison behavior should be implemented in shared functionality where practical.

4.1.14 Testing and verification

OpenSTREAM-database does not currently include an automated test suite.

Until automated testing is implemented, contributors should verify changes using reproducible checks appropriate to the contribution.

These checks should include, where applicable:

- Confirming that the source-data files can be loaded correctly.
- Confirming that required fields are present.
- Confirming that all numerical source data use SI units.
- Confirming that absolute temperatures use kelvin.
- Reviewing unit conversions against the original experimental source.
- Comparing manually transcribed, character-recognized, or digitized values against the original source.
- Performing relevant geometry, power, unit, and energy-balance checks.
- Verifying that the intended experimental cases can be selected.
- Reviewing the generated OpenSTREAM input files.
- Running the affected application cases.
- Confirming that the selected OpenSTREAM calculations complete and converge as expected.
- Comparing calculated and measured quantities.
- Confirming that missing measurements are not interpreted as zeros.
- Confirming that direct measurements and derived quantities are distinguished.
- Reviewing generated figures and post-processing results.
- Confirming that existing application workflows are not adversely affected.
- Comparing results with previous approved executions, when available.

The verification procedure and its results should be described in the pull request.

Automated testing is a planned improvement. Future tests may cover:

- Source-data parsing.
- SI-unit conversion.
- Dataset metadata.
- Required-field validation.
- Case selection.
- OpenSTREAM input generation.

- Dataset-specific plotting and comparison methods.
- Application workflows.
- Numerical regression against approved results.

4.1.15 Documentation requirements

Each new dataset contribution should include a package README describing:

- The original experimental source and complete bibliographic reference.
- The scope of the implemented data.
- The experimental facility and relevant geometry.
- The implemented experimental conditions and measured quantities.
- The mandatory and dataset-specific fields.
- Confirmation that all numerical data use SI units.
- Confirmation that absolute temperatures use kelvin.
- Any unit conversions, manual transcription, character recognition, digitization, interpolation, filtering, reconstruction, or other data processing.
- Any sanity checks or consistency checks performed.
- The reported measurement uncertainties, where available.
- Known limitations, ambiguities, or missing information.
- Supported OpenSTREAM application workflows.
- A minimal reproducible usage example.
- Citation, licensing, attribution, and redistribution requirements.

User-visible changes should also be reflected in the OpenSTREAM Applications documentation when they affect:

- The OpenSTREAM-database description.
- Available datasets.
- Publication companion workflows.
- Exported project workflows.
- Publications.
- Installation or path requirements.
- Recommended application procedures.

4.1.16 Commit messages

Commit messages must follow the conventions documented in the [OpenSTREAM commit message style guide](#).

Contributors should organize changes into focused commits. Source-data corrections should normally be committed separately from code and documentation changes so that their provenance and verification remain clear in the repository history.

4.1.17 Submitting a contribution

Before submitting a contribution:

1. Confirm that OpenSTREAM and OpenSTREAM-database are both available on the MATLAB path.
2. Perform the relevant reproducible verification checks.
3. Run the affected application projects.
4. Review generated OpenSTREAM inputs and calculated results.
5. Confirm that all numerical source data use SI units.
6. Confirm that absolute temperatures use kelvin.
7. Review unit conversions and retained numerical precision.
8. Verify bibliographic references and dataset provenance.
9. Review applicable licensing, attribution, and redistribution conditions.
10. Identify one authoritative machine-readable source representation for each dataset package.
11. Confirm that generated outputs are not included in the commit.
12. Confirm that source PDFs and other copyrighted publications are not included unless redistribution is explicitly permitted.
13. Remove MATLAB autosave files, obsolete copies, generated outputs, local sandbox scripts, and temporary processing files.
14. Update the documentation and project exports when required.
15. Describe intentional numerical changes in the pull request.
16. Organize changes into focused commits using the required message style.

The pull-request description should identify:

- The purpose and scope of the contribution.
- The original experimental source.
- Added or modified cases and quantities.
- Data-processing and unit-conversion steps.
- Affected OpenSTREAM solvers and models.
- The verification procedure performed.
- The results of the verification.
- Numerical differences from previous results.
- Remaining limitations or unresolved questions.

4.1.18 Contributor checklist

Before submitting a pull request, confirm that:

- The contribution follows the established dataset package organization.
- The dataset template was used or the equivalent required structure was followed.
- The strict naming convention is followed.
- `getSourceFilePath` is used to locate the dataset source file.

- One authoritative machine-readable source representation is identified.
- The original experimental source is identified and cited.
- Applicable licensing and redistribution conditions have been reviewed.
- Source PDFs and other copyrighted publications are not unintentionally committed.
- Every implemented case defines the fields required by its supported workflow.
- Generic field names use the exact required spelling and capitalization.
- All mandatory and dataset-specific numerical fields use SI units.
- Absolute temperatures use kelvin.
- `WallMesh` and `WallPower` are mutually consistent.
- Dataset-specific fields are documented in the dataset README.
- Dataset-specific fields do not replace or rename mandatory fields.
- Measured, prescribed, transcribed, digitized, reconstructed, derived, and calculated quantities are distinguished.
- Unit conversions are documented.
- Source-data preparation methods are documented.
- Measurement uncertainties are included where available.
- Dataset assumptions and processing steps are documented.
- Relevant sanity checks and consistency checks have been performed.
- Missing numerical information is not interpreted as zero.
- The affected dataset and application workflows have been verified.
- The verification procedure and results are documented.
- Generated OpenSTREAM inputs and calculated results have been reviewed.
- Any numerical differences from previous results are understood and justified.
- Documentation and application projects have been updated.
- Publication companion workflows are not presented as complete research records.
- Generated input, result, session, and output directories are not committed.
- Temporary files, MATLAB autosaves, obsolete copies, local scripts, and processing files are not committed.
- Commit messages follow the OpenSTREAM style guide.

4.2 OpenSTREAM-database

OpenSTREAM-database provides an application and validation environment for **OpenSTREAM**, enabling users to configure, run, and evaluate OpenSTREAM solvers using publicly available experimental datasets.

OpenSTREAM-database complements the core OpenSTREAM repository:

- **OpenSTREAM** provides the solver frameworks, physical and closure models, numerical methods, input handling, visualization capabilities, tutorials, documentation, and core automated tests.
- **OpenSTREAM-database** provides dataset implementations, application workflows, calculated-versus-measured comparisons, publication companion workflows, and an environment for model assessment and validation.

The two repositories are maintained separately. OpenSTREAM is not included as a Git submodule of OpenSTREAM-database. A compatible OpenSTREAM working copy must be installed independently and made available on the MATLAB path.

4.2.1 Purpose

OpenSTREAM-database supports:

- Application of OpenSTREAM solvers to experimental cases.
- Reproducible generation of OpenSTREAM input files from dataset definitions.
- Execution of individual cases or selected groups of cases.
- Comparison of calculated and measured quantities.
- Development and assessment of physical and closure models.
- Numerical sensitivity and uncertainty studies.
- Documentation of dataset provenance, assumptions, and processing.
- Publication companion workflows illustrating selected calculations, figures, and results.
- Application- and validation-oriented research and solver development.

Separating application datasets from the core solver repository allows OpenSTREAM and OpenSTREAM-database to evolve independently while retaining a common application interface.

4.2.2 Repository organization

The principal repository structure is:

```
openstream-database/  
├── +DatasetName/  
├── +Template/  
├── @Dataset/  
├── projects/  
├── sandbox/  
├── LICENSE  
├── README.md  
└── functionSignatures.json
```

Each dataset is implemented as a MATLAB package that generally contains:

```
+DatasetName/  
├── +src/  
│   └── DatasetName.xml  
├── @DatasetName/  
│   ├── DatasetName.m  
│   └── plotResults.m  
└── README.md
```

Additional dataset-specific methods may be provided when required, for example for post-processing or trend analysis.

The package components have the following roles:

+src

Contains the structured experimental data and the information required to define OpenSTREAM cases.

@DatasetName

Contains the dataset-specific class and associated methods.

plotResults.m

Provides dataset-specific visualization and calculated-versus-measured comparison functionality.

README.md

Documents the original experimental source, implemented data, units, uncertainties, assumptions, processing methods, supported workflows, and known limitations.

The generic `Dataset` class provides functionality shared by the individual dataset implementations, including source-data handling, case selection, OpenSTREAM input generation, case execution, convergence review, and result storage.

The `+Template` package provides the recommended starting structure for new dataset contributions.

The `projects` folder contains publication companion workflows and other reviewed application projects.

The `sandbox` folder provides a local workspace for exploratory OpenSTREAM-database calculations. Temporary sandbox scripts, generated inputs, results, logs, figures, and saved objects are not maintained as repository content.

4.2.3 Dataset source files

Experimental cases and associated information are normally stored in a structured XML or JSON file under the dataset package's `+src` directory.

XML and JSON are supported source formats. A dataset package should normally maintain only one authoritative machine-readable source file. If equivalent XML and JSON representations are both retained, the dataset README must identify the authoritative source, explain how the alternative representation is generated, and describe how consistency is verified.

Each case can contain two categories of information:

1. **OpenSTREAM case-definition information**, whose field names follow a strict naming convention so that the generic `Dataset` functionality can identify geometry, boundary conditions, wall heating, and other information passed to OpenSTREAM.
2. **Dataset-specific information**, whose field names and contents can be defined according to the needs of the dataset implementation.

Dataset-specific information can include experimental measurements, metadata, classifications, uncertainties, derived quantities, and other information used by the dataset class or application workflows. For example, a dataset-specific `plotResults` method can use these fields to construct calculated-versus-measured comparisons.

The reserved OpenSTREAM field names are case-sensitive and must use the spelling expected by the input-generation implementation. Other fields can be added freely, provided that they do not conflict with reserved names and are interpreted consistently by the corresponding dataset class.

Scalar quantities are stored as individual values. Array quantities are represented by repeated XML elements or JSON arrays. Related arrays must have compatible lengths, ordering, and physical meaning.

Missing numerical measurements must be represented consistently and must not be interpreted as zero. The convention used for a dataset should be documented in its README and handled explicitly by dataset-specific methods.

Detailed source-file conventions should be documented in the corresponding dataset README and class implementation.

4.2.4 Lightweight in-memory datasets

OpenSTREAM-database can also be used without an XML or JSON source-data file.

A case can be defined directly using a one-row MATLAB table and passed to a lightweight `Dataset` object:

```
data = Dataset( ...  
    1, ...  
    'isLightWeight',true, ...  
    'lightWeightEntryData',entryData);
```

This in-memory workflow is useful for:

- publication demonstration cases;
- concise examples and tutorials;
- exploratory calculations;
- user-defined cases that do not require a maintained dataset package;
- comparisons of multiple OpenSTREAM solver frameworks using a common case definition.

The table must contain the geometry, boundary-condition, wall-heating, and fluid fields required by the selected Open-STREAM workflow. Numerical values must follow the same SI-unit conventions as file-based datasets.

A lightweight case is not a substitute for a maintained dataset package when experimental provenance, uncertainty, repeated case selection, or long-term reuse must be documented.

4.2.5 Units

OpenSTREAM uses SI units for all inputs, calculated quantities, and stored results.

OpenSTREAM-database follows the same convention. All numerical data stored in dataset source files or supplied through lightweight in-memory records must use SI units. Absolute temperatures are expressed in kelvin.

Data converted from other unit systems must be checked and documented. Units may be stated explicitly for individual fields or established through a clear dataset-wide convention. Units must be stated whenever clarification is useful, particularly for quantities whose representation could otherwise be ambiguous.

For example, a field called `InletSubcooling` may represent an enthalpy difference in J/kg rather than a temperature difference. Such conventions must be stated explicitly in the dataset README.

The original publications should be consulted for the reported quantities, measurement definitions, original units, and experimental uncertainties.

4.2.6 Data preparation and provenance

Dataset values may be obtained from publication tables, supplementary data, public data repositories, or digitized figures.

The dataset README must document the preparation method, including, as applicable:

- manual transcription from tables;
- character recognition from scanned reports;
- figure digitization;
- unit conversion;
- interpolation or reconstruction;
- corrections applied after review;
- derived quantities added for OpenSTREAM input generation;
- sanity checks and consistency checks.

Relevant checks can include geometry consistency, mass-flow and mass-flux consistency, power reconstruction, energy balances, array-length checks, unit checks, and comparison of related experimental cases.

The original publication or public data source remains authoritative. Machine-readable dataset implementations can contain transcription, recognition, digitization, conversion, or processing errors even after review.

Direct measurements must be distinguished from quantities derived during source-data preparation, calculation, or post-processing.

4.2.7 Application projects

The `projects` folder contains reviewed MATLAB Live Scripts that accompany OpenSTREAM publications and illustrate selected computational workflows, figures, and results presented in those publications.

A companion Live Script can include:

- explanatory text;
- selected experimental cases;
- physical-model and numerical-option selections;
- executable OpenSTREAM calculations;
- convergence information;
- calculated-versus-measured comparisons;
- figures and post-processing;
- interpretation of the selected results.

The companion workflows do not represent complete records of the research activities supporting the publications. Research and model-development activities generally include additional experimental cases, closure-model evaluations, sensitivity studies, uncertainty analyses, numerical investigations, intermediate results, and unsuccessful model variants that are not included in the published figures or companion workflows.

The MATLAB Live Scripts are the authoritative executable versions of the publication companion workflows. Completed executions may also be exported as static HTML documents containing explanatory text, code, calculated output, and figures. HTML exports can be viewed without MATLAB, although interactive Live Editor features may not be preserved.

Further information about the available projects is provided on the [application projects](#) page.

4.2.8 Requirements

OpenSTREAM-database requires:

- A compatible MATLAB installation.
- A working OpenSTREAM installation.
- The Python and CoolProp environment required by OpenSTREAM.
- Access to the OpenSTREAM-database repository.

OpenSTREAM is maintained as a separate repository and must be installed independently. Consult the [OpenSTREAM installation guide](#) for the current MATLAB, Python, CoolProp, and installation requirements.

4.2.9 Installation

Clone OpenSTREAM and OpenSTREAM-database as separate repositories.

A convenient local organization is:

```
projects/  
├── openstream/  
└── openstream-database/
```

The repositories do not need to share the same parent directory, provided that both repository roots are available on the MATLAB path.

4.2.10 MATLAB setup

Add both repository roots to the MATLAB path:

```
addpath('<path-to-openstream>')  
addpath('<path-to-openstream-database>')
```

Verify that MATLAB can locate functionality from both repositories:

```
which Inputs.InputSet  
which Dataset
```

The returned paths should refer to the intended OpenSTREAM and OpenSTREAM-database working copies.

4.2.11 Typical workflow

A typical file-based dataset workflow consists of:

1. Constructing a dataset-specific object from its source-data file.
2. Selecting one or more experimental cases.
3. Initializing the corresponding OpenSTREAM inputs.
4. Selecting an OpenSTREAM solver and model configuration.
5. Generating the OpenSTREAM input files.
6. Running the selected calculations.
7. Reviewing solver convergence.
8. Extracting calculated quantities.
9. Comparing calculated and measured results.
10. Visualizing and interpreting the comparison.

A lightweight workflow instead constructs a generic `Dataset` object from an in-memory MATLAB table and then follows the same input-generation, execution, convergence-review, and post-processing sequence.

The exact workflow depends on the selected dataset and solver. Consult the dataset README and any corresponding project file before running an application.

4.2.12 Generated files

OpenSTREAM-database may generate input packages and output folders containing:

- OpenSTREAM input files;
- solver session files;
- solver results;
- convergence logs;
- saved solver objects;
- processed comparisons;
- figures;
- other post-processing outputs.

Generated artifacts are excluded from version control through the repository `.gitignore` rules. Source-data files, MATLAB classes, reviewed project source files, templates, and documentation remain version-controlled.

Generated files should not be used as the only record of a calculation. Reproducible work should retain the source dataset, model and numerical configuration, software version or commit, and workflow required to regenerate the results.

4.2.13 Data provenance and citation

The datasets implemented in OpenSTREAM-database are derived from publicly available experimental sources.

When using a dataset:

- Consult and cite the original experimental publication or public data source.
- Review the experimental conditions, measurement definitions, original units, and reported uncertainties.
- Review any conversion, transcription, character recognition, digitization, processing, interpolation, filtering, reconstruction, or assumptions introduced by the implementation.
- Distinguish direct measurements from derived or calculated quantities.
- Identify the OpenSTREAM and OpenSTREAM-database versions or commits used.
- Cite OpenSTREAM and OpenSTREAM-database as appropriate.

Use of an OpenSTREAM-database implementation does not replace citation of the original experimental source.

Licensing of OpenSTREAM-database software and documentation does not supersede rights, licenses, attribution requirements, or redistribution conditions associated with the original publications and experimental data.

4.2.14 Validation and interpretation

Inclusion of a dataset, comparison method, or application workflow does not imply that every OpenSTREAM solver, closure model, model combination, or calculated quantity has been comprehensively verified or validated against that dataset.

Users remain responsible for evaluating:

- The applicability and quality of the experimental data.
- The assumptions and validity ranges of the selected models.
- The effects of dataset processing and derived quantities.
- Numerical convergence and discretization sensitivity.

- Sensitivity to closure-model and numerical-option selections.
- Experimental, model, parameter, and numerical uncertainty.
- The suitability of the selected comparison quantities and metrics.
- The interpretation of calculated-versus-measured differences.

A favorable comparison for one case, quantity, dataset, or model selection must not be interpreted as general validation outside the conditions examined.

4.2.15 Testing and verification

OpenSTREAM-database does not currently include a repository-wide automated test suite.

Dataset implementations and application workflows should therefore be verified through documented and reproducible execution of the affected cases. Verification should include, as applicable:

- successful loading of the source-data records;
- review of source-data provenance and processing;
- consistency checks on geometry, boundary conditions, units, and power;
- review of generated OpenSTREAM inputs;
- confirmation of solver convergence;
- comparison of calculated and experimental quantities;
- confirmation that missing measurements are not interpreted as zeros;
- confirmation that derived quantities are identified as derived;
- sensitivity to relevant physical models and numerical settings.

Detailed review and contribution requirements are provided in the [contribution guidelines](#).

4.2.16 Development status

OpenSTREAM-database and its application workflows are under active development. Dataset implementations, source-file formats, project workflows, comparison methods, documentation, and verification capabilities may continue to evolve.

Users reproducing earlier calculations should review:

- the dataset-specific README;
- the corresponding publication companion, when available;
- the OpenSTREAM and OpenSTREAM-database repository histories;
- the versions or commits used for the original calculation.

4.2.17 Contributing

Contributions of new datasets, source-data corrections, application workflows, publication companions, documentation, comparison methods, and verification capabilities are welcome.

Before contributing, review the [OpenSTREAM-database contribution guidelines](#).

4.3 Application projects

OpenSTREAM-database includes MATLAB Live Script projects that accompany OpenSTREAM publications and illustrate the principal computational workflows and selected results presented in those publications.

The project source files are located in the OpenSTREAM-database `projects` folder:

```
projects/
├── NURETH21.mlx
├── NUTHOS15_MRM.mlx
```

Each Live Script combines explanatory text, model selections, executable OpenSTREAM calculations, convergence information, figures, and interpretation. The objective is to provide a transparent and reproducible workflow for the selected calculations and results presented in the corresponding publication.

The companion workflows do not represent the complete research activities supporting the publications. Model development and qualification generally involve additional experimental cases, closure-model evaluations, sensitivity studies, numerical investigations, and intermediate analyses that are not included in the published figures or companion Live Scripts.

The MATLAB Live Scripts are the authoritative executable versions of the published companion workflows. Completed executions may also be provided as static HTML exports containing the calculated output and figures. The HTML exports can be viewed without MATLAB, although interactive Live Editor features may not be preserved.

Before running or extending a project:

- Install OpenSTREAM and OpenSTREAM-database separately and make both repository roots available on the MATLAB path.
- Review the corresponding publication and associated dataset documentation.
- Review the selected experimental cases and generated OpenSTREAM inputs.
- Review the physical-model and numerical-option selections.
- Confirm that the calculations converge as documented.
- Distinguish direct experimental measurements from derived quantities.
- Record the OpenSTREAM and OpenSTREAM-database versions or commits used.

Further installation and configuration information is provided on the [OpenSTREAM-database](#) page.

OpenSTREAM-database does not currently include an automated test suite. Project workflows are therefore verified through reproducible execution, review of convergence, and documented comparison of calculated and experimental quantities.

4.3.1 NURETH21

The NURETH21 project is provided as a MATLAB Live Script:

```
projects/NURETH21.mlx
```

The Live Script is a reproducible computational companion to the NURETH-21 publication:

The workflow reproduces selected OpenSTREAM calculations and results presented in the paper:

- Figures 3 and 4: comparison of the mixture, two-fluid, three-field, and four-field solver frameworks.
- Figure 6: validation of fully developed base-film and disturbance-wave properties using the Wurtz1978 dataset.
- Figure 7: validation of developing disturbance-wave behavior using the Sawai1989 dataset.

The demonstration case also illustrates the lightweight OpenSTREAM-database workflow. The geometry, boundary conditions, and heating conditions are defined in a one-row MATLAB table and passed directly to lightweight `Dataset` objects. No XML or JSON source-data file is required.

The validation calculations use the dataset packages to select experimental cases, generate OpenSTREAM inputs, run the FourField solver, check convergence, and compare calculated and measured annular-flow properties.

The Live Script focuses on the calculations required to illustrate the figures and conclusions presented in the publication. It does not include all cases, model-development activities, closure-model evaluations, sensitivity studies, or numerical investigations performed during the underlying research.

Figure 5 is not reproduced because the TRACE reference results required for that comparison are not included in the companion workflow.

A completed execution of the workflow is available as an HTML export:

[Open the completed NURETH21 companion workflow](#)

4.3.2 NUTHOS15_MRM

The NUTHOS15_MRM project is provided as a MATLAB Live Script:

```
projects/NUTHOS15_MRM.mlx
```

The Live Script is intended to provide a reproducible computational companion to the corresponding NUTHOS-15 publication. It will document the selected OpenSTREAM calculations and results used to illustrate the application and evaluation of the MRM model, including:

- the selected experimental cases;
- the physical-model and numerical-option selections;
- the executable OpenSTREAM calculations;
- convergence information;
- calculated-versus-measured comparisons;
- interpretation of the principal results.

The companion workflow will focus on the calculations and figures presented in the publication. It will not constitute a complete record of the model-development process, sensitivity studies, closure-model evaluations, or additional cases considered during the research.

The workflow and its HTML export will be linked from this page after the project has been completed and reviewed.

4.4 Publications

This page features a curated list of publications where the **OpenSTREAM solvers** have been documented, developed, or applied to research activities in two-phase flow simulation and thermal-hydraulic modeling.

4.4.1 Overviews

1. Jean-Marie Le Corre, Jason Chan, Elisabeth Walter, Evan T. Hurlburt, and Roman W. Morse. Open-STREAM: an open-source platform for two-phase flow modeling and simulation. In *Proceedings of the 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*. Busan, Korea, August 31–September 5 2025a. URL: https://openstream-solvers.github.io/openstream/_static/publications/LeCorre2025OpenSTREAM.pdf.

2. Jean-Marie Le Corre, Jason Chan, Elisabeth Walter, Evan T. Hurlburt, and Roman W. Morse. OpenSTREAM: a new open-source platform for two-phase flow model development. In *Proceedings of the 12th International Conference on Multiphase Flow (ICMF 2025)*. Toulouse, France, May 12–16 2025b. URL: https://openstream-solvers.github.io/openstream/_static/publications/LeCorre2025ICMF.pdf.
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4.4.2 Mixture model

1. Jean-Marie Le Corre, Fabrice Francois, and Henryk Anglart. A mixture relaxation model for thermal non-equilibrium in convective flow boiling. In *Proceedings of the 15th International Topical Meeting on Nuclear Reactor Thermal-Hydraulic, Operation and Safety (NUTHOS-15)*. Kashiwa-no-ha, Japan, September 2026.
-

4.4.3 Two-fluid model

1. Elisabeth Walter, Jason Chan, Roman W. Morse, Haipeng Li, and Jean-Marie Le Corre. One-dimensional, two-fluid, two-phase flow simulation framework in the open-source platform OpenSTREAM. In *Proceedings of the 12th International Conference on Multiphase Flow (ICMF 2025)*. Toulouse, France, May 12–16 2025. URL: https://openstream-solvers.github.io/openstream/_static/publications/Walter2025.pdf.
 2. Elisabeth K. Walter. Derivation and open source implementation of a simplified one-dimensional two-fluid model. Master's thesis, KTH Royal Institute of Technology, Stockholm, Sweden, 2024. URL: <https://www.diva-portal.org/smash/record.jsf?pid=diva2:1888131>.
-

4.4.4 Three-field model

1. Jason Chan, Roman W. Morse, Nathan Hogan, Evan T. Hurlburt, and Jean-Marie Le Corre. One-dimensional, advanced annular two-phase flow simulations with the open-source platform OpenSTREAM. In *Proceedings of the 12th International Conference on Multiphase Flow (ICMF 2025)*. Toulouse, France, May 12–16 2025. URL: https://openstream-solvers.github.io/openstream/_static/publications/Chan2025.pdf.
 2. Cole Dunbar, Juliana P. Duarte, Roman W. Morse, and Jean-Marie Le Corre. Multi-field simulations of dry-out and rewet cycles in rod bundle geometry. In *Proceedings of the 21st International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH-21)*. Busan, Korea, August 31–September 5 2025. URL: https://openstream-solvers.github.io/openstream/_static/publications/Dunbar2025.pdf.
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4.4.5 Four-field model

1. Jason Chan, Roman W. Morse, Nathan Hogan, Evan T. Hurlburt, and Jean-Marie Le Corre. One-dimensional, advanced annular two-phase flow simulations with the open-source platform OpenSTREAM. In *Proceedings of the 12th International Conference on Multiphase Flow (ICMF 2025)*. Toulouse, France, May 12–16 2025. URL: https://openstream-solvers.github.io/openstream/_static/publications/Chan2025.pdf.
2. Howard Y. K. Li, Ketan Ajay, David R. Novog, and Jean-Marie Le Corre. Four-field modeling of disturbance waves at high pressure near the onset of annular two-phase flow. In *Proceedings of the 15th International Topical Meeting on Nuclear Reactor Thermal-Hydraulic, Operation and Safety (NUTHOS-15)*. Kashiwa-no-ha, Japan, September 2026.

PACKAGES

5.1 InputEnums

The *InputEnums* package defines a set of enumeration classes used to specify valid options for physical models in **OpenSTREAM**. These enums help ensure consistency and clarity when selecting model configurations, making the simulation setup more robust and less error-prone.

Typical examples include choosing between void fraction models, interfacial transfer options, or flow regime assumptions, each represented by a clearly defined enum.

class InputEnums.BTMTM

BTMTM Enumeration of post-boiling transition wall momentum transfer models

This class defines the available *Inputs.Model.BTMTM* models for calculating the post-boiling transition wall momentum transfer, used by the mixture solver.

Models:

- TPFM — Wall momentum transfer based on two-phase friction multiplier, using multiplier selected in *Inputs.Model.TPFM*
- VAPOR — Wall momentum transfer based on vapor phase only, using single-phase momentum transfer model selected in *Inputs.Model.SPMTM*

class InputEnums.EQSTROUHAL

EQSTROUHAL Enumeration of wave equilibrium Strouhal number models

This class defines the available *Inputs.Model.EQSTROUHAL* models for calculating the wave equilibrium Strouhal number, used by the four-field solver.

Models:

- RISO — Based on RISO dataset (Le Corre [2022a])
- SAWAI — Based on SAWAI dataset (Le Corre [2022a])
- MFVAL — MFVAL model (under development)
- CUSTOM — Custom model using user-defined *Inputs.Model.EQSTROUHALCOEF*
- CSTFRQ — Custom constant frequency using user-defined *Inputs.Model.CSTWAVEFREQ*

class InputEnums.INTAREA

INTAREA Enumeration of volumetric interfacial area calculation models

This class defines the available *Inputs.Model.INTAREA* models for calculating the volumetric interfacial area, used by the two-fluid solver.

Models:

- DISPGAS2DISPLIQ — Transition from spherical bubbles to spherical droplets (Walter [2024])

class InputEnums.FLUIDPROPERTIES

FLUIDPROPERTIES Enumeration of fluid property calculation assumptions

This class defines the available [Inputs.Model.PROPERTIES](#) assumptions to calculate fluid properties, applicable to all solvers.

Models:

- SATURATED — Properties based on saturated conditions at system pressure
- PSYSTEM — Properties based on system pressure and local phase enthalpy

class InputEnums.WAVEBASEINT

WAVEBASEINT Enumeration of wave to base film interfacial momentum transfer models

This class defines the available [Inputs.Model.WAVEBASEINT](#) models for calculating the interfacial momentum transfer between wave and base film regions, used by the four-field solver.

Models:

- VAPORSHEAR — Vapor shear across wave area (consistent with Le Corre [2022a], Eq. 46).
- VAPORSHEARDROPMASS — Also includes drop deposition momentum transfer, ensuring consistency between wave and base film velocities for thin films.

class InputEnums.SPMTM

SPMTM Enumeration of single-phase wall momentum transfer models

This class defines the available [Inputs.Model.SPMTM](#) models for calculating the single-phase wall momentum transfer (liquid or gas), used by the mixture solver.

Models:

- BLASIUS — Blasius wall momentum transfer model ($f = C(1)Re^{C(2)} + C(3)$) using user-defined [Inputs.Model.FRCTION](#) coefficients

class InputEnums.MFBT

MFBT Enumeration of Minimum Film Boiling Transition (MFBT) models

This class defines the available [Inputs.Model.MFBT](#) models for evaluating the Minimum Film Boiling Transition, used by the mixture solver.

Models:

- NONE — No MFBT transition applied
- CONSTANT — Constant deviation from the saturation temperature using user-defined [Inputs.Model.DTMFB](#)

class InputEnums.NEARWALLH

NEARWALLH Enumeration of near-wall equilibrium enthalpy models

This class defines the available [Inputs.Model.NEARWALLH](#) models for setting the near-wall enthalpy to mixture enthalpy ratio, used by the mixture solvers.

Models:

- RATIO — Constant near-wall enthalpy to mixture enthalpy ratio using user-defined [Inputs.Model.NEARWALLHRATIO](#)
- FILM — Mass flux-dependent input for annular two-phase flow

class InputEnums.**INTTRANSH**

INTTRANSH Enumeration of interfacial enthalpy transfer models

This class defines the available [Inputs.Model.INTTRANSH](#) models for calculating the phase enthalpy associated with interfacial mass transfer, used by the mixture and two-fluid solvers.

Models:

- BULK — Uses bulk phase enthalpy
- SATURATED — Uses saturated phase enthalpy

class InputEnums.**DEPOSITION**

DEPOSITION Enumeration of droplet deposition models

This class defines the available [Inputs.Model.DEPOSITION](#) models for calculating the droplet deposition mass flux, used by the three-field and four-field solvers.

NOTE: Entrainment and deposition correlations are coupled. It is recommended to use the same model for both entrainment and deposition mass flux to ensure consistency.

Models:

- NONE — No deposition applied
- GOVAN — Hewitt and Govan correlation (Hewitt and Govan [1990])
- OKAWA — Okawa et al. correlation (Okawa *et al.* [2003])

class InputEnums.**NEARWALLRELAX**

NEARWALLRELAX Enumeration of near-wall energy transfer time relaxation models

This class defines the available [Inputs.Model.NEARWALLRELAX](#) models for calculating the near-wall energy transfer time relaxation, used in mixture solver.

Models:

- QUALITY — Simple model relating relaxation time to equilibrium quality using user-defined [Inputs.Model.NEARWALLRELAXX](#) and [Inputs.Model.NEARWALLRELAXT](#)
- VOID — Physical model based on local phase volumetric fraction using user-defined [Inputs.Model.NEARWALLRELAXCOEF](#)
- FILM — Model applicable to annular two-phase flow

class InputEnums.**OAFENTRAINED**

OAFENTRAINED Enumeration of entrained drop models at onset of annular flow

This class defines the available [Inputs.Model.OAFENTRAINED](#) models for splitting the droplet and film mass flow rates at the onset of annular flow, used by the three-field and four-field solvers.

Models:

- RATIO — Ratio of droplet to liquid mass flow rates using user-defined [Inputs.Model.OAFDROPRATIO](#)
- EQUILIBRIUM — Equilibrium assumption model ($M_{\text{ent}} = M_{\text{dep}}$) (Adamsson and Le Corre [2011], Equations 15 and 16)

class InputEnums.**BTHTM**

BTHTM Enumeration of post-boiling transition wall heat transfer models

This class defines the available [Inputs.Model.BTHTM](#) models for calculating the post-boiling transition wall heat transfer, used by the mixture solver.

Models:

- VAPOR — Wall heat transfer to the vapor phase only, based on single-phase heat transfer model selected in [Inputs.Model.SPHTM](#)
- DOUGALL — Dougall-Rohsenow wall heat transfer model (Dougall and Rohsenow [1963])
- BISHOP — Bishop-Sandberg-Tong wall heat transfer model (Bishop *et al.* [1965])
- MOECK — Groeneveld-Moeck wall heat transfer model (Groeneveld and Moeck [1969])
- DELORME — Groeneveld-Delorme wall heat transfer model (Groeneveld and Delorme [1976])
- CONDIEIV — Condie-Bengston IV wall heat transfer model (Morris *et al.* [1982])

class InputEnums.**WAVEFREQUENCY**

WAVEFREQUENCY Enumeration of wave number density transport models

This class defines the available [Inputs.Model.WAVEFREQUENCY](#) models for solving the wave number density (or wave frequency) transport equation, used in the four-field solver.

Models:

- EQUILIBRIUM — Assumes instantaneous equilibrium between wave generation and dissipation.
- RELAXATION — Full transport model including time relaxation effects set by user-defined RELAXTW (Le Corre [2022a])

class InputEnums.**THINFILMFRIC**

THINFILMFRIC Enumeration of thin film friction models

This class defines the available [Inputs.Model.THINFILMFRIC](#) models for calculating wall friction under thin film conditions, used by the three-field and four-field solvers.

Models:

- LAMINAR — Laminar flow model ($16/Re_{film}$)
- TURBULENT — Turbulent flow model

class InputEnums.**TPFM**

TPFM Enumeration of two-phase wall friction multiplier models

This class defines the available [Inputs.Model.TPFM](#) models for calculating the two-phase wall friction multiplier used by the mixture solver.

Models:

- HOMOGENEOUS — Assumes no slip between phases; both phases move at the same velocity
- SLIP — Accounts for velocity differences (slip) between liquid and vapor phases using user-defined [Inputs.Model.SLIP](#)
- EPRI — Empirical model developed by the Electric Power Research Institute (EPRI) (Reddy and others [1982])

class InputEnums.**MOMENTWAVE**

MOMENTWAVE Enumeration of film momentum conservation models

This class defines the available [Inputs.Model.MOMENTWAVE](#) models for solving the momentum conservation equation for the wave field, used to compute wave velocity in the four-field solver.

Models:

- ALGEBRAIC — Simple algebraic model (Adamsson and Le Corre [2014], Equations 21 or 23)
- EQUILIBRIUM — Equilibrium model

- FULL — Full non-equilibrium momentum model (Le Corre [2022a])

class InputEnums.TPHTM

TPHTM Enumeration of two-phase wall heat transfer models

This class defines the available `Inputs.Model.TPHTM` models for calculating the two-phase wall heat transfer, used by the mixture solver.

Models:

- THOM — Thom wall heat transfer model (Rohsenow *et al.* [1998])

class InputEnums.MOMENTDROP

MOMENTDROP Enumeration of droplet momentum conservation models

This class defines the available `Inputs.Model.MOMENTDROP` models for solving the droplet momentum conservation equation, used to compute droplet velocity in the three-field and four-field solvers.

Models:

- ALGEBRAIC — Simple algebraic model consistent with mixture solver
- SLIP — Slip ratio model (droplet/gas velocity)
- EQUILIBRIUM — Equilibrium model
- EQUILIBRIUMS — Simplified equilibrium model based on force balance ($F_{\text{drag}} + F_{\text{gravity}} + F_{\text{buoyancy}} = 0$)
- FULL — Full non-equilibrium momentum model (Le Corre *et al.* [2025a])

class InputEnums.BUBBLEDRAG

BUBBLEDRAG Enumeration of bubble drag coefficient models

This class defines the available `Inputs.Model.BUBBLEDRAG` models for calculating the bubble drag coefficient, used by the two-fluid solver.

Models:

- CONSTANT — Constant drag coefficient using user-defined `Inputs.Model.BUBBLEDRAGCOEF`
- STOKES — Stokes flow regime model ($\frac{24}{Re_p}$)
- VISCOUS — Viscous regime model (Walter [2024])
- DISTORTED — Model for distorted bubbles (non-spherical) (Walter [2024])

class InputEnums.MOMENTBASE

MOMENTBASE Enumeration of base film momentum conservation models

This class defines the available `Inputs.Model.MOMENTBASE` models for solving the base film momentum conservation equation, used to compute the base film velocity in the four-field solver.

Models:

- ALGEBRAIC — Simple algebraic model (Adamsson and Le Corre [2014], Equations 21 or 23)
- EQUILIBRIUM — Equilibrium model
- EQUILIBRIUMS — Simplified equilibrium model based on force balance ($F_{\text{wall}} + F_{\text{vapor}} = 0$)
- FULL — Full non-equilibrium model (Le Corre [2022a])
- FULLNOP — Full non-equilibrium model excluding pressure terms (gravity and buoyancy)

class InputEnums.**INTLENGTH**

INTLENGTH Enumeration of interfacial length scale models

This class defines the available [Inputs.Model.INTLENGTH](#) models for calculating the interfacial length scale, used by the two-fluid solver.

Models:

- **CONSTANT** — Assumes a constant interfacial length scale using user-defined [Inputs.Model.INTLENGTHVCS](#) and [Inputs.Model.INTLENGTHLCST](#) for the dispersed gas and liquid phases, respectively

class InputEnums.**SPHTM**

SPHTM Enumeration of single-phase wall heat transfer models

This class defines the available [Inputs.Model.SPHTM](#) models for calculating the single-phase wall heat transfer (liquid or gas), used by the mixture solver.

Models:

- **DITTUSBOELTER** — Dittus-Boelter wall heat transfer model for heating ($Nu = 0.023 \cdot Re^{0.8} \cdot Pr^{0.4}$) (Dittus and Boelter [1930], McAdams [1942])
- **DITTUSBOELTERGEN** — Generalized Dittus-Boelter wall heat transfer model using user-defined [Inputs.Model.DITTUSBOELTERCOEF](#)
- **SIEDERTATE** — Sieder-Tate wall heat transfer model ($Nu = 0.027 \cdot Re^{0.8} \cdot Pr^{1/3} \cdot \left(\frac{\mu_b}{\mu_w}\right)^{0.14}$) (Sieder and Tate [1936])
- **GNIELINSKI** — Gnielinski wall heat transfer model (Gnielinski [1976])

class InputEnums.**DEPENHANCEMENT**

DEPENHANCEMENT Enumeration of droplet deposition enhancement models

This class defines the available [Inputs.Model.DEPENHANCEMENT](#) models for calculating local droplet deposition enhancement, used by the three-field and four-field solvers.

Models:

- **NONE** — No enhancement applied
- **WINDECKER** — Model adapted from Windecker [1999], as documented in Le Corre [2024], using user-defined [Inputs.Model.KTUNING](#)

class InputEnums.**BASEEQTHICK**

BASEEQTHICK Enumeration of base film equilibrium thickness models

This class defines the available [Inputs.Model.BASEEQTHICK](#) models for calculating the equilibrium base film thickness, used by the four-field solver.

Models:

- **RISO** — Based on RISO dataset (Le Corre [2022a])
- **COEFS** — Generalized RISO model using user-defined [Inputs.Model.BASEEQTHICKCOEF](#)
- **YPLUS** — Based on dimensionless wall distance (y^+) using user-defined [Inputs.Model.BASEYPLUS](#)
- **MFVAL** — MFVAL model (under development)

class InputEnums.DROPDRAG

DROPDRAG Enumeration of droplet drop drag coefficient models

This class defines the available [Inputs.Model.DROPDRAG](#) models for calculating the droplet drag coefficient, used by the two-fluid solver.

Models:

- CONSTANT — Constant drag coefficient using user-defined [Inputs.Model.DROPDRAGCOEF](#)
- STOKES — Stokes flow regime model ($\frac{24}{Re_p}$)
- VISCOUS — Viscous regime model (Walter [2024])
- DISTORTED — Model for distorted (non-spherical) droplets (Walter [2024])

class InputEnums.CBT

CBT Enumeration of Critical Boiling Transition (CBT) models

This class defines the available [Inputs.Model.CBT](#) models for evaluating the Critical Boiling Transition, used by the mixture solver.

Models:

- NONE — No CBT transition applied
- BIASI — Biasi correlation model (Biasi *et al.* [1966])
- BEZRUKOV — Bezrukov correlation model (Bezrukov *et al.* [1976])
- ELEVATION — CBT determined from specified elevation using user-defined [Inputs.Model.CBTELEVATION](#)

class InputEnums.ENTRAINMENT

ENTRAINMENT Enumeration of droplet entrainment models

This class defines the available [Inputs.Model.ENTRAINMENT](#) models for calculating the droplet entrainment mass flux, used by the three-field and four-field solvers.

NOTE: Entrainment and deposition correlations are coupled. It is recommended to use the same model for both entrainment and deposition mass flux to ensure consistency.

Models:

- NONE — No entrainment
- GOVAN — Hewitt and Govan correlation (Hewitt and Govan [1990])
- OKAWA2003 — Okawa *et al.* correlation (Okawa *et al.* [2003])
- OKAWA2004 — Okawa *et al.* correlation (Okawa *et al.* [2004])
- OKAWA2004MOD — Modified Okawa (2004) model from (Adamsson and Le Corre [2011])
- OKAWAGEN — Generic Okawa-based model using user-defined [Inputs.Model.OKAWACOEFS](#)

class InputEnums.TPKM

TPKM Enumeration of two-phase local loss multiplier models

This class defines the available [Inputs.Model.TPKM](#) models for calculating the two-phase local loss multiplier used by the mixture model

Models:

- HOMOGENEOUS — Assumes no slip between phases; both phases move at the same velocity.

- SLIP — Accounts for velocity differences (slip) between liquid and vapor phases using user-defined [*Inputs.Model.SLIP*](#).
- ROMIE — Romie model; behaves like the homogeneous model when a homogeneous void fraction model is applied.

class InputEnums.OAF

OAF Enumeration of onset of annular flow models

This class defines the available [*Inputs.Model.OAF*](#) models for predicting the onset of annular flow, used by the three-field and four-field solvers.

Models:

- WALLIS — Full Wallis correlation (Wallis [1969], Equation 11.1019)
- WALLIS_SIMP — Simplified Wallis correlation (Sanmiguel Gimeno *et al.* [2015], Equation 6), used for compatibility with codes employing this approach
- LEVITAN — Levitan correlation (Levitan and Borevsky [1989])

class InputEnums.VOID

VOID Enumeration of void fraction models

This class defines the available [*Inputs.Model.VOID*](#) models to calculate the void fraction in the mixture solver.

Models:

- HOMOGENEOUS — Assumes vapor and liquid phases move at the same velocity.
- SLIP — Assumes a slip ratio between vapor and liquid phases using user-defined [*Inputs.Model.SLIP*](#).
- BESTION — Drift-flux model developed by Bestion (Bestion [1990]).
- EPRI — Drift-flux model developed by the Electric Power Research Institute (EPRI) (Lellouche and Zolotar [1982]).

class InputEnums.OAFFILMSPLIT

OAFFILMSPLIT Enumeration of film mass flow ratio models at onset of annular flow

This class defines the available [*Inputs.Model.OAFFILMSPLIT*](#) models for splitting the base film and wave mass flow rates at the onset of annular flow, used by the four-field solver.

Models:

- RATIO — Ratio of base film to film mass flow rates using user-defined [*Inputs.Model.OAFBASERATIO*](#)
- EQUILIBRIUM — Equilibrium model (film thickness = equilibrium film thickness) (Le Corre [2022a])

class InputEnums.MOMENTFILM

MOMENTFILM Enumeration of film momentum conservation models

This class defines the available [*Inputs.Model.MOMENTFILM*](#) models for solving the momentum conservation equation for the liquid film, used to compute film velocity in the three-field solver.

Models:

- ALGEBRAIC — Simple algebraic model
- EQUILIBRIUM — Equilibrium model
- EQUILIBRIUMS — Simplified equilibrium model based on force balance ($F_{\text{wall}} + F_{\text{vapor}} = 0$)
- FULL — Full non-equilibrium momentum model (Le Corre *et al.* [2025a])

class InputEnums.**MRMTIMEINT**

MRMTIMEINT Enumeration of time integration options for the MRM vapor mass/energy conservation equations

This class defines the available [Inputs.Options.MRMTIMEINT](#) options for solving the vapor mass and energy conservation equations when using the MRM model in the mixture solver.

Models:

- EULER — Backward Euler
- EXPONENTIAL — Exponential

class InputEnums.**LOCRELVEL**

LOCRELVEL Enumeration of local relative velocity models

This class defines the available [Inputs.Model.LOCRELVEL](#) models for calculating the local relative velocity between liquid and vapor phases, used by the two-fluid solver.

Models:

- AREAMEAN — Difference between area-averaged phase velocities (Walter [2024])
- SCALED — Scaled difference using user-defined [Inputs.Model.RELVELCST](#) (Walter [2024])
- DRIFT — Uses drift velocity as the local relative velocity (Walter [2024])
- SIMPLE — Simplified assumptions for bubbly and annular flow regime (Walter [2024])

class InputEnums.**VAPORFRIC**

VAPORFRIC Enumeration of vapor to film interfacial friction models

This class defines the available [Inputs.Model.VAPORFRIC](#) models for calculating the vapor-to-film interfacial friction coefficient used by the three-field and four-field solvers.

Models:

- CONSTANT — Uses a fixed, friction coefficient using user-defined [Inputs.Model.VAPORFRICCST](#)
- WALLIS — Wallis model based on void fraction (Wallis [1969]).
- WALLISTHICK — Wallis model based on film thickness (Adamsson and Le Corre [2011], Equation 35).
- SOLVER_DEPENDENT — Friction model determined by the solver configuration.

class InputEnums.**THERMALRELAX**

THERMALRELAX Enumeration of MRM time relaxation models

This class defines the available [Inputs.Model.THERMALRELAX](#) models for calculating relaxation times in the MRM model, used when [Inputs.Model.THERMALNONEQ](#) is set to *MRM*.

Models:

- FOURIERX — Simple model relating the Fourier number to equilibrium quality using user-defined [Inputs.Model.RELAXX](#), [Inputs.Model.RELAXCONDFO](#) and [Inputs.Model.RELAXEVAPFO](#)
- TIMEX — Simple model relating relaxation time to equilibrium quality using user-defined [Inputs.Model.RELAXX](#), [Inputs.Model.RELAXCONDT](#) and [Inputs.Model.RELAXEVAPT](#)
- VOID — Physical model simplified to Fourier number dependence on phase volumetric fraction and input characteristic length of the dispersed phase at CBT using user-defined [Inputs.Model.RELAXCONDCOE](#) and [Inputs.Model.RELAXEVAPCOEF](#)
- HOMOGENEOUS — Physical model based on homogeneous assumptions and spherical dispersed phase

- FOURIER — Empirical model based on Fourier number dependence on Reynolds number, etc, using user-defined [Inputs.Model.RELAXCONDFOCOEF](#) and [Inputs.Model.RELAXEVAPFOCOEF](#)

class InputEnums.THERMALNONEQ

THERMALNONEQ Enumeration of thermal non-equilibrium models

This class defines the available [Inputs.Model.THERMALNONEQ](#) models for calculating phase thermal non-equilibrium effects, used by the mixture solver.

Models:

- EQUILIBRIUM — Assumes thermal equilibrium ($X = \max(0, X_{EQ})$)
- SAHAZUBER — Saha-Zuber subcooled boiling model (Saha and Zuber [1973])
- EPRI — EPRI subcooled boiling model (Lellouche and Zolotar [1982])
- MRM — Mixture Relaxation Model for subcooled boiling and post-CHF conditions

class InputEnums.INTNU

INTNU Enumeration of interfacial Nusselt number models

This class defines the available [Inputs.Model.INTNU](#) models for calculating the interfacial Nusselt number, used by the two-fluid solver.

Models:

- CONSTANT — Assumes a constant Nusselt number using user-defined [Inputs.Model.INTNUVCST](#) and [Inputs.Model.INTNULCST](#) for the dispersed gas and liquid phases, respectively
- RANZMARSHALL — Ranz-Marshall correlation-based model (Ranz and Marshall [1952]) using user-defined [Inputs.Model.RANZMARSHALLVCST](#) and [Inputs.Model.RANZMARSHALLLCST](#) for the dispersed gas and liquid phases, respectively
- RELAXATION — Time relaxation-based model (Walter [2024])

class InputEnums.MOMENTLIQUID

MOMENTLIQUID Enumeration of liquid momentum conservation models

This class defines the available [Inputs.Model.MOMENTLIQUID](#) models for solving the momentum conservation equation for the liquid phase, used to compute liquid velocity in the two-fluid solver.

Models:

- MIXTURE — Uses the same velocity as in the mixture model
- SLIP — Slip ratio model (gas/liquid velocity) using user-defined [Inputs.Model.SLIP](#)
- FULL — Full non-equilibrium momentum model (Le Corre *et al.* [2025a], Walter [2024])

class InputEnums.MOMENTGAS

MOMENTGAS Enumeration of gas momentum conservation models

This class defines the available [Inputs.Model.MOMENTGAS](#) models for solving the momentum conservation equation for the gas phase, used to compute gas velocity in the two-fluid solver.

Models:

- MIXTURE — Uses the same gas velocity as in the mixture model
- SLIP — Slip ratio model (gas/liquid velocity) using user-defined [Inputs.Model.SLIP](#)
- FULL — Full non-equilibrium momentum model (Le Corre *et al.* [2025a], Walter [2024])

5.2 Inputs

The *Inputs* package defines a set of classes used to configure and import all the necessary parameters for running OpenSTREAM simulations. This includes:

- Boundary conditions: *Inputs.BoundaryConditions*
- System geometry: *Inputs.Geometry*
- Physical models: *Inputs.Model*
- Numerical options: *Inputs.Options*
- Fluid thermophysical properties: *Inputs.FluidProperties*
- Input handling: *Inputs.Input*
- Complete input set management: *Inputs.InputSet*

These input definitions form the backbone of each simulation setup, ensuring consistency, flexibility, and clarity across different solver frameworks.

class *Inputs.Model*(filePath, modelID)

Bases: *Inputs.Input*

MODEL Class for defining and managing physical model configuration

This class encapsulates all physical modeling parameters used in the simulations, including fluid properties, regime transitions, wall and interfacial exchange models for all solvers. It reads model data from input files and handles default values as well as input validation.

Constructor Summary

Model(filePath, modelID)

MODEL Constructor for *Model()* class

Parses model input file and initializes properties. Applies default values and validates entries.

Inputs:

- filePath — Path to model input file
- modelID — Identifier for model configuration

Property Summary

BASEEQTHICK = 'RISO'

Equilibrium base film thickness model selected from *InputEnums.BASEEQTHICK*

BASEEQTHICKCOEF = [5.37E-5-0.641.21]

Equilibrium base film thickness coefficients [-]

BASEYPLUS = 15

Equilibrium base film y+ value [-]

BTHTM = 'VAPOR'

Boiling transition wall heat transfer model selected from *InputEnums.BTHTM*

BTMTM = 'TPFM'

Boiling transition wall momentum transfer model selected from *InputEnums.BTMTM*

BUBBLEDRAG = 'STOKES'

Bubble drag model selected from *InputEnums.BUBBLEDRAG*

BUBBLEDRAGCOEF = 0.45

Bubble constant drag coefficient [-]

CBT = 'NONE'

Critical Boiling Transition model selected from *InputEnums.CBT*

CBTELEVATION = 1

Critical Boiling Transition Elevation [m]

CBTKEFFECT = [00]

Grid effect coefficients ($1 + C(1) \exp(C(2)z)$) [-]

CBTMULT = @z1

Critical boiling Heat flux multiplier function

CSTWAVEFREQ = 100

Imposed constant wave frequency [Hz]

DEPENHANCEMENT = 'NONE'

Drop deposition enhancement model due to local perturbations selected from *InputEnums.DEPENHANCEMENT*

DEPOSITION = 'OKAWA'

Drop deposition model selected from *InputEnums.DEPOSITION*

DITTUSBOELTERCOEF = [0.0230.80.4]

Dittus-Boelter coefficients [-]

DROPDIA = 1E-3

Drop diameter [m]

DROPDRAG = 'VISCOUS'

Drop drag model selected from *InputEnums.DROPDRAG*

DROPDRAGCOEF = 0.45

Drop constant drag coefficient [-]

DROPSLIP = 1.0

Drop/vapor velocity ratio [-]

DTMFB = 100

Minimum film boiling temperature from saturation [K]

ENTRAINMENT = 'OKAWA2003'

Film entrainment model selected from *InputEnums.ENTRAINMENT*

EQSTROUHAL = 'RISO'

Equilibrium wave Strouhal number model selected from *InputEnums.EQSTROUHAL*

EQSTROUHALCOEF = [1.1236E-40.50.0]

Equilibrium wave Strouhal number coefficients [-]

FLUID = 'WATER'

CoolProp fluid identifier

FRICITION = [0.2-0.20]

Wall friction coefficients [-]

G = 9.81

Gravitational acceleration [m/s²]

ID

Identifier for the model configuration

INTAREA = 'DISPGAS2DISPLIQ'

Interfacial area model selected from *InputEnums.INTAREA*

INTLENGTH = 'CONSTANT'

Interfacial length scale model selected from *InputEnums.INTLENGTH*

INTLENGTHLCST = 2E-3

Dispersed liquid constant interfacial length scale [m]

INTLENGTHVCST = 2E-3

Dispersed gas constant interfacial length scale [m]

INTNU = 'RELAXATION'

Interfacial heat transfer model selected from *InputEnums.INTNU*

INTNULCST = 2.0

Dispersed liquid constant interfacial Nusselt number [-]

INTNUVCST = 2.0

Dispersed gas constant interfacial Nusselt number [-]

INTTRANSH = 'BULK'

Interfacial enthalpy transfer model selected from *InputEnums.INTTRANSH*

KBLOCKRATIO = [00]

Blockage ratios of local perturbations [-]

KLOC = [00]

Elevation of local perturbations [m]

KLOSS = [00]

Pressure loss coefficients of local perturbations [-]

KTRELAX = NaN

Thermal relaxation time at local perturbations [s]

KTUNING = [00]

Drop deposition enhancement tuning coefficients [-]

LOCRELVEL = 'AREAMEAN'

Local relative phase velocity model selected from *InputEnums.LOCRELVEL*

MFBT = 'NONE'

Minimum Film Boiling Transition model selected from *InputEnums.MFBT*

MOECKCOEF = [1.09E-30.9891.41-1.15]

Groeneveld-Moeck post-boiling transition model coefficients [-]

MOMENTBASE = 'FULLNOP'

Base film momentum conservation model selected from *InputEnums.MOMENTBASE*

MOMENTDROP = 'SLIP'

Drop momentum conservation model selected from *InputEnums.MOMENTDROP*

MOMENTFILM = 'ALGEBRAIC'

Film momentum conservation model selected from *InputEnums.MOMENTFILM*

MOMENTGAS = 'MIXTURE'

Gas momentum conservation model selected from *InputEnums.MOMENTGAS*

MOMENTLIQUID = 'MIXTURE'

Liquid momentum conservation model selected from *InputEnums.MOMENTLIQUID*

MOMENTWAVE = 'FULL'

Wave momentum conservation model selected from *InputEnums.MOMENTWAVE*

NEARWALLEQOAF = false

Near-wall equilibrium at onset of annular two-phase flow

NEARWALLH = 'RATIO'

Near-wall equilibrium enthalpy model selected from *InputEnums.NEARWALLH*

NEARWALLHRATIO = 1.0

Near-wall equilibrium enthalpy ratio [-]

NEARWALLRATIO = 0.5

Near-wall mass flow distribution ratio [-]

NEARWALLRELAX = 'QUALITY'

Near-wall energy transfer time relaxation model selected from *InputEnums.NEARWALLRELAX*

NEARWALLRELAXCOEF = [0.1E-31/30.05]

Near-wall energy transfer time relaxation coefficients for void option

NEARWALLRELAXT = [1.00.50.30.10.1]

Near-wall energy transfer relaxation time array [s]

NEARWALLRELAXX = [-0.5-0.25-0.10.01.0]

Near-wall exchange relaxation time thermodynamic quality [-]

NNODES = 100

Number of axial nodes used in the simulation

OAF = 'WALLIS'

Onset of annular flow model selected from *InputEnums.OAF*

OAFBASERATIO = 0.5

Base/Film mass ratio at onset of annular flow [-]

OAFDROPRATIO = 0.7

Drop/Liquid mass ratio at onset of annular flow [-]

OAFENTRAINED = 'EQUILIBRIUM'

Entrained drop model at onset of annular flow selected from *InputEnums.OAFENTRAINED*

OAFFILMSPLIT = 'EQUILIBRIUM'

Film mass flow split model at onset of annular flow selected from *InputEnums.OAFFILMSPLIT*

OAFTRANSITION = [0.100.0]

Annular flow transition function parameters (sigmoid width/location wrt OAF) [m]

OKAWACOEFS = [3200.1114.79E-41]

Coefficients of Okawa entrainment model [-]

POSFILM = true

Positive film flow rate/thickness model

PROPERTIES = 'SATURATED'

Assumption model for fluid properties selected from *InputEnums.FLUIDPROPERTIES*

RANZMARSHALLCST = [20.61/21/3]

Dispersed liquid Ranz-Marshall coefficients [-]

RANZMARSHALLVCST = [20.61/21/3]

Dispersed gas Ranz-Marshall coefficients [-]

RELAXCONDCOEF = [0.1E-31/30.050.0]

Interfacial condensation time relaxation coefficients for THERMALRELAX HOMOGENEOUS model

RELAXCONDFO = [1.01.01.01.01.0].*1E-4

Interfacial condensation Fourier number array for THERMALRELAX FOURIERX model [-]

RELAXCONDFOCOEF = [5E-55E500]

Interfacial condensation time relaxation coefficients for THERMALRELAX FOURIER model

RELAXCONDT = [1.00.50.30.10.1]

Interfacial condensation relaxation time array for THERMALRELAX TIMEX model [s]

RELAXEVAPCOEF = [0.1E-31/31E-50.0]

Interfacial evaporation time relaxation coefficients for THERMALRELAX HOMOGENEOUS model

RELAXEVAPFO = [1.01.01.01.01.0].*5E-5

Interfacial evaporation Fourier number array for THERMALRELAX FOURIERX model [-]

RELAXEVAPFOCOEF = [1E-45E52210E-31]

Interfacial evaporation time relaxation coefficients for THERMALRELAX FOURIER model

RELAXEVAPT = [0.30.30.30.30.3]

Interfacial evaporation relaxation time array for THERMALRELAX TIMEX model [s]

RELAXTB = 0.2

Base film / wave mass exchange relaxation time [s]

RELAXTW = 0.2

Wave number density relaxation time [s]

RELAXX = [-0.5-0.25-0.10.01.0]

Interfacial phase change relaxation time thermodynamic quality [-]

RELVELCST = 1.0

Multiplication factor to the local relative velocity [-]

SHAPEFACTORCOEF = [1.325E5210]

Wave shape factor coefficients [-]

SLIP = 1

Phase velocity ratio [-]

SOLVERDEPENDENTPROPS = struct("VAPORFRIC", ...)

Mapping of solver-dependent properties and their values

SPHTM = 'DITTUSBOELTER'

Single-phase wall heat transfer model selected from *InputEnums.SPHTM*

SPMTM = 'BLASIUS'

Single-phase wall momentum transfer model selected from *InputEnums.SPMTM*

THERMALNONEQ = 'EQUILIBRIUM'

Thermal non-equilibrium model selected from *InputEnums.THERMALNONEQ*

THERMALRELAX = 'TIMEX'

Thermal non-equilibrium time relaxation model selected from *InputEnums.THERMALRELAX*

THINFILMFRIC = 'TURBULENT'

Thin film wall friction model selected from *InputEnums.THINFILMFRIC*

THINFILMTHICK = 1E-4

Minimum thin film thickness [m]

THINWAVETHICK = 1E-5

Minimum thin wave thickness [m]

TPFM = 'HOMOGENEOUS'

Two-phase friction multiplier model selected from *InputEnums.TPFM*

TPHTM = 'THOM'

Two-phase wall heat transfer model selected from *InputEnums.TPHTM*

TPKM = 'HOMOGENEOUS'

Two-phase local loss multiplier model selected from *InputEnums.TPKM*

VAPORFRIC = 'SOLVER_DEPENDENT'

Vapor friction model selected from *InputEnums.VAPORFRIC*

VAPORFRICCST = 0.005

Vapor friction constant [-]

VOID = 'HOMOGENEOUS'

Void fraction model selected from *InputEnums.VOID*

WAVEBASEINT = 'VAPORSHEAR'

Wave / base film interfacial momentum transfer model selected from *InputEnums.WAVEBASEINT*

WAVEDRAGCOEF = [0.021.350E50.437]

Wave drag coefficient [-]

WAVEFREQUENCY = 'RELAXATION'

Wave number density model selected from *InputEnums.WAVEFREQUENCY*

WAVEMIXCOEF = 2

Base film / wave turbulent mixing coefficient

WBOILINGN = 2

Wall boiling function exponent [-]

WBOILINGXSAT = 0

Equilibrium thermodynamic quality at onset of saturated wall boiling [-]

WBOILINGXSUB = -0.2

Equilibrium thermodynamic quality at onset of subcooled wall boiling [-]

Method Summary

static writeInputFile(filePathName, ID, varargin)

WRITEINPUTFILE Writes model configuration data to input file

Inputs: - filePathName — Path to output file - ID — Model identifier - varargin — Additional name-value pairs for model properties

class Inputs.InputSet(opts)

INPUTSET Class for managing a complete set of input objects

This class initializes and stores all input components required for a simulation run. It handles model, options, geometry, and boundary condition inputs, and sets up logging via a Session object. It also supports applying solver-dependent property modifications.

Constructor Summary

InputSet(opts)

INPUTSET Constructor for [InputSet\(\)](#) class

Initializes all input objects and sets up logging session. Parses input files and handles warnings.

Inputs:

- opts — Struct with fields:
 - modelFilePath, modelID
 - optionsFilePath, optionsID
 - geometryFilePath, geometryID
 - bcFilePath
 - LOGMODE, sessionName, sessionDirName
 - sessionParentDir, overwriteSessionFiles

Property Summary

bc

BoundaryConditions input object

geometry

Geometry input object

model

Model input object

options

Options input object

session

Session object for logging and file management

Method Summary**applySolverDependentProps**(*inputSet*, *solverName*)

APPLYSOLVERDEPENDENTPROPS Applies solver-dependent property modifications to all input objects

Useful for customizing inputs based on selected solver

Inputs:

- *inputSet* — InputSet object
- *solverName* — Name of the solver to apply dependencies for

class Inputs.**FluidProperties**(*P*, *modelObj*)

FLUIDPROPERTIES Class for managing thermophysical properties of a fluid

This class encapsulates saturated and generic fluid properties used in thermal-hydraulic simulations. Saturated properties are computed based on system pressure and stored as class properties. Generic properties (e.g., temperature, density, viscosity) are computed dynamically using enthalpy and pressure.

Constructor Summary**FluidProperties**(*P*, *modelObj*)

FLUIDPROPERTIES Constructor for *FluidProperties()* class

Initializes fluid properties at given pressure(s) using CoolProp. Saturated properties are computed and assigned to each object instance.

Inputs:

- *P* — Vector of system pressures [Pa]
- *modelObj* — Model object containing fluid name and property assumptions

Property Summary**ALPHA_F = 1**

Saturated liquid thermal diffusivity [m²/s]

ALPHA_G = 1

Saturated vapor thermal diffusivity [m²/s]

CPF = 1

Saturated liquid constant pressure specific heat [J/kg/K]

CPG = 1

Saturated vapor constant pressure specific heat [J/kg/K]

FLUID = 'WATER'

Coolprop fluid identifier

HF = 1

Saturated liquid enthalpy [J/kg]

HFG = 0

Latent heat of evaporation [J/kg]

HG = 1

Saturated vapor enthalpy [J/kg]

KF = 1
 Saturated liquid thermal conductivity [W/m/K]

KG = 1
 Saturated vapor thermal conductivity [W/m/K]

MUF = 1
 Saturated liquid viscosity [Pa.s]

MUG = 1
 Saturated vapor viscosity [Pa.s]

PCRIT = 1
 Critical pressure [Pa]

PRANDTLF = 1
 Saturated liquid Prandtl number [-]

PRANDTLG = 1
 Saturated vapor Prandtl number [-]

PRESSURE = 1
 System pressure [Pa]

PROPERTIES = 'SATURATED'
 Fluid property assumptions from *Inputs.Model.PROPERTIES*

PTRIPLE = 1
 Triple point pressure [Pa]

QK = 1
 Kutateladze characteristic heat flux [W/m²]

RHOF = 1
 Saturated liquid mass density [kg/m³]

RHOG = 1
 Saturated vapor mass density [kg/m³]

SIGMA = 1
 Surface tension [N/m]

TSAT = 1
 Saturated fluid temperature [K]

UC = 1
 Kutateladze vapor critical velocity [m/s]

Method Summary

ALPHAL(*H*)
ALPHAL() Liquid thermal diffusivity [m²/s]
 Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

ALPHAV(*H*)
ALPHAV() Vapor thermal diffusivity [m²/s]
 Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

CPL(*H*)

CPL() Liquid constant pressure specific heat [J/kg/K]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

CPV(*H*)

CPV() Vapor constant pressure specific heat [J/kg/K]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

H(*T*)

H() Fluid enthalpy at given temperature and system pressure [J/kg]

KL(*H*)

KL() Liquid thermal conductivity [W/m/K]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

KV(*H*)

KV() Vapor thermal conductivity [W/m/K]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

MUL(*H*)

MUL() Liquid dynamic viscosity [Pa.s]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

MUV(*H*)

MUV() Vapor dynamic viscosity [Pa.s]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

PRANDTLL(*H*)

PRANDTLL() Liquid Prandtl number [-]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

PRANDTLV(*H*)

PRANDTLV() Vapor Prandtl number [-]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

RHOL(*H*)

RHOL() Liquid mass density [kg/m³]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

RHOV(*H*)

RHOV() Vapor mass density [kg/m³]

Behavior depends on property assumption model (*Inputs.Model.PROPERTIES*)

T(*H*)

T() Fluid temperature at given enthalpy and system pressure [K]

Enthalpy is clamped between HMIN and HMAX to ensure valid CoolProp input

plot(*H*)

PLOT Plot properties for given enthalpy vector

transient(*param, opt*)

TRANSIENT Generate *transient()* distribution array for parameter param

class Inputs.**Geometry**(*filePath, geometryID*)

Bases: *Inputs.Input*

GEOMETRY Class for defining and managing geometrical input parameters

This class reads and stores geometrical parameters from an input file. It supports calculations of derived quantities such as hydraulic diameter, area-based diameter, and wall perimeter ratios.

Constructor Summary

Geometry(*filePath, geometryID*)

GEOMETRY Constructor for *Geometry()* class

Parses geometry input file and initializes properties. Applies default values if no input is provided.

Inputs:

- *filePath* — Path to geometry input file
- *geometryID* — Identifier for geometry configuration

Property Summary

ANGLE = 0

Inclination angle from vertical [deg]

AREA = 1

Coolant area [m²]

ID

Channel identifier

LENGTH = 1

Axial length [m]

PERIM = 1

Perimeter(s) of the channel walls [m]

Method Summary

ADIAM()

ADIAM() Diameter based on coolant cross-section area [m]

Formula: $2\sqrt{\frac{AREA}{\pi}}$

HDIAM()

HDIAM() Hydraulic diameter [m]

Formula: $4\frac{Area}{Totalperimeter}$

NWALL()

NWALL() Number of wall segments defined by PERIM

RWALL()

RWALL() Relative contribution of each wall segment to total perimeter [-]

static writeInputFile(filePathName, ID, varargin)

Writes geometry input data to file

Inputs:

- filePathName — Path to output file
- ID — Geometry identifier
- varargin — Additional name-value pairs for geometry properties

class Inputs.BoundaryConditions(filePath, geometryObjInput)

Bases: [Inputs.Input](#)

BOUNDARYCONDITIONS Class for defining and managing boundary condition parameters

This class provides access to boundary condition inputs and saturated fluid properties at system pressure. It is used by the solvers to initialize and manage inlet conditions, power distributions, and fluid state calculations.

Key Features:

- Parses input files and assigns values to boundary condition fields.
- Interfaces with fluid property libraries to compute saturation and inlet thermodynamic states.
- Validates consistency between geometry and power mesh definitions.
- Supports plotting of boundary condition time series.

Constructor Summary

BoundaryConditions(filePath, geometryObjInput)

BOUNDARYCONDITIONS Constructor for [BoundaryConditions\(\)](#) class

Parses input file and initializes boundary condition parameters. Validates consistency with geometry and applies default values if needed.

Inputs:

- filePath — Path to input file (optional)
- geometryObjInput — Geometry object for validation

Property Summary

HIN = []

Inlet enthalpy [J/kg]

MFLOW = []

Mass flow rate [kg/s]

POWER = 0

Total power [W]

PRESSURE = []

System pressure [Pa]

TIME

Simulation time [s]

WMESH = 1

Node size distribution [m]

WPOWER = 1

Relative power distribution(s) [-]

Method Summary

DHIN(*fluidObj*)

DHIN() Inlet subcooling enthalpy (HF - HIN) [J/kg]

DTIN(*fluidObj*)

DTIN() Inlet subcooling temperature (TSAT - TIN) [K]

HF(*fluidObj*)

HF() Liquid saturated enthalpy at system pressure using fluid object [J/kg]

HG(*fluidObj*)

HG() Vapor saturated enthalpy at system pressure using fluid object [J/kg]

TIN(*fluidObj*)

TIN() Inlet temperature based on inlet enthalpy and pressure [K]

TSAT(*fluidObj*)

TSAT() Saturation temperature at system pressure using fluid object [K]

XIN(*fluidObj*)

XIN() Inlet equilibrium quality [-]

plot(*fluidObj*, *opt*)

PLOT Generates plots of boundary condition parameters over time

Inputs:

- *fluidObj* — FluidProperties object
- *opt* — Struct with fields:
 - *display* — Cell array of parameters to *plot()*
 - *tIdx* — Time indices to include
 - *unitTemp* — Temperature unit ('K' or 'C')

static writeInputFile(*filePathName*, *TIME*, *PRESSURE*, *HIN*, *MFLOW*, *varargin*)

WRITEINPUTFILE Writes boundary condition data to input file

Inputs:

- *filePathName* — Path to output file
- *TIME* — Time value [s]
- *PRESSURE* — Pressure value [Pa]
- *HIN* — Enthalpy value [J/kg]
- *MFLOW* — Mass flow rate [kg/s]
- *varargin* — Additional name-value pairs

class Inputs.Options(*filePath*, *optionsID*)

Bases: *Inputs.Input*

OPTIONS Class for defining and managing numerical solver options

This class encapsulates all numerical parameters used in the simulations, including time stepping, convergence criteria, and relaxation factors for all solvers. It reads options data from input files and handles default values as well as input validation.

Constructor Summary

Options(*filePath*, *optionsID*)

OPTIONS Constructor for *Options()* class

Parses options input file and initializes properties. Applies default values and validates entries.

Inputs:

- *filePath* — Path to options input file
- *optionsID* — Identifier for options configuration

Property Summary

AXIALINTERP = 'next'

Method for interpolating axial power distribution

ENTNUMMAXITER = 100

Maximum number of iterations to find film entrainment number

ENTNUMTOL = 1E-2

Target velocity slip tolerance to find film entrainment number [-]

ERRORFW = 1E-1

Wave frequency error target in inner iterations [Hz]

ERRORH = 1E-1

Enthalpy error target in inner iterations [J/kg]

ERRORP = 1E-1

Pressure error target in inner iterations [Pa]

ERRORU = 1E-4

Velocity error target in inner iterations [m/s]

ERRORUD = 1E-2

Drop velocity error target in inner iterations [m/s]

ERRORUF = 1E-2

Film velocity error target in inner iterations [m/s]

ERRORW = 1E-3

Mass flow rate error target in inner iterations [kg/s]

ERRORWF = 1E-4

Film mass flow rate error target in inner iterations [kg/s/m]

ID

Identifier for the options configuration

MAXITER = 100

Maximum number of inner iterations per time step

MRMTIMEINT = 'EXPONENTIAL'

Time integration for vapor mass/energy conservation equations

OAFEQUILMAXITER = 100

Maximum number of iterations to find equilibrium at onset of annular flow

OAFEQUILTOL = 1E-4

Target film mass flow rate tolerance to find equilibrium at onset of annular flow [kg/s/m]

RELAXFW = 0.5

Relaxation factor for the wave number conservation equation [-]

RELAXHL = 1

Relaxation factor for the liquid energy conservation equation [-]

RELAXHM = 1.0

Relaxation factor for the mixture energy conservation equation [-]

RELAXHV = 0.7

Relaxation factor for the vapor energy conservation equation [-]

RELAXPM = 1.0

Relaxation factor for the mixture momentum conservation equation [-]

RELAXUB = 0.2

Relaxation factor for the base momentum conservation equation [-]

RELAXUD = 0.2

Relaxation factor for the drop momentum conservation equation [-]

RELAXUF = 0.2

Relaxation factor for the film momentum conservation equation [-]

RELAXUL = 0.5

Relaxation factor for the liquid momentum conservation equation [-]

RELAXUV = 0.5

Relaxation factor for the vapor momentum conservation equation [-]

RELAXUW = 0.2

Relaxation factor for the wave momentum conservation equation [-]

RELAXWB = 0.5

Relaxation factor for the base mass conservation equation [-]

RELAXWF = 0.5

Relaxation factor for the film mass conservation equation [-]

RELAXWL = 1

Relaxation factor for the liquid mass conservation equation [-]

RELAXWM = 1.0

Relaxation factor for the mixture mass conservation equation [-]

RELAXWV = 0.7

Relaxation factor for the vapor mass conservation equation [-]

RELAXWW = 0.5

Relaxation factor for the wave mass conservation equation [-]

SSCONVFW = 1E-1

Wave frequency steady-state convergence criterion [Hz]

SSCONVH = 1E+0

Enthalpy steady-state convergence criterion [J/kg]

SSCONVP = 1E+0

Pressure steady-state convergence criterion [Pa]

SSCONVU = 1E-3

Velocity steady-state convergence criterion [m/s]

SSCONVUD = 1E-2

Drop velocity steady-state convergence criterion [m/s]

SSCONVUF = 1E-2

Film velocity steady-state convergence criterion [m/s]

SSCONVW = 1E-3

Mass flow rate steady-state convergence criterion [kg/s]

SSCONVWF = 1E-4

Film mass flow rate steady-state convergence criterion [kg/s/m]

SSMAXITER = 30

Maximum number of steady-state iterations

SSTSTEP = 1.0

Time step size for steady-state iterations [s]

TIMEINTERP = 'linear'

Method for interpolating time-dependent boundary conditions

TSTEP = 0.1

Time step size for transient simulations [s]

UDEQUILMAXITER = 100

Maximum number of iterations to find droplet equilibrium velocity

UDEQUILTOL = 1E-3

Target drop force tolerance to find droplet equilibrium velocity [N/m³]

UFEQUILMAXITER = 100

Maximum number of iterations to find film equilibrium velocity

UFEQUILTOL = 1E-3

Target film force tolerance to find film equilibrium velocity [N/m³]

Method Summary

static writeInputFile(filePathName, ID, varargin)

Writes numerical options configuration to input file

Inputs:

- filePathName — Path to output file
- ID — Options identifier
- varargin — Additional name-value pairs for options properties

class Inputs.Input(inputFilePath, key, val)

Bases: dynamicprops, matlab.mixin.Copyable

INPUT Abstract superclass for input-handling classes

Provides core functionality for reading, validating, and managing structured input data. Supports dynamic property creation, default value handling, and solver-dependent customization.

Constructor Summary

Input (*inputFilePath*, *key*, *val*)

INPUT Constructor for [Input\(\)](#) class

Parses input file and optionally filters entries by key-value pair. Adds dynamic property 'inputStruct' to hold parsed data.

Inputs:

- *inputFilePath* — Path to input file (string)
- *key* — Field name to filter by (optional)
- *val* — Value to match for filtering (optional)

Property Summary

extra = struct.empty()

Stores unused input entries for reference or debugging

warnings = struct.empty()

Stores warning messages generated during input parsing

Method Summary

applySolverDependentProperties (*solverName*)

APPLYSOLVERDEPENDENTPROPERTIES Applies solver-specific property values based on dependency flags

Returns a copy of the object with updated properties

static convert2JSON (*inputFilePath*)

CONVERT2JSON Converts input file contents to JSON-formatted string

defaultValueUsedReport (*propNames*, *propValues*)

DEFAULTVALUEUSEDREPORT Generates report of properties for which default values were used

Can print or return formatted string array

listInputProperties (*opts*)

LISTINPUTPROPERTIES Returns list of protected property names for the object

Optionally excludes specified properties

static validateFunctionHandleInput (*handleString*)

VALIDATEFUNCTIONHANDLEINPUT Validates safety of function handle string

Throws error if restricted keywords are detected

validateInputEntry (*objPropname*, *opts*)

VALIDATEINPUTENTRY Validates whether a property entry is specified or defaulted

Throws error if required entry is missing or empty Returns flags and default value if applicable

static writeInputFile (*filePathName*, *fdMode*, *varargin*)

WRITEINPUTFILE Writes input data to file in standard format

Accepts name-value pairs and optional file mode

5.3 Session

The `Session` package in **OpenSTREAM** provides tools for managing session environments, logging, and file structure during simulations. It helps ensure organized workflows and consistent runtime behavior. This includes:

- Session logging manager `Session.Log`
 - Logging mode: `Session.LogMode`
 - Session management: `Session.Session`
-

`Session.isLegalPath(pathname)`

ISLEGALPATH Checks if the given pathname is valid on the current operating system

This function attempts to convert the input pathname to a Java Path object. If the conversion succeeds, the path is considered legal and the function returns true. If an exception is thrown during conversion, the path is considered illegal and the function returns false.

Input:

- `pathname` — String representing the file or folder path to validate

Output:

- `bool` — Logical true if the path is valid, false otherwise

`class Session.Session(opt)`

Bases: `handle`

SESSION Class for managing simulation session metadata and logging

This class handles session naming, directory creation, and logging setup. It ensures that session directories are valid and manages file overwriting behavior. It also integrates with the `Log` class to control output and warnings.

Constructor Summary

`Session(opt)`

SESSION Constructor for `Session()` class

Initializes session metadata including name, directory, and overwrite behavior.

Inputs:

- `opt.name` — `Session()` name
- `opt.dirName` — Directory name
- `opt.parentDir` — Parent directory path
- `opt.overwriteFiles` — Flag to allow overwriting existing session files

Property Summary

`dirName` = ""

Name of the session directory

`directory`

Full path to the session *directory*

`log`

Log object for managing output and warnings

`name` = ""

Name of the session (used for log file naming)

overwriteFiles = false

Flag to allow overwriting existing session files

parentDir = ""

Parent directory where session folder will be created

showWarnings = true

Flag to control display of warnings

Method Summary

makeSessionDirectory()

MAKESESSIONDIRECTORY Creates the session directory

Validates path legality, handles existing directories, closes open files and diaries if necessary, and creates the directory.

setupLog(session, LOGMODE, opts)

SETUPLOG Initializes the Log object for the session

Inputs:

- LOGMODE — Logging mode (console, file, both, none)
- opts.LOGFID — Optional file ID for logging

class Session.Log(LOGMODE, opts)

Bases: `handle`

LOG Console and file-based logging manager

This class manages logging of messages and warnings to console and/or file. It supports multiple logging modes, integrates with session management, and provides diary functionality for persistent logs.

Constructor Summary

Log(LOGMODE, opts)

LOG Constructor for `Log()` class

Initializes logging mode, session context, and warning settings. Creates log file and session directory if needed.

Inputs:

- LOGMODE — Determines where logs go (console, file, both, or none), using `Session.LogMode`.
- opt.session — Session object for directory and naming context
- opt.LOGFID — File ID for the open log file.
- opt.showWarnings — Whether to display warnings in the console.

Property Summary

LOGFID = -1

log file

LOGMODE

Logging mode (console only, file only, both, or none)

diaryFilePath

Full path to the diary file

diaryIsOn = false

Flag indicating whether MATLAB diary is active

logFileName

Name of the log file (without extension)

logFilePath

Full path to the log file

showWarnings = true

Warnings

Method Summary**closeLog()**

CLOSELOG Closes log file and resets file ID

delete()

DELETE Destructor for log class

Ensures log file and diary are properly closed

diaryOff()

DIARYOFF Turn off diary logging

diaryOn()

DIARYON Turn on diary logging if applicable

TODO: re-evaluate if using the diary is needed. What does the diary do that the log doesn't? If we find a use case, describe it here.

log(varargin)

LOG Logs messages to console and/or file depending on LOGMODE

openLog(opts)

OPENLOG Opens log file for writing

Optionally keeps it open between writes

warning(varargin)

WARNING Logs warnings using MATLAB's built-in *warning()* mechanism

Optionally writes to file if warnings are suppressed

class Session.LogMode

Bases: uint16

LOGMODE Enumeration of supported logging modes

This enumeration defines the available logging configurations for the simulation framework. It is used to control whether messages and warnings are printed to the console, written to a file, both, or suppressed entirely.

5.4 Solver

The *Solvers* package defines the **OpenSTREAM** core solver modules and their associated superclasses. Each solver is tailored to simulate specific regimes of one-dimensional, two-phase flow, with increasing levels of physical detail and complexity.

Explore the available solver frameworks:

5.4.1 Four-field solver

The `Solvers.FourField` module in **OpenSTREAM** provides tools and classes for simulating annular two-phase flows using a four-field approach. In this method, the liquid phase is represented by three distinct fields (base film, disturbance waves, and droplets), each governed by its own set of conservation equations for mass, momentum, and energy. The vapor phase is solved using the `Solvers.Mixture` module. This framework enables detailed modeling of liquid film structures, interfacial interactions, slip velocities, and hydrodynamic non-equilibrium effects, making it particularly suitable for annular boiling two-phase flow regimes where field separation and wave dynamics play a significant role.

This module includes:

- Solver class for four-field-based flow simulations: `Solvers.FourField.FourFieldSolver`
- Base film, wave and drop field classes: `Solvers.FourField.Base`, `Solvers.FourField.Wave`, `Solvers.FourField.Drop`
- Film class: `Solvers.FourField.Film`
- Four-field primary properties: `Solvers.FourField.Base.W`, `Solvers.FourField.Base.U`, `Solvers.FourField.Base.H`, `Solvers.FourField.Wave.W`, `Solvers.FourField.Wave.U`, `Solvers.FourField.Wave.H`
- Key solver class methods: `Solvers.FourField.FourFieldSolver.initializeSolver`, `Solvers.FourField.FourFieldSolver.solve()`, `Solvers.FourField.FourFieldSolver.plotz`
- Field class methods for calculating secondary four-field parameters.

class `Solvers.FourField.FourFieldSolver`(*inputSet*, *mixSolver*)

Bases: `Solvers.ThreeField.ThreeFieldSolver.ThreeFieldSolver`

FOURFIELDSOLVER Solver for initializing, solving, and visualizing four-field flow.

The `FourFieldSolver` class extends `Solvers.ThreeField.ThreeFieldSolver` to handle thermal-hydraulic simulations using a four-field approach.

Responsibilities:

- Initializes solver parameters from an `Inputs.InputSet` object
- Constructs `Solvers.FourField.Film`, `Solvers.FourField.Wave`, `Solvers.FourField.Base`, and `Solvers.FourField.Drop` objects for transient and steady-state analysis
- Provides plotting utilities for spatial and temporal distributions

Key Components:

- Liquid film (base and wave) and drop field construction
- Field mass/momentum/energy transport modeling
- Pressure drop gradient obtained from mixture solver
- Vapor solution obtained from mixture solver
- Visualization of results across time and axial domains

Constructor Summary

FourFieldSolver(*inputSet, mixSolver*)

FOURFIELDSOLVER Constructor for the FourFieldSolver() class.

Creates a new instance of the FourFieldSolver() class using the provided *Inputs.InputSet*. This constructor initializes the solver by calling the *Solvers.ThreeField.ThreeFieldSolver* constructor and then sets up all necessary solver parameters and internal data structures via *Solvers.FourField.FourFieldSolver.initializeSolver*

Input:

- *inputSet* — An *Inputs.InputSet* object containing model configuration, geometry, boundary conditions, and solver options.
- *mixSolver* — An optional solved *Solvers.Mixture.MixtureSolver* object

Notes:

- This constructor assumes *Inputs.InputSet* is fully validated.
- A solved mixture-model solution is required for initialization. If a mixture solver is supplied, its solution is reused. If no mixture solver is supplied, one is created and solved automatically.
- Initialization includes film, wave, base, and droplet objects.
- Time/space discretization and boundary condition interpolation initialized by *Solvers.Mixture.MixtureSolver* are used

Method Summary**EQUIL**(*ffSolver, flm, drp, mix, zIdx*)

EQUIL() Find entrained ratio at film/drop equilibrium state [-]

Iteratively computes the entrained liquid ratio at the onset of annular flow by balancing droplet deposition and film entrainment rates.

Inputs:

- *ffSolver* — *Solvers.FourField.FourFieldSolver* object
- *flm* — Film field object (*Solvers.FourField.Film*)
- *drp* — Droplet field object (*Solvers.FourField.Drop*)
- *mix* — Mixture object (*Solvers.Mixture.Mixture*)
- *zIdx* — Axial index for evaluation

Algorithm:

- Starts with an initial guess for droplet mass flow (50% of liquid mass)
- Updates droplet and film flow rates iteratively
- Computes difference between deposition and entrainment rates
- Stops when error < 1e-4 kg/s/m or after 100 iterations

Notes:

- Enhanced drop deposition is disabled
- Uses interpolation for improved convergence
- Falls back to ad-hoc update if interpolation fails
- Method update film and drop mass flow rates. Use *flm.copy()* and *drp.copy()* as input arguments if used in post-process

initializeSolver(*ffSolver*)

INITIALIZESOLVER Initializes solver parameters and constructs film, wave, base, and drop objects.

Sets up the solver's internal state using the provided *Inputs.InputSet* and mixture solver solutions. Includes initialization of data structures for transient and steady-state simulations.

Operations performed:

- Initializes film, wave, and base arrays for each time step
- Initializes fluid property objects
- Sets up mixture object
- Assigns initial conditions for field mass flow rates, velocities, enthalpies

Notes:

- Assumes uniform axial discretization.
- Wall heat flux computed from interpolated boundary conditions.

plott(*ffSolver*, *zIdx*, *opt*)

PLOTT Plots temporal distributions of four-field parameters at a given axial location.

Generates time-series plots of selected four-field parameters at a specified axial index. The method supports multiple display modes, wall selections, and plot arrangements.

Inputs:

- *ffSolver* — [Solvers.FourField.FourFieldSolver](#) object containing simulation data
- *zIdx* — Axial index (scalar, positive integer)
- *opt.display* — Parameters to display (e.g., 'HFLUX', 'W', 'U', etc.)
- *opt.solveMode* — Solve mode: 'REAL' or 'NULL'
- *opt.wall* — Wall index(es) to plot
- *opt.tIdx* — Time indices to include in the plot
- *opt.reverseTime* — Logical flag to reverse time axis
- *opt.unitTemp* — Temperature unit: 'K' or 'C'
- *opt.arrangement* — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- *opt.resize* — Resize factor for plot scaling

Notes:

- If *opt.display* is set to 'ALL', all supported parameters are plotted.
- The function validates that at least two time indices are provided.
- Temperature unit conversion is applied if 'C' is selected.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.

plotz(*ffSolver*, *tIdx*, *opts*)

PLOTZ Plots spatial (axial) distributions of four-field parameters.

Generates axial plots of selected four-field parameters at specified time indices. The function supports multiple display modes, wall selections, and optional animation over time.

Inputs:

- *ffSolver* — [Solvers.FourField.FourFieldSolver](#) object containing simulation data
- *tIdx* — Time index or indices (vector of positive integers)
- *opts.display* — Parameters to display (e.g., 'HFLUX', 'W', 'U', etc.)
- *opts.solveMode* — Solve mode: 'REAL' or 'NULL'
- *opts.wall* — Wall index(es) to plot
- *opts.zIdx* — Axial indices to include in the plot
- *opts.obstructions* — Logical flag to display obstruction locations
- *opts.annular* — Logical flag to plot annular two-phase flow parameters only from the onset of annular flow
- *opts.unitTemp* — Temperature unit: 'K' or 'C'
- *opts.arrangement* — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- *opts.resize* — Resize factor for plot scaling

Notes:

- If multiple time indices are provided, the function creates an animated plot.
- At least two axial indices must be specified to enable spatial plotting.
- Temperature unit conversion is applied if 'C' is selected.
- Obstruction locations are plotted if *opts.obstructions* is true.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.

plotzt(*ffSolver*, *opt*)

PLOTZT Plots 2D time/elevation distributions of four-field parameters.

Generates surface plots of selected four-field related parameters over time and axial (elevation) positions. It supports plotting for base and wave fields, and can handle both real and null (initial) transient

data.

Inputs:

- `ffSolver` — [Solvers.FourField.FourFieldSolver](#) object containing simulation data
- `opt.display` — Parameter(s) to display (e.g., 'HFLUX', 'U', etc.)
- `opt.label` — Corresponding labels for display parameters
- `opt.unit` — Units for each parameter
- `opt.field` — Field to plot: 'film' or 'drop'
- `opt.solveMode` — Solve mode: 'REAL' or 'NULL'
- `opt.wall` — Wall index(es) to plot
- `opt.zIdx` — Axial indices (must include at least 2)
- `opt.tIdx` — Time indices (must include at least 2)
- `opt.reverseTime` — Logical flag to reverse time axis
- `opt.shading` — Surface shading style: 'faceted', 'flat', or 'interp'
- `opt.view` — View angle for 3D plot [azimuth elevation]

Notes:

- If 'ALL' is passed to `opt.display`, all supported parameters are plotted.
- The function validates that at least two axial and time indices are provided to enable meaningful 2D plotting.
- Each wall is plotted in a separate figure window with appropriate titles and subplot grouping.

`solve(ffSolver)`

SOLVE Executes the four-field solver for steady-state and transient simulations.

Runs the full solution process for the [Solvers.FourField.FourFieldSolver](#) object. It manages initialization, time stepping, axial sweeps, inner iterations, convergence checks, and logging.

Workflow:

- Enables logging and opens persistent log file
- Validates solver state before execution
- Runs steady-state initialization (`solveINIT = true`)
- If initial step converges, proceeds with transient simulation
- Handles exceptions and ensures proper log closure

Notes:

- Uses internal *solver* function to handle both steady-state and transient modes
- Applies relaxation factors for phase flows, velocities, and enthalpies
- Supports thermal non-equilibrium modeling
- Logs progress and outputs to session directory

class Solvers.FourField.Drop

Bases: [Solvers.ThreeField.Drop](#)

DROP Class for modeling drop field in four-field solver

This class encapsulates the physical and numerical properties of the drop field, including flow variables and phase interactions. It supports multiple solver models and provides methods for computing derived quantities.

All methods are inherited from [Solvers.ThreeField.Drop](#)

class Solvers.FourField.Film(inputSet, fluid)

Bases: [Solvers.AbstractFilm](#)

FILM Class for modeling liquid film in four-field solver

Represents the physical and numerical properties of the liquid film in a four-field thermal-hydraulic solver. The film consists of two sub-fields: base film and wave film, which interact with vapor and droplet phases.

This class manages initialization, mass flow distribution, velocity and enthalpy calculations, and provides methods for splitting film flow at the onset of annular flow (OAF) according to model options.

Key responsibilities:

- Store solver state (time, axial positions, heat flux, evaporation flux)
- Aggregate flow properties from base and wave components
- Initialize base and wave fields and apply OAF split logic
- Support equilibrium calculations for film thickness and flow ratios

Constructor Summary

Film(*inputSet*, *fluid*)

FILM Creates a *Film*() object

Initializes the film object with input set and fluid properties required by the four-field solver.

Inputs:

- *inputSet* — *Inputs.InputSet* object containing geometry and model options
- *fluid* — *Inputs.FluidProperties* object containing thermophysical properties

Property Summary

DT = 0

Time step size [s] from *Inputs.options.TSTEP*

H

) double {mustBeNumeric} = 1E6 % Enthalpy [J/kg]

Type

(

Type

,

HFLUX = 1.

Film heat flux [W/m^2]

ITR

Iteration tracking

MEVAP = -1.

Evaporation mass flux [kg/s/m^2]

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIDX = 1

Time step index [-]

TIME = 0

Time series [s]

U

) double {mustBeNumeric} = 1. % Velocity [m/s]

Type

(

```

    Type
    ,
W
    ) double {mustBeNumeric} = 1. % Mass flow rate [kg/s]
    Type
    (
    Type
    ,
Z = 1.
    Elevation [m]
base = NaN
    Solvers.FourField.Base object
wave = NaN
    Solvers.FourField.Wave object
```

Method Summary

EQUIL(*film*, *Wf*, *zIdx*)

EQUIL() Equilibrium base/wave split ratio [-]

Calculates the equilibrium split ratio of film mass flow between base and wave using base equilibrium thickness, base velocity, perimeter, and liquid density.

Inputs:

- *film* — *Solvers.FourField.Film* object
- *Wf* — Film mass flow rate [kg/s]
- *zIdx* — Axial indices to evaluate (optional)

copyFlowProperties(*srcObj*, *targetObj*, *opts*)

COPYFLOWPROPERTIES Copy flow properties between Film objects

Copies flow-related properties (base and wave fields) from a source Film object to one or more target Film objects. Ensures that spatial meshes match before copying.

Inputs:

- *srcObj* — Source *Solvers.FourField.Film* object
- *targetObj* — Target *Solvers.FourField.Film* object(s)
- *opts.all* — Logical flag to copy all properties (optional, default = false)

distributeOAFW(*film*, *Wf*, *zIdx*)

DISTRIBUTEOAFW Distribute film mass flow at OAF [kg/s]

Distributes the film mass flow rate at the onset of annular flow (OAF) between base and wave fields according to the *Inputs.Model.OAFFILMSPLIT* model option.

Inputs:

- *film* — *Solvers.FourField.Film* object
- *Wf* — Film mass flow rate to distribute [kg/s] (per wall or per z-index)
- *zIdx* — Axial indices to apply distribution (optional)

initializeBaseAndWave(*film*, *WIN*, *ITR*)

INITIALIZEBASEANDWAVE Initialize base and wave fields

Initializes base and wave arrays given inlet film mass flow rate and iteration settings. Performs OAF film split, computes velocities, enthalpies, and sets initial wave frequency using equilibrium correlations.

Inputs:

- film — *Solvers.FourField.Film* object
- WIN — Inlet film mass flow rate [kg/s]
- ITR — Iteration control struct

class *Solvers.FourField.Wave*(film)

Bases: *Solvers.AbstractFilm*

WAVE Class for modeling liquid waves in four-field solver

This class represents the physical and numerical properties of the wave field in a four-field thermal-hydraulic solver. It handles flow variables, heat transfer, and phase interaction mechanisms between the waves and other fields (base film, drop, vapor).

The class supports initialization from an existing film object and provides methods for computing derived quantities such as mass flow fractions, interfacial fractions, and deposition/evaporation fluxes.

Constructor Summary

Wave(film)

WAVE Constructor for *Wave()* class

Initializes the wave object using properties from an existing film object. Copies solver configuration and geometry data, and allocates flow property arrays for multiple walls.

Inputs:

- film — *Solvers.AbstractFilm* object representing the parent film

Property Summary

DT = 0

Time step size [s] from *Inputs.options.TSTEP*

FREQUENCY = 1

Wave frequency [Hz]

H = 1E6

Enthalpy [J/kg]

HFLUX = 1.

Film heat flux [W/m^2]

ITR

Iteration tracking

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIDX = 1

Time step index [-]

TIME = 0

Time series [s]

U = 1.

Velocity [m/s]

W = 1.

Mass flow rate [kg/s]

Z = 1.

Elevation [m]

film

Parent *Solvers.AbstractFilm* object

Method Summary

AMPLITUDE(*wave*, *zIdx*)

AMPLITUDE() Wave amplitude [m]

Computes wave amplitude based on mass flow, shape factor, fluid density, and frequency. Limited by half the hydraulic diameter.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

BETA(*wave*, *zIdx*)

BETA() Wave film interfacial fraction [-]

Calculates the fraction of the interface occupied by waves.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

BETAP(*wave*, *zIdx*)

BETAP() Wave film heat flux fraction [-]

Determines the fraction of heat flux directed to the waves.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

DELTA_FREQ(*wave*, *zIdx*)

DELTA_FREQ() Wave frequency source/sink [Hz/m]

Calculates the exchange term for wave frequency relaxation toward equilibrium frequency.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

DRAGCOEF(*wave*, *zIdx*)

DRAGCOEF() Wave drag coefficient [-]

Calculates the drag coefficient for vapor-wave interaction based on Reynolds number and *Inputs.Model.WAVEDRAGCOEF* model coefficients.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

EPSILON(*wave*, *zIdx*)*EPSILON()* Wave mass flow fraction [-]

Computes the fraction of total film mass flow carried by waves.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

EQFREQUENCY(*wave*, *zIdx*)*EQFREQUENCY()* Equilibrium wave frequency [Hz]

Computes the equilibrium wave frequency based on Strouhal number and vapor velocity.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

EQPERIOD(*wave*, *zIdx*)*EQPERIOD()* Equilibrium wave period [s]

Calculates the inverse of equilibrium wave frequency.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

EQSTROUHAL(*wave*, *zIdx*)*EQSTROUHAL()* Equilibrium Strouhal number [-]Calculates the equilibrium Strouhal number using selected *Inputs.Model.EQSTROUHAL* model.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

ETA(*wave*, *zIdx*)*ETA()* Wave film deposition fraction [-]

Computes the fraction of deposition attributed to waves.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

FBASE(*wave*, *drop*, *zIdx*)*FBASE()* Base interfacial force [N/m²]

Computes the interfacial force between wave and base film.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *drop* — *Solvers.FourField.Drop* object
- *zIdx* — Axial indices to evaluate (optional)

FBASEMASS(*wave*, *drop*, *zIdx*)*FBASEMASS()* Wave mass exchange force [N/m²]

Calculates the force due to mass exchange between wave and base film.

Inputs:

- *wave* — *Solvers.FourField.Wave* object

- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FDEP(*wave, drop, zIdx*)

FDEP() Droplet interaction force [N/m²]

Calculates the force due to deposition of droplets onto the wave field.

Inputs:

- wave — *Solvers.FourField.Wave* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FDRAG(*wave, zIdx*)

FDRAG() Vapor drag force on wave [N/m²]

Calculates the shear stress due to vapor drag acting on the wave field.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

FSHEAR(*wave, zIdx*)

FSHEAR() Vapor shear stress [N/m²]

Computes the shear stress exerted by vapor on the wave field using friction factor.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

FTOT(*wave, drop, zIdx*)

FTOT() Total interfacial force [N/m²]

Computes the total interfacial force acting on the wave field, including vapor shear, buoyancy, gravity, droplet interaction, and base film forces.

Inputs:

- wave — *Solvers.FourField.Wave* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FVAPOR(*wave, zIdx*)

FVAPOR() Wave vapor shear stress [N/m²]

Computes the total shear stress exerted by vapor on the wave field.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

FWALL(*wave, zIdx*)

FWALL() Wave wall shear stress [N/m²]

Placeholder method. Wave does not implement wall shear stress.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

MBASE(*wave, drop, zIdx*)

MBASE() Mass flux interaction with base film [kg/m²/s]

Computes the net mass exchange between wave and base film, including turbulent mixing.

Inputs:

- wave — *Solvers.FourField.Wave* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

MDEP(*wave, drop, zIdx*)

MDEP() Wave deposition mass flux [kg/m²/s]

Computes the deposition mass flux from waves to droplets.

Inputs:

- wave — *Solvers.FourField.Wave* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

MENT(*wave, zIdx*)

MENT() Wave entrainment mass flux [kg/m²/s]

Computes the mass flux associated with entrainment from the wave field.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

MEVAP(*wave, zIdx*)

MEVAP() Wave evaporation mass flux [kg/m²/s]

Calculates the evaporation mass flux from the wave field based on heat flux fraction.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

MTOT(*wave, drop, zIdx*)

MTOT() Total mass flux [kg/m²/s]

Calculates the total mass flux for the wave field, including evaporation, entrainment, deposition, and interactions with base film.

Inputs:

- wave — *Solvers.FourField.Wave* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

MTURB(*wave, zIdx*)

MTURB() Turbulent mass exchange [kg/m²/s]

Returns the turbulent mixing mass flux between wave and base film.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

N(*wave, zIdx*)

N() Wave number density [1/m]

Computes the number of waves per unit length (inverse of wave spacing).

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

PERIOD(wave, zIdx)

PERIOD() Wave time period [s]

Calculates the inverse of wave frequency.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

REV(wave, zIdx)

REV() Vapor Reynolds number (relative to waves) [-]

Computes the Reynolds number for vapor flow interacting with waves.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

SHAPEFACTOR(wave, zIdx)

SHAPEFACTOR() Wave shape factor [-]

Computes the dimensionless shape factor for waves based on Reynolds number and *Inputs.Model.SHAPEFACTORCOEF* model.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

SPACING(wave, zIdx)

SPACING() Wave spacing [m]

Computes the axial spacing between waves based on velocity and frequency.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

THICK(wave, zIdx)

THICK() Wave thickness [m]

Computes the wave thickness as the product of amplitude and interfacial fraction. This parameter represents the thickness that the wave field would have if fully overlayed on top of the base film.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

WIDTH(wave, zIdx)

WIDTH() Wave width [m]

Calculates wave width as a function of amplitude and shape factor, limited by wave spacing.

Inputs:

- wave — *Solvers.FourField.Wave* object
- zIdx — Axial indices to evaluate (optional)

WL(*wave*, *zIdx*)

WL() Wave mass flow rate per unit perimeter [kg/s/m]

Calculates the mass flow carried by the wave field at specified axial locations.

Inputs:

- *wave* — *Solvers.FourField.Wave* object
- *zIdx* — Axial indices to evaluate (optional)

class Solvers.FourField.Base(*film*)

Bases: *Solvers.AbstractFilm*

BASE Class for modeling liquid base film in four-field solver

This class represents the physical and numerical properties of the base film field in a four-field thermal-hydraulic solver. It handles flow variables, heat transfer, and phase interaction mechanisms between the base film and other fields (wave, drop, vapor).

The class supports initialization from an existing film object and provides methods for computing derived quantities such as mass flow fractions, interfacial fractions, and deposition/evaporation fluxes.

Constructor Summary

Base(*film*)

BASE Constructor for *Base()* class

Initializes the base film object using properties from an existing film object. Copies solver configuration and geometry data, and allocates flow property arrays for multiple walls.

Inputs:

- *film* — *Solvers.AbstractFilm* object representing the parent film

Property Summary

DT = 0

Time step size [s] from *Inputs.options.TSTEP*

H = 1E6

Enthalpy [J/kg]

HFLUX = 1.

Film heat flux [W/m^2]

ITR

Iteration tracking

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIDX = 1

Time step index [-]

TIME = 0

Time series [s]

U = 1.

Velocity [m/s]

W = 1.

Mass flow rate [kg/s]

Z = 1.

Elevation [m]

Method Summary

BETA(*base, zIdx*)

BETA() Base film interfacial fraction [-]

Computes the fraction of interfacial area attributed to the base film.

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

BETAP(*base, zIdx*)

BETAP() Base film heat flux fraction [-]

Determines the fraction of heat flux directed to the base film. Applies suppression in pre-annular region to enforce target base film thickness at onset of annular flow (OAF).

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Applies suppression in pre-annular region to enforce target base film thickness at onset of annular flow (OAF). The entire film evaporation hence occurs in the wave field.

EPSILON(*base, zIdx*)

EPSILON() Base film mass flow fraction [-]

Calculates the fraction of total film mass flow carried by the base film at specified axial locations.

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

EQTHICK(*base, zIdx*)

EQTHICK() Base equilibrium thickness [m]

Computes the equilibrium thickness of the base film based on *Inputs.Model.BASEEQTHICK* model.

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

ETA(*base, zIdx*)

ETA() Base film deposition fraction [-]

Computes the fraction of drop deposition allocated to the base film based on mass flow and interfacial fractions.

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

FBASEVAPOR(*base, zIdx*)*FBASEVAPOR()* Vapor shear stress on base film [N/m²]

Computes the shear force exerted by the vapor phase on the base film, scaled by interfacial fraction.

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

FDEP(*base, drop, zIdx*)*FDEP()* Droplet deposition force [N/m²]

Calculates the momentum transfer to the base film from droplet deposition, based on velocity difference and deposition flux.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FDRY(*base, drop, zIdx*)*FDRY()* Dry fraction [-]

Computes the fraction of the exposure period during which the base film is expected to be dry.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FTOT(*base, drop, zIdx*)*FTOT()* Total interfacial force on base film [N/m²]

Aggregates all force contributions acting on the base film, including wall shear, wave interaction, vapor shear, buoyancy, gravity, and droplet deposition based on *Inputs.Model.MOMENTBASE* model.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FWAVE(*base, drop, zIdx*)*FWAVE()* Wave interfacial force [N/m²]

Computes the interfacial shear force exerted by the wave field on the base film based on *Inputs.Model.WAVEBASEINT* model.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FWAVEMASS(*base, drop, zIdx*)*FWAVEMASS()* Base-wave mass exchange force [N/m²]

Calculates the momentum exchange force due to mass transfer between base and wave films, based on velocity difference and MWAVE flux.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object

- `zIdx` — Axial indices to evaluate (optional)

MDEP(*base, drop, zIdx*)

MDEP() Base deposition mass flux [kg/m²/s]

Calculates deposition flux to the base film from drops, scaled by deposition fraction.

Inputs:

- `base` — *Solvers.FourField.Base* object
- `zIdx` — Axial indices to evaluate (optional)

MENT(*base, zIdx*)

MENT() Base entrainment mass flux [kg/m²/s]

Placeholder for future implementation if needed. Currently returns zero entrainment.

Inputs:

- `base` — *Solvers.FourField.Base* object
- `zIdx` — Axial indices to evaluate (optional)

MEVAP(*base, zIdx*)

MEVAP() Base evaporation mass flux [kg/m²/s]

Computes evaporation flux from the base film using heat flux fraction and parent film evaporation rate.

Inputs:

- `base` — *Solvers.FourField.Base* object
- `zIdx` — Axial indices to evaluate (optional)

MTOT(*base, drop, zIdx*)

MTOT() Total mass flux of the base film [kg/m²/s]

Calculates the total mass flux exchange associated with the base film, including evaporation, entrainment, deposition, and wave interaction.

Inputs:

- `base` — *Solvers.FourField.Base* object
- `drop` — *Solvers.FourField.Drop* object
- `zIdx` — Axial indices to evaluate (optional)

MTURB(*base, zIdx*)

MTURB() Turbulent mass exchange [kg/m²/s]

Estimates turbulent mass exchange between base and wave films using wave mixing coefficient *Inputs.Model.WAVEMIXCOEF*.

Inputs:

- `base` — *Solvers.FourField.Base* object
- `zIdx` — Axial indices to evaluate (optional)

MWAVE(*base, drop, zIdx*)

MWAVE() Base film mass flux interaction with wave [kg/m²/s]

Computes the mass flux exchange between the base film and wave field. Includes net exchange term based on relaxation toward equilibrium thickness and turbulent mixing contribution.

Inputs:

- `base` — *Solvers.FourField.Base* object
- `drop` — *Solvers.FourField.Drop* object
- `zIdx` — Axial indices to evaluate (optional)

Outputs:

- Mwave — Mass flux transferred to wave field [kg/m²/s]
- Mnet — Net exchange term before turbulent adjustment [kg/m²/s]

TBASE(*base, zIdx*)

TBASE() Base film exposure time [s]

Calculates the time during which the base film is exposed to vapor between successive waves, based on wave spacing and relative velocity (eq 24 of Le Corre [2022a]).

Inputs:

- base — *Solvers.FourField.Base* object
- zIdx — Axial indices to evaluate (optional)

TDRY(*base, drop, zIdx*)

TDRY() Base film dry-out time [s]

Estimates the time of base film dry out under current evaporation and deposition conditions. Returns zero if dry-out does not occur.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

THICKMIN(*base, drop, zIdx*)

THICKMIN() Minimum base film thickness [m]

Calculates the minimum thickness of the base film at the end of its exposure period, based on mass flow rate, liquid density, velocity, and channel perimeter (eq. 28 of Le Corre [2022a]).

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

WMIN(*base, drop, zIdx*)

WMIN() Minimum base film mass flow rate [kg/s]

Calculates the minimum mass flow rate of the base film at the end of its exposure period (TBASE), accounting for evaporation and deposition.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

WMINL(*base, drop, zIdx*)

WMINL() Minimum base film mass flow rate per perimeter [kg/m-s]

Computes the minimum base film mass flow rate normalized by channel perimeter.

Inputs:

- base — *Solvers.FourField.Base* object
- drop — *Solvers.FourField.Drop* object
- zIdx — Axial indices to evaluate (optional)

copyFlowProperties(*srcObj, targetObj, opts*)

COPYFLOWPROPERTIES Copy flow property arrays between Base objects

Copies selected flow properties (W, U, H) from a source object to one or more target Base objects. Supports full copy or partial copy (excluding first axial node) based on options.

Inputs:

- srcObj — Source object containing flow properties
- targetObj — Array of *Solvers.FourField.Base* objects to update
- opts.all — Logical flag; if true, copies all axial nodes, otherwise skips the first node (default: false)

5.4.2 Mixture solver

The *Solvers.Mixture* module in **OpenSTREAM** provides classes and tools for simulating two-phase flows using a mixture approach. In this method, all phases are treated as a single continuum with averaged properties, making it suitable for cases where phase separation is minimal or not explicitly resolved. Additionally, the solver supports a Mixture Relaxation Model (MRM), which introduces thermal non-equilibrium between phases by solving separate mass and energy conservation equations for the vapor phase.

This module includes:

- Solver class for mixture-based flow simulations: *Solvers.Mixture.MixtureSolver*
- Mixture field class: *Solvers.Mixture.Mixture*
- Phase classes: *Solvers.Mixture.Liquid*, *Solvers.Mixture.Vapor*
- Mixture primary properties: *Solvers.Mixture.Mixture.W*, *Solvers.Mixture.Mixture.P*, *Solvers.Mixture.Mixture.H*
- Key solver class methods: *Solvers.Mixture.MixtureSolver.initializeSolver*, *Solvers.Mixture.MixtureSolver.solve()*, *Solvers.Mixture.MixtureSolver.plotz*
- Field class methods for calculating secondary mixture and phase parameters.

class *Solvers.Mixture.MixtureSolver*(*inputSet*)

Bases: *Solvers.AbstractSolver*

MIXTURESOLVER Solver for initializing, solving, and visualizing mixture flow.

The *MixtureSolver* class handles the setup, execution, and visualization of thermal-hydraulic simulations using a two-phase mixture approach or a Mixture Relaxation Model (MRM).

Responsibilities:

- Initializes solver parameters from an *Inputs.InputSet* object
- Interpolates boundary conditions in time and space
- Constructs *Solvers.Mixture.Mixture* objects for transient and steady-state analysis
- Provides plotting utilities for spatial and temporal distributions

Key Components:

- Mixture construction and phase separation (liquid/vapor)
- Mixture mass/momentum/energy transport modeling
- Support for relaxation models and near-wall phenomena
- Visualization of results across time and axial domains

Constructor Summary

MixtureSolver(*inputSet*)

MIXTURESOLVER Constructor for the MixtureSolver() class.

Creates a new instance of the MixtureSolver() class using the provided *Inputs.InputSet*. This constructor initializes the solver by calling the abstract *Solvers.AbstractSolver* superclass constructor and then sets up all necessary solver parameters and internal data structures via *Solvers.Mixture.MixtureSolver.initializeSolver*

Input:

- inputSet — An *Inputs.InputSet* object containing model configuration, geometry, boundary conditions, and solver options.

Notes:

- This constructor assumes *Inputs.InputSet* is fully validated.
- Solver initialization includes time/space discretization, boundary condition interpolation, and mixture object setup.

Property Summary

DT = 0

Time step size [s] from *Inputs.Options.TSTEP*

DZ = 0

Axial step size [m]

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIME = 0

Time series [s]

Z = 1.

Elevation [m]

boundaryConditions

Boundary conditions object *Inputs.BoundaryConditions*

fluid

Fluid object *Inputs.FluidProperties*

inputSet

Input set object *Inputs.InputSet*

mixture

Mixture object

mixtureInit

Null transient mixture object

Method Summary

axialInterpolate(*mix, x, y*)

AXIALINTERPOLATE Performs axial interpolation of data using a specified method.

Interpolates the values in y at positions defined by mix.Z using the interpolation method specified in *Inputs.options.AXIALINTERP*. The interpolation is performed over the domain defined by x.

If the interpolation returns NaN (e.g., due to points slightly outside the interpolation range when using methods like ‘next’), linear extrapolation is applied to those specific points to ensure continuity.

This function ensures robust handling of out-of-bound values by using extrapolation and fallback strategies.

initializeSolver(*mixSolver*)

INITIALIZESOLVER Initializes solver parameters and constructs mixture objects.

Sets up the solver’s internal state using the provided [Inputs.InputSet](#). This includes discretization of time and space, interpolation of boundary conditions, and initialization of data structures for transient and steady-state simulations.

Operations performed:

- Computes time steps and axial node positions
- Interpolates boundary conditions in time and space
- Initializes mixture array for each time step
- Constructs data structures for:
 - Pressure drops (DP, DPSUM)
 - Momentum and energy derivatives (MDER)
 - Mixture Relaxation Model (MRM)
 - Near-wall transport (NEARWALL)
 - Iteration tracking (ITR)
- Initializes fluid property objects
- Sets up phase-specific objects (Liquid, Vapor)
- Assigns initial conditions for mass flow rate, pressure, enthalpy

Notes:

- The function assumes uniform axial discretization.
- Wall heat flux is computed from interpolated boundary conditions.
- MRM fields are initialized after enthalpy to ensure correct vapor quality (XEQ) computation.

interpBoundaryConditions(*mixSolver*)

INTERBOUNDARYCONDITIONS Interpolates boundary conditions in time and space.

Expands user-defined [Inputs.BoundaryConditions](#) to match the solver’s spatial grid and time steps. It performs interpolation of relevant boundary condition parameters both temporally and axially, and computes derived quantities such as wall heat flux.

Operations performed:

- Interpolates time-dependent boundary condition parameters: ‘TIME’, ‘PRESSURE’, ‘HIN’, ‘MFLOW’, ‘POWER’
- Interpolates wall power distribution in space (axially)
- Interpolates wall power in time across all walls
- Computes wall heat flux at each axial node and time step

Notes:

- Uses `Solvers.Mixture.MixtureSolver.timeInterpolate` and `Solvers.Mixture.MixtureSolver.axialInterpolate` for interpolation.
- Throws an error if any expected boundary condition parameter is missing.
- The calculation of HFLUX is based on normalized wall power and geometric properties.

plott(*mixSolver*, *zIdx*, *opt*)

PLOTT Plots temporal distributions of mixture parameters at a given axial location.

Generates time-series plots of selected mixture parameters at a specified axial index. The method supports multiple display modes, wall selections, and plot arrangements.

Inputs:

- mixSolver — [Solvers.Mixture.MixtureSolver](#) object containing simulation data

- `zIdx` — Axial index (scalar, positive integer)
- `opt.display` — Parameters to display (e.g., 'HFLUX', 'W', 'DP', etc.)
- `opt.solveMode` — Solve mode: 'REAL' or 'NULL'
- `opt.wall` — Wall index(es) to plot
- `opt.tIdx` — Time indices to include in the plot
- `opt.reverseTime` — Logical flag to reverse time axis
- `opt.unitTemp` — Temperature unit: 'K' or 'C'
- `opt.arrangement` — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- `opt.resize` — Resize factor for plot scaling
- `opt.nearWall` — Logical flag to include near-wall parameters

Notes:

- If `opt.display` is set to 'ALL', all supported parameters are plotted.
- The function validates that at least two time indices are provided.
- Temperature unit conversion is applied if 'C' is selected.
- Near-wall data is included if `opt.nearWall` is true.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.
- In the pressure drop plot, the temporal acceleration includes the slip-induced contribution (if any)

plotz(*mixSolver*, *tIdx*, *opts*)

PLOTZ Plots spatial (axial) distributions of mixture parameters.

Generates axial plots of selected mixture parameters at specified time indices. The function supports multiple display modes, wall selections, near-wall data, and optional animation over time.

Inputs:

- `mixSolver` — [Solvers.Mixture.MixtureSolver](#) object containing simulation data
- `tIdx` — Time index or indices (vector of positive integers)
- `opts.display` — Parameters to display (e.g., 'HFLUX', 'W', 'DP', etc.)
- `opts.solveMode` — Solve mode: 'REAL' or 'NULL'
- `opts.wall` — Wall index(es) to plot
- `opts.zIdx` — Axial indices to include in the plot
- `opts.obstructions` — Logical flag to display obstruction locations
- `opts.unitTemp` — Temperature unit: 'K' or 'C'
- `opts.arrangement` — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- `opts.resize` — Resize factor for plot scaling
- `opts.nearWall` — Logical flag to include near-wall parameters

Notes:

- If multiple time indices are provided, the function creates an animated plot.
- At least two axial indices must be specified to enable spatial plotting.
- Temperature unit conversion is applied if 'C' is selected.
- Obstruction locations are plotted if `opts.obstructions` is true.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.
- In the pressure drop plot, the temporal acceleration includes the slip-induced contribution (if any)

plotzt(*mixSolver*, *opt*)

PLOTZT Plots 2D time/elevation distributions of mixture parameters.

Generates surface plots of selected mixture-related parameters over time and axial (elevation) positions. It supports plotting for mixture, liquid and vapor fields, and can handle both real and null (initial) transient data.

Inputs:

- `mixSolver` — [Solvers.Mixture.MixtureSolver](#) object containing simulation data
- `opt.display` — Parameter(s) to display (e.g., 'HFLUX', 'U', etc.)
- `opt.label` — Corresponding labels for display parameters
- `opt.unit` — Units for each parameter

- `opt.field` — Field to plot: ‘mixture’, ‘liquid’, or ‘vapor’
- `opt.solveMode` — Solve mode: ‘REAL’ or ‘NULL’
- `opt.wall` — Wall index(es) to plot
- `opt.zIdx` — Axial indices (must include at least 2)
- `opt.tIdx` — Time indices (must include at least 2)
- `opt.reverseTime` — Logical flag to reverse time axis
- `opt.shading` — Surface shading style: ‘faceted’, ‘flat’, or ‘interp’
- `opt.view` — View angle for 3D plot [azimuth elevation]

Notes:

- If ‘ALL’ is passed to `opt.display`, all supported parameters are plotted.
- The function validates that at least two axial and time indices are provided to enable meaningful 2D plotting.
- Each wall is plotted in a separate figure window with appropriate titles and subplot grouping.

`solve`(*mixSolver*)

SOLVE Executes the mixture solver for steady-state and transient simulations.

Runs the full solution process for the [Solvers.Mixture.MixtureSolver](#) object. It manages initialization, time stepping, axial sweeps, inner iterations, convergence checks, and logging.

Workflow:

- Enables logging and opens persistent log file
- Validates solver state before execution
- Runs steady-state initialization (`solveINIT = true`)
- If initial step converges, proceeds with transient simulation
- Handles exceptions and ensures proper log closure

Notes:

- Uses internal *solver* function to handle both steady-state and transient modes
- Applies relaxation factors for flow, pressure, enthalpy, and vapor properties
- Supports thermal non-equilibrium modeling via vapor relaxation
- Logs progress and outputs to session directory

`timeInterpolate`(*mix*, *y*)

TIMEINTERPOLATE Performs temporal interpolation of data using a specified method.

Interpolates the values in *y* at the time points defined by *mix.TIME* using the interpolation method specified in `Inputs.options.TIMEINTERP`.

If the boundary condition time vector (*mix.inputSet.bc.TIME*) is scalar, interpolation is skipped and the original data *y* is returned unchanged.

This function supports flexible interpolation strategies and handles cases where interpolation is unnecessary.

`class Solvers.Mixture.Vapor`(*mix*)

Bases: [Solvers.AbstractPhase](#)

VAPOR Represents the vapor phase in a mixture solver simulation.

The Vapor class provides access to vapor-specific properties and calculations derived from the mixture solution. It encapsulates methods for computing flow, thermodynamic, and transport properties of the vapor phase at each axial node.

Responsibilities:

- Compute vapor mass fraction and void fraction
- Calculate vapor velocity, enthalpy, and temperature

- Evaluate wall heat flux and wall evaporation contributions
- Support derived quantities such as Reynolds number and mass flux

Notes:

- All methods assume access to a valid `Solvers.Mixture.Mixture` object
- Axial indexing is optional; defaults to full axial domain
- Enthalpy calculations adapt to thermal non-equilibrium models
- Velocity and enthalpy calculations include fallback logic to handle single-phase liquid regions and numerical stability

Constructor Summary

Vapor(*mix*)

VAPOR Constructor of the `Vapor()` class

Initializes the `Vapor()` object from one or more instances of the `Solvers.Mixture.Mixture` class. Each `Vapor()` instance stores a reference to its corresponding Mixture object and the number of axial nodes (NZ).

Input:

- *mix* — Array of `Solvers.Mixture.Mixture` objects representing the simulation state

Notes:

- Supports vectorized initialization for transient simulations
- Assumes each `Solvers.Mixture.Mixture` object is fully initialized

Method Summary

C(*vapor*, *zIdx*)

`C()` Vapor mass fraction [-]

FW(*vapor*, *zIdx*)

`FW()` Vapor wall friction factor [-]

Computes the Darcy wall friction factor based on `Inputs.Model.SPMTM` model.

Supported models

- BLASIUS: Blasius model ($f = C(1)Re^{C(2)} + C(3)$) using user-defined `Inputs.Model.FRICTION` coefficients

H(*vapor*, *zIdx*)

`H()` Vapor enthalpy [J/kg]

Calculation depends on `Inputs.Model.THERMALNONEQ` model.

HFLUX(*vapor*, *zIdx*)

`HFLUX()` Wall heat flux to vapor [W/m^2]

HFLUXWALEVAP(*vapor*, *zIdx*)

HFLUX Wall evaporation heat flux [W/m^2]

HWALL(*vapor*, *twall*, *zIdx*)

`HWALL()` Single-phase vapor wall heat transfer coefficient [W/m^2/K]

Computes the wall heat transfer coefficient using vapor thermal conductivity and Nusselt number.

MFLUX(*vapor*, *zIdx*)

`MFLUX()` Vapor mass flux [kg/m^2-s]

NU(*vapor*, *twall*, *zIdx*)

NU() Vapor wall Nusselt number [-]

Computes the Nusselt number for wall heat transfer to vapor based on *Inputs.Model.SPHTM* model.

Q(*vapor*, *zIdx*)

Q() Vapor volumetric flow rate [m³/s]

RE(*vapor*, *zIdx*)

RE() Vapor Reynolds number [-]

T(*vapor*, *zIdx*)

T() Vapor temperature [K]

TAUW(*vapor*, *zIdx*)

TAUW() Vapor wall shear stress [N/m²]

TIME(*vapor*)

TIME() Time series [s]

U(*vapor*, *zIdx*)

U() Vapor velocity [m/s]

VF(*vapor*, *zIdx*)

VF() Void fraction [-]

W(*vapor*, *zIdx*)

W() Vapor mass flow rate [kg/s]

X(*vapor*, *zIdx*)

X() Vapor mass flow fraction [-]

Z(*vapor*, *zIdx*)

Z() Axial nodes [m]

class *Solvers.Mixture.Mixture*(*inputSet*, *fluid*)

Bases: *Solvers.AbstractField*

MIXTURE Class for modeling two-phase mixture field in mixture solver

This class encapsulates the physical and numerical properties of a fluid mixture, including flow variables, phase interactions, pressure losses, heat transfer, and relaxation models. It supports multiple solver models and provides methods for computing derived quantities and handling transitions such as boiling and annular flow.

Constructor Summary

Mixture(*inputSet*, *fluid*)

MIXTURE Constructor for *Mixture()* class

Initializes the mixture object with input configuration and fluid properties. Sets up flow property tracking for simulation.

Inputs:

- *inputSet* — *Inputs.InputSet* object containing model, geometry, and boundary conditions
- *fluid* — *Inputs.FluidProperties* object containing thermophysical fluid data

Property Summary

DP
 Saved detailed pressure drops [Pa]

DPSUM
 Saved detailed cumulative pressure drops [Pa]

DT = 0
 Time step size [s] from `Inputs.options.TSTEP`

DZ = 0
 Axial step size [m]

H = 1E6
 Enthalpy [J/kg]

HFLUX = 1.
 Wall heat flux [W/m²]

ITR
 Iteration tracking

MDER
 Saved detailed material derivative terms

MRM
 Mixture relaxation terms

NEARWALL
 Near-wall terms

NTIME = 0
 Number of time steps [-]

NZ = 0
 Number of axial steps [-] from `Inputs.Model.NNODES`

P = 7E6
 Pressure [Pa]

TIDX = 1
 Time step index [-]

TIME = 0
 Time series [s]

W = 1.
 Mass flow rate [kg/s]

Z = 1.
 Elevation [m]

cbt = false
 Critical Boiling Transition flag [-]

fluid
[`Inputs.FluidProperties`](#) object

inputSet

Inputs.InputSet object

liquid

Liquid phase object

mfbt = false

Minimum Film Boiling Transition flag [-]

vapor

Vapor phase object

Method Summary**AFDISTR**(*mix, param1, param2, zIdx*)

AFDISTR() Computes annular flow distribution over axial positions.

Evaluates a weighted distribution between param1 and param2 using the annular flow fraction defined in *Solvers.Mixture.Mixture.AFFNC*.

Inputs:

- mix — *Solvers.Mixture.Mixture* object containing axial grid and flow fractions
- param1 — First flow property
- param2 — Second flow property
- zIdx — Axial indices to evaluate (optional; defaults to full axial range)

Notes:

- The annular flow fraction is extracted from *Solvers.Mixture.Mixture.AFFNC* at zIdx
- The output is a linear interpolation: (1 - affnc) * param1 + affnc * param2
- mix(1).NZ is used for default indexing, assuming mix may be an array

AFFNC(*mix, zIdx*)

AFFNC() Computes annular flow fraction over axial positions.

Evaluates the annular flow fraction using a sigmoid function parameterized by the transition model and geometry. The sigmoid is evaluated via the cached *sigm* method.

Inputs:

- mix — *Solvers.Mixture.Mixture* object containing model and geometry data
- zIdx — Axial indices to evaluate (optional; defaults to full axial range)

Notes:

- Sigmoid parameters are scaled by node density (NNODES / LENGTH)
- The center of the transition is offset by mix.OAFIDX()
- Uses mix.sigm(...) to evaluate the sigmoid function

CBT(*mix, zIdx*)

CBT() Critical Boiling Transition flag [-], wall dependent

Retrieves precomputed *CBT()* flags by *Solvers.Mixture.Mixture.CBT_CALC*, updated when mix.H is set.

Inputs:

- mix — *Solvers.Mixture.Mixture* object
- zIdx — Axial indices to retrieve (optional)

CBTIDX(*mix*)

CBTIDX() Index of first CBT occurrence [-]

Compute the index of first CBT occurrence. If no occurrence is found, NaN is returned.

Inputs

- `mix` — *Solvers.Mixture.Mixture* object

CBTZ(*mix*, *zIdx*)

CBTZ() Length from upstream CBT [m]

Compute the distance from the upstream critical boiling transition. Support multiple transitions.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to evaluate (optional)

CHF(*mix*, *zIdx*)

CHF() Critical Heat Flux [W/m²], wall dependent

Retrieves precomputed *CHF()* values by *Solvers.Mixture.Mixture.CBT_CALC*, updated when *mix.H* is set.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to retrieve (optional)

DPACCS(*mix*, *MCTold*, *zIdx*)

DPACCS() Slip-induced temporal acceleration pressure drop [Pa]

Computes the pressure drop due to the slip-induced temporal acceleration between time steps.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- *MCTold* — Previous slip-induced momentum correction [kg/s]
- `zIdx` — Axial indices to evaluate (optional)

DPACCT(*mix*, *UPold*, *zIdx*)

DPACCT() Temporal acceleration pressure drop [Pa]

Computes the pressure drop due to temporal acceleration between time steps.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- *UPold* — Previous axial (advection) velocity [m/s]
- `zIdx` — Axial indices to evaluate (optional)

DPACCZ(*mix*, *zIdx*)

DPACCZ() Spatial acceleration pressure drop [Pa]

Computes the pressure drop due to spatial acceleration of the flow.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to evaluate (optional)

DPGRAV(*mix*, *zIdx*)

DPGRAV() Gravitational pressure loss [Pa]

Computes the pressure loss due to gravity along the axial direction.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to evaluate (optional)

DPK(*mix*, *zIdx*)

DPK() Local pressure loss [Pa]

Computes the pressure loss due to localized effects (e.g., flow mixers).

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

DPPARTS(*mix*, *UPold*, *MCTold*, *zIdx*)

DPPARTS() All pressure loss components [Pa]

Returns a structure containing all individual pressure loss components.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *UPold* — Previous axial (advection) velocity [m/s]
- *zIdx* — Axial indices to evaluate (optional)

Returns:

- *dpparts* — Struct with fields:
 - GRAV : Gravitational loss [Pa]
 - WALL : Wall friction loss [Pa]
 - ACCZ : Spatial acceleration loss [Pa]
 - ACCT : Temporal acceleration loss [Pa]
 - ACCS : Slip-induced temporal acceleration loss [Pa]
 - K : Local loss [Pa]
 - TOT : Total pressure loss [Pa]

DPTOT(*mix*, *UPold*, *MCTold*, *zIdx*)

DPTOT() Total pressure loss [Pa]

Computes the total pressure loss by summing all contributing components.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *UPold* — Previous axial (advection) velocity [m/s]
- *MCTold* — Previous slip-induced momentum correction [kg/s]
- *zIdx* — Axial indices to evaluate (optional)

DPWALL(*mix*, *zIdx*)

DPWALL() Wall friction pressure drop [Pa]

Computes the pressure drop due to wall friction.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

ECT(*mix*, *zIdx*)

ECT() Slip-induced energy correction term [J/m]

Computes the slip-induced energy correction term per unit length.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

FOURIERCOND(*mix*, *zIdx*)

FOURIERCOND() Calculates the non-dimensional relaxation time for wall-dependent interfacial condensation.

Computes the non-dimensional relaxation time (akin to a Fourier number) associated with interfacial condensation at the specified axial indices, based on the thermal relaxation model defined in [Inputs.Model.THERMALRELAX](#).

Inputs:

- *mix* — [Solvers.Mixture.Mixture](#) object containing fluid and model data
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial domain)

Supported Models:

- FOURIERX: Interpolate Fourier number from pre-defined quality-time pairs
- TIMEX: Interpolate relaxation time from pre-defined quality-time pairs
- VOID: Computes Fourier number based on void fraction and fluid properties
- HOMOGENEOUS: Computes Fourier number based on homogeneous approach
- FOURIER: Empirical Fourier number correlation

FOURIEREVAP(*mix*, *zIdx*)

[FOURIEREVAP\(\)](#) Calculates the non-dimensional relaxation time for wall-dependent interfacial evaporation.

Computes the non-dimensional relaxation time (akin to a Fourier number) associated with interfacial evaporation at the specified axial indices, based on the thermal relaxation model defined in [Inputs.Model.THERMALRELAX](#).

Inputs:

- *mix* — [Solvers.Mixture.Mixture](#) object containing fluid and model data
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial domain)

Supported Models:

- FOURIERX: Interpolate Fourier number from pre-defined quality-time pairs
- TIMEX: Interpolate relaxation time from pre-defined quality-time pairs
- VOID: Computes relaxation time based on void fraction and fluid properties
- HOMOGENEOUS: Computes Fourier number based on homogeneous approach
- FOURIER: Empirical Fourier number correlation

FW(*mix*, *zIdx*)

[FW\(\)](#) Darcy wall friction factor [-]

Computes the Darcy wall friction factor based on [Inputs.Model.SPMTM](#) model.

Inputs:

- *mix* — [Solvers.Mixture.Mixture](#) object
- *zIdx* — Axial indices to evaluate (optional)

Supported models

- BLASIUS: Blasius model ($f = C(1)Re^{C(2)} + C(3)$) using user-defined [Inputs.Model.FRICTION](#) coefficients

FWL(*mix*, *zIdx*)

[FWL\(\)](#) Liquid-equivalent Darcy wall friction factor [-]

Computes the Darcy friction factor, using liquid-equivalent Reynolds number, based on [Inputs.Model.SPMTM](#) model.

Inputs:

- *mix* — [Solvers.Mixture.Mixture](#) object
- *zIdx* — Axial indices to evaluate (optional)

Supported models

- BLASIUS: Blasius model ($f = C(1)Re^{C(2)}$) using user-defined [Inputs.Model.FRICTION](#) coefficients

HINT(*mix*, *zIdx*)

HINT() Linear interfacial heat transfer rate [W/m]

Computes the interfacial heat transfer rates due to condensation and evaporation, based on interfacial mass transfer and enthalpy differences.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with flow and enthalpy data
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial range)

Notes:

- Uses *Solvers.Mixture.Mixture.MINT* for mass transfer and *Solvers.Mixture.Mixture.MRM.HV* for vapor enthalpy
- Behavior depends on *Inputs.Model.INTTRANSH*:
 - ‘*BULK*’: Consideration of bulk enthalpy
 - ‘*SATURATED*’: Consideration of saturated enthalpy

HINTCOND(*mix*, *zIdx*)

HINTCOND() Linear interfacial heat condensation rate [W/m]

Extracts the condensation component from *Solvers.Mixture.Mixture.HINT*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

HINTEVAP(*mix*, *zIdx*)

HINTEVAP() Linear interfacial heat evaporation rate [W/m]

Extracts the evaporation component from *Solvers.Mixture.Mixture.HINT*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

HM(*mix*, *zIdx*)

H Static enthalpy [J/kg]

Computes the mass-centered mean mixture enthalpy.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

HTOT(*mix*, *zIdx*)

HTOT() Total linear vapor heat rate [W/m]

Computes the total vapor heat transfer rate by summing interfacial and wall contributions.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

HVTOT(*mix*, *zIdx*)

HVTOT() Total linear vapor specific enthalpy transfer [J/kg/m]

Computes the specific enthalpy input to the vapor per unit mass flow. Uses a lumped wall approach for now.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Uses lumped wall approach: sums WV and Htot across walls
- Set to zero when WV is negligible

HWALEVAP(*mix*, *zIdx*)

HWALEVAP() Linear wall heat evaporation (boiling) rate [W/m]

Computes the heat transfer rate associated with wall boiling.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with boiling and enthalpy data
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Behavior depends on *Inputs.Model.INTTRANSH*:
 - BULK: Consideration of bulk enthalpy
 - SATURATED: Consideration of saturated enthalpy

HWALHEAT(*mix*, *zIdx*)

HWALHEAT() Linear wall heat to vapor rate [W/m]

Returns the wall heat generation rate (LHGR) at boiling transition nodes.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Applies only at nodes flagged by *Solvers.Mixture.Mixture.CBT* or *Solvers.Mixture.Mixture.MFBT*.

HWALL(*mix*, *twall*, *zIdx*)

HWALL() Wall heat transfer coefficient [W/m²/K]

Computes the effective wall heat transfer coefficient by combining single-phase, boiling and post-CBT models. Supports multiple walls

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

HWALLBOIL(*mix*, *zIdx*)

HWALLBOIL() Boiling wall heat transfer coefficient [W/m²/K]

Computes the boiling wall heat transfer coefficient using selected *Inputs.Model.TPHTM* model.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

Supported models

- THOM: Thom's correlation

HWALLPOSTCBT(*mix*, *twall*, *zIdx*)

HWALLPOSTCBT() Post-CBT wall heat transfer coefficient [W/m²/K]

Computes the effective wall heat transfer coefficient under post-CBT conditions. based on *Inputs.Model.BTHTM* model.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *twall* — Wall temperature
- *zIdx* — Axial indices to evaluate (optional)

Supported wall heat transfer models

- VAPOR: Heat transfer to vapor phase
- DOUGALL: Dougall-Rohsenow model (Dougall and Rohsenow [1963])
- BISHOP: Bishop-Sandberg-Tong model (Bishop *et al.* [1965])
- MOECK: Groeneveld-Moeck model (Groeneveld and Moeck [1969])
- DELORME: Groeneveld-Delorme model (Groeneveld and Delorme [1976])
- CONDIEIV: Condie-Bengston IV model (Morris *et al.* [1982])

JG(*mix*, *zIdx*)

JG() Superficial vapor velocity [m/s]

Computes the superficial velocity of the vapor phase.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

JL(*mix*, *zIdx*)

JL() Superficial liquid velocity [m/s]

Computes the superficial velocity of the liquid phase.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

KDIST(*mix*, *zIdx*)

KDIST() Distance from upstream obstruction [m]

Computes the axial distance from the nearest upstream obstruction or inlet, based on *Inputs.Model.KLOC*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with axial grid and obstruction location
- *zIdx* — Axial indices to evaluate (optional)

KIDX(*mix*, *zIdx*)

KIDX() Obstruction index

Returns the axial node index of the obstruction closest to *Inputs.Model.KLOC*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with axial grid
- *zIdx* — Axial indices to evaluate (optional)

KLOCZ(*mix*, *zIdx*)

KLOCZ() Obstruction positions [m]

Returns the axial position of the obstruction closest to *Inputs.Model.KLOC*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with axial grid
- *zIdx* — Axial indices to evaluate (optional)

KLOSS(*mix*, *zIdx*)

KLOSS() Local pressure loss coefficient [-]

Returns the pressure loss coefficient at the elevation closest to *Inputs.Model.KLOC*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object

- `zIdx` — Axial indices to evaluate (optional)

KTRELAX(*mix*, *zIdx*)

KTRELAX() Local relaxation time [s]

Returns the relaxation time at the elevation closest to *Inputs.Model.KLOC*. All other axial positions are set to NaN.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with axial grid and model parameters
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial range)

LHGR(*mix*, *zIdx*)

LHGR() Linear heat generation rate [W/m]

Computes the linear heat generation rate by multiplying wall perimeter with local heat flux.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with geometry and heat flux data
- *zIdx* — Axial indices to evaluate (optional)

MCT(*mix*, *zIdx*)

MCT() Slip-induced momentum correction term [kg/s]

Computes the slip-induced momentum correction term per unit length.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

MFBT(*mix*, *zIdx*)

MFBT() Minimum Film Boiling Transition flag [-], wall dependent

Retrieves precomputed **MFBT()** flags by *Solvers.Mixture.Mixture.MFBT_CALC*, updated when *mix.H* is set.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to retrieve (optional)

MFLUX(*mix*, *zIdx*)

MFLUX() Mass flux [kg/m²/s]

Retrieves precomputed mass flux values by *Solvers.Mixture.Mixture.MFLUX_CALC*, updated when *mix.W* is set.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to retrieve (optional)

MINT(*mix*, *zIdx*)

MINT() Linear interfacial mass transfer rates [kg/s/m]

Computes condensation and evaporation mass transfer rates based on deviation from equilibrium vapor flow and time relaxation.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with flow and relaxation data
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial range)

Notes:

- Uses *mix.MRM.WV* for equilibrium deviation

- U_{Veq} is the approximated vapor velocity

MINTCOND(*mix*, *zIdx*)

MINTCOND() Linear interfacial condensation rate [kg/s/m]

Extracts the condensation component from *Solvers.Mixture.Mixture.MINT*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

MINTEVAP(*mix*, *zIdx*)

MINTEVAP() Linear interfacial evaporation rate [kg/s/m]

Extracts the evaporation component from *Solvers.Mixture.Mixture.MINT*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

MTOT(*mix*, *zIdx*)

MTOT() Total linear vapor mass transfer rate [kg/s/m]

Computes the total vapor mass transfer rate by summing interfacial condensation, evaporation, and wall boiling contributions.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

MTRANSV(*mix*, *zIdx*)

MTRANSV() Linear transversal mass exchange [kg/s/m]

Models transverse vapor mass exchange between wall regions using a time relaxation approach. Currently not used.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Uses a fixed relaxation time for now (RELAXT = 0.01 s)

MU(*mix*, *zIdx*)

MU() Dynamic viscosity [Pa·s]

Computes the mixture dynamic viscosity using vapor and liquid phase contributions weighted by vapor quality.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with fluid properties
- *zIdx* — Axial indices to evaluate (optional)

MWALEVAP(*mix*, *zIdx*)

MWALEVAP() Linear wall mass evaporation rate [kg/s/m]

Computes the wall boiling mass transfer rate using latent heat and wall heat generation rate.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with heat transfer and boiling data
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Latent heat is computed based on *Inputs.Model.INTTRANSH* setting

OAFIDX(*mix*)

OAFIDX() Onset of annular flow node.

Returns the axial node index corresponding to the onset of annular flow, determined by comparing the equilibrium quality to the onset threshold. The result is cached in *mix.oafidx_const* to avoid recomputation.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object containing flow quality and onset threshold

Notes:

- If no node satisfies the condition, the most upstream node (1) is returned

OAFWL(*mix*)

OAFWL() Liquid mass flow rate at onset of annular flow.

Returns the liquid mass flow rate at the axial index corresponding to the onset of annular flow, as defined by *Solvers.Mixture.Mixture.OAFIDX*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object containing liquid flow data

OAFX(*mix*, *zIdx*)

OAFX() Onset of annular flow equilibrium quality.

Computes the equilibrium quality at the onset of annular flow using a model selected in *Inputs.Model.OAF*. The quality is evaluated over the specified axial indices.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object containing model, geometry, and fluid data
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial range)

Supported Models:

- WALLIS: Full Wallis model
- WALLIS_SIMP: Simplified Wallis model
- LEVITAN: Levitan model

OAFZ(*mix*)

OAFZ() Onset of annular flow elevation.

Returns the axial elevation (in meters) corresponding to the onset of annular flow, as defined by *Solvers.Mixture.Mixture.OAFIDX*.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object containing axial grid data

PHI2F(*mix*, *zIdx*)

PHI2F() Two-phase wall friction multiplier [-]

Computes the two-phase wall friction multiplier using selected *Inputs.Model.TPFM* model.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

Supported models

- HOMOGENEOUS: Homogeneous model
- SLIP: Constant velocity slip ratio
- EPRI: EPRI multiplier

PHI2K(*mix*, *zIdx*)

PHI2K() Two-phase local pressure drop multiplier [-]

Computes the multiplier for local pressure drop in two-phase flow using selected *Inputs.Model.TPKM* model.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

Supported models

- HOMOGENEOUS: Homogeneous model
- SLIP: Constant velocity slip ratio
- ROMIE: Romie multiplier

RE(*mix*, *zIdx*)

RE() Reynolds number [-]

Computes the Reynolds number based on total mass flow rate and mixture viscosity.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

REL(*mix*, *zIdx*)

REL() Liquid-equivalent Reynolds number [-]

Computes the Reynolds number using liquid viscosity and total mass flow.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to evaluate (optional)

RELAXT(*mix*, *zIdx*)

RELAXT() Interfacial time relaxation [s]

Computes interfacial time relaxation based on sign of deviation from equilibrium vapor flow.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object with flow and relaxation data
- *zIdx* — Axial indices to evaluate (optional; defaults to full axial range)

Notes:

- Uses *mix.MRM.WV* for equilibrium deviation
- Vapor mass deviation from equilibrium, *Wint*, must be consistent with MINT method

RELAXTCOND(*mix*, *zIdx*)

RELAXTCOND() Time relaxation for interfacial condensation [s]

Retrieves precomputed time relaxation values for condensation by *Solvers.Mixture.Mixture.RELAXTCOND_CALC*, updated when *mix.H* is set.

Inputs:

- *mix* — *Solvers.Mixture.Mixture* object
- *zIdx* — Axial indices to retrieve (optional)

RELAXTEVAP(*mix*, *zIdx*)

RELAXTEVAP() Time relaxation for interfacial evaporation [s]

Retrieves precomputed time relaxation values for evaporation by *Solvers.Mixture.Mixture.RELAXTEVAP_CALC*, updated when *mix.H* is set.

Inputs:

- `mix` — [Solvers.Mixture.Mixture](#) object
- `zIdx` — Axial indices to retrieve (optional)

RHO(*mix*, *zIdx*)

[RHO\(\)](#) Density [kg/m³]

Retrieves precomputed mixture density values by `Solvers.Mixture.Mixture.RHO_CALC`, updated when `mix.H` is set.

Inputs:

- `mix` — [Solvers.Mixture.Mixture](#) object
- `zIdx` — Axial indices to retrieve (optional)

T(*mix*, *zIdx*)

[T\(\)](#) Temperature [K]

Returns the fluid temperature corresponding to the local enthalpy.

Inputs:

- `mix` — [Solvers.Mixture.Mixture](#) object with enthalpy data
- `zIdx` — Axial indices to evaluate (optional)

TAUW(*mix*, *zIdx*)

[TAUW\(\)](#) Wall shear stress [N/m²], wall dependent

Computes the wall shear stress for each wall segment using the friction factor, mass flux and two-phase multiplier. Supports multiple walls and adjusts for boiling transition conditions using [Inputs.Model.BTMTM](#) model.

Inputs:

- `mix` — [Solvers.Mixture.Mixture](#) object containing flow and geometry data
- `zIdx` — Axial indices to evaluate (optional)

Supported boiling transition models

- TPFM: Two-phase friction multiplier
- VAPOR: Shear stress to vapor phase

TWALL(*mix*, *zIdx*)

[TWALL\(\)](#) Wall temperature [K]

Computes the wall temperature based on heat flux and wall heat transfer coefficient. Uses liquid bulk temperature pre-CBT and vapor bulk temperature post-CBT.

Inputs:

- `mix` — [Solvers.Mixture.Mixture](#) object
- `zIdx` — Axial indices to evaluate (optional)

U(*mix*, *zIdx*)

[U\(\)](#) Static velocity [m/s]

Computes the mass-centered mean mixture velocity.

Inputs:

- `mix` — [Solvers.Mixture.Mixture](#) object
- `zIdx` — Axial indices to evaluate (optional)

UP(*mix*, *zIdx*)

[UP\(\)](#) Advection velocity [m/s]

Retrieves precomputed mixture (advection) velocity values by `Solvers.Mixture.Mixture.UP_CALC`, updated when `mix.H` is set.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to retrieve (optional)

VF(*mix*, *zIdx*)

VF() Void fraction [-]

Retrieves precomputed void fraction values by `Solvers.Mixture.Mixture.VF_CALC`, updated when `mix.H` is set.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to retrieve (optional)

WALEVAPRATIO(*mix*, *zIdx*)

WALEVAPRATIO() Wall mass evaporation ratio [-]

Computes the fraction of liquid mass undergoing wall boiling based on equilibrium quality and boiling onset thresholds *Inputs.Model.WBOILINGXSUB* and *Inputs.Model.WBOILINGXSAT*.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object with boiling model parameters
- `zIdx` — Axial indices to evaluate (optional)

Notes:

- Ratio is clamped between 0 and 1
- CBT and MFBT flags suppress boiling at transition nodes

WWALL(*mix*, *zIdx*)

WWALL() Mixture mass flow distribution per wall [kg/s]

Computes the wall-distributed mass flow rate by scaling the total flow with the wall perimeter ratio.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object with geometry and flow data
- `zIdx` — Axial indices to evaluate (optional; defaults to full axial range)

X(*mix*, *zIdx*)

X() Vapor quality [-]

Retrieves precomputed vapor quality values by `Solvers.Mixture.Mixture.X_CALC`, updated when `mix.H` is set.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to retrieve (optional)

XEQ(*mix*, *zIdx*)

XEQ() Equilibrium quality [-]

Retrieves precomputed equilibrium quality values by `Solvers.Mixture.Mixture.XEQ_CALC`, updated when `mix.H` is set.

Inputs:

- `mix` — *Solvers.Mixture.Mixture* object
- `zIdx` — Axial indices to retrieve (optional)

class `Solvers.Mixture.Liquid`(*mix*)

Bases: *Solvers.AbstractPhase*

LIQUID Represents the liquid phase in a mixture solver simulation.

The `Liquid` class provides access to liquid-specific properties and calculations derived from the mixture solution. It encapsulates methods for computing flow, thermodynamic, and transport properties of the liquid phase at each axial node.

Responsibilities:

- Compute liquid mass fraction and volumetric fraction
- Calculate liquid velocity, enthalpy, and temperature
- Evaluate wall heat flux contributions to the liquid phase
- Support derived quantities such as Reynolds number and mass flux

Notes:

- All methods assume access to a valid `Solvers.Mixture.Mixture` object
- Axial indexing is optional; defaults to full axial domain
- Velocity and enthalpy calculations include fallback logic to handle single-phase vapor regions and numerical stability

Constructor Summary

Liquid(*mix*)

LIQUID Constructor

Initializes the `Liquid()` object from one or more instances of the `Solvers.Mixture.Mixture` class. Each `Liquid()` instance stores a reference to its corresponding Mixture object and the number of axial nodes (NZ).

Input:

- *mix* — Array of `Solvers.Mixture.Mixture` objects representing the simulation state

Notes:

- Supports vectorized initialization for transient simulations
- Assumes each `Solvers.Mixture.Mixture` object is fully initialized

Method Summary

C(*liquid*, *zIdx*)

`C()` Liquid mass fraction [-]

H(*liquid*, *zIdx*)

`H()` Liquid enthalpy [J/kg]

Calculated based on mixture and vapor enthalpies, and vapor quality.

HFLUX(*liquid*, *zIdx*)

`HFLUX()` Wall heat flux to liquid phase [W/m²]

HWALL(*liquid*, *twall*, *zIdx*)

`HWALL()` Single-phase liquid wall heat transfer coefficient [W/m²/K]

Computes the wall heat transfer coefficient using liquid thermal conductivity and Nusselt number.

MFLUX(*liquid*, *zIdx*)

`MFLUX()` Liquid mass flux [kg/m²/s]

NU(*liquid*, *twall*, *zIdx*)

`NU()` Liquid wall Nusselt number [-]

Computes the Nusselt number for wall heat transfer to liquid based on `Inputs.Model.SPHTM` model.

Q(*liquid*, *zIdx*)
Q() Liquid volumetric flow rate [m³/s]

RE(*liquid*, *zIdx*)
RE() Liquid Reynolds number [-]

T(*liquid*, *zIdx*)
T() Liquid temperature [K]

TIME(*liquid*)
TIME() Time series [s]

U(*liquid*, *zIdx*)
U() Liquid velocity [m/s]

VF(*liquid*, *zIdx*)
VF() Liquid volumetric fraction [-]

W(*liquid*, *zIdx*)
W() Liquid mass flow rate [kg/s]

X(*liquid*, *zIdx*)
X() Liquid mass flow fraction [-]

Z(*liquid*, *zIdx*)
Z() Axial node positions [m]

5.4.3 Three-field solver

The *Solvers.ThreeField* module in **OpenSTREAM** provides classes and tools for simulating annular two-phase flows using a three-field approach. In this method, the two liquid fields (film and droplets) are treated separately, each governed by its own set of conservation equations for mass, momentum, and energy. The vapor phase is solved using the *Solvers.Mixture* module. This framework enables explicit modeling of interactions between the film and droplet liquid fields, slip velocities, and hydrodynamic non-equilibrium effects, making it well-suited for applications involving annular boiling two-phase flow regimes where field separation and interfacial dynamics are significant.

This module includes:

- Solver class for three-field-based flow simulations: *Solvers.ThreeField.ThreeFieldSolver*
- Film and drop field classes: *Solvers.ThreeField.Film*, *Solvers.ThreeField.Drop*
- Three-field primary properties: *Solvers.ThreeField.Film.W*, *Solvers.ThreeField.Film.U*, *Solvers.ThreeField.Film.H*, *Solvers.ThreeField.Drop.W*, *Solvers.ThreeField.Drop.U*, *Solvers.ThreeField.Drop.H*
- Key solver class methods: *Solvers.ThreeField.ThreeFieldSolver.initializeSolver*, *Solvers.ThreeField.ThreeFieldSolver.solve()*, *Solvers.ThreeField.ThreeFieldSolver.plotz*
- Field class methods for calculating secondary three-field parameters.

class *Solvers.ThreeField.ThreeFieldSolver*(*inputSet*, *mixSolver*)

Bases: *Solvers.AbstractSolver*

THREEFIELDSOLVER Solver for initializing, solving, and visualizing three-field flow.

The *ThreeFieldSolver* class handles the setup, execution, and visualization of thermal-hydraulic simulations using a two-phase three-field approach.

Responsibilities:

- Initializes solver parameters from an *Inputs.InputSet* object
- Constructs *Solvers.ThreeField.Film* and *Solvers.ThreeField.Drop* objects for transient and steady-state analysis
- Provides plotting utilities for spatial and temporal distributions

Key Components:

- Liquid field construction (film/drop)
- Field mass/momentum/energy transport modeling
- Pressure drop gradient obtained from mixture solver
- Vapor solution obtained from mixture solver
- Visualization of results across time and axial domains

Constructor Summary

ThreeFieldSolver(*inputSet*, *mixSolver*)

THREEFIELDSOLVER Constructor for the ThreeFieldSolver() class.

Creates a new instance of the ThreeFieldSolver() class using the provided *Inputs.InputSet*. This constructor initializes the solver by calling the abstract *Solvers.AbstractSolver* superclass constructor and then sets up all necessary solver parameters and internal data structures via *Solvers.ThreeField.ThreeFieldSolver.initializeSolver*

Input:

- *inputSet* — An *Inputs.InputSet* object containing model configuration, geometry, boundary conditions, and solver options.
- *mixSolver* — An optional solved *Solvers.Mixture.MixtureSolver* object

Notes:

- This constructor assumes *Inputs.InputSet* is fully validated.
- A solved mixture-model solution is required for initialization. If a solved mixture solver is supplied, its solution is reused. If unsolved or if no mixture solver is supplied, one is created and solved automatically.
- Solver initialization includes liquid and vapor objects setup.
- Time/space discretization and boundary condition interpolation initialized by *Solvers.Mixture.MixtureSolver* are used

Property Summary

DT = 0

Time step size [s] from *Inputs.Options.TSTEP*

DZ = 0

Axial step size [m]

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIME = 0

Time series [s]

Z = 1.

Elevation [m]

boundaryConditions

Boundary conditions object [Inputs.BoundaryConditions](#)

drop

Drop object

dropInit

Null transient drop object

film

Film object

filmInit

Null transient film object

fluid

Fluid object [Inputs.FluidProperties](#)

inputSet

Input set object [Inputs.InputSet](#)

mixSolver

Mixture object [Solvers.Mixture.Mixture](#)

Method Summary**EQUIL**(*tfSolver, flm, drp, mix, zIdx*)

EQUIL() Find entrained ratio at film/drop equilibrium state [-]

Iteratively computes the entrained liquid ratio at the onset of annular flow by balancing droplet deposition and film entrainment rates.

Inputs:

- *tfSolver* — [Solvers.ThreeField.ThreeFieldSolver](#) object
- *flm* — Film field object ([Solvers.ThreeField.Film](#))
- *drp* — Droplet field object ([Solvers.ThreeField.Drop](#))
- *mix* — Mixture object ([Solvers.Mixture.Mixture](#))
- *zIdx* — Axial index for evaluation

Algorithm:

- Starts with an initial guess for droplet mass flow (50% of liquid mass)
- Updates droplet and film flow rates iteratively
- Computes difference between deposition and entrainment rates
- Stops when error < 1e-4 kg/s/m or after 100 iterations

Notes:

- Enhanced drop deposition is disabled
- Uses interpolation for improved convergence
- Falls back to ad-hoc update if interpolation fails
- Method update film and drop mass flow rates. Use `flm.copy()` and `drp.copy()` as input arguments if used in post-process

initializeSolver(*tfSolver*)

INITIALIZESOLVER Initializes solver parameters and constructs film/drop objects.

Sets up the solver's internal state using the provided [Inputs.InputSet](#) and [Solvers.ThreeField.ThreeFieldSolver](#). This includes initialization of data structures for transient and steady-state simulations based on the mixture solver solutions.

Operations performed:

- Initializes film and drop arrays for each time step

- Initializes fluid property objects
- Sets up mixture object
- Assigns initial conditions for phase mass flow rates, velocities, enthalpies

Notes:

- The function assumes uniform axial discretization.
- Wall heat flux is computed from interpolated boundary conditions.

plott(*tfSolver, zIdx, opt*)

PLOTT Plots temporal distributions of three-field parameters at a given axial location.

Generates time-series plots of selected three-field parameters at a specified axial index. The method supports multiple display modes, wall selections, and plot arrangements.

Inputs:

- *tfSolver* — [Solvers.ThreeField.ThreeFieldSolver](#) object containing simulation data
- *zIdx* — Axial index (scalar, positive integer)
- *opt.display* — Parameters to display (e.g., 'HFLUX', 'W', 'U', etc.)
- *opt.solveMode* — Solve mode: 'REAL' or 'NULL'
- *opt.wall* — Wall index(es) to plot
- *opt.tIdx* — Time indices to include in the plot
- *opt.reverseTime* — Logical flag to reverse time axis
- *opt.unitTemp* — Temperature unit: 'K' or 'C'
- *opt.arrangement* — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- *opt.resize* — Resize factor for plot scaling

Notes:

- If *opt.display* is set to 'ALL', all supported parameters are plotted.
- The function validates that at least two time indices are provided.
- Temperature unit conversion is applied if 'C' is selected.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.

plotz(*tfSolver, tIdx, opts*)

PLOTZ Plots spatial (axial) distributions of three-field parameters.

Generates axial plots of selected three-field parameters at specified time indices. The function supports multiple display modes, wall selections, and optional animation over time.

Inputs:

- *tfSolver* — [Solvers.ThreeField.ThreeFieldSolver](#) object containing simulation data
- *tIdx* — Time index or indices (vector of positive integers)
- *opts.display* — Parameters to display (e.g., 'HFLUX', 'W', 'U', etc.)
- *opts.solveMode* — Solve mode: 'REAL' or 'NULL'
- *opts.wall* — Wall index(es) to plot
- *opts.zIdx* — Axial indices to include in the plot
- *opts.obstructions* — Logical flag to display obstruction locations
- *opts.annular* — Logical flag to plot annular two-phase flow parameters only from the onset of annular flow
- *opts.unitTemp* — Temperature unit: 'K' or 'C'
- *opts.arrangement* — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- *opts.resize* — Resize factor for plot scaling

Notes:

- If multiple time indices are provided, the function creates an animated plot.
- At least two axial indices must be specified to enable spatial plotting.
- Temperature unit conversion is applied if 'C' is selected.
- Obstruction locations are plotted if *opts.obstructions* is true.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.

plotzt(*tfSolver, opt*)

PLOTZT Plots 2D time/elevation distributions of three-field parameters.

Generates surface plots of selected three-field related parameters over time and axial (elevation) positions. It supports plotting for film and drop fields, and can handle both real and null (initial) transient data.

Inputs:

- *tfSolver* — [Solvers.ThreeField.ThreeFieldSolver](#) object containing simulation data
- *opt.display* — Parameter(s) to display (e.g., 'HFLUX', 'U', etc.)
- *opt.label* — Corresponding labels for display parameters
- *opt.unit* — Units for each parameter
- *opt.field* — Field to plot: 'film' or 'drop'
- *opt.solveMode* — Solve mode: 'REAL' or 'NULL'
- *opt.wall* — Wall index(es) to plot
- *opt.zIdx* — Axial indices (must include at least 2)
- *opt.tIdx* — Time indices (must include at least 2)
- *opt.reverseTime* — Logical flag to reverse time axis
- *opt.shading* — Surface shading style: 'faceted', 'flat', or 'interp'
- *opt.view* — View angle for 3D plot [azimuth elevation]

Notes:

- If 'ALL' is passed to *opt.display*, all supported parameters are plotted.
- The function validates that at least two axial and time indices are provided to enable meaningful 2D plotting.
- Each wall is plotted in a separate figure window with appropriate titles and subplot grouping.

solve(*tfSolver*)

SOLVE Executes the three-field solver for steady-state and transient simulations.

Runs the full solution process for the [Solvers.ThreeField.ThreeFieldSolver](#) object. It manages initialization, time stepping, axial sweeps, inner iterations, convergence checks, and logging.

Workflow:

- Enables logging and opens persistent log file
- Validates solver state before execution
- Runs steady-state initialization (*solveINIT* = true)
- If initial step converges, proceeds with transient simulation
- Handles exceptions and ensures proper log closure

Notes:

- Uses internal *solver* function to handle both steady-state and transient modes
- Applies relaxation factors for phase flows, velocities, and enthalpies
- Supports thermal non-equilibrium modeling
- Logs progress and outputs to session directory

class [Solvers.ThreeField.Drop](#)(*inputSet, fluid*)

Bases: [Solvers.AbstractField](#)

DROP Class for modeling drop field in three-field solver

This class encapsulates the physical and numerical properties of the drop field, including flow variables and phase interactions. It supports multiple solver models and provides methods for computing derived quantities.

Constructor Summary

Drop(*inputSet*, *fluid*)

DROP Constructor for *Drop()* class

Initializes the drop field object with input configuration and fluid properties.

Inputs:

- *inputSet* — *Inputs.InputSet* object containing model, geometry, and boundary conditions
- *fluid* — *Inputs.FluidProperties* object containing thermophysical fluid data

Property Summary

DT = 0

Time step size [s] from *Inputs.Options.TSTEP*

H = 1E6

Enthalpy [J/kg]

ITR

Iteration tracking

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIDX = 1

Time step index [-]

TIME = 0

Time series [s]

U = 1.

Velocity [m/s]

W = 1.

Mass flow rate [kg/s]

Z = 1.

Elevation [m]

mix = NaN

Solvers.Mixture.Mixture object

Method Summary

AI(*drop*, *zIdx*)

AI() Drop volumetric interfacial area [m⁻¹]

Calculates the interfacial area per unit volume for drops.

Inputs:

- *drop* — *Solvers.ThreeField.Drop* object
- *zIdx* — Axial indices to evaluate (optional)

AREA(*drop*, *zIdx*)

AREA() Drop interfacial area [m²]

Calculates the surface area of a spherical drop.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

CONC(drop, zIdx)

CONC() Drop concentration [kg/m³]

Calculates the local drop-field concentration based on mass flow and vapor-phase properties.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

DENSITY(drop, zIdx)

DENSITY() Drop number density [m⁻³]

Calculates the number of drops per unit volume based on flow properties and geometry.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

DIAM(drop, zIdx)

DIAM() Drop diameter [m]

Returns the drop diameter based on *Inputs.Model.DROPDIA*.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

DRAG(drop, zIdx)

DRAG() Drop drag coefficient [-]

Calculates drag coefficient using selected correlation based on *Inputs.Model.DROPDRAG*.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FBUOY(drop, zIdx)

FBUOY() Drop buoyancy force [N/m³]

Calculates buoyancy force based on pressure gradient.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FDRAG(drop, zIdx)

FDRAG() Vapor drag force on drops [N/m³]

Calculates shear force exerted by vapor on drops.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FENT(drop, film, zIdx)

FENT() Film entrainment shear force [N/m³]

Calculates shear force due to film entrainment acting on drops.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- film — *Solvers.ThreeField.Film* object
- zIdx — Axial indices to evaluate (optional)

FGRV(drop, zIdx)

FGRV() Gravitational force on drops [N/m³]

Calculates gravity force acting on drops based on channel inclination angle.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

FTOT(drop, film, zIdx, opt)

FTOT() Total force acting on drops [N/m³]

Calculates combined forces (drag, buoyancy, gravity, entrainment) based on selected option.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- film — *Solvers.ThreeField.Film* object
- zIdx — Axial indices to evaluate (optional)
- opt — Option flag (0: full balance, 1: simplified)

KENH(drop, zIdx)

KENH() Drop deposition enhancement factor [-]

Calculates axial variation of deposition enhancement using Windecker correlation or other models based on *Inputs.Model.DEPENHANCEMENT*.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

MDEP(drop, zIdx, enhanced)

MDEP() Drop deposition mass flux [kg/m²/s]

Implements various deposition models selected based on *Inputs.Model.DEPOSITION*. Applies enhancement factors, and restricts deposition to annular flow region. The enhancement flag allow to enable/disable deposition enhancement

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)
- enhanced — Flag indicating enhanced deposition downstream obstructions (optional, default = true)

RE(drop, zIdx)

RE() Drop Reynolds number [-]

Calculates the Reynolds number for the drop field based on mass flow rate, viscosity, and channel perimeter.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

REV(drop, zIdx)

REV() Drop Reynolds number with respect to vapor properties [-]

Calculates Reynolds number using vapor density, viscosity, and relative phase velocity.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

UALGEBR(*drop, film, zIdx*)

UALGEBR() Algebraic drop velocity [m/s]

Calculates drop velocity consistent with mixture model based on void fraction and geometry.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- film — *Solvers.ThreeField.Film* object
- zIdx — Axial indices to evaluate (optional)

UEQUIL(*drop, film, zIdx, opt*)

UEQUIL() Drop velocity based on equilibrium model ($F_{tot} = 0$) [m/s]

Iteratively solves for drop velocity such that the total force acting on the drop phase is zero (force balance). Uses secant method with relaxation for convergence.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- film — *Solvers.ThreeField.Film* object
- zIdx — Axial indices to evaluate (optional)
- **opt** — **Option flag for force balance:**
0 = full force balance (drag, buoyancy, gravity, entrainment) 1 = simplified (drag + buoyancy only)

USLIP(*drop, zIdx*)

USLIP() Slip velocity model for drops [m/s]

Calculates drop velocity using slip ratio model based on *Inputs.Model.DROPSLIP*.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

VOLUME(*drop, zIdx*)

VOLUME() Drop volume [m³]

Calculates the volume of a spherical drop.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

VR(*drop, zIdx*)

VR() Drop local relative velocity [m/s]

Calculates the velocity difference between vapor and drop fields.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Only AREAMEAN option is implemented.

X(*drop, zIdx*)

X() Vapor quality in the bulk (drop) region [-]

Vapor quality in the bulk region, considering only the droplet for the liquid phase. The bulk region is complementary of the near-wall region, defined in the mixture solver.

Inputs:

- drop — *Solvers.ThreeField.Drop* object
- zIdx — Axial indices to evaluate (optional)

class *Solvers.ThreeField.Film*

Bases: *Solvers.AbstractFilm*

FILM Class for modeling liquid film in three-field solver

This class encapsulates the physical and numerical properties of the film field, including flow variables and phase interactions.

Property Summary

DT = 0

Time step size [s] from *Inputs.Options.TSTEP*

H = 1E6

Enthalpy [J/kg]

HFLUX = 1.

Film heat flux [W/m²]

ITR

Iteration tracking

MEVAP = -1.

Evaporation mass flux [kg/s/m²]

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIDX = 1

Time step index [-]

TIME = 0

Time series [s]

U = 1.

Velocity [m/s]

W = 1.

Mass flow rate [kg/s]

Z = 1.

Elevation [m]

5.4.4 Two-fluid solver

The *Solvers.TwoFluid* module in **OpenSTREAM** provides classes and tools for simulating two-phase flows using a two-fluid approach. In this method, each phase (liquid and vapor) is modeled as an interpenetrating continuum with its own set of conservation equations for mass, momentum, and energy. This framework enables explicit representation

of phase interactions, slip velocities, and non-equilibrium effects, making it well-suited for applications where phase separation, interfacial dynamics, and transient boiling phenomena are significant.

This module includes:

- Solver class for two-fluid-based flow simulations: `Solvers.TwoFluid.TwoFluidSolver`
 - Liquid and vapor field classes: `Solvers.TwoFluid.Liquid`, `Solvers.TwoFluid.Vapor`
 - Mixture class: `Solvers.TwoFluid.Mixture`
 - Two-fluid primary properties: `Solvers.TwoFluid.Liquid.W`, `Solvers.TwoFluid.Liquid.U`, `Solvers.TwoFluid.Liquid.H`, `Solvers.TwoFluid.Vapor.W`, `Solvers.TwoFluid.Vapor.U`, `Solvers.TwoFluid.Vapor.H`
 - Key solver class methods: `Solvers.TwoFluid.TwoFluidSolver.initializeSolver`, `Solvers.TwoFluid.TwoFluidSolver.solve()`, `Solvers.TwoFluid.TwoFluidSolver.plotz`
 - Field class methods for calculating secondary two-fluid parameters.
-

class `Solvers.TwoFluid.TwoFluidSolver`(*inputSet*, *mixSolver*)

Bases: `Solvers.AbstractSolver`

TWOFLUIDSOLVER Solver for initializing, solving, and visualizing two-fluid flow.

The `TwoFluidSolver` class handles the setup, execution, and visualization of thermal-hydraulic simulations using a two-phase two-fluid approach.

Responsibilities:

- Initializes solver parameters from an `Inputs.InputSet` object
- Constructs `Solvers.TwoFluid.Liquid` and `Solvers.TwoFluid.Vapor` objects for transient and steady-state analysis
- Provides plotting utilities for spatial and temporal distributions

Key Components:

- Phase construction (liquid/vapor)
- Phase mass/momentum/energy transport modeling
- Pressure drop gradient obtained from mixture solver
- Support for relaxation models
- Visualization of results across time and axial domains

Constructor Summary

TwoFluidSolver(*inputSet*, *mixSolver*)

TWOFLUIDSOLVER Constructor for the `TwoFluidSolver()` class.

Creates a new instance of the `TwoFluidSolver()` class using the provided `Inputs.InputSet`. This constructor initializes the solver by calling the abstract `Solvers.AbstractSolver` superclass constructor and then sets up all necessary solver parameters and internal data structures via `Solvers.TwoFluid.TwoFluidSolver.initializeSolver`

Input:

- *inputSet* — An `Inputs.InputSet` object containing model configuration, geometry, boundary conditions, and solver options.
- *mixSolver* — An optional solved `Solvers.Mixture.MixtureSolver` object

Notes:

- This constructor assumes *Inputs.InputSet* is fully validated.
- A solved mixture-model solution is required for initialization. If a solved mixture solver is supplied, its solution is reused. If unsolved or if no mixture solver is supplied, one is created and solved automatically.
- Solver initialization includes liquid and vapor objects setup.
- Time/space discretization and boundary condition interpolation initialized by *Solvers.Mixture.MixtureSolver* are used

Property Summary

DT = 0

Time step size [s] from *Inputs.Options.TSTEP*

DZ = 0

Axial step size [m]

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from *Inputs.Model.NNODES*

TIME = 0

Time series [s]

Z = 1.

Elevation [m]

boundaryConditions

Boundary conditions object *Inputs.BoundaryConditions*

fluid

Fluid object *Inputs.FluidProperties*

inputSet

Input set object *Inputs.InputSet*

liquid

Liquid object

liquidInit

Null transient liquid object

mixSolver

Mixture object *Solvers.Mixture.Mixture*

vapor

Vapor object

vaporInit

Null transient vapor object

Method Summary

initializeSolver(*twfSolver*)

INITIALIZESOLVER Initializes solver parameters and constructs liquid/vapor objects.

Sets up the solver's internal state using the provided *Inputs.InputSet* and *Solvers.TwoFluid.TwoFluidSolver*. This includes initialization of data structures for transient and steady-state simulations based on the mixture solver solutions.

Operations performed:

- Initializes liquid and vapor arrays for each time step
- Initializes fluid property objects
- Sets up mixture object
- Assigns initial conditions for phase mass flow rates, velocities, enthalpies

Notes:

- The function assumes uniform axial discretization.
- Wall heat flux is computed from interpolated boundary conditions.

plott(*twfSolver*, *zIdx*, *opt*)

PLOTT Plots temporal distributions of two-fluid parameters at a given axial location.

Generates time-series plots of selected two-fluid parameters at a specified axial index. The method supports multiple display modes, wall selections, and plot arrangements.

Inputs:

- twfSolver — *Solvers.TwoFluid.TwoFluidSolver* object containing simulation data
- zIdx — Axial index (scalar, positive integer)
- opt.display — Parameters to display (e.g., 'HFLUX', 'W', 'U', etc.)
- opt.solveMode — Solve mode: 'REAL' or 'NULL'
- opt.wall — Wall index(es) to plot
- opt.tIdx — Time indices to include in the plot
- opt.reverseTime — Logical flag to reverse time axis
- opt.unitTemp — Temperature unit: 'K' or 'C'
- opt.arrangement — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- opt.resize — Resize factor for plot scaling

Notes:

- If opt.display is set to 'ALL', all supported parameters are plotted.
- The function validates that at least two time indices are provided.
- Temperature unit conversion is applied if 'C' is selected.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.

plotz(*twfSolver*, *tIdx*, *opts*)

PLOTZ Plots spatial (axial) distributions of two-fluid parameters.

Generates axial plots of selected two-fluid parameters at specified time indices. The function supports multiple display modes, wall selections, and optional animation over time.

Inputs:

- twfSolver — *Solvers.TwoFluid.TwoFluidSolver* object containing simulation data
- tIdx — Time index or indices (vector of positive integers)
- opts.display — Parameters to display (e.g., 'HFLUX', 'W', 'U', etc.)
- opts.solveMode — Solve mode: 'REAL' or 'NULL'
- opts.wall — Wall index(es) to plot
- opts.zIdx — Axial indices to include in the plot
- opts.obstructions — Logical flag to display obstruction locations
- opts.unitTemp — Temperature unit: 'K' or 'C'
- opts.arrangement — Plot arrangement: 'flow', 'vertical', or 'horizontal'
- opts.resize — Resize factor for plot scaling

Notes:

- If multiple time indices are provided, the function creates an animated plot.
- At least two axial indices must be specified to enable spatial plotting.
- Temperature unit conversion is applied if 'C' is selected.
- Obstruction locations are plotted if `opts.obstructions` is true.
- Each wall is plotted in a separate tile with appropriate legends and axis scaling.

plotzt(*twfSolver*, *opt*)

PLOTZT Plots 2D time/elevation distributions of two-fluid parameters.

Generates surface plots of selected two-fluid related parameters over time and axial (elevation) positions. It supports plotting for liquid and vapor fields, and can handle both real and null (initial) transient data.

Inputs:

- *twfSolver* — *Solvers.TwoFluid.TwoFluidSolver* object containing simulation data
- *opt.display* — Parameter(s) to display (e.g., 'HFLUX', 'U', etc.)
- *opt.label* — Corresponding labels for display parameters
- *opt.unit* — Units for each parameter
- *opt.field* — Field to plot: 'liquid' or 'vapor'
- *opt.solveMode* — Solve mode: 'REAL' or 'NULL'
- *opt.wall* — Wall index(es) to plot
- *opt.zIdx* — Axial indices (must include at least 2)
- *opt.tIdx* — Time indices (must include at least 2)
- *opt.reverseTime* — Logical flag to reverse time axis
- *opt.shading* — Surface shading style: 'faceted', 'flat', or 'interp'
- *opt.view* — View angle for 3D plot [azimuth elevation]

Notes:

- If 'ALL' is passed to *opt.display*, all supported parameters are plotted.
- The function validates that at least two axial and time indices are provided to enable meaningful 2D plotting.
- Each wall is plotted in a separate figure window with appropriate titles and subplot grouping.

solve(*twfSolver*)

SOLVE Executes the two-fluid solver for steady-state and transient simulations.

Runs the full solution process for the *Solvers.TwoFluid.TwoFluidSolver* object. It manages initialization, time stepping, axial sweeps, inner iterations, convergence checks, and logging.

Workflow:

- Enables logging and opens persistent log file
- Validates solver state before execution
- Runs steady-state initialization (`solveINIT = true`)
- If initial step converges, proceeds with transient simulation
- Handles exceptions and ensures proper log closure

Notes:

- Uses internal *solver* function to handle both steady-state and transient modes
- Applies relaxation factors for phase flows, velocities, and enthalpies
- Supports thermal non-equilibrium modeling
- Logs progress and outputs to session directory

class *Solvers.TwoFluid.REGIMES*

Bases: `double`

REGIMES Enumeration of two-phase flow regimes

This class defines the available two-phase flow regimes for constitutive model selection, used by the two-fluid solver.

The flow regimes are defined as an enum class to increase computational efficiency while increasing readability of code, particularly when different calculations are carried out for different flow regimes.

Flow regimes:

- LIQUID — Single-phase liquid
- BUBBLY_SUBCOOLED — Bubbly flow in subcooled region
- BUBBLY_SATURATED — Bubbly flow in saturated region
- INTERMEDIATE — Intermediate flow region
- ANNULAR — Annular flow
- DFFB — Dispersed Flow Film Boiling

class `Solvers.TwoFluid.Vapor`(*inputSet*, *fluid*)

Bases: [`Solvers.AbstractField`](#)

VAPOR Class for modeling vapor field in two-fluid solver

This class encapsulates the physical and numerical properties of the vapor phase, including flow variables, phase interactions, heat transfer, and relaxation models. It supports multiple solver models and provides methods for computing derived quantities and handling flow regime transitions.

Constructor Summary

Vapor(*inputSet*, *fluid*)

VAPOR Constructor for [`Vapor\(\)`](#) class

Initializes the vapor field object with input configuration and fluid properties.

Inputs:

- *inputSet* — [`Inputs.InputSet`](#) object containing model, geometry, and boundary conditions
- *fluid* — [`Inputs.FluidProperties`](#) object containing thermophysical fluid data

Property Summary

DT = 0

Time step size [s] from [`Inputs.Options.TSTEP`](#)

H = 1E6

Enthalpy [J/kg]

ITR

Iteration properties

NTIME = 0

Number of time steps [-]

NZ = 0

Number of axial steps [-] from [`Inputs.Model.NNODES`](#)

TIDX = 1

Time step index [-]

TIME = 0

Time series [s]

U = 1.

Velocity [m/s]

W = 1.

Mass flow rate [kg/s]

Z = 1.

Elevation [m]

mix = NaN

Solvers.TwoFluid.Mixture object

Method Summary

AREA(*vapor, liquid, zIdx*)

AREA() Vapor cross-section area [m²]

Computes the effective cross-sectional area occupied by the vapor phase.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

CD(*vapor, liquid, zIdx*)

CD() Drag coefficient for dispersed gas [-]

Computes the drag coefficient for dispersed vapor based on *Inputs.Model.DROPDRAG* model.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Values are clamped to a maximum of 1

FBUOY(*vapor, liquid, zIdx*)

FBUOY() Vapor buoyancy force [N/m]

Computes the buoyancy force acting on the vapor phase due to pressure gradient.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- The pressure gradient from the mixture solver is used

FDRAG(*vapor, liquid, zIdx*)

FDRAG() Interfacial drag force [N/m]

Computes the interfacial drag force acting on the vapor phase by reversing the sign of the liquid-phase drag (*Solvers.TwoFluid.Liquid.FDRAG()*).

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

FGRAV(*vapor, liquid, zIdx*)*FGRAV()* Vapor gravity force [N/m]

Computes the gravitational force acting on the vapor phase.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

FINTEVAP(*vapor, liquid, zIdx*)*FINTEVAP()* Momentum exchange through interfacial evaporation [N/m]

Computes the momentum exchange due to interfacial evaporation.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

FLOWREGIME(*vapor, liquid, zIdx*)*FLOWREGIME()* Categorical two-phase flow regimesReturns the flow regime classification based on liquid-phase criteria (*Solvers.TwoFluid.Liquid.FLOWREGIME()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

Returns:

- flowregime — Flow regime category (from *Solvers.TwoFluid.REGIMES*)

FTOT(*vapor, liquid, zIdx*)*FTOT()* Total force [N/m]

Computes the total force acting on the vapor phase, including gravity, buoyancy, wall shear, drag, and interfacial momentum exchanges.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

FWALEVAP(*vapor, liquid, zIdx*)*FWALEVAP()* Momentum exchange through wall evaporation [N/m]

Computes the momentum exchange due to wall boiling evaporation.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

FWALL(*vapor, liquid, zIdx*)*FWALL()* Wall shear force [N/m]

Computes the wall shear force acting on the vapor phase, weighted by void fraction.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

HFLUX(vapor, zIdx)

HFLUX() Wall heat flux to vapor phase [W/m²]

Computes the net wall heat flux transferred to the vapor phase.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Uses *Solvers.TwoFluid.Vapor.HWALHEAT()* divided by wall perimeter

HFLUXWALEVAP(vapor, liquid, zIdx)

HFLUXWALEVAP() Wall boiling heat flux [W/m²]

Computes the portion of wall heat flux attributed to boiling (evaporation) for the vapor phase, based on liquid-phase calculations (*Solvers.TwoFluid.Liquid.HFLUXWALEVAP()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

HINT(vapor, liquid, zIdx)

HINT() Linear interfacial heat transfer rate [W/m]

Computes the interfacial heat transfer rate for the vapor phase based on mass transfer and enthalpy difference (based on *Inputs.Model.INTTRANS* model).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

Returns:

- Hcond — Condensation heat transfer rate [W/m]
- Hevap — Evaporation heat transfer rate [W/m]

HINTCOND(vapor, liquid, zIdx)

HINTCOND() Linear interfacial heat condensation rate [W/m]

Extracts the condensation component from *Solvers.TwoFluid.Vapor.HINT()*.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

HINTEVAP(vapor, liquid, zIdx)

HINTEVAP() Linear interfacial heat evaporation rate [W/m]

Extracts the evaporation component from *Solvers.TwoFluid.Vapor.HINT()*.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

HTOT(*vapor, liquid, zIdx*)

HTOT() Total linear vapor heat rate [W/m]

Computes the total heat transfer to the vapor phase, including interfacial and wall contributions.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

HWALEVAP(*vapor, liquid, zIdx*)

HWALEVAP() Linear wall heat evaporation (boiling) rate [W/m]

Computes the wall boiling heat transfer rate using latent heat (based on *Inputs.Model.INTTRANSH* model) and wall mass evaporation.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

HWALHEAT(*vapor, zIdx*)

HWALHEAT() Linear wall heat to vapor rate [W/m]

Returns the wall heat generation rate transferred to the vapor phase.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Set to zero before CBT

INTAREA(*vapor, liquid, zIdx*)

INTAREA() Volumetric interfacial area [m⁻¹]

Computes the interfacial area concentration using liquid-phase logic (from *Solvers.TwoFluid.Liquid.INTAREA()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

INTHFLUX(*vapor, liquid, zIdx*)

INTHFLUX() Interfacial heat flux [W/m²]

Computes the interfacial heat flux due to evaporation and condensation (delegated to liquid-phase calculation in *Solvers.TwoFluid.Liquid.INTHFLUX()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

Returns:

- inthflux_evap — Evaporation heat flux [W/m²]
- inthflux_cond — Condensation heat flux [W/m²]

Notes:

- Positive toward liquid phase

INTNU(*vapor, liquid, zIdx*)

INTNU() Interfacial Nusselt number for dispersed vapor [-]

Computes the interfacial Nusselt number based on *Inputs.Model.INTNU* model.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

J(*vapor, zIdx*)

J() Vapor superficial velocity [m/s]

Computes the superficial velocity of the vapor phase.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

L(*vapor, zIdx*)

L() Dispersed vapor interfacial length scale [m]

Returns the characteristic length scale for dispersed vapor (e.g., bubble diameter).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Currently uses a constant model

MINT(*vapor, liquid, zIdx*)

MINT() Linear interfacial mass transfer rates [kg/s/m]

Computes condensation and evaporation mass transfer rates for the vapor phase by reversing the sign of liquid-phase values (*Solvers.TwoFluid.Liquid.MINT()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

Returns:

- Mcond — Condensation mass transfer rate [kg/s/m]
- Mevap — Evaporation mass transfer rate [kg/s/m]

MINTCOND(*vapor, liquid, zIdx*)

MINTCOND() Linear interfacial condensation mass transfer [kg/s/m]

Extracts the condensation component from *Solvers.TwoFluid.Vapor.MINT()*.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

MINTEVAP(*vapor, liquid, zIdx*)

MINTEVAP() Linear interfacial evaporation mass transfer [kg/s/m]

Extracts the evaporation component from *Solvers.TwoFluid.Vapor.MINT()*.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

MTOT(*vapor, liquid, zIdx*)

MTOT() Total linear mass transfer [kg/s/m]

Computes the total vapor mass transfer rate by reversing the sign of the liquid-phase total (*Solvers.TwoFluid.Liquid.MTOT()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

MWALEVAP(*vapor, liquid, zIdx*)

MWALEVAP() Linear wall mass evaporation rate [kg/s/m]

Computes the wall boiling mass transfer rate for vapor by reversing the sign of the liquid-phase value (*Solvers.TwoFluid.Liquid.MWALEVAP()*).

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

RE(*vapor, zIdx*)

RE() Vapor Reynolds number [-]

Computes the vapor-phase Reynolds number based on flow rate and viscosity.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

REL(*vapor, liquid, zIdx*)

REL() Dispersed vapor Reynolds number [-]

Computes the Reynolds number for dispersed vapor using liquid properties and relative velocity.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

S(*vapor, liquid, zIdx*)

S() Phase slip ratio [-]

Returns the slip ratio between vapor and liquid phases.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

T(*vapor, zIdx*)

T() Vapor temperature [K]

Computes the vapor temperature from enthalpy.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object

- *zIdx* — Axial indices to evaluate (optional)

TAUW(*vapor, zIdx*)

TAUW() Wall shear stress [N/m²]

Returns the wall shear stress acting on the vapor phase.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Uses mixture-level shear stress for consistency

USLIP(*vapor, liquid, zIdx*)

USLIP() Vapor velocity based on input phase slip [m/s]

Computes the vapor velocity using slip ratio and liquid velocity.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

VF(*vapor, liquid, zIdx*)

VF() Vapor volumetric fraction [-]

Computes the vapor volumetric (i.e., void) fraction using slip ratio and densities.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

VISC(*vapor, liquid, zIdx*)

VISC() Viscosity number for dispersed gas [-]

Computes the dimensionless viscosity number using liquid and vapor properties.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

WALEVAPRATIO(*vapor, liquid, zIdx*)

WALEVAPRATIO() Wall evaporation mass ratio [-]

Returns the fraction of liquid mass undergoing wall boiling.

Inputs:

- *vapor* — *Solvers.TwoFluid.Vapor* object
- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

Returns:

- *walevapratio* — Wall boiling mass ratio [-]

X(*vapor, zIdx*)

X() Vapor mass fraction [-]

Computes the vapor mass fraction relative to the mixture flow.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

XTR_ANN(vapor, liquid)

XTR_ANN() Quality at onset of annular flow region

XTR_CBT(vapor, liquid)

XTR_CBT() Quality at Critical Boiling Transition [-]

Returns the equilibrium quality threshold for the onset of critical boiling transition.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object

XTR_ITM(vapor, liquid)

XTR_ITM() Quality at onset of intermediate region [-]

Returns the equilibrium quality threshold for the onset of the intermediate flow regime.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object

XTR_SAT(vapor, liquid)

XTR_SAT() Quality at end of subcooled boiling transition [-]

Returns the equilibrium quality threshold for the onset of saturated boiling.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object

XTR_SUB(vapor, liquid)

XTR_SUB() Quality at onset of subcooled boiling transition [-]

Returns the equilibrium quality threshold for the onset of subcooled boiling.

Inputs:

- vapor — *Solvers.TwoFluid.Vapor* object
- liquid — *Solvers.TwoFluid.Liquid* object

class *Solvers.TwoFluid.Mixture*(mixture, liquid, vapor)

Bases: *Solvers.AbstractPhase*

MIXTURE Class for two-fluid mixture properties in thermal-hydraulic solver

Represents combined properties of liquid and vapor phases in a two-fluid solver. Provides methods to compute mixture-level quantities such as mass flow, density, velocity, heat flux, and thermodynamic quality.

Responsibilities:

- Aggregate liquid and vapor properties into mixture-level results
- Compute derived quantities (void fraction, equilibrium quality, CHF)
- Provide access to geometry and fluid properties for calculations

Key features:

- Supports axial indexing for spatially resolved calculations
- Interfaces with liquid and vapor sub-objects for phase-specific data

Constructor Summary

Mixture(*mixture, liquid, vapor*)

MIXTURE Constructor for *Mixture()* class

Initializes the mixture object by linking liquid and vapor phase objects and copying solver configuration and fluid properties.

Inputs:

mixture — *Solvers.Mixture.Mixture* object liquid — *Solvers.TwoFluid.Liquid* object

vapor — *Solvers.TwoFluid.Vapor* object

Method Summary

AFDISTR(*mix, param1, param2, zIdx*)

AFDISTR() Annular flow distribution function [-]

Computes the annular flow distribution function for given parameters.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object param1 — First parameter for distribution function

param2 — Second parameter for distribution function zIdx — Axial indices to evaluate (optional)

CBT(*mix, zIdx*)

CBT() Critical Boiling Transition flag [-]

Returns the *CBT()* flag for each axial location.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

CHF(*mix, zIdx*)

CHF() Critical Heat Flux [W/m²]

Returns the *CHF()* value for each axial location.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

DPDZ(*mix, zIdx*)

DPDZ() Pressure gradient [Pa/m]

Computes the axial pressure gradient from mixture pressure drop.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

H(*mix, zIdx*)

H() Mixture enthalpy [J/kg]

Calculates the mixture enthalpy as a mass-weighted average of phase enthalpies.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

HFLUX(*mix, zIdx*)

HFLUX() Wall heat flux [W/m²]

Returns the wall heat flux distribution along the axial domain.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

LHGR(*mix, zIdx*)

LHGR() Linear heat generation rate [W/m]

Computes the linear heat generation rate from wall heat flux and perimeter.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

MFLUX(mix, zIdx)

MFLUX() Mixture mass flux [kg/m²/s]

Calculates the mixture mass flux using total flow and channel area.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

OAFX(mix, zIdx)

OAFX() Onset of annular flow equilibrium quality [-]

Returns the OAF quality from the mixture object.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

RELAXTCOND(mix, zIdx)

RELAXTCOND() Time relaxation for interfacial condensation [s]

Returns the relaxation time applied to interfacial condensation processes in the two-fluid mixture model.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

RELAXTEVAP(mix, zIdx)

RELAXTEVAP() Time relaxation for interfacial evaporation [s]

Returns the relaxation time applied to interfacial evaporation processes in the two-fluid mixture model.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

RHO(mix, zIdx)

RHO() Mixture density [kg/m³]

Computes the mixture density based on void fraction and phase densities.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

TAUW(mix, zIdx)

TAUW() Wall shear stress [N/m²]

Returns the wall shear stress from the mixture object.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

TIME(mix)

TIME() Time series [s]

Returns the time array from the mixture object.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object

U(mix, zIdx)

U() Mixture velocity [m/s]

Calculates the mixture velocity as a mass-weighted average of phase velocities.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object zIdx — Axial indices to evaluate (optional)

VF(*mix*, *zIdx*)

VF() Void fraction [-]

Computes the vapor volume fraction in the mixture.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object *zIdx* — Axial indices to evaluate (optional)

W(*mix*, *zIdx*)

W() Mixture mass flow rate [kg/s]

Computes the total mass flow rate as the sum of liquid and vapor flows.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object *zIdx* — Axial indices to evaluate (optional)

WALEVAPRATIO(*mix*, *zIdx*)

WALEVAPRATIO() Wall mass evaporation ratio [-]

Returns the ratio of wall heat flux converted to evaporation.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object *zIdx* — Axial indices to evaluate (optional)

XEQ(*mix*, *zIdx*)

XEQ() Equilibrium thermodynamic quality [-]

Returns the equilibrium quality from the mixture object.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object *zIdx* — Axial indices to evaluate (optional)

Z(*mix*, *zIdx*)

Z() Axial nodes [m]

Returns the axial positions for the mixture domain.

Inputs:

mix — *Solvers.TwoFluid.Mixture* object *zIdx* — Axial indices to evaluate (optional)

class *Solvers.TwoFluid.Liquid*(*inputSet*, *fluid*)

Bases: *Solvers.AbstractField*

LIQUID Class for modeling liquid field in two-fluid solver

This class encapsulates the physical and numerical properties of the liquid phase, including flow variables, phase interactions, heat transfer, and relaxation models. It supports multiple solver models and provides methods for computing derived quantities and handling flow regime transitions.

Constructor Summary

Liquid(*inputSet*, *fluid*)

LIQUID Constructor for *Liquid()* class

Initializes the liquid field object with input configuration and fluid properties.

Inputs:

- *inputSet* — *Inputs.InputSet* object containing model, geometry, and boundary conditions
- *fluid* — *Inputs.FluidProperties* object containing thermophysical fluid data

Property Summary

DT = 0

Time step size [s] from *Inputs.Options.TSTEP*

H = 1E6
Enthalpy [J/kg]

ITR
Iteration tracking

NTIME = 0
Number of time steps [-]

NZ = 0
Number of axial steps [-] from *Inputs.Model.NNODES*

TIDX = 1
Time step index [-]

TIME = 0
Time series [s]

U = 1.
Velocity [m/s]

W = 1.
Mass flow rate [kg/s]

Z = 1.
Elevation [m]

mix = NaN
Solvers.TwoFluid.Mixture object

Method Summary

AREA(*liquid, vapor, zIdx*)
AREA() Liquid cross-section area [m²]
Computes the effective cross-sectional area occupied by the liquid phase.
Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

CD(*liquid, vapor, zIdx*)
CD() Drag coefficient for dispersed liquid [-]
Computes the drag coefficient for dispersed liquid based on *Inputs.Model.DROPDRAG* model.
Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Values are clamped to a maximum of 1

FBUOY(*liquid, vapor, zIdx*)
FBUOY() Liquid buoyancy force [N/m]
Computes the buoyancy force acting on the liquid phase due to pressure gradient.
Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- The pressure gradient from the mixture solver is used

FDRAG(*liquid, vapor, zIdx*)

FDRAG() Interfacial drag force [N/m]

Computes the interfacial drag force acting on the liquid phase.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Uses smooth transition between dispersed gas and liquid drag based on *Inputs.Model.DISPGAS2DISPLIQ* model.

FGRAV(*liquid, vapor, zIdx*)

FGRAV() Liquid gravity force [N/m]

Computes the gravitational force acting on the liquid phase.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

FINTCOND(*liquid, vapor, zIdx*)

FINTCOND() Momentum exchange through interfacial condensation [N/m]

Computes the momentum exchange due to interfacial condensation.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

FLOWID(*liquid, zIdx*)

FLOWID() Two-phase flow regime ID

Returns the integer ID corresponding to the flow regime classification.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

FLOWREGIME(*liquid, zIdx*)

FLOWREGIME() Categorical two-phase flow regimes

Classifies flow regime based on equilibrium quality and boiling transition flags.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

Returns:

- flowregime — Categorical flow regime (from *Solvers.TwoFluid.REGIMES*)
- id — Integer regime ID

FTOT(*liquid*, *vapor*, *zIdx*)*FTOT*() Total force [N/m]

Computes the total force acting on the liquid phase, including gravity, buoyancy, wall shear, drag, and interfacial condensation.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

FWALL(*liquid*, *vapor*, *zIdx*)*FWALL*() Liquid wall shear force [N/m]

Computes the wall shear force acting on the liquid phase, weighted by void fraction.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

HFLUX(*liquid*, *zIdx*)*HFLUX*() Wall heat flux to liquid phase [W/m²]

Computes the net wall heat flux transferred to the liquid phase, combining direct wall heating and boiling contributions.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Combines *Solvers.TwoFluid.Liquid.HWALHEAT* and *Solvers.TwoFluid.Liquid.HFLUXWALEVAP*
- Uses wall perimeter from geometry

HFLUXWALEVAP(*liquid*, *zIdx*)*HFLUXWALEVAP*() Wall evaporation (boiling) heat flux [W/m²]

Computes the portion of wall heat flux attributed to evaporation (e.g., boiling), based on the wall evaporation mass ratio.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

HINT(*liquid*, *vapor*, *zIdx*)*HINT*() Linear interfacial heat transfer rate [W/m]

Computes the interfacial heat transfer rate using mass transfer and enthalpy difference (based on *Inputs.Model.INTTRANSH* model).

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

Returns:

- *Hcond* — Condensation heat transfer rate [W/m]
- *Hevap* — Evaporation heat transfer rate [W/m]

HINTCOND(*liquid, vapor, zIdx*)*HINTCOND()* Linear interfacial heat condensation rate [W/m]Extracts the condensation component from *Solvers.TwoFluid.Liquid.HINT*.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

HINTEVAP(*liquid, vapor, zIdx*)*HINTEVAP()* Linear interfacial heat evaporation rate [W/m]Extracts the evaporation component from *Solvers.TwoFluid.Liquid.HINT*.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

HTOT(*liquid, vapor, zIdx*)*HTOT()* Total linear vapor heat rate [W/m]

Computes the total heat transfer to the vapor phase, including interfacial and wall contributions.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

HWALEVAP(*liquid, vapor, zIdx*)*HWALEVAP()* Linear wall heat evaporation (boiling) rate [W/m]Computes the wall boiling heat transfer rate using latent heat (based on *Inputs.Model.INTTRANSH* model) and wall mass evaporation.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

HWALHEAT(*liquid, zIdx*)*HWALHEAT()* Linear wall heat to liquid rate [W/m]

Returns the wall heat generation rate transferred to the liquid phase.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Set to zero beyond CBT

INTAREA(*liquid, vapor, zIdx*)*INTAREA()* Volumetric interfacial area [m⁻¹]

Computes the interfacial area concentration based on flow regime and phase topology.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

Notes:

- Uses a smooth transition between dispersed gas and dispersed liquid models

INTHFLUX(*liquid*, *vapor*, *zIdx*)

INTHFLUX() Interfacial heat flux [W/m²]

Computes the interfacial heat flux due to evaporation and condensation based on *Inputs.Model.INTAREA* interfacial area model.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

Returns:

- *inthflux_evap* — Evaporation heat flux [W/m²]
- *inthflux_cond* — Condensation heat flux [W/m²]

Notes:

- Uses interfacial Nusselt number and length scale
- Driven by deviation from saturated temperature

INTNU(*liquid*, *vapor*, *zIdx*)

INTNU() Interfacial Nusselt number for dispersed liquid [-]

Computes the interfacial Nusselt number based on *Inputs.Model.INTNU* model.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

J(*liquid*, *zIdx*)

J() Liquid superficial velocity [m/s]

Computes the superficial velocity of the liquid phase.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

L(*liquid*, *zIdx*)

L() Dispersed liquid interfacial length scale [m]

Returns the characteristic length scale for dispersed liquid (e.g., drop Sauter mean diameter).

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *zIdx* — Axial indices to evaluate (optional)

Notes:

- Currently uses a constant model

MINT(*liquid*, *vapor*, *zIdx*)

MINT() Linear interfacial mass transfer rates [kg/s/m]

Computes condensation and evaporation mass transfer rates using either heat flux-based or relaxation-based models.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object
- *vapor* — *Solvers.TwoFluid.Vapor* object
- *zIdx* — Axial indices to evaluate (optional)

Returns:

- Mcond — Condensation mass transfer rate [kg/s/m]
- Mevap — Evaporation mass transfer rate [kg/s/m]

Notes:

- Supports ‘CONSTANT’, ‘RANZMARSHALL’, and ‘RELAXATION’ models based on [Inputs.Model.INTNU](#) model
- Applies wall-wise distribution

MINTCOND(*liquid, vapor, zIdx*)

[MINTCOND\(\)](#) Linear interfacial condensation mass transfer [kg/s/m]

Extracts the condensation component from [Solvers.TwoFluid.Liquid.MINT](#).

Inputs:

- liquid — [Solvers.TwoFluid.Liquid](#) object
- vapor — [Solvers.TwoFluid.Vapor](#) object
- zIdx — Axial indices to evaluate (optional)

MINTEVAP(*liquid, vapor, zIdx*)

[MINTEVAP\(\)](#) Linear interfacial evaporation mass transfer [kg/s/m]

Extracts the evaporation component from [Solvers.TwoFluid.Liquid.MINT](#).

Inputs:

- liquid — [Solvers.TwoFluid.Liquid](#) object
- vapor — [Solvers.TwoFluid.Vapor](#) object
- zIdx — Axial indices to evaluate (optional)

MTOT(*liquid, vapor, zIdx*)

[MTOT\(\)](#) Total linear mass transfer [kg/s/m]

Computes the total vapor mass transfer rate by summing interfacial condensation, evaporation, and wall boiling contributions.

Inputs:

- liquid — [Solvers.TwoFluid.Liquid](#) object
- vapor — [Solvers.TwoFluid.Vapor](#) object
- zIdx — Axial indices to evaluate (optional)

MWALEVAP(*liquid, vapor, zIdx*)

[MWALEVAP\(\)](#) Linear wall mass evaporation rate [kg/s/m]

Computes the wall boiling mass transfer rate using latent heat (based on [Inputs.Model.INTTRANSH](#) model) and wall heat flux.

Inputs:

- liquid — [Solvers.TwoFluid.Liquid](#) object
- vapor — [Solvers.TwoFluid.Vapor](#) object
- zIdx — Axial indices to evaluate (optional)

RE(*liquid, zIdx*)

[RE\(\)](#) Liquid Reynolds number [-]

Computes the liquid-phase Reynolds number based on flow rate and viscosity.

Inputs:

- liquid — [Solvers.TwoFluid.Liquid](#) object
- zIdx — Axial indices to evaluate (optional)

REV(*liquid, vapor, zIdx*)

REV() Dispersed liquid Reynolds number [-]

Computes the Reynolds number for dispersed liquid using vapor properties and relative velocity.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

S(*liquid, vapor, zIdx*)

S() Phase vapor/liquid slip ratio [-]

Computes the slip ratio between vapor and liquid velocities.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

T(*liquid, zIdx*)

T() Liquid temperature [K]

Computes the temperature of the liquid phase from enthalpy.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

TAUW(*liquid, zIdx*)

TAUW() Liquid wall shear stress [N/m²]

Returns the wall shear stress acting on the liquid phase.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

USLIP(*liquid, vapor, zIdx*)

USLIP() Liquid velocity based on input phase slip [m/s]

Computes the liquid velocity using a slip-based void fraction model.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

VF(*liquid, vapor, zIdx*)

VF() Liquid volumetric fraction [-]

Computes the liquid volumetric fraction as the complement of vapor **VF()**.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

VISC(*liquid, vapor, zIdx*)

VISC() Viscosity number for dispersed liquid [-]

Computes the dimensionless viscosity number using vapor properties and interfacial tension.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

VR(*liquid, vapor, zIdx*)

VR() Local relative velocity [m/s]

Computes the relative velocity between vapor and liquid phases using various models (AREAMEAN, SCALED, DRIFT, FLOWREGIME) based on *Inputs.Model.LOCRELVEL* model.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- vapor — *Solvers.TwoFluid.Vapor* object
- zIdx — Axial indices to evaluate (optional)

WALEVAPRATIO(*liquid, zIdx*)

WALEVAPRATIO() Wall evaporation mass ratio [-]

Returns the fraction of liquid mass undergoing wall boiling.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

X(*liquid, zIdx*)

X() Liquid mass fraction [-]

Computes the liquid mass fraction relative to the mixture flow.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object
- zIdx — Axial indices to evaluate (optional)

XTR_ANN(*liquid*)

XTR_ANN() Quality at onset of annular flow region [-]

Returns the equilibrium quality threshold for the onset of annular flow.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object

Notes:

- Uses minimum value from mix.OAFX

XTR_ITM(*liquid*)

XTR_ITM() Quality at onset of intermediate region [-]

Returns the equilibrium quality threshold for the onset of the intermediate flow regime.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object

XTR_SAT(*liquid*)

XTR_SAT() Quality at saturated boiling transition [-]

Returns the equilibrium quality threshold for the onset of saturated boiling.

Inputs:

- liquid — *Solvers.TwoFluid.Liquid* object

XTR_SUB(*liquid*)

XTR_SUB() Quality at subcooled boiling transition [-]

Returns the equilibrium quality threshold for the onset of subcooled boiling.

Inputs:

- *liquid* — *Solvers.TwoFluid.Liquid* object

The following is a list of associated superclasses, each with its respective properties and methods. These foundational components provide shared functionality and structure across solver implementations.

- Abstract solver: *Solvers.AbstractSolver*
 - Abstract field: *Solvers.AbstractField*
 - Abstract phase: *Solvers.AbstractPhase*
 - Abstract film: *Solvers.AbstractFilm*
 - Solver plotter: *Solvers.SolverPlotter*
 - Solver state: *Solvers.SolverState*
-

class *Solvers.AbstractFilm*(*inputSet*, *fluid*)

Bases: *Solvers.AbstractField*

ABSTRACTFILM Base class for liquid film models in three- and four-field solvers

This abstract class defines shared methods and properties for liquid film modeling. It includes calculations for film thickness, entrainment, shear forces, and velocity models. Specific entrainment models (e.g., Okawa, Govan) and friction models are implemented.

Constructor Summary

AbstractFilm(*inputSet*, *fluid*)

AbstractFilm() Constructor for *AbstractFilm()* class

Initializes *Inputs.InputSet* and *Inputs.FluidProperties*

Property Summary

DZ = 0

Axial step size [m]

fluid

Inputs.FluidProperties object containing *fluid* thermophysical properties

inputSet

Inputs.InputSet object containing geometry, model, and boundary conditions

mix = NaN

Solvers.Mixture.Mixture object of the mixture solver

Method Summary

CV(*absfilm*, *zIdx*)

CV() Film/vapor interfacial friction factor [-]

Based on selected *Inputs.Model.VAPORFRIC* model.

CV_WALLISTHICK_CALC(*absfilm*, *thick*, *C*)

CV_WALLISTHICK_CALC() Private method to calculate the interfacial shear factor [-]

Based on the *Inputs.Model.VAPORFRIC* = *WALLISTHICK* model.

CW(*absfilm*, *zIdx*)

CW() Film wall friction factor [-]

Based on selected *Inputs.Model.THINFILMFRIC* model.

CW_LAM_CALC(*absfilm*, *zIdx*, *C*)

CW_LAM_CALC() Private method to calculate the laminar wall friction factor [-]

CW_TURB_CALC(*absfilm*, *zIdx*, *C*)

CW_TURB_CALC() Private method to calculate the turbulent wall friction factor [-]

ENTNUM(*absfilm*, *zIdx*)

ENTNUM() Private method to calculate the entrainment number [-]

Based on Okawa model assumptions.

FBUOY(*absfilm*, *zIdx*)

FBUOY() Film buoyancy force [N/m^2]

FDEP(*absfilm*, *drop*, *zIdx*)

FDEP() Drop deposition shear force [N/m^2]

FGRAV(*absfilm*, *zIdx*)

FGRAV() Gravitational force on film [N/m^2]

FTOT(*absfilm*, *drop*, *zIdx*)

FTOT() Total film forces per unit wall area [N/m^2]

FVAPOR(*absfilm*, *zIdx*)

FVAPOR() Vapor shear stress on film [N/m^2]

FWALL(*absfilm*, *zIdx*)

FWALL() Film wall shear stress [N/m^2]

MENT(*absfilm*, *zIdx*)

MENT() Film entrainment mass flux [kg/m^2/s]

Based on selected *Inputs.Model.ENTRAINMENT* model.

MTOT(*absfilm*, *drop*, *zIdx*)

MTOT() Total film mass transfer [kg/m^2/s]

OKAWACOEFS(*absfilm*, *entnum*, *coefs*)

OKAWACOEFS() Private method to determine the coefficients used in *Inputs.Model.ENTRAINMENT* = *OKAWA* entrainment models

Based on the calculated entrainment number

OKAWAMENT(*absfilm*, *zIdx*, *coefs*)

OKAWAMENT() Private method to calculate the entrainment mass flux [kg/m^2/s]

Based on Okawa models.

RE(*absfilm*, *zIdx*)

RE() Film Reynolds number [-]

THICK(*absfilm*, *zIdx*)

THICK() Film thickness [m]

UALGEBR(*absfilm*, *zIdx*)

UALGEBR() Film velocity using simple algebraic model [m/s]

From Adamsson and Le Corre [2014] (eq. 21 or 23).

UEQUIL(*absfilm*, *drop*, *zIdx*)

UEQUIL() Film velocity using full force equilibrium model [m/s]

UEQUILS(*absfilm*, *zIdx*)

UEQUILS() Film velocity assuming equilibrium between wall and vapor shear [m/s]

UWALL(*absfilm*, *zIdx*)

UWALL() Film wall friction velocity [m/s]

WL(*absfilm*, *zIdx*)

WL() Film mass flow rate per unit perimeter [kg/s/m]

X(*absfilm*, *zIdx*)

X() Vapor quality in the near-wall region [-]

Vapor quality in the near-wall region, considering only the film for the liquid phase. The near-wall region is defined in the mixture solver.

Inputs:

- *absfilm* — *Solvers.AbstractFilm* object
- *zIdx* — Axial indices to evaluate (optional)

YPLUS2THICK(*absfilm*, *yplus*, *zIdx*)

YPLUS2THICK() Thickness based on wall unit value [m]

class *Solvers.SolverPlotter*(*titles*, *WallIdxs*, *opts*)

Bases: *handle*

SOLVERPLOTTER Visualization framework for solver results.

This class provides a flexible plotting interface for visualizing solver results. It supports tiled layouts, animated series, and customizable plot styles. It also includes UI controls for animation playback and export.

Constructor Summary

SolverPlotter(*titles*, *WallIdxs*, *opts*)

SOLVERPLOTTER Constructor method to initialize plotter objects and layout

Property Summary

FontSize = 13

Font size of text in plots

Grid = 'on'

Grid display option

Title = ""

Title of the figure

Ts = []

Time series [s]

WallIdx = 1

Wall index for multi-wall simulations

Zs = []

Axial positions [m]

currentAhIdx = NaN

Index of current active axes

Method Summary

addTile(*plotters, opts*)

ADDTILE Method to add a new tile to the layout or retrieve existing one by title

static fieldName2plotStyle(*fieldName*)

FIELDNAME2PLOTSTYLE Static method to map field names to plot styles (color, line, marker)

gca(*plotters*)

GCA Get Method to retrieve current axes handle

gcf(*plotters*)

GCF Method to retrieve current plotter handle

legend(*plotters, varargin*)

LEGEND Control legends on current tile, a specific tile, or all tiles

Usage:

plotter.legend('Location','northwest') plotter.legend('Tile',2,'Location','best') plotter.legend('AllTiles',true,'Location','south') plotter.legend('Axes',ah,'off')

newTile(*plotters, opts*)

NEWTILE Method to create a new tile with labels and formatting

plotK(*plotters, klocZ, ah*)

PLOTK Method to plot obstruction elevation indicators

plotOAF(*plotters, oafZ*)

PLOTTOAF Method to plot onset of annular flow boundary indicator

plotz(*plotters, YData, fieldName, opts*)

PLOTZ Method to plot spatial or temporal data (YData) on the current axes

resizeFigure(*plotters, scaleFactor*)

[resizeFigure\(\)](#) Resizes the figure based on tiledlayout rows/columns Keeps top-left fixed and goes full-screen if size exceeds screen resolution.

Usage:

- `plotters.resizeFigure()` — Default `scaleFactor` = 1
- `plotters.resizeFigure(1.2)` — Tiles 20% larger
- `plotters.resizeFigure(0.8)` — Tiles 20% smaller

setZs(*plotters, Zs*)

SETZS Method to assign the parameter(space or time) on the plotter x-axis

xlim(*plotters, newLim*)

[xlim\(\)](#) Method to get or set x-axis limits for current axes

ylabel(*plotters, labels*)

YLABEL Method to set y-axis tick labels

ylim(*plotters, newLim*)

YLIM Method to get or set y-axis limits for current axes

class Solvers.**AbstractSolver**(*inputSet*)

Bases: handle

ABSTRACTSOLVER Defines the base interface and shared functionality for all solver classes.

Subclasses must implement the abstract methods and properties defined here.

Constructor Summary

AbstractSolver(*inputSet*)

ABSTRACTSOLVER Constructor for *AbstractSolver()*

Initializes the solver with a given *Inputs.InputSet* and applies solver-specific properties.

Property Summary

STATE

Solver state, defined by *Solvers.SolverState* enumeration

inputSet

Input configuration object of type *Inputs.InputSet*

Method Summary

static CreateITR(*NZ, ITRFields*)

CreateITR() Create and initialize the inner iteration storage structure

Inputs:

- *NZ* — Number of axial nodes (scalar)
- *ITRFields* — (string array) Names of fields to include in the struct

Output:

- *ITR* — Struct with fields initialized to zero vectors of length *NZ*

log(*solver, varargin*)

LOG Log messages to the session *log()*

save(*solver, opts*)

SAVE Save the solver object to a .mat file with a customizable name.

Inputs:

- *solver* — The solver object to be saved.
- *opts.name* — (string) Name of the variable under which to *save()* the solver. Default set to solver name
- *opts.showpath* — (logical) Whether to display the *save()* path. Default: true.

solverName(*solver*)

SOLVERNAME Returns the name of the solver

warning(*solver, varargin*)

LOG Log *warning()* messages to the session log

class Solvers.**AbstractField**

Bases: matlab.mixin.Copyable

ABSTRACTFIELD Base class for field definitions across solvers

This abstract class defines shared properties and methods for all field-type classes used in solver implementations. It provides mechanisms for transient data extraction, plotting, memoization, struct conversion, and property copying.

Constructor Summary

AbstractField()

ABSTRACTFIELD Constructor for *AbstractField()* class

Property Summary

DT

Time step size [s]

ITR

Iteration properties

NTIME

Number of time steps [-]

NZ

Number of axial steps [-]

TIDX

Time step index [-]

TIME

Time series [s]

Z

Elevation [m]

Method Summary

copyFlowProperties(*srcObj, targetObj, opts*)

COPYFLOWPROPERTIES Copies flow properties from source to target field object

Supports full or partial copying depending on 'opts.all' flag.

memoizeFunction(*methodStr, methodHandle, varargin*)

MEMOIZEDMETHOD Registers and retrieves memoized function handles

Avoids redundant computation by caching results.

Adapted from <https://stackoverflow.com/a/75037451>

plotzt(*param, ylabelText, ylabelUnit, k, opt, subobj, annular, time*)

PLOTZT 2D Generates a 2D space-time distribution plot for a given parameter

struct()

STRUCT Converts field object to a structured array

Includes TIME, ITR, and flow properties.

transient(*param, subobj, opt*)

TRANSIENT Extracts *transient()* distribution array for a given parameter

Supports wall-wise and axial/time slicing with optional subobject access.

class Solvers.AbstractPhase

Bases: handle

ABSTRACTPHASE Base class for phase definitions across solvers

This abstract class serves as a common interface for all phase-type classes used in solver implementations. It is intended to provide a foundation for shared behavior and structure.

class Solvers.SolverState

Bases: uint16

SOLVERSTATE Enumeration of solver execution states

This enumeration defines the possible states a solver can be in during or after execution. It is used to track solver execution and convergence status.

States:

- UNSOLVED — Solver has not yet run
- SOLVEDCONVERGED — Solver completed and converged successfully
- SOLVEDNOTCONVERGED — Solver completed but did not converge
- INITIALSTEPCONVERGED — Initial steady-state step completed and converged
- INITIALSTEPNOTCONVERGED — Initial steady-state step completed but did not converge

EXTERNAL PACKAGES

6.1 CoolPropWrapper

The CoolPropWrapper package is designed to serve as a MATLAB interface to the CoolProp thermophysical property library, which is a Python-based library for calculating fluid properties.

More information about the CoolPropWrapper can be found at <https://github.com/mfval/CoolPropWrapper>.

Information about CoolProp can be found at <https://coolprop.org>.

COMMUNITY

7.1 Authors

The **OpenSTREAM** solvers have been developed and validated through a dynamic and growing international collaboration that began in 2023 between **Westinghouse Electric Sweden** (WSE) and the **University of Wisconsin–Madison** (UWM). Since its inception, the initiative has expanded to include partnerships with the **Naval Nuclear Laboratory** (NNL), the **Royal Institute of Technology** (KTH), the **Massachusetts Institute of Technology** (MIT), the **Swiss Federal Institute of Technology Zurich** (ETHZ), and the **French Alternative Energies and Atomic Energy Commission** (CEA). In addition, the solvers are being utilized in research programs at other leading institutions, including the **Paul Scherrer Institut** (PSI) and **McMaster University**.

This effort is driven by the OpenSTREAM code development team, with active contributions from MSc and PhD students as part of their research and coursework under the guidance of their academic supervisors. The collaboration continues to grow, with the goal of engaging more academic and research institutions working on advanced two-phase flow modeling and thermal-hydraulic simulation.

Maintainers and contributors/users listed alphabetically by first name, last name.

7.1.1 Maintainers

These are the primary maintainers of OpenSTREAM, actively supporting development, reviewing contributions, and guiding the project forward:

- [Jason Chan](#) (UWM)
- [Jean-Marie Le Corre](#) (WSE)
- [Roman Morse](#) (MIT)

7.1.2 Contributors and users

OpenSTREAM is built through the collaborative efforts of many talented individuals. Here are some of the contributors and users who have helped shape the project:

- [Allison Mahvi](#) (UWM)
- [Cole Dunbar](#) (UWM)
- [Elisabeth Walter](#) (ETHZ)
- [Evan Hurlburt](#) (NNL)
- [Fabrice Francois](#) (CEA)
- [Haipeng Li](#) (KTH)

- [Juliana Duarte](#) (UWM)
- [Tyler Lee](#) (UWM)
- You! — Contributions are welcome. Join us and help improve OpenSTREAM!

7.2 Contribute to the code

Contributions to the **OpenSTREAM** solvers, documentation, tutorials, tests, and examples are welcome. Following the guidelines below helps ensure that changes can be reviewed, tested, and integrated efficiently.

7.2.1 How to contribute

1. Fork the repository

Open the [OpenSTREAM repository](#) and select **Fork** to create a copy under your GitHub account.

2. Clone your fork

```
git clone https://github.com/<your-username>/openstream.git
cd openstream
```

3. Create a development branch

Create a branch with a concise name describing the intended change:

```
git checkout -b feat/my-new-feature
```

Other appropriate prefixes may include `fix/`, `docs/`, `test/`, and `refactor/`.

4. Make your changes

- Follow the coding and documentation standards described below.
- Add or update tests when functionality or numerical behavior changes.
- Update user documentation when the change affects visible behavior, inputs, outputs, workflows, or examples.
- Avoid committing generated output files.

5. Run the automated tests

From the `tests` folder, run:

```
runOpenSTREAMTests
```

Confirm that the complete available test suite passes before pushing the change.

6. Commit and push the changes

Review the staged files before committing to ensure that generated outputs, temporary files, and unrelated changes are not included:

```
git status
git diff --cached
```

Commit messages must follow the [OpenSTREAM commit message style guide](#).

Organize the changes into focused commits. For example:

```
git add <files>
git commit -m "feat(scope): describe the change"
git push origin feature/my-new-feature
```

Avoid using `git add .` without first reviewing all modified and untracked files. Separate functional changes from formatting, documentation, test-reference, and unrelated maintenance changes where practical.

7. Open a pull request

Open a pull request against the original OpenSTREAM repository.

The pull-request description should:

- Explain the purpose of the change.
- Summarize the implementation.
- Summarize the focused commits included in the pull request.
- Identify affected models, solvers, inputs, tests, or documentation.
- Describe any intentional numerical changes.
- Explain and justify any changes to approved reference solutions.
- Reference related issues when applicable.

7.2.2 Coding standards

MATLAB

- Use clear and descriptive variable, property, method, and class names.
- Follow the existing OpenSTREAM package and class organization.
- Document public classes, properties, functions, and methods.
- Add comments where they clarify numerical methods, physical assumptions, or non-obvious implementation choices.
- Avoid hard-coded repository paths.
- Use `fullfile` when constructing file-system paths.
- Preserve the existing coding style unless a broader refactoring is part of the proposed change.
- Avoid unrelated formatting changes in the same commit as a functional modification.

Python

- Follow PEP 8 conventions where practical.
- Document externally visible functions and classes.
- Keep interactions between MATLAB and Python explicit and reproducible.
- Update dependency specifications when adding or modifying Python package requirements.

General practices

- Keep commits focused and descriptive.
- Follow the [OpenSTREAM commit message style guide](#).
- Prefer automated tests over manual verification whenever practical.

- Avoid duplicating existing functionality.
- Preserve backward compatibility unless an incompatible change is intentional, documented, and reviewed.
- Separate functional, documentation, test-reference, formatting, and repository-maintenance changes where practical.
- Review staged and untracked files before every commit.
- Do not commit generated outputs, temporary files, local sandbox scripts, MATLAB autosave files, or obsolete copies.

7.2.3 Documentation

Update the documentation when a contribution changes:

- User-visible functionality.
- Input parameters or accepted values.
- Default model or numerical selections.
- Solver behavior or calculated outputs.
- Public classes, methods, or properties.
- Installation or dependency requirements.
- Tutorials or recommended workflows.
- File names, links, or repository organization.

Documentation changes should be reviewed in both the generated website and PDF documentation when the affected content is included in both formats.

7.2.4 Testing

OpenSTREAM includes automated environment, unit, integration, and regression verification. The tests are organized under:

```
tests/environment/  
tests/unit/  
tests/integration/
```

The complete available test suite can be run from the `tests` folder:

```
runOpenSTREAMTests
```

The runner recursively discovers the available test classes, displays detailed diagnostics, produces a summary table, reports the total test-suite execution time, and raises an error if any test fails or does not complete.

After execution, the MATLAB test results and generated summary table are available in the workspace as `testResults` and `testSummary`.

The current test categories include:

- Environment tests that verify the MATLAB-Python interface, required Python packages, and basic CoolProp functionality.
- Unit tests that verify focused OpenSTREAM components, currently including input handling.
- Integration tests that construct and solve representative OpenSTREAM cases, perform physical-consistency checks, compare selected numerical results with approved references, and verify solver persistence.

The automated test suite is under active development and does not yet exercise every OpenSTREAM class, method, physical model, numerical option, solver path, error condition, or post-processing capability. A successful test run confirms that the currently implemented tests passed. It does not constitute complete verification or validation of OpenSTREAM.

Testing contributions

Contributors should:

- Add focused unit tests for new testable classes, functions, and methods.
- Add integration tests when a change affects solver workflows, physical consistency, or numerical results.
- Add environment tests when introducing new external dependencies.
- Verify that all existing tests continue to pass.
- Review numerical differences from approved references.
- Update tests when expected behavior changes intentionally.
- Update the testing documentation when the framework, test organization, or execution procedure changes.

Reference solutions

Approved solver-reference objects are stored below:

```
tests/integration/references/<session-directory>/
```

These files form part of the numerical regression baseline and are version-controlled.

Reference solutions should be updated only when:

- A defect has been corrected.
- A model formulation has intentionally changed.
- Numerical behavior has intentionally changed and the new behavior has been reviewed.
- A justified change to the approved test environment changes the calculated reference results.

Reference solutions should not be replaced merely to make a failing test pass. Before updating a reference, identify and understand the numerical differences and document why the new result is correct.

Generated outputs

Some tests generate temporary files, logs, session directories, CSV files, and saved solver objects under category-specific outputs directories.

Generated outputs are excluded from version control and should not normally be committed. Approved reference solutions are stored separately and should be committed only as part of a justified and reviewed update to the regression baseline.

Continuous integration

The complete available test suite is executed automatically by the MATLAB CI workflow when changes are pushed to the repository.

After pushing a change:

1. Open the corresponding GitHub Actions run.
2. Confirm that the MATLAB CI workflow completes successfully.

3. Review the diagnostic output if a test fails or does not complete.
4. Reproduce failures locally using `runOpenSTREAMTests`.
5. Push a correction only after rerunning the complete local test suite.

The documentation build and MATLAB CI are separate workflows. A successful documentation build or GitHub Pages deployment does not indicate that the MATLAB tests passed. Review both workflow results independently.

7.2.5 Reporting issues

Use the [OpenSTREAM issue tracker](#) to report defects, request enhancements, or propose changes.

An issue report should include, where applicable:

- A clear description of the problem.
- Steps required to reproduce the behavior.
- Expected and actual behavior.
- The relevant OpenSTREAM version or commit.
- MATLAB and Python versions.
- Relevant dependency versions.
- Input files or a minimal reproducible case.
- Solver output, logs, screenshots, or error messages.
- Any temporary workaround that has been identified.

Before opening a new issue, review existing issues to determine whether the topic has already been reported or discussed.

7.2.6 Code of conduct

Contributors should communicate respectfully and constructively. Technical disagreements should focus on the implementation, evidence, assumptions, and expected behavior rather than on individuals.

7.2.7 Communication

Use [GitHub Discussions](#) for general questions, ideas, and community discussions.

For major changes, open an issue before beginning implementation. Early discussion is particularly useful for:

- New solver frameworks.
- New conservation equations or closure-model families.
- Changes to established model formulations.
- Changes to public interfaces or input formats.
- Changes that affect backward compatibility.
- Changes that require substantial updates to regression references.

7.2.8 Contributor checklist

Before submitting a pull request, confirm that:

- The change is limited to the intended scope.
- The code follows the existing OpenSTREAM style and organization.

- Changes are organized into focused commits.
- Commit messages follow the [OpenSTREAM commit message style guide](#)

<https://github.com/OpenSTREAM-solvers/openstream/wiki> - Staged and untracked files have been reviewed before committing. - New or modified functionality includes appropriate tests. - The complete local test suite passes. - Numerical differences from approved references have been reviewed. - Reference solutions have been updated only when justified. - User-visible behavior is documented. - Relevant tutorials or examples have been updated. - Documentation links and cross-references remain valid. - Generated outputs and temporary files are not committed. - The MATLAB CI workflow passes after the changes are pushed. - Documentation workflows pass when documentation has been modified.

7.3 Contribute to the documentation

This documentation is made with [Sphinx](#) using the [Read the Docs](#) theme. The goal of this section is to help you implement a workflow for contributing to this documentation.

- Setup Sublime Text as text editor
- Configure Python to compile the documentation

The contents of this guide is partially taken from [here](#).

7.3.1 Setup Sublime Text as text editor

Any text editor, such as Vi, Nano, Sublime Text, and Visual Studio Code, or integrated development environment(IDE) software, such as Visual Studio, can be used to write and manage the files needed to create this documentation. For convenience, Sublime Text will be used for this guide.

1. Download and install [Sublime Text](#).
2. Open Sublime Text.
3. Install *Package Control*. On the top menu bar, select *Tools > Command Palette*, or press *Ctrl+shift+p* (Win, Linux), *cmd+shift+p* (Mac). In the search bar that pops up, type *Install Package Control*. Select the matching item that is listed, or press *Enter*. The installation progress will be shown on the bottom left corner of the window.
4. Package Control is driven by the Command Palette. To open the palette, press *ctrl+shift+p* (Win, Linux) or *cmd+shift+p* (Mac). All Package Control commands begin with Package Control:, so start by typing *Package*. Select *Install Package*, and install the following packages:
 - *RestructuredText Improved*: syntax highlighting
 - *OmniMarkupPreviewer*: preview RST in a web browser
 - *Sublime RST Completion*: convenient group of snippets and commands to facilitate writing restructuredText with SublimeText

7.3.2 Configure Python and Sphinx

You need Python and Sphinx (a Python package) to build this documentation.

Setup Python

To build your document in HTML (or other formats) on your computer, you must install Sphinx.

See [Sphinx Overview](#) for background reading. Then follow the steps below.

1. If you are using Windows, you might need to [Install Python](#).

Depending on your Windows setup, after installation you might need to manually add the Python directory to your path. Try the [Python Windows Install](#) for help.

If you are using a Mac, it's probably installed.

2. Install PIP.

Depending on your Windows setup, after installation you might need to manually add the PIP directory to your path.

3. Install *virtualenv*

It is generally a good idea to use a *Virtual environment* to contain all the required dependencies of the project. This way, different versions of the same Python package can be used for each project contained within their respective *Virtual environment*. On the command line, enter:

```
$ pip install virtualenv
```

On Windows, *virtualenv.exe* will likely now be found in your python installation directory under the Scripts subdirectory.

4. Create a Virtual Python Environment

cd to the *docs* folder:

```
$ cd C:\Path\to\openstream\docs
```

Use the *virtualenv* command to create a virtual environment directory called *venv*:

```
$ virtualenv venv
```

Note

A *.gitignore* file should be automatically added to the *venv* folder. This way everything within it should be ignored by Git later on.

1. Activate the Environment

Now that we have a virtual environment, we need to activate it.:

```
$ source .\venv\Scripts\activate
```

After you activate the environment, your command prompt will be modified to reflect the change.

Install Sphinx and dependencies

Use PIP to install Sphinx and dependencies. On the command line, enter:

```
$ pip install -r ./source/requirements.txt
```

This process may take a minute.

Note

If you are using Windows, this might be a frustrating task. If you get stuck, share the error messages and ask for help.

7.3.3 Start writing

This documentation is written in reStructuredText and compiled using Sphinx, both of which offers extensive options and customizabilities that can be difficult to master quickly. reStructuredText is similar to Markdown, which is the format used for GitHub *README.md* files. It is recommended that you go through [this quick tutorial](#) to familiarize yourself with the syntax. Sphinx-doc also offers a [great tutorial](#). Some useful cheatsheets can be found [here](#).

7.3.4 Build the documentation

This section assumes you are using the *build* function of Sublime Text to build the documentation.

Quick preview

For a quick preview of the document you are currently working on, use `ctrl+alt+o` (Win/Linux) or `cmd+opt+o` (Mac). Alternatively, open the *Command Palette* and select *OmniMarkupPreviewer: Preview Current Markup in Browser*. This functionality requires the *OmniMarkupPreviewer* package mentioned in *Setup Sublime Text as text editor*, and does not use *Sphinx*.

Full build

For a full build of the project, *Sphinx* needs to be called. More specifically, *sphinx-build* is the program that is run.

As to why this combination of schemes are used to create this documentation, here's an excerpt from the Sphinx website:

Sphinx makes it easy to create intelligent and beautiful documentation.

Here are some of Sphinx's major features:

- Output formats: HTML (including Windows HTML Help), LaTeX (for printable PDF versions), ePub, Texinfo, manual pages, plain text
- Extensive cross-references: semantic markup and automatic links for functions, classes, citations, glossary terms and similar pieces of information
- Hierarchical structure: easy definition of a document tree, with automatic links to siblings, parents and children
- Automatic indices: general index as well as a language-specific module indices
- Code handling: automatic highlighting using the Pygments highlighter
- Extensions: automatic testing of code snippets, inclusion of docstrings from Python modules (API docs) via built-in extensions, and much more functionality via third-party extensions.
- Themes: modify the look and feel of outputs via creating themes, and reuse many third-party themes.
- Contributed extensions: dozens of extensions contributed by users; most of them installable from PyPI.

Sphinx uses the reStructuredText markup language by default, and can read MyST markdown via third-party extensions. Both of these are powerful and straightforward to use, and have functionality for complex documentation and publishing workflows. They both build upon Docutils to parse and write documents.

7.4 OpenSTREAM FAQ

This page provides concise answers to common questions about the purpose, distinctive features, capabilities, assumptions, simplifications, and limitations of **OpenSTREAM**. It also provides guidance on installation, configuration, solver selection, simulation setup, numerical convergence, result interpretation, post-processing, troubleshooting, model development, and contribution to the project.

Select a question to display its answer.

7.4.1 Getting started

What is OpenSTREAM?

OpenSTREAM, short for *Open Solvers for Two-phase flow Research, Engineering Analysis and Modeling*, is an open-source, object-oriented MATLAB environment for simulating one-dimensional, multi-field, liquid-vapor two-phase flows with phase change in straight channels.

OpenSTREAM includes four solver frameworks:

- a mixture solver with thermal non-equilibrium capabilities;
- a two-fluid solver that represents the liquid and vapor phases separately;
- a three-field solver for annular two-phase flow;
- a four-field solver that separates the liquid film into base-film and disturbance-wave fields.

Depending on the selected solver and closure models, OpenSTREAM can represent wall boiling, interfacial evaporation and condensation, and the associated exchanges of mass, momentum, and energy between phases and fields.

The solver frameworks provide different levels of physical detail and support education, model development, performance evaluation, and validation.

Why should I use OpenSTREAM instead of another available code?

OpenSTREAM is not intended as a replacement for established thermal-hydraulic codes. It is designed primarily as an open, transparent, and extensible environment for education, fundamental model development, closure-model implementation, controlled numerical experiments, and validation of one-dimensional two-phase flow formulations.

OpenSTREAM may be particularly useful when you need:

- **Full access to the source code.** The governing equations, closure models, numerical algorithms, input processing, and post-processing methods can be inspected and modified directly.
- **A transparent modeling environment.** Primary variables, secondary quantities, source terms, field exchanges, and convergence histories are accessible through the MATLAB solver objects. This facilitates interpretation, debugging, and conservation checks.
- **Several solver frameworks within a common architecture.** OpenSTREAM includes mixture, two-fluid, three-field, and four-field formulations. The same geometry, boundary conditions, fluid-property interface, and general workflow can therefore be used to compare different levels of physical resolution.
- **State-of-the-art annular-flow modeling.** The three-field solver separates the liquid into wall-film and entrained-droplet fields. The four-field solver further separates the liquid film into base-film and disturbance-wave fields and includes wave-frequency transport.
- **A practical platform for closure-model development.** Physical models are selected through documented input enumerations and implemented in dedicated phase or field classes. A developer can add, modify, test, and compare closure models without working within a large proprietary code base.
- **Rapid prototyping in MATLAB.** MATLAB provides an interactive environment for inspecting objects, modifying equations, visualizing results, performing parameter studies, and developing new models.
- **Reproducible research and education.** Input sets, tutorials, documentation, references, and source code can be distributed together, allowing users to inspect the assumptions behind a calculation and reproduce the workflow.
- **A computationally efficient one-dimensional model.** OpenSTREAM is useful when cross-section-averaged axial behavior is sufficient and the computational expense and geometric detail of a multidimensional CFD calculation are not required.

Other codes may be more appropriate when the application requires:

- qualified or extensively validated models for a specific industrial application;
- complete reactor-system or plant-network simulation;
- closed-loop systems, including natural-circulation loops;
- simultaneous solution of multiple connected flow paths and their flow distribution;
- dedicated component models for pumps, valves, vessels, separators, or heat exchangers;
- multidimensional flow resolution;
- complex geometries, crossflow, or connected flow networks;
- compressible pressure-wave dynamics;
- multi-component fluids or non-condensable gases;
- safety or licensing analysis.

The appropriate code therefore depends on the purpose of the analysis. OpenSTREAM is especially well suited to understanding, developing, and evaluating models in a transparent one-dimensional framework. A more established application code may be preferable when a broad component library, an application-specific validation basis, or a qualified engineering workflow is required.

Selecting OpenSTREAM does not remove the need for numerical verification, closure-model assessment, applicability review, and validation against appropriate reference data.

What is unique about OpenSTREAM, and what are its main strengths?

OpenSTREAM combines advanced one-dimensional two-phase flow models with an open, modern, and extensible software architecture. Its purpose is not only to perform simulations, but also to make the governing equations, closure models, numerical methods, and calculated source terms accessible for inspection, modification, and collaborative development.

Its main distinctive features and strengths include:

- **Advanced models rarely available in open-source thermal-hydraulic codes.** To the best of the developers' knowledge, OpenSTREAM provides the only openly available implementations of its four-field annular-flow model and Mixture Relaxation Model within a general two-phase flow simulation environment.

The four-field model explicitly represents vapor, entrained droplets, the base liquid film, and disturbance waves. It also includes transport of wave frequency, allowing the evolution of disturbance-wave behavior to be studied. The model therefore provides a level of annular-flow detail that is not commonly available in one-dimensional thermal-hydraulic codes.

The Mixture Relaxation Model extends the conventional mixture formulation by solving additional vapor mass and energy conservation equations. It provides a computationally efficient framework for representing thermal non-equilibrium through relaxation of interfacial mass and energy transfer.

- **Several solver frameworks within one consistent environment.** OpenSTREAM includes mixture, two-fluid, three-field, and four-field solvers. These frameworks share the same general input system, geometry definition, fluid-property interface, session management, and post-processing approach.

This common structure allows users to compare different levels of physical resolution without moving between unrelated codes or reconstructing the complete simulation workflow.

- **A modern object-oriented architecture.** OpenSTREAM is implemented using MATLAB classes that represent solvers, phases, fields, inputs, fluid properties, sessions, and plotting utilities. Shared behavior is organized through inheritance and common base classes, while solver-specific behavior is implemented in dedicated modules.

This architecture supports modular development and helps isolate changes to a particular physical model, field, or solver. It also makes primary variables, secondary quantities, source terms, and solver methods directly accessible through the corresponding objects.

- **Direct access to primary and secondary variables.** Users can inspect the primary solution variables, such as mass flow rate, pressure, velocity, and enthalpy, together with derived quantities such as void fraction, film thickness, interfacial area, flow regime, and disturbance-wave frequency, where applicable to the selected solver.

The stored pseudo-time convergence histories also allow users to examine how the steady-state solution was obtained. This level of access is useful for interpretation, debugging, numerical diagnostics, and verification.

- **Direct access to field exchange terms.** The mass, momentum, and energy exchange terms calculated by the selected solver are readily accessible through the corresponding phase and field objects.

This includes wall and interfacial transfer terms, such as evaporation, condensation, entrainment, deposition, drag, shear, and heat transfer, where applicable to the selected solver. Access to the individual contributions facilitates interpretation of the governing equations, conservation checks, closure-model evaluation, and debugging.

- **Direct access to pressure-drop contributions.** The individual pressure-drop components calculated by the mixture solver are readily available for inspection and post-processing.

These contributions can be examined separately to determine the relative effects of gravity, wall friction, spatial acceleration, temporal acceleration, and local pressure losses, as applicable to the simulated case. This decomposition helps users understand the physical origin of the calculated pressure distribution.

- **Integrated plotting and animation capabilities.** Each solver provides dedicated methods for visualizing its principal variables and calculated exchange terms. Axial distributions, time histories, and, where implemented, time-elevation distributions can be generated directly from the solver object.

The plotting methods support solver-specific quantities, selected axial and time indices, multiple plot arrangements, and animation of transient results. This allows users to inspect a calculation interactively without first developing separate post-processing routines.

- **Designed for model development and comparison.** Physical models are selected through documented input options and implemented in dedicated phase or field classes. New closure models can therefore be introduced, verified, and compared with existing models without modifying an opaque or monolithic code base.

The shared solver architecture is particularly useful for studying how assumptions made at the mixture, phase, or field level influence the calculated results.

- **Comprehensive and integrated documentation.** The OpenSTREAM documentation includes installation instructions, theory descriptions, governing equations, notation, a glossary, package and class references, tutorials, publications, contribution guidance, and this FAQ.

Solver-specific MATLAB Live Scripts provide executable examples that users can inspect and modify. The documentation is generated from the same repository as the source code, helping to keep implementation details and user guidance together.

- **Interactive MATLAB workflow.** MATLAB provides an environment in which users can run a case, inspect solver objects, modify models, visualize intermediate quantities, perform parameter studies, and develop post-processing routines interactively.

MATLAB Live Scripts combine formatted explanations, executable code, figures, and calculated output in a single document, making them useful for education, demonstrations, and reproducible technical studies.

- **Open and modifiable source code.** OpenSTREAM is distributed under the MIT License. Users can inspect, use, modify, and redistribute the code in accordance with the license terms.

Open development enables independent scrutiny of the modeling basis and supports collaboration among researchers and institutions. The source code, documentation, input cases, and references can be considered together when evaluating a model.

- **Computational efficiency.** The one-dimensional formulation and streamlined numerical methods provide a computationally efficient framework for steady-state and transient calculations.

This efficiency facilitates rapid model development, repeated calculations, closure-model comparisons, parameter studies, sensitivity analyses, and validation exercises.

These strengths do not mean that OpenSTREAM is more appropriate than other thermal-hydraulic codes. Established system, subchannel, and CFD codes may provide broader component libraries, more complex geometries, multidimensional resolution, or a more extensive validation basis for particular applications.

OpenSTREAM is most distinctive when transparency, access to the governing models, advanced annular-flow modeling, thermal non-equilibrium model development, and controlled comparison of solver frameworks are central to the objective of the study.

What should I read first?

A suggested learning path is:

1. Follow the installation instructions in the Usage section.
2. Run **Tutorial #1: OpenSTREAM Quick Start.**
3. Continue with the solver-specific tutorial that matches your needs:
 - Tutorial #2 for the mixture solver;
 - Tutorial #3 for the two-fluid solver;
 - Tutorial #4 for the three-field solver;
 - Tutorial #5 for the four-field solver.
4. Consult the theory pages for governing equations and assumptions.
5. Use the package reference to inspect model options, object properties, and methods.

How do I make OpenSTREAM available in MATLAB?

Add the OpenSTREAM root folder to the MATLAB search path. For example, when running a tutorial from the tutorials folder:

```
osp = './..';
addpath(osp);
```

If MATLAB cannot find a class such as `InputSet` or `MixtureSolver`, verify that the OpenSTREAM root folder is correct and present on the MATLAB path.

You can inspect the resolved location of a class with:

```
which Inputs.InputSet
which Solvers.Mixture.MixtureSolver
```

Why does OpenSTREAM require Python?

OpenSTREAM uses the Python version of **CoolProp** to calculate fluid thermophysical properties. `CoolPropWrapper` provides the interface between MATLAB and Python.

MATLAB must therefore use a compatible Python installation. Check the active Python environment with:

```
pyenv
```

You can test the property interface with:

```
cp = CoolPropWrapper.CoolPropWrapper();
```

If Python or CoolProp cannot be loaded, follow the MATLAB-Python compatibility and CoolPropWrapper setup instructions in the installation section.

Which MATLAB and Python versions should I use?

Use one of the MATLAB/Python combinations listed in the tested compatibility table. Other combinations may work but are not necessarily exercised by the continuous-integration workflow.

When reproducing or reporting a result, record the MATLAB release with:

```
version('-release')
```

Also record the Python and CoolProp versions when fluid-property behavior is relevant.

Which CoolProp version should I install?

The CI-tested environment currently installs CoolProp==8.0.0 and nanobind. However, other CoolProp versions may work.

Where can I find working examples?

The tutorials folder contains MATLAB Live Scripts that can be run and modified directly. The documentation also provides completed HTML versions for reference.

Start with **Tutorial #1: OpenSTREAM Quick Start**, then continue with the solver-specific tutorial that matches your application.

7.4.2 Running simulations

What is the minimum workflow for running a case?

A basic mixture-solver calculation consists of four steps:

1. import the required packages;
2. construct an InputSet;
3. construct the solver;
4. call solve.

For example:

```
import Inputs.*
import Solvers.*
import Solvers.Mixture.*

% Load the inputSet.
inputSet = InputSet( ...
    modelFilePath      = './inputs/models.inp', ...
    modelID            = 'TUTORIAL1', ...
    optionsFilePath    = './inputs/options.inp', ...
    optionsID          = 'DEFAULT', ...
    geometryFilePath   = './inputs/geom.inp', ...
    geometryID         = 'TUTORIAL1', ...
    bcFilePath         = './inputs/tutorial1.inp', ...
```

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```

sessionParentDir      = fullfile(pwd,'outputs'), ...
overwriteSessionFiles = true, ...
LOGMODE               = 'BOTH');

% Create and solve the three-field solver.
mixSolver = MixtureSolver(inputSet);
mixSolver.solve();

% Generate defaults axial distribution plots
mixSolver.plotz();

```

How are OpenSTREAM inputs organized?

An OpenSTREAM case is assembled from four separate input categories. Each category is normally defined in a dedicated input file:

- **Model inputs** select the physical models and closure models.
- **Numerical options** control time steps, iteration limits, relaxation factors, and convergence criteria.
- **Geometry inputs** define the channel length, cross-sectional flow area, wall perimeters, and orientation.
- **Boundary conditions** define the system pressure, inlet enthalpy, inlet mass flow rate, total power, and axial power distribution.

The paths to the four input files are provided separately when constructing an `InputSet`, for example:

```

inputSet = InputSet( ...
    modelFilePath      = './inputs/models.inp', ...
    modelID            = 'TUTORIAL1', ...
    optionsFilePath    = './inputs/options.inp', ...
    optionsID          = 'DEFAULT', ...
    geometryFilePath   = './inputs/geom.inp', ...
    geometryID         = 'TUTORIAL1', ...
    bcFilePath         = './inputs/tutorial1.inp');

```

The model, numerical-option, and geometry files may each contain several input sets. An identifier such as `TUTORIAL1` or `DEFAULT` selects the required set from the corresponding file.

A boundary-condition file may contain the information required for a particular steady-state or transient case. Depending on how the study is organized, separate boundary-condition files can be used for different operating conditions or transients.

This separation allows physical models, numerical options, geometries, and boundary conditions to be combined and reused without duplicating complete case definitions. For example, one geometry can be evaluated using several physical-model sets, or one physical-model set can be applied to several boundary-condition cases.

Can I specify inlet temperature instead of inlet enthalpy?

No. The boundary-condition input interface requires the inlet specific enthalpy, H_{IN} , in J/kg. Direct specification of inlet temperature is not supported and is not currently planned.

Specific enthalpy provides a more general definition of the inlet thermodynamic state. At a specified pressure, temperature can define a single-phase subcooled-liquid or superheated-vapor state, but it cannot uniquely define a saturated two-phase state. At saturation, the liquid, vapor, and all intermediate two-phase states have the same saturation temperature but different specific enthalpies.

In contrast, inlet specific enthalpy can represent:

- a subcooled-liquid inlet;
- a saturated-liquid inlet;
- a saturated two-phase inlet with a specified vapor quality;
- a saturated-vapor inlet;
- a superheated-vapor inlet.

If the inlet condition is known as temperature and pressure for a single-phase state, first calculate the corresponding specific enthalpy for the selected fluid. Enter the resulting value as HIN in the boundary-condition file:

```
PRESSURE    ! System pressure [Pa]           > ...  
HIN         ! Inlet specific enthalpy [J/kg] > ...
```

For a saturated two-phase inlet, determine HIN from the system pressure and the required inlet vapor quality. A temperature-only input would not contain enough information to distinguish among saturated liquid, a two-phase mixture, and saturated vapor.

What input-file formats are supported?

OpenSTREAM supports its native `.inp` format and the JSON format. The examples and tutorials generally use the native `.inp` format.

What happens if I do not specify a model or numerical option?

Unspecified entries use the default values defined by the corresponding input class, principally `Inputs.Model` and `Inputs.Options`.

Defaults make basic cases easier to configure, but they should not be treated as universally appropriate. For research or validation work, review and document all influential physical and numerical selections.

Why does MATLAB print many default-value warnings?

The warnings identify properties that were not explicitly specified and therefore use their class defaults.

These messages are useful while developing or reviewing a case. Tutorials may temporarily suppress them to keep the displayed output concise:

```
warning off  
inputSet = InputSet(...);  
warning on
```

Do not suppress warnings routinely until the configuration has been reviewed. An unexpected warning can reveal an omitted or unsupported input.

Why does OpenSTREAM report that an input entry was not used?

Common causes include:

- a misspelled keyword;
- a keyword that does not belong to the selected input class;
- an obsolete keyword;
- an entry intended for another model configuration.

Check the keyword against `Inputs.Model`, `Inputs.Options`, `Inputs.Geometry`, `Inputs.BoundaryConditions`, or the relevant package-reference page.

An entry reported as unused is not applied to the corresponding input object. Whether this affects the simulation depends on whether the corresponding valid property is used by the selected solver and model configuration. If it is required, OpenSTREAM may instead use the default value, and the simulation may differ from what was intended. If the property is not used by the selected solver or active models, the unused entry may have no effect on the results.

How do I define a steady-state calculation?

A boundary-condition file containing a single time entry defines a steady-state case.

The solver obtains the steady-state solution through pseudo-time advancement. The pseudo-time sequence is a numerical procedure used to approach a stationary solution and should not be interpreted as a physical transient.

How do I define a transient calculation?

Provide boundary-condition values at multiple physical times. Depending on the boundary-condition interface, pressure, inlet enthalpy, inlet mass flow, power, and axial power distribution can vary with time.

The solver first calculates the initial steady state. The physical transient proceeds only after the initial solution has converged.

The physical time step is controlled through the numerical options. A time-step sensitivity study is recommended when transient timing or peak values are important.

Why is the mixture solver run before the other solvers?

The two-fluid, three-field, and four-field solvers require a solved mixture solution for initialization. They also reuse information from the mixture solution, including the pressure-gradient solution.

A solved mixture solver can be passed explicitly:

```
mixSolver = MixtureSolver(inputSet);
mixSolver.solve();

twfSolver = TwoFluidSolver(inputSet, mixSolver);
tfSolver = ThreeFieldSolver(inputSet, mixSolver);
ffSolver = FourFieldSolver(inputSet, mixSolver);
```

If a mixture solver is not supplied, the advanced solver constructor creates and solves one when required. Passing an existing solved object avoids repeating the same mixture calculation.

Can I call solve more than once on the same solver object?

A solver object represents a specific initialized calculation. Calling solve again on an already solved object produces an error stating that the solver must be reinitialized.

For a new calculation, construct a new `InputSet` and solver object. This keeps each simulation configuration and its stored results clearly separated.

How do I reduce the amount of information printed in the MATLAB Command Window?

Set `LOGMODE` to `'LOGTOFILEONLY'` when constructing the `InputSet`:

```
inputSet = InputSet( ...
    ...
    LOGMODE = 'LOGTOFILEONLY');
```

Solver progress remains available in the session log while most Command Window output is suppressed.

7.4.3 Choosing a solver

Which solver should I use?

The appropriate solver depends on the physical phenomena and level of detail required.

Mixture solver

Use for robust initialization, efficient calculations with mixture quantities, hydrodynamically coupled phases, and mixture-based thermal non-equilibrium models.

Two-fluid solver

Use when separate liquid and vapor mass, momentum, and energy behavior is needed. The solver can represent different phase velocities, enthalpies, and temperatures.

Three-field solver

Use for annular two-phase flow when the liquid distribution between a wall film and entrained droplets is important.

Four-field solver

Use for annular two-phase flow when the liquid film must be separated into base-film and disturbance-wave fields and wave transport is part of the analysis.

Increasing the number of fields does not automatically make a simulation more accurate. The selected solver assumptions and closure models must be appropriate for the application.

What is the difference between a phase and a field?

A **phase** is a thermodynamic state of matter, such as liquid or vapor.

A **field** is a computationally resolved constituent with its own transported variables or conservation equations. One phase can be represented by more than one field. For example:

- The three-field solver represents the liquid phase through a wall-film field and a droplet field.
- The four-field solver further divides the wall film into a base-film field and a disturbance-wave field.

Are the three-field and four-field solvers thermal non-equilibrium models?

The current three-field and four-field formulations assume thermal equilibrium. Their additional detail concerns the hydrodynamic distribution and transport of liquid fields.

Use the mixture relaxation model or the two-fluid solver when separate thermal behavior is required, subject to the assumptions and implemented closure models of the selected framework.

Where are the three-field and four-field equations applied?

Although the separate annular-flow field equations are numerically solved from the channel inlet, their results are physically meaningful only from the onset of annular flow. Upstream of this location, the conservation equations are formulated so that the annular-flow field variables evolve smoothly toward the prescribed conditions at the onset of annular flow.

Solving the field equations upstream of the onset of annular flow ensures that preceding-time-step values are available for all required field variables if the onset location moves upstream during a transient calculation.

The onset-of-annular-flow model, the initial film-droplet split, and, for the four-field solver, the initial base-film and disturbance-wave split can therefore influence the solution near the transition.

Why can results near the onset of annular flow be sensitive?

The initial liquid distribution among film, droplets, base film, and waves depends on the selected onset and field-splitting models. Entrainment and deposition then redistribute liquid as the solution develops downstream.

When the region near annular-flow onset is important, examine sensitivity to:

- the onset-of-annular-flow model;
- the initial entrained-droplet fraction;
- the initial base-film and wave split;
- entrainment and deposition models;
- spatial resolution.

Should entrainment and deposition models be selected together?

Yes. Entrainment and deposition are coupled processes that exchange liquid between the film and droplet fields. The corresponding models are generally developed and calibrated together to provide a consistent representation of this exchange.

When available, use entrainment and deposition models from the same model family unless there is a documented justification for combining models from different families.

7.4.4 Numerical convergence

What is the difference between point convergence and steady-state convergence?

At each axial node and time step, OpenSTREAM performs local point iterations for the nonlinear equations. **Point convergence** refers to the change in solved variables between these local iterations.

For a steady-state calculation, OpenSTREAM also compares solutions between successive pseudo-time steps. **Steady-state convergence** refers to these temporal changes becoming smaller than the specified steady-state criteria.

A calculation therefore needs adequately converged local iterations and an adequately converged pseudo-time solution.

What does solveMode='NULL' mean in plotting methods?

The NULL solve mode displays the pseudo-time history used to obtain the initial steady-state solution. It does not represent the physical transient.

For example:

```
solver.plott( ...
    solver.NZ, ...
    'display', {'W','U'}, ...
    'solveMode', 'NULL');
```

The default REAL mode is used to display the stored physical solution.

What should I do if the steady-state calculation does not converge?

Start with the information printed by the solver:

- identify the node with the largest point-iteration count;
- identify which variable has the largest residual;
- inspect the final solver state;
- plot the pseudo-time history with solveMode='NULL';

- check model, geometry, and boundary-condition warnings;
- verify that input values and units are physically reasonable.

Numerical options that may affect convergence include:

- maximum point iterations;
- maximum steady-state pseudo-time iterations;
- pseudo-time step;
- variable-specific relaxation factors;
- pointwise convergence criteria;
- steady-state convergence criteria.

Change one numerical setting at a time and document each modification. After convergence is achieved, check that the converged solution is not excessively sensitive to the selected relaxation factors, iteration limits, pseudo-time step, or convergence criteria. Do not obtain apparent convergence merely by relaxing the convergence criteria without verifying the resulting residuals and physical solution.

Why was my transient skipped?

The physical transient is not calculated when the initial steady-state solution fails to converge. The solver log indicates that the transient is being skipped.

Inspect the initial pseudo-time solution, correct any input or model problems, and obtain a converged initial state before interpreting transient results.

Can a converged calculation still be inaccurate?

Yes. Numerical convergence indicates that the implemented discrete equations have been solved to the selected tolerances. It does not establish that:

- the spatial mesh is sufficiently refined;
- the physical time step is sufficiently small;
- the selected closure models are valid for the conditions;
- the one-dimensional assumptions are appropriate;
- the model has been validated for the application.

Mesh sensitivity, time-step sensitivity, conservation checks, and comparison with appropriate reference data remain necessary.

Is OpenSTREAM limited in terms of mesh size or time-step size?

OpenSTREAM does not impose a hard-coded maximum number of axial nodes, `NNODES`, or a predefined set of allowable physical time-step values, `TSTEP`. Within the applicable input requirements, users can select the spatial and temporal discretizations appropriate for their calculations.

The absence of hard-coded limits does not mean that any mesh or time step is numerically appropriate. Practical limits arise from available memory, computational effort, convergence behavior, numerical accuracy, and the spatial and temporal scales of the modeled problem.

The axial mesh is defined by `NNODES` in the physical-model inputs. Increasing `NNODES` improves the spatial resolution of the one-dimensional solution but also increases the number of calculated values, memory use, and computational effort.

A coarse axial mesh may adequately represent smooth distributions but can smooth localized variations, shift the predicted location of a transition, or fail to capture a local maximum or minimum. Finer meshes may be required near:

- abrupt changes in wall heat flux;
- local geometric perturbations or pressure losses;
- boiling or flow-regime transitions;
- the onset of annular flow;
- film dryout or regeneration;
- other regions containing strong axial gradients.

The physical time step is defined by TSTEP in the numerical-option inputs. OpenSTREAM does not impose a hard-coded minimum or maximum physical time-step size. However, TSTEP must be selected according to the duration and shortest relevant time scale of the physical transient.

OpenSTREAM uses fully implicit backward Euler time integration, which provides robust time advancement. Robustness or numerical stability does not, however, guarantee temporal accuracy.

A large physical time step may reproduce the overall transient trend but smooth rapid changes, underestimate or miss peak values, or shift the calculated timing of an event. A smaller time step provides greater temporal resolution but requires more calculated time steps and greater computational effort.

Important prescribed changes in the boundary conditions should also be represented with sufficient temporal resolution. A time step that is adequate for slowly varying quantities may still be too large for the accurate calculation of peak values, peak timing, or short-duration behavior.

The physical time step, TSTEP, should not be confused with the steady-state pseudo-time step, SSTSTEP. TSTEP controls advancement of the physical transient, whereas SSTSTEP controls the numerical progression from the solver-assigned initialization state toward the steady-state solution.

A smaller pseudo-time step can make the point iterations easier to converge but may require more pseudo-time steps. A larger pseudo-time step may accelerate progression toward steady state but can make the local nonlinear iterations more difficult or produce less regular convergence.

Mesh and time-step adequacy should be demonstrated through sensitivity studies. Successive spatial or temporal refinements should produce progressively smaller changes in the quantities of interest. The finest mesh or smallest time step used in a study is a reference solution and should not automatically be treated as exact.

Can OpenSTREAM use a non-uniform axial mesh?

No. The current OpenSTREAM implementation uses a uniform axial mesh.

The user specifies the total channel length, LENGTH, and the number of axial nodes, NNODES. OpenSTREAM then distributes the nodes uniformly along the channel. The current input interface does not provide an option for specifying individual axial-node locations or variable node lengths.

Consequently, local mesh refinement cannot currently be applied only in selected regions, such as near:

- an abrupt change in wall heat flux;
- a local geometric perturbation or pressure loss;
- boiling or flow-regime transitions;
- the onset of annular flow;
- film dryout or regeneration;
- another region containing a strong axial gradient.

When additional spatial resolution is required, `NNODES` must be increased for the complete channel. This reduces the uniform axial node spacing but also increases the number of calculated values and the computational effort throughout the domain.

The absence of a non-uniform mesh option does not necessarily prevent accurate calculations. A uniform mesh can be adequate when the axial behavior is sufficiently smooth or when enough nodes are used to resolve the shortest relevant axial length scale. Mesh adequacy should be assessed through a spatial-sensitivity study.

Support for user-defined axial-node locations or variable node lengths may be considered in a future development, but it is not currently implemented.

Which numerical options control steady-state convergence?

Relevant options include the pseudo-time step, the maximum number of steady-state iterations, variable-specific relaxation factors, local point-iteration tolerances, and steady-state temporal convergence criteria.

The exact property names and defaults are listed in `Inputs.Options`. Use the solver-specific tutorials and package reference when changing these values.

7.4.5 Results and post-processing

How do I display results?

The solver classes provide plotting methods for common result types.

plotz

Plots axial distributions. Multiple physical time indices can produce an animated axial plot.

plott

Plots time histories at a selected axial location.

plotzt

Plots time-elevation distributions where implemented.

For example:

```
mixSolver.plotz( ...  
    'display', {'W','VR'}, ...  
    'arrangement', 'horizontal', ...  
    'resize', 1);
```

Available display names depend on the selected solver. Consult the corresponding solver class in the package reference.

Can I access results without using the built-in plots?

Yes. Solver objects retain their solution objects and arrays. Examples include:

- mixture variables in `mixSolver.mixture`;
- liquid and vapor variables in `twfSolver.liquid` and `twfSolver.vapor`;
- film and droplet variables in `tfSolver.film` and `tfSolver.drop`;
- base-film and wave variables through the four-field film object.

The package reference lists the available properties and methods. Check property dimensions before processing because arrays may vary over axial position, time, and wall index.

How do I save a solver object?

Use the solver save method:

```
mixSolver.save( ...
    'name', 'mixSolver', ...
    'showpath', true);
```

The solver object contains the imported inputs, solution objects, and associated simulation information. Saving it allows later post-processing without repeating the calculation.

How do I access the base-film and disturbance-wave objects in a four-field solution?

In the four-field solver, the base-film and disturbance-wave objects are contained within the liquid-film object. They are therefore accessed through the `film` property of the four-field solver:

```
base = ffSolver.film.base;
wave = ffSolver.film.wave;
```

For a transient calculation, `ffSolver.film` is an array containing one film object for each stored physical time step. A specific time step can therefore be accessed using its time index:

```
tIdx = 1;

base = ffSolver.film(tIdx).base;
wave = ffSolver.film(tIdx).wave;
```

The base-film and disturbance-wave objects contain their separately calculated field quantities. For example:

```
baseMassFlow = ffSolver.film(tIdx).base.W;
baseVelocity = ffSolver.film(tIdx).base.U;

waveMassFlow = ffSolver.film(tIdx).wave.W;
waveVelocity = ffSolver.film(tIdx).wave.U;
waveFrequency = ffSolver.film(tIdx).wave.FREQUENCY;
```

The enclosing `film` object represents the total liquid film and also provides aggregate film quantities derived from the solved base-film and disturbance-wave fields. The droplet field is stored separately in `ffSolver.drop`, while the vapor-field solution is retained through the mixture solver.

The base-film and disturbance-wave objects used during the pseudo-time calculation of the initial steady state are similarly contained in the stored initialization film objects:

```
baseInit = ffSolver.filmInit(tIdx).base;
waveInit = ffSolver.filmInit(tIdx).wave;
```

What is the minimum base film in the four-field solver?

In the four-field model, disturbance waves pass over a continuous and comparatively slow-moving base liquid film. The calculated base-film mass flow rate represents the condition immediately after a disturbance wave has passed and interacted with the base film.

Between two successive disturbance waves, the base film evolves without direct replenishment from a wave. During this wave-waiting period, wall evaporation removes liquid from the base film, while droplet deposition can add liquid to it. The **minimum base-film mass flow rate** represents the condition reached at the end of this period, immediately before the next disturbance wave arrives and replenishes the base film.

The minimum base-film mass flow rate is therefore a local, periodically varying quantity associated with the passage of disturbance waves. It should not be interpreted as the minimum value of the calculated base-film mass flow rate over the channel or over the complete transient.

Assuming that the base-film velocity remains constant during the wave-waiting period, the corresponding minimum base-film thickness is calculated from:

$$\delta_{b,\min} = \frac{W_{b,\min}}{\rho_{l,s} u_b \Pi}$$

where:

- $\delta_{b,\min}$ is the minimum base-film thickness;
- $W_{b,\min}$ is the minimum base-film mass flow rate;
- $\rho_{l,s}$ is the saturated-liquid density;
- u_b is the base-film velocity;
- Π is the corresponding wall perimeter.

The minimum base film is useful for evaluating intermittent base-film depletion between disturbance waves. If the minimum base-film mass flow rate or thickness reaches zero, the model predicts that the base film becomes locally dry before being reached by the next disturbance wave. This is referred to as **intermittent base-film dryout**.

The minimum base film differs from the equilibrium base film. The **equilibrium base-film thickness** is the target condition used by the base-film and disturbance-wave mass-exchange model, whereas the **minimum base-film thickness** describes the depletion of the base film during the interval between successive waves.

What does it mean when the calculated liquid-film mass flow rate become negative?

The treatment of negative liquid-film quantities is controlled by the POSFILM physical-model input:

POSFILM	! Positive film flow-rate/thickness model	> TRUE
---------	---	--------

POSFILM is set to TRUE by default. With this setting, the film mass flow rate and film thickness are restricted to non-negative values.

When POSFILM is set to FALSE, OpenSTREAM allows the calculated film mass flow rate and corresponding film thickness to become negative after the film has been completely depleted.

A negative film mass flow rate is not physically meaningful and does not represent liquid flowing in the opposite direction. It is an intentionally extended numerical quantity indicating that the integrated film mass balance has progressed beyond the predicted dryout condition.

The liquid-film mass conservation equation accounts for:

- liquid gain through droplet deposition;
- liquid loss through film entrainment;
- liquid loss through wall evaporation.

When the cumulative film losses exceed the available film inventory, the physical film mass flow rate reaches zero and film dryout occurs. With POSFILM set to FALSE, the film mass-balance equation continues to be integrated beyond this point, producing a negative value.

This continuation provides a signed measure of the margin to dryout:

- a positive film mass flow rate indicates that liquid film remains;
- a value close to zero indicates proximity to complete film dryout;
- a negative value indicates that the imposed conditions exceed the predicted dryout condition.

This behavior is useful in critical-power calculations and model-tuning studies. For example, power or a closure-model coefficient can be varied until the minimum film mass flow rate is sufficiently close to zero. The corresponding condition represents the predicted dryout threshold according to the selected models.

Allowing negative values also improves numerical robustness during parameter studies and power iterations. A trial calculation can continue after exceeding the dryout condition instead of terminating when the film first reaches zero.

Negative film values must be interpreted only as numerical dryout-margin indicators. They must not be interpreted as physical post-dryout film mass flow rates, negative film thicknesses, or reverse liquid-film flow. OpenSTREAM does not use this continuation as a physical post-dryout film model.

What is the difference between calculated and equilibrium base-film and wave quantities?

The four-field solver distinguishes between the locally calculated base-film thickness and wave frequency and their corresponding equilibrium values.

The **calculated base-film thickness** is obtained from the transported base-film mass flow rate, base-film velocity, liquid density, and wall perimeter. It represents the current local state of the base-film field and may vary with axial position and time.

The **equilibrium base-film thickness** is obtained from the selected equilibrium base-film-thickness model. It represents the base-film thickness toward which the local base-film state tends under the current flow conditions. It is a closure-model target rather than a separately transported field variable.

The difference between the calculated and equilibrium base-film thicknesses is used in the relaxation model governing mass exchange between the base-film and disturbance-wave fields. In simplified form, the net source driving the base film toward equilibrium is proportional to:

$$\frac{\rho_{l,s} (\delta_{b,eq} - \delta_b)}{t_{b,relax}}$$

where:

- δ_b is the calculated base-film thickness;
- $\delta_{b,eq}$ is the equilibrium base-film thickness;
- $\rho_{l,s}$ is the saturated-liquid density;
- $t_{b,relax}$ is the base-film relaxation time.

If the calculated base-film thickness is smaller than its equilibrium value, the relaxation term tends to transfer liquid from the disturbance waves to the base film. If the calculated base-film thickness is larger than its equilibrium value, the relaxation term tends to transfer liquid from the base film to the disturbance waves.

The complete base-film and wave mass balances also include other mechanisms, such as droplet deposition, wall evaporation, film entrainment, and turbulent mixing. Consequently, the final base-film and wave mass-exchange rates are not determined by the thickness-relaxation term alone.

Similarly, the **calculated wave frequency** characterizes the current local disturbance-wave field. It is related to the transported wave number density and may differ from the frequency corresponding to local equilibrium conditions because of wave transport and hydrodynamic non-equilibrium.

The **equilibrium wave frequency** is obtained from the selected equilibrium wave-frequency model. It represents the wave frequency toward which the local disturbance-wave population tends under the current flow conditions.

The four-field solver transports the wave number density using a Boltzmann-type transport equation. Its source term represents the net effect of disturbance-wave interactions, including wave creation, merging, splitting, and dissipation. Using a relaxation-time approximation, the source term is expressed as:

$$\Phi_w = \frac{N_{w,eq} - N_w}{t_{w,relax}}$$

where:

- N_w is the calculated wave number density;
- $N_{w,eq}$ is the equilibrium wave number density;
- $t_{w,relax}$ is the wave relaxation time;
- Φ_w is the net wave-number source or sink rate.

Wave frequency is often used as a more practical representation of wave number density. The corresponding frequency form similarly drives the calculated wave frequency toward its equilibrium value.

If the calculated wave frequency is smaller than its equilibrium value, the relaxation source is positive and tends to increase the local wave population. If the calculated wave frequency is larger than its equilibrium value, the source is negative and tends to reduce the local wave population.

The base-film and wave relaxation processes are related but distinct:

- the base-film-thickness relaxation controls mass redistribution between the base-film and disturbance-wave fields;
- the wave-frequency relaxation controls the source term in the wave number-density transport equation and therefore the evolution of the disturbance-wave population.

The equilibrium quantities should therefore be interpreted as local closure-model targets. Differences between calculated and equilibrium values are expected in developing or transient annular flow and represent the hydrodynamic non-equilibrium modeled by the four-field formulation.

What is wave number density, and why is it transported in the four-field solver?

In the four-field model, disturbance waves are represented as a dispersed field of liquid structures moving over the continuous base film. Rather than tracking every individual disturbance wave, the model describes the wave population using cross-section-averaged macroscopic quantities.

One of these quantities is the **wave number density**, N_w . It is the number of disturbance waves per unit axial distance and is therefore a spatial measure of the local wave population.

The wave number density is defined as the inverse of the average wave spacing:

$$N_w = \frac{1}{\lambda_w}$$

where λ_w is the average axial distance between two successive disturbance waves.

Wave spacing is related to the wave velocity and wave frequency by:

$$\lambda_w = \frac{u_w}{f_w}$$

Consequently:

$$N_w = \frac{f_w}{u_w}$$

where u_w is the disturbance-wave velocity and f_w is the frequency at which disturbance waves pass a fixed axial location.

Wave number density and wave frequency describe related but different quantities. Wave number density describes how many waves are present per unit axial distance, whereas wave frequency describes how many waves pass a fixed location per unit time. Their relationship therefore depends on the wave velocity.

The four-field solver does not assume that the local wave population is always equal to its equilibrium value. Instead, it transports the wave number density using a one-dimensional wave number-density transport equation.

This equation is based on the concept of a Boltzmann transport equation applied to the number density of dispersed fluid particles. In the four-field model, disturbance waves are treated as the dispersed fluid structures. The equation is integrated over the possible wave sizes, so it transports the total wave number density rather than tracking separate populations for each wave size.

The transport equation can be written as:

$$\frac{\partial N_w}{\partial t} + \frac{\partial (u_w N_w)}{\partial z} = \Phi_w$$

The left-hand side represents:

- the local temporal change in wave number density; and
- the axial transport of waves at the disturbance-wave velocity.

The right-hand side, Φ_w , represents the net source or sink of disturbance waves caused by interactions such as wave formation, merging, splitting, and dissipation.

In the current model, these interactions are represented using a relaxation-time approximation:

$$\Phi_w = \frac{N_{w,eq} - N_w}{t_{w,relax}}$$

where:

- N_w is the calculated wave number density;
- $N_{w,eq}$ is the equilibrium wave number density obtained from the selected closure model;
- $t_{w,relax}$ is the wave relaxation time.

If the calculated wave number density is smaller than its equilibrium value, the source term is positive and increases the local wave population. This represents a net effect of processes that create or split waves.

If the calculated wave number density is larger than its equilibrium value, the source term is negative and decreases the local wave population. This represents a net effect of processes such as wave merging or dissipation.

The transport equation allows the disturbance-wave population to respond progressively to changing flow conditions instead of changing instantaneously to its local equilibrium value. It therefore represents hydrodynamic non-equilibrium associated with developing annular flow, phase change, inlet effects, geometric perturbations, and transient conditions.

Wave frequency is generally more convenient for interpretation and comparison with measurements. OpenSTREAM therefore commonly expresses and plots the transported wave population in terms of calculated and equilibrium wave frequencies. However, the underlying physical concept remains the transport of the spatial wave population represented by the wave number density.

The wave number-density transport equation is separate from the wave mass and momentum conservation equations:

- the wave mass equation determines how much liquid mass is carried by the disturbance-wave field;
- the wave momentum equation determines the disturbance-wave velocity;
- the wave number-density transport equation determines how the number of disturbance waves evolves.

Together with the wave shape model, these quantities are used to characterize wave spacing, width, amplitude, and other properties of the disturbance-wave field.

Where are simulation outputs stored?

The `sessionParentDir` argument defines the parent output folder. Each solver uses the session infrastructure to store logs and requested results.

For example:

```
sessionParentDir = fullfile(pwd, 'outputs');
```

Use a separate session directory or case name for parameter studies. Be careful with:

```
overwriteSessionFiles = true
```

because existing files for the corresponding session may be replaced.

Why do my plots show results only from the onset of annular flow?

The three-field and four-field plotting functions can restrict field-specific results to the annular-flow region. This is appropriate because separate film, droplet, base-film, and wave equations apply from the onset of annular flow.

Where supported, set the annular plotting option to `false` to inspect initialization outside that region:

```
solver.plotz( ...  
    'display', 'W', ...  
    'annular', false);
```

Interpret pre-annular field values as initialization quantities rather than as a complete pre-annular multi-field model.

How do I plot only part of the channel or selected times?

By default, `plotz` includes all axial elevations and all stored physical time steps. Similarly, `plott` includes all stored physical time steps at the selected axial location.

The plotting methods also allow selected axial and time indices to be specified. The same general syntax is used by all OpenSTREAM solvers.

To plot only part of the channel, first identify the corresponding axial indices and pass them using `zIdx`. For example, the following code plots results between elevations of 1 and 4 m:

```
zIdx = find(solver.Z >= 1 & solver.Z <= 4);  
  
solver.plotz( ...  
    'display', {'W', 'U'}, ...  
    'zIdx', zIdx, ...  
    'arrangement', 'horizontal', ...  
    'resize', 1);
```

For a transient calculation, the first positional argument of `plotz` selects one or more stored time indices. For example, the following code plots the first, middle, and final stored states over the selected axial range:

```
tIdx = unique([1, round(solver.NTIME/2), solver.NTIME]);  
zIdx = find(solver.Z >= 1 & solver.Z <= 4);  
  
solver.plotz( ...  
    tIdx, ...  
    'display', {'W', 'U'}, ...  
    'zIdx', zIdx, ...  
    'arrangement', 'horizontal', ...  
    'resize', 1);
```

When several time indices are selected, `plotz` generates an animated axial plot containing the selected states.

To plot a time history at one axial location, pass the axial index as the first argument of `plott` and select the stored time indices using `tIdx`. For example, the following code plots only the first half of the transient at the axial node closest to 3.5 m:

```
zTarget = 3.5;
[~, zIdx] = min(abs(solver.Z - zTarget));

tIdx = 1:ceil(solver.NTIME/2);

solver.plott( ...
    zIdx, ...
    'display', {'W','U'}, ...
    'tIdx', tIdx, ...
    'arrangement', 'horizontal', ...
    'resize', 1);
```

The same index-selection approach applies to the mixture, two-fluid, three-field, and four-field solvers. However, the available `display` values depend on the selected solver because each solver stores different physical quantities.

How do I compare several model configurations?

Create one `InputSet` and solver object per model ID, store each object separately, and extract the same comparison quantities from every case.

Use distinct session directories to avoid overwriting results. A compact MATLAB table is useful for comparing scalar metrics, while solver objects can be retained separately for detailed post-processing.

7.4.6 Physical scope and limitations

What are the main simplifications in OpenSTREAM, and why are they considered reasonable?

OpenSTREAM uses several deliberate simplifications to provide a transparent, computationally efficient, and numerically robust environment for developing and evaluating one-dimensional two-phase flow models. These simplifications define the intended scope of the code and must be considered when interpreting its results.

The main simplifications are:

- **One-dimensional, cross-section-averaged flow.** Flow variables vary only along the channel axis. Radial, azimuthal, and other local distributions are not resolved explicitly and must instead be represented through averaged quantities and closure models.

This approximation is reasonable for straight channels when axial transport dominates and detailed multidimensional effects, such as crossflow, flow separation, or strongly asymmetric mixing, are not essential to the quantity being studied. It also makes OpenSTREAM suitable for rapid model development, sensitivity studies, and comparison of closure models.

- **Straight channels with constant cross-sectional area.** The current geometry represents a straight flow path with constant flow area. Multiple wall perimeters and non-uniform wall heating can be defined, and local perturbations can be represented through dedicated models, but continuous geometric variation is not resolved.

This approximation covers many experimental channels and idealized thermal-hydraulic problems, including tubes, annuli, rectangular channels, and simplified small rod-bundle representations. It enables investigation of axial two-phase flow behavior without introducing additional geometric effects.

- **Mixture-based pressure-gradient solution.** The pressure-gradient solution obtained from the mixture solver is reused by the two-fluid, three-field, and four-field solvers rather than independently solving a fully coupled pressure-velocity system in each framework.

This treatment significantly improves numerical stability and provides a consistent pressure field for comparisons among the solver frameworks. It introduces some inconsistency between the pressure solution and the separate field momentum equations, but the available OpenSTREAM publications report that this inconsistency is negligible for many applications. Its acceptability must nevertheless be assessed for each new application.

- **Fluid properties evaluated at the system pressure.** Fluid thermophysical properties are currently evaluated using the prescribed system pressure rather than the calculated local pressure along the channel.

Consequently, an axial pressure decrease does not modify the local saturation properties and cannot, by itself, produce flashing. OpenSTREAM therefore does not currently represent flashing caused by local pressure drop along the channel.

Time-dependent changes in the prescribed system pressure are included in the fluid-property evaluation. OpenSTREAM can therefore represent flashing caused by a decrease in system pressure, subject to the assumptions and capabilities of the selected thermal non-equilibrium and phase-change models.

This simplification is considered reasonable for the high-pressure, relatively short channel applications targeted by the current OpenSTREAM frameworks, when the axial pressure variation is small relative to the system pressure. Under these conditions, the resulting variation in saturation properties along the channel is generally expected to be limited.

The approximation must be reassessed when the channel pressure drop is sufficiently large to affect the local saturation temperature, saturation enthalpies, phase densities, or phase-change behavior. It is not appropriate when pressure-drop-induced flashing is an important part of the physical problem.

- **Neglect of surface-tension forces in the conservation equations.** Surface tension may be used in closure quantities, but explicit surface-tension force contributions are neglected in the governing momentum equations.

This simplification is considered reasonable for the high-pressure channel-flow conditions targeted by the current implementations, where the retained pressure, inertia, gravity, and wall and interfacial momentum-transfer terms are generally more important at the one-dimensional field scale. It may not be appropriate for capillary-dominated flows, very small channels, or problems controlled by interface curvature.

- **Neglect of heating caused by friction.** Mechanical energy dissipated by wall or interfacial friction is not added explicitly to the fluid energy equations.

This contribution is considered small relative to the imposed wall heating and phase-change energy transfer for the high-pressure boiling applications for which the current models were developed. It should not be neglected without assessment in applications involving exceptionally large pressure losses or little external heating.

- **Neglect of temporal pressure-gradient contributions.** The current formulations omit minor energy or momentum contributions associated with the temporal pressure gradient.

This approximation is considered reasonable for the operational transients targeted by the current high-pressure boiling applications. It is not appropriate for rapid pressure-wave, depressurization, water-hammer, choking, or shock-wave problems, which are currently outside the intended scope of OpenSTREAM.

- **Neglect of spatial gradients of saturated-fluid enthalpies.** Spatial changes in saturated liquid and vapor enthalpies associated with the pressure distribution are neglected in the simplified conservation equations.

This approximation is considered reasonable for the high-pressure conditions and operational transients targeted by the current OpenSTREAM frameworks. Its adequacy should be reconsidered for cases with large pressure variations or strong depressurization.

- **Simplified flow-regime and interfacial-topology transitions.** The current two-fluid solver uses simplified assumptions to identify flow regimes and select the corresponding closure models.

This approach provides a practical framework for implementing and testing separate-phase conservation equations, but it limits the physical realism of complex transients involving repeated or strongly evolving changes in interfacial topology. The two-fluid results should therefore be interpreted according to the maturity and validation range of the selected transition and closure models.

- **Thermal equilibrium in the three-field and four-field solvers.** The film, droplet, base-film, disturbance-wave, and vapor fields are assumed to share the applicable saturation thermodynamic state. The solvers resolve hydrodynamic non-equilibrium among the fields but do not solve separate field energy equations for thermal non-equilibrium.

This approximation is reasonable for saturated annular-flow studies in which liquid-field mass and momentum transport, entrainment, deposition, film depletion, and disturbance-wave behavior are the primary phenomena of interest. It is not appropriate when separate field temperatures (e.g., post-dryout thermal non-equilibrium) are essential.

- **First-order numerical discretization.** The conservation equations are solved using first-order upwind spatial discretization and fully implicit backward Euler time integration. Nonlinear source terms and field couplings are treated through fixed-point iterations.

These methods are comparatively simple and numerically robust, which is useful for an open model-development platform and for obtaining steady-state solutions through pseudo-time advancement. Their numerical diffusion and first-order accuracy make mesh- and time-step-sensitivity studies necessary when spatial gradients, transient timing, or peak values are important.

These simplifications are reasonable only within the intended application domain and for quantities that are not controlled by the neglected phenomena. They make the governing equations easier to understand, modify, test, and compare while reducing computational cost and improving numerical robustness.

A simplification must not be interpreted as universally negligible. Before using OpenSTREAM for a new fluid, geometry, flow regime, or transient, assess whether the neglected terms and unresolved phenomena could materially affect the quantities of interest. Numerical convergence, sensitivity studies, conservation checks, and comparison with suitable reference data remain necessary.

What are the current practical and technical limitations of OpenSTREAM?

OpenSTREAM is under active development. The current implementation is intended for one-dimensional, thermally expandable, single-component, liquid-vapor boiling flows in straight channels. The thermally expandable formulation accounts for changes in fluid properties with the local thermodynamic state, but it is not a general compressible-flow formulation.

The principal current limitations include:

- **No general compressible-flow or pressure-wave formulation.** OpenSTREAM accounts for thermodynamic density changes associated with heating, cooling, system pressure variation, and phase change. This thermally expandable treatment is suitable for the intended boiling-flow applications, but it is not a fully compressible-flow formulation.

In particular, the current implementation is not intended to resolve acoustic-wave propagation, rapid pressure waves, shock waves, expansion waves, choked flow, water hammer, or other phenomena for which compressible pressure-wave dynamics are essential.

- **No wall condensation model.** OpenSTREAM can represent interfacial condensation within the fluid, where supported by the selected solver and closure models, but it does not currently calculate condensation caused by heat transfer to a cooled wall.
- **No counter-current flow.** The current solution algorithms and boundary-condition treatment are intended for co-current flow in a single axial direction.

In addition, OpenSTREAM does not currently support:

- multi-component fluid mixtures;
- non-condensable gases;
- solid-particle flows, including liquid-solid, gas-solid, and liquid-gas-solid flows;

- pressure-drop-induced flashing. Fluid properties are evaluated at the prescribed system pressure rather than the calculated local pressure;
- wall heat conduction or conjugate heat transfer;
- continuously varying channel area;
- bends, junctions, plena, or connected flow networks;
- crossflow between neighboring channels;
- multidimensional flow resolution;
- general-purpose component models such as pumps, valves, tanks, or heat exchangers;
- thermal non-equilibrium in the three-field and four-field solvers;
- fully coupled pressure-velocity solution in the advanced solvers;

Some currently unsupported capabilities may be considered for future development.

Is OpenSTREAM a CFD code?

No. OpenSTREAM is not a Computational Fluid Dynamics (CFD) code in the conventional engineering sense.

OpenSTREAM solves one-dimensional, cross-section-averaged conservation equations for steady-state and transient two-phase flows. Depending on the selected solver, these equations describe mixture, phase, or field mass, momentum, and energy transport. Physical processes that are not resolved explicitly are represented through closure models.

In contrast, a conventional CFD code discretizes a multidimensional computational domain and resolves spatial variations of the flow within that domain. OpenSTREAM does not resolve radial, azimuthal, or other multidimensional distributions of velocity, pressure, temperature, void fraction, or phase structure.

OpenSTREAM also does not explicitly resolve:

- local velocity and temperature profiles across the channel;
- multidimensional turbulence structures;
- flow separation, recirculation, or secondary flow;
- circumferential transport around the channel;
- crossflow between neighboring channels;
- detailed three-dimensional geometrical effects.

Instead, OpenSTREAM uses cross-section-averaged quantities and closure models for wall friction, interfacial transfer, phase change, entrainment, deposition, and other unresolved processes.

This one-dimensional formulation is intentional. OpenSTREAM was developed to support the formulation, implementation, assessment, and validation of the types of one-dimensional thermal-hydraulic models used in established system codes and subchannel-analysis codes. Its purpose is to provide an open and transparent environment in which these models can be inspected, modified, compared, and extended.

The one-dimensional approach also provides a computationally efficient environment for model development, sensitivity studies, and validation. It is particularly suitable for straight-channel problems in which axial transport is the principal behavior of interest and multidimensional effects are not essential to the quantities being studied.

OpenSTREAM should therefore be described as a **one-dimensional, multi-field, two-phase flow simulation environment**, rather than as a CFD code.

Can OpenSTREAM simulate flashing caused by a pressure decrease?

OpenSTREAM can represent flashing caused by a time-dependent decrease in the prescribed system pressure, subject to the assumptions and capabilities of the selected phase-change model.

However, OpenSTREAM does not currently represent flashing caused by the calculated local pressure drop along the channel. Fluid thermophysical properties, including saturation properties, are evaluated using the prescribed system pressure rather than the calculated local pressure. Therefore, the axial pressure decrease does not locally modify the saturation state used by the fluid-property calculations.

This approximation is considered reasonable for the high-pressure, relatively short channel applications targeted by the current OpenSTREAM frameworks when the axial pressure variation is small relative to the system pressure. Under these conditions, the corresponding axial variation in saturation properties is expected to remain limited.

The approximation is not suitable when pressure-drop-induced flashing or axial variations in saturation properties are important to the physical problem.

What geometries can OpenSTREAM represent?

OpenSTREAM represents one-dimensional flow in straight channels. Cross-sectional geometry is described through averaged quantities such as flow area and wall perimeter.

The formulation supports multi-wall channel representations, allowing different wall surfaces and heating distributions to be considered. Examples may include tubes, annuli, rectangular channels, and simplified small rod-bundle representations when the one-dimensional approximation is appropriate.

OpenSTREAM does not resolve bends, crossflow, or detailed three-dimensional velocity and temperature distributions.

How are multi-wall geometries represented in OpenSTREAM?

OpenSTREAM represents a multi-wall geometry as a single one-dimensional flow channel bounded by two or more wall surfaces. The channel has one common cross-sectional flow area, while the perimeter of each wall is specified separately using the PERIM geometry input.

For example, an annular channel can be represented by defining separate perimeters for the inner and outer walls. Other idealized configurations, such as rectangular channels or simplified subchannels, can be represented in the same way when the one-dimensional approximation is appropriate.

A simplified annular-channel geometry could, for example, use:

```
ID      ! Geometry identifier           > ANNULUS
LENGTH ! Channel length [m]           > ...
AREA    ! Common cross-sectional flow area [m2] > ...
PERIM   ! Inner and outer wall perimeters [m] > ... ...
END
```

The number of walls is determined automatically from the number of perimeter values specified in PERIM. OpenSTREAM also calculates the contribution of each wall to the total channel perimeter.

When several walls are present, the boundary-condition power distribution must be consistent with the number of wall perimeters. Wall-specific results can be inspected using the wall-selection options provided by the applicable solver plotting methods.

Wall-dependent quantities are stored separately for each wall where supported by the selected solver. These quantities may include:

- imposed wall heat flux;
- liquid-film mass flow rate and velocity;
- film thickness;

- wall boiling or film evaporation;
- wall shear stress;
- film entrainment and droplet deposition;
- base-film and disturbance-wave quantities in the four-field solver.

This wall-level formulation allows different walls to have different heating conditions. Heated and unheated walls, or walls with different power distributions, can therefore be represented within the same one-dimensional channel.

The bulk or core flow remains shared by all walls. Depending on the selected solver, this includes mixture quantities, vapor quantities, and the pressure distribution. Wall-specific fields interact through this common bulk-flow solution and the implemented closure models.

Multi-wall treatment does not create separate connected flow channels. OpenSTREAM does not currently calculate crossflow between neighboring channels, independent flow redistribution among parallel channels, or multidimensional transport around the channel cross-section. These configurations require a different system or subchannel representation.

Can the wall-specific solutions interact with one another?

Yes, but only indirectly through the common bulk-flow solution.

In a multi-wall geometry, OpenSTREAM represents all walls as bounding surfaces of the same one-dimensional flow channel. Each wall can have its own perimeter, heating conditions, and wall-specific quantities. In the three-field and four-field solvers, the liquid-film equations are formulated separately for each wall. The four-field solver similarly calculates separate base-film and disturbance-wave quantities for each wall.

The wall-specific fields do not exchange liquid directly with one another. For example, there is no model that transfers liquid laterally from the film on one wall directly to the film on another wall.

However, the wall solutions are coupled indirectly through the shared mixture, vapor, and droplet fields. A wall can therefore influence the other walls through mechanisms such as:

- wall heating and evaporation, which affect the common vapor and mixture solution;
- film entrainment, which transfers liquid from a wall film to the common entrained-droplet field;
- droplet deposition, which redistributes liquid from the common droplet field back to the wall films;
- wall and interfacial momentum exchanges, which contribute to the common flow solution;

Consequently, changing the heat flux or film behavior on one wall can affect the bulk-flow conditions and thereby modify the conditions experienced by the other walls. This is an indirect interaction through the shared flow, rather than a direct exchange between neighboring wall fields. The multi-wall formulation is specifically intended to represent straight channels with multiple independent wall surfaces and potentially different heating rates.

In the three-field and four-field formulations, entrainment and deposition provide an important indirect coupling mechanism. Liquid entrained from one wall enters the common droplet population and may subsequently deposit on another wall, depending on the implemented deposition-partition model. The wall equations themselves nevertheless remain wall-specific. The four-field equations additionally include direct exchanges between the base-film and disturbance-wave fields on the same wall, but these are not exchanges between different walls.

This treatment does not resolve circumferential transport, lateral liquid-film flow, or multidimensional mixing between walls. OpenSTREAM also does not represent separate neighboring channels or crossflow between them. When these effects are important, a multidimensional or subchannel formulation is required.

What units should I use?

OpenSTREAM uses SI units unless a specific model or input explicitly states otherwise. Common units include:

- length in metres;
- time in seconds;
- pressure in pascals;
- temperature in kelvin;
- mass flow rate in kilograms per second;
- velocity in metres per second;
- specific enthalpy in joules per kilogram;
- wall heat flux in watts per square metre.

Check comments in the example input files and the relevant property documentation before entering a value.

Can OpenSTREAM be used for safety or licensing analysis?

No. OpenSTREAM must not be used for safety or licensing analysis.

OpenSTREAM is intended for research, education, model development, performance evaluation, and validation activities. It is not a licensed safety-analysis code and has not been qualified for use in regulatory or licensing applications.

Results produced with OpenSTREAM must not be used as the basis for safety decisions, licensing submissions, regulatory compliance demonstrations, or the operation of safety-related systems.

Which fluids can be simulated with OpenSTREAM?

OpenSTREAM is designed for single-component, liquid-vapor fluids whose required thermophysical properties are available through CoolProp. The fluid is selected using its CoolProp identifier in the FLUID model input.

The CoolProp interface can be tested in MATLAB before constructing a complete simulation:

```
cp = CoolPropWrapper.CoolPropWrapper('WATER');
```

CoolProp compatibility alone does not demonstrate that an OpenSTREAM calculation is physically applicable or validated. The selected closure models must also be appropriate for the fluid and operating conditions. Many current models and example cases have been developed or evaluated primarily for boiling water and steam.

OpenSTREAM does not currently support multi-component fluid mixtures or non-condensable gases. Pseudo-pure fluids, predefined mixtures, and incompressible solutions available in CoolProp must not be assumed to be compatible without verification of the full property and model requirements.

How are fluid properties calculated?

Fluid properties are calculated using CoolProp through CoolPropWrapper. The fluid is selected using the FLUID model input, while the PROPERTIES model input defines how the phase properties are evaluated.

Two fluid-property options are available:

- **SATURATED:** Phase properties are evaluated at saturated conditions using the prescribed system pressure. Saturated liquid properties are used for the liquid phase, and saturated vapor properties are used for the vapor phase. The properties therefore do not vary with the local phase enthalpy.
- **PSYSTEM:** Phase properties are evaluated using the prescribed system pressure and the local phase enthalpy. This allows liquid properties to vary between subcooled and saturated-liquid conditions and vapor properties to vary between saturated-vapor and superheated conditions.

SATURATED is the default option and provides a simple property treatment that is appropriate when variations within the subcooled liquid and superheated vapor regions are not important. PSYSTEM is more appropriate when thermal non-equilibrium causes the liquid or vapor enthalpy to differ significantly from its saturation value.

The option is selected in the physical-model input file. For example:

FLUID	! CoolProp fluid identifier	> WATER
PROPERTIES	! Fluid property assumptions	> PSYSTEM

For both options, the saturation temperature, saturated liquid and vapor enthalpies, latent heat, saturated phase densities, and other saturation properties are calculated at the prescribed system pressure.

The current implementation does not evaluate properties using the calculated local pressure along the channel. Consequently, selecting PSYSTEM accounts for variations with local phase enthalpy, but it does not enable flashing caused by the local axial pressure drop.

Does a more detailed solver always give a better result?

No. Greater field resolution introduces additional equations, closure models, numerical parameters, and applicability limits. A more detailed solver is useful only when its modeled phenomena and validation basis match the application.

Select the simplest solver that represents the required physics and document the sensitivity to influential assumptions.

What should I verify before trusting a result?

At minimum, check:

- input values, IDs, units, and warnings;
- point and steady-state convergence;
- mass and energy consistency;
- spatial-mesh sensitivity;
- physical time-step sensitivity for transients;
- sensitivity to influential closure models;
- applicability and validation of the selected models.

7.4.7 Extending and contributing

How do I select another closure model?

Closure models are selected in the physical-model input file. Valid values are defined by classes in the `InputEnums` package.

Model selections may control, for example:

- void fraction;
- wall friction;
- interfacial heat transfer;
- phase or field momentum;
- entrainment and deposition;
- onset of annular flow;
- base-film thickness;
- disturbance-wave frequency.

Consult the relevant `InputEnums` class and `Inputs.Model` property before adding an entry. If a selection is not explicitly provided, the documented default is used.

How do I add a new closure model?

Adding a closure model normally requires coordinated changes to:

1. the relevant `InputEnums` class;
2. any new model properties and defaults in `Inputs.Model`;
3. the solver or field method that evaluates the closure;
4. model documentation and references;
5. input examples;
6. verification and regression tests.

The implementation should document equations, units, assumptions, validity range, and source references. Verify the closure independently before using it in validation or engineering analysis.

How should I report a bug?

Use the OpenSTREAM GitHub issue tracker. Before opening an issue:

- search for an existing report;
- provide the OpenSTREAM revision;
- provide the MATLAB, Python, and CoolProp versions;
- include the smallest input case that reproduces the problem;
- include the complete error message and stack trace;
- distinguish unexpected code behavior from a model-validity question.

Issue tracker:

- <https://github.com/OpenSTREAM-solvers/openstream/issues>

How can I request a feature?

Open a GitHub issue describing:

- the use case and physical or workflow need;
- the proposed behavior;
- which solver or package is affected;
- any relevant equations, references, or examples;
- how the feature could be verified.

Search existing issues first to avoid duplicating an active request.

How can I contribute?

Fork the repository, create a focused branch, implement and test the change, and submit a pull request. Follow the coding, testing, documentation, and code-of-conduct guidance in the Community section.

Keep changes focused and include documentation and tests when behavior or interfaces change.

How do I contribute to the documentation?

Edit the reStructuredText source files, build the documentation locally when possible, and verify both HTML and PDF outputs. Add or update links, references, examples, and package documentation as needed.

Where can I get additional help?

Consult, in this order:

1. the Quick Start and solver-specific tutorials;
2. the theory overview;
3. the glossary and notation guide;
4. the package reference;
5. existing GitHub issues;
6. the GitHub issue tracker for a new question or reproducible problem.

When requesting help, include enough information for another user to reproduce the calculation.

7.5 Support

Need help with **OpenSTREAM**? The best way to get assistance is by using the issue [tracker](#) on GitHub. *Maintainers* are notified of new issues and will do their best to respond promptly.

Ask a question, report a bug, or suggest a feature here:

<https://github.com/OpenSTREAM-solvers/openstream/issues>

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MATLAB MODULE INDEX

i

[InputEnums](#), 65

[Inputs](#), 75

S

[Session](#), 92

[Solvers](#), 168

[Solvers.FourField](#), 98

[Solvers.Mixture](#), 116

[Solvers.ThreeField](#), 138

[Solvers.TwoFluid](#), 147

A

AbstractField (*class in Solvers*), 172
 AbstractField() (*Solvers.AbstractField method*), 173
 AbstractFilm (*class in Solvers*), 168
 AbstractFilm() (*Solvers.AbstractFilm method*), 168
 AbstractPhase (*class in Solvers*), 173
 AbstractSolver (*class in Solvers*), 172
 AbstractSolver() (*Solvers.AbstractSolver method*), 172
 addTile() (*Solvers.SolverPlotter method*), 171
 ADIAM() (*Inputs.Geometry method*), 85
 AFDISTR() (*Solvers.Mixture.Mixture method*), 120
 AFDISTR() (*Solvers.TwoFluid.Mixture method*), 157
 AFFNC() (*Solvers.Mixture.Mixture method*), 120
 AI() (*Solvers.ThreeField.Drop method*), 139
 ALPHAF (*Inputs.FluidProperties attribute*), 82
 ALPHAG (*Inputs.FluidProperties attribute*), 82
 ALPHAL() (*Inputs.FluidProperties method*), 83
 ALPHAV() (*Inputs.FluidProperties method*), 83
 AMPLITUDE() (*Solvers.FourField.Wave method*), 102
 ANGLE (*Inputs.Geometry attribute*), 85
 Annular flow, 24
 applySolverDependentProperties() (*Inputs.Input method*), 91
 applySolverDependentProps() (*Inputs.InputSet method*), 82
 AREA (*Inputs.Geometry attribute*), 85
 AREA() (*Solvers.ThreeField.Drop method*), 139
 AREA() (*Solvers.TwoFluid.Liquid method*), 160
 AREA() (*Solvers.TwoFluid.Vapor method*), 149
 AXIALINTERP (*Inputs.Options attribute*), 88
 axialInterpolate() (*Solvers.Mixture.MixtureSolver method*), 113

B

Base (*class in Solvers.FourField*), 107
 base (*Solvers.FourField.Film attribute*), 100
 Base() (*Solvers.FourField.Base method*), 107
 BASEEQTHICK (*class in InputEnums*), 70
 BASEEQTHICK (*Inputs.Model attribute*), 75
 BASEEQTHICKCOEF (*Inputs.Model attribute*), 75
 BASEYPLUS (*Inputs.Model attribute*), 75
 bc (*Inputs.InputSet attribute*), 81
 BETA() (*Solvers.FourField.Base method*), 108

BETA() (*Solvers.FourField.Wave method*), 102
BETAP() (*Solvers.FourField.Base method*), 108
BETAP() (*Solvers.FourField.Wave method*), 102
Boundary condition, 24
BoundaryConditions (*class in Inputs*), 86
boundaryConditions (*Solvers.Mixture.MixtureSolver attribute*), 113
boundaryConditions (*Solvers.ThreeField.ThreeFieldSolver attribute*), 135
boundaryConditions (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
BoundaryConditions() (*Inputs.BoundaryConditions method*), 86
BHTM (*class in InputEnums*), 67
BHTM (*Inputs.Model attribute*), 75
BTMT (*class in InputEnums*), 65
BTMT (*Inputs.Model attribute*), 75
BUBBLEDRAE (*class in InputEnums*), 69
BUBBLEDRAE (*Inputs.Model attribute*), 75
BUBBLEDRAECOEF (*Inputs.Model attribute*), 76

C

C() (*Solvers.Mixture.Liquid method*), 133
C() (*Solvers.Mixture.Vapor method*), 117
CBT (*class in InputEnums*), 71
CBT (*Inputs.Model attribute*), 76
cbt (*Solvers.Mixture.Mixture attribute*), 119
CBT() (*Solvers.Mixture.Mixture method*), 120
CBT() (*Solvers.TwoFluid.Mixture method*), 157
CBTELEVATION (*Inputs.Model attribute*), 76
CBTIDX() (*Solvers.Mixture.Mixture method*), 120
CBTKEFFECT (*Inputs.Model attribute*), 76
CBTMULT (*Inputs.Model attribute*), 76
CBTZ() (*Solvers.Mixture.Mixture method*), 121
CD() (*Solvers.TwoFluid.Liquid method*), 160
CD() (*Solvers.TwoFluid.Vapor method*), 149
CHF() (*Solvers.Mixture.Mixture method*), 121
CHF() (*Solvers.TwoFluid.Mixture method*), 157
closeLog() (*Session.Log method*), 94
Closure model, 24
CONC() (*Solvers.ThreeField.Drop method*), 140
Conservation equation, 24
convert2JSON() (*Inputs.Input static method*), 91
copyFlowProperties() (*Solvers.AbstractField method*), 173
copyFlowProperties() (*Solvers.FourField.Base method*), 111
copyFlowProperties() (*Solvers.FourField.Film method*), 100
CPF (*Inputs.FluidProperties attribute*), 82
CPG (*Inputs.FluidProperties attribute*), 82
CPL() (*Inputs.FluidProperties method*), 83
CPV() (*Inputs.FluidProperties method*), 84
CreateITR() (*Solvers.AbstractSolver static method*), 172
CSTWAVEFREQ (*Inputs.Model attribute*), 76
currentAhIdx (*Solvers.SolverPlotter attribute*), 171
CV() (*Solvers.AbstractFilm method*), 168
CV_WALLISTHICK_CALC() (*Solvers.AbstractFilm method*), 168
CW() (*Solvers.AbstractFilm method*), 169
CW_LAM_CALC() (*Solvers.AbstractFilm method*), 169
CW_TURB_CALC() (*Solvers.AbstractFilm method*), 169

D

defaultUsedReport() (*Inputs.Input* method), 91
 delete() (*Session.Log* method), 94
 DELTAFREQ() (*Solvers.FourField.Wave* method), 102
 DENSITY() (*Solvers.ThreeField.Drop* method), 140
 DEPENDANCE (class in *InputEnums*), 70
 DEPENDANCE (*Inputs.Model* attribute), 76
 Deposition, 24
 DEPOSITION (class in *InputEnums*), 67
 DEPOSITION (*Inputs.Model* attribute), 76
 DHIN() (*Inputs.BoundaryConditions* method), 87
 DIAM() (*Solvers.ThreeField.Drop* method), 140
 diaryFilePath (*Session.Log* attribute), 93
 diaryIsOn (*Session.Log* attribute), 93
 diaryOff() (*Session.Log* method), 94
 diaryOn() (*Session.Log* method), 94
 directory (*Session.Session* attribute), 92
 dirName (*Session.Session* attribute), 92
 distributeOAFW() (*Solvers.FourField.Film* method), 100
 Disturbance wave, 24
 DITTUSBOELTERCOEF (*Inputs.Model* attribute), 76
 DP (*Solvers.Mixture.Mixture* attribute), 118
 DPACCS() (*Solvers.Mixture.Mixture* method), 121
 DPACCT() (*Solvers.Mixture.Mixture* method), 121
 DPAC CZ() (*Solvers.Mixture.Mixture* method), 121
 DPDZ() (*Solvers.TwoFluid.Mixture* method), 157
 DPGRV() (*Solvers.Mixture.Mixture* method), 121
 DPK() (*Solvers.Mixture.Mixture* method), 121
 DPPARTS() (*Solvers.Mixture.Mixture* method), 122
 DPSUM (*Solvers.Mixture.Mixture* attribute), 119
 DPTOT() (*Solvers.Mixture.Mixture* method), 122
 DPWALL() (*Solvers.Mixture.Mixture* method), 122
 DRAG() (*Solvers.ThreeField.Drop* method), 140
 DRAGCOEF() (*Solvers.FourField.Wave* method), 102
 Drop (class in *Solvers.FourField*), 98
 Drop (class in *Solvers.ThreeField*), 138
 drop (*Solvers.ThreeField.ThreeFieldSolver* attribute), 136
 Drop() (*Solvers.ThreeField.Drop* method), 138
 DROPDIAM (*Inputs.Model* attribute), 76
 DROPDRAG (class in *InputEnums*), 70
 DROPDRAG (*Inputs.Model* attribute), 76
 DROPDRAGCOEF (*Inputs.Model* attribute), 76
 dropInit (*Solvers.ThreeField.ThreeFieldSolver* attribute), 136
 Droplet field, 24
 DROPSLIP (*Inputs.Model* attribute), 76
 DT (*Solvers.AbstractField* attribute), 173
 DT (*Solvers.FourField.Base* attribute), 107
 DT (*Solvers.FourField.Film* attribute), 99
 DT (*Solvers.FourField.Wave* attribute), 101
 DT (*Solvers.Mixture.Mixture* attribute), 119
 DT (*Solvers.Mixture.MixtureSolver* attribute), 113
 DT (*Solvers.ThreeField.Drop* attribute), 139
 DT (*Solvers.ThreeField.Film* attribute), 143
 DT (*Solvers.ThreeField.ThreeFieldSolver* attribute), 135

DT (*Solvers.TwoFluid.Liquid attribute*), 159
DT (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
DT (*Solvers.TwoFluid.Vapor attribute*), 148
DTIN() (*Inputs.BoundaryConditions method*), 87
DTMFB (*Inputs.Model attribute*), 76
DZ (*Solvers.AbstractFilm attribute*), 168
DZ (*Solvers.Mixture.Mixture attribute*), 119
DZ (*Solvers.Mixture.MixtureSolver attribute*), 113
DZ (*Solvers.ThreeField.ThreeFieldSolver attribute*), 135
DZ (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145

E

ECT() (*Solvers.Mixture.Mixture method*), 122
ENTNUM() (*Solvers.AbstractFilm method*), 169
ENTNUMMAXITER (*Inputs.Options attribute*), 88
ENTNUMTOL (*Inputs.Options attribute*), 88
Entrainment, 24
ENTRAINMENT (*class in InputEnums*), 71
ENTRAINMENT (*Inputs.Model attribute*), 76
EPSILON() (*Solvers.FourField.Base method*), 108
EPSILON() (*Solvers.FourField.Wave method*), 102
EQFREQUENCY() (*Solvers.FourField.Wave method*), 103
EQPERIOD() (*Solvers.FourField.Wave method*), 103
EQSTROUHAL (*class in InputEnums*), 65
EQSTROUHAL (*Inputs.Model attribute*), 76
EQSTROUHAL() (*Solvers.FourField.Wave method*), 103
EQSTROUHALCOEF (*Inputs.Model attribute*), 76
EQTHICK() (*Solvers.FourField.Base method*), 108
EQUIL() (*Solvers.FourField.Film method*), 100
EQUIL() (*Solvers.FourField.FourFieldSolver method*), 96
EQUIL() (*Solvers.ThreeField.ThreeFieldSolver method*), 136
ERRORFW (*Inputs.Options attribute*), 88
ERRORH (*Inputs.Options attribute*), 88
ERRORP (*Inputs.Options attribute*), 88
ERRORU (*Inputs.Options attribute*), 88
ERRORUD (*Inputs.Options attribute*), 88
ERRORUF (*Inputs.Options attribute*), 88
ERRORW (*Inputs.Options attribute*), 88
ERRORWF (*Inputs.Options attribute*), 88
ETA() (*Solvers.FourField.Base method*), 108
ETA() (*Solvers.FourField.Wave method*), 103
extra (*Inputs.Input attribute*), 91

F

FBASE() (*Solvers.FourField.Wave method*), 103
FBASEMASS() (*Solvers.FourField.Wave method*), 103
FBASEVAPOR() (*Solvers.FourField.Base method*), 108
FBUOY() (*Solvers.AbstractFilm method*), 169
FBUOY() (*Solvers.ThreeField.Drop method*), 140
FBUOY() (*Solvers.TwoFluid.Liquid method*), 160
FBUOY() (*Solvers.TwoFluid.Vapor method*), 149
FDEP() (*Solvers.AbstractFilm method*), 169
FDEP() (*Solvers.FourField.Base method*), 109
FDEP() (*Solvers.FourField.Wave method*), 104

FDRAG() (*Solvers.FourField.Wave method*), 104
 FDRAG() (*Solvers.ThreeField.Drop method*), 140
 FDRAG() (*Solvers.TwoFluid.Liquid method*), 161
 FDRAG() (*Solvers.TwoFluid.Vapor method*), 149
 FDRY() (*Solvers.FourField.Base method*), 109
 FENT() (*Solvers.ThreeField.Drop method*), 140
 FGRAV() (*Solvers.AbstractFilm method*), 169
 FGRAV() (*Solvers.ThreeField.Drop method*), 141
 FGRAV() (*Solvers.TwoFluid.Liquid method*), 161
 FGRAV() (*Solvers.TwoFluid.Vapor method*), 149
 Field, [24](#)
 fieldName2plotStyle() (*Solvers.SolverPlotter static method*), 171
 Film (*class in Solvers.FourField*), [98](#)
 Film (*class in Solvers.ThreeField*), [143](#)
 film (*Solvers.FourField.Wave attribute*), [102](#)
 film (*Solvers.ThreeField.ThreeFieldSolver attribute*), [136](#)
 Film dryout, [24](#)
 Film() (*Solvers.FourField.Film method*), [99](#)
 filmInit (*Solvers.ThreeField.ThreeFieldSolver attribute*), [136](#)
 FINTCOND() (*Solvers.TwoFluid.Liquid method*), [161](#)
 FINTEVAP() (*Solvers.TwoFluid.Vapor method*), [150](#)
 FLOWID() (*Solvers.TwoFluid.Liquid method*), [161](#)
 FLOWREGIME() (*Solvers.TwoFluid.Liquid method*), [161](#)
 FLOWREGIME() (*Solvers.TwoFluid.Vapor method*), [150](#)
 FLUID (*Inputs.FluidProperties attribute*), [82](#)
 FLUID (*Inputs.Model attribute*), [76](#)
 fluid (*Solvers.AbstractFilm attribute*), [168](#)
 fluid (*Solvers.Mixture.Mixture attribute*), [119](#)
 fluid (*Solvers.Mixture.MixtureSolver attribute*), [113](#)
 fluid (*Solvers.ThreeField.ThreeFieldSolver attribute*), [136](#)
 fluid (*Solvers.TwoFluid.TwoFluidSolver attribute*), [145](#)
 FLUIDPROPERTIES (*class in InputEnums*), [66](#)
 FluidProperties (*class in Inputs*), [82](#)
 FluidProperties() (*Inputs.FluidProperties method*), [82](#)
 FontSize (*Solvers.SolverPlotter attribute*), [170](#)
 Four-field model, [25](#)
 FourFieldSolver (*class in Solvers.FourField*), [95](#)
 FourFieldSolver() (*Solvers.FourField.FourFieldSolver method*), [95](#)
 FOURIERCOND() (*Solvers.Mixture.Mixture method*), [122](#)
 FOURIEREVAP() (*Solvers.Mixture.Mixture method*), [123](#)
 FREQUENCY (*Solvers.FourField.Wave attribute*), [101](#)
 FRICTION (*Inputs.Model attribute*), [76](#)
 FSHEAR() (*Solvers.FourField.Wave method*), [104](#)
 FTOT() (*Solvers.AbstractFilm method*), [169](#)
 FTOT() (*Solvers.FourField.Base method*), [109](#)
 FTOT() (*Solvers.FourField.Wave method*), [104](#)
 FTOT() (*Solvers.ThreeField.Drop method*), [141](#)
 FTOT() (*Solvers.TwoFluid.Liquid method*), [161](#)
 FTOT() (*Solvers.TwoFluid.Vapor method*), [150](#)
 FVAPOR() (*Solvers.AbstractFilm method*), [169](#)
 FVAPOR() (*Solvers.FourField.Wave method*), [104](#)
 FW() (*Solvers.Mixture.Mixture method*), [123](#)
 FW() (*Solvers.Mixture.Vapor method*), [117](#)
 FWALEVAP() (*Solvers.TwoFluid.Vapor method*), [150](#)

FWALL() (*Solvers.AbstractFilm method*), 169
FWALL() (*Solvers.FourField.Wave method*), 104
FWALL() (*Solvers.TwoFluid.Liquid method*), 162
FWALL() (*Solvers.TwoFluid.Vapor method*), 150
FWAVE() (*Solvers.FourField.Base method*), 109
FWAVEMASS() (*Solvers.FourField.Base method*), 109
FWL() (*Solvers.Mixture.Mixture method*), 123

G

G (*Inputs.Model attribute*), 77
gca() (*Solvers.SolverPlotter method*), 171
gcf() (*Solvers.SolverPlotter method*), 171
Geometry (*class in Inputs*), 85
geometry (*Inputs.InputSet attribute*), 81
Geometry() (*Inputs.Geometry method*), 85
Governing equation, 25
Grid (*Solvers.SolverPlotter attribute*), 170

H

H (*Solvers.FourField.Base attribute*), 107
H (*Solvers.FourField.Film attribute*), 99
H (*Solvers.FourField.Wave attribute*), 101
H (*Solvers.Mixture.Mixture attribute*), 119
H (*Solvers.ThreeField.Drop attribute*), 139
H (*Solvers.ThreeField.Film attribute*), 143
H (*Solvers.TwoFluid.Liquid attribute*), 159
H (*Solvers.TwoFluid.Vapor attribute*), 148
HC() (*Inputs.FluidProperties method*), 84
HC() (*Solvers.Mixture.Liquid method*), 133
HC() (*Solvers.Mixture.Vapor method*), 117
HC() (*Solvers.TwoFluid.Mixture method*), 157
HDIAM() (*Inputs.Geometry method*), 85
HF (*Inputs.FluidProperties attribute*), 82
HF() (*Inputs.BoundaryConditions method*), 87
HFG (*Inputs.FluidProperties attribute*), 82
HFLUX (*Solvers.FourField.Base attribute*), 107
HFLUX (*Solvers.FourField.Film attribute*), 99
HFLUX (*Solvers.FourField.Wave attribute*), 101
HFLUX (*Solvers.Mixture.Mixture attribute*), 119
HFLUX (*Solvers.ThreeField.Film attribute*), 143
HFLUX() (*Solvers.Mixture.Liquid method*), 133
HFLUX() (*Solvers.Mixture.Vapor method*), 117
HFLUX() (*Solvers.TwoFluid.Liquid method*), 162
HFLUX() (*Solvers.TwoFluid.Mixture method*), 157
HFLUX() (*Solvers.TwoFluid.Vapor method*), 151
HFLUXWALEVAP() (*Solvers.Mixture.Vapor method*), 117
HFLUXWALEVAP() (*Solvers.TwoFluid.Liquid method*), 162
HFLUXWALEVAP() (*Solvers.TwoFluid.Vapor method*), 151
HG (*Inputs.FluidProperties attribute*), 82
HG() (*Inputs.BoundaryConditions method*), 87
HIN (*Inputs.BoundaryConditions attribute*), 86
HINT() (*Solvers.Mixture.Mixture method*), 123
HINT() (*Solvers.TwoFluid.Liquid method*), 162
HINT() (*Solvers.TwoFluid.Vapor method*), 151

HINTCOND() (*Solvers.Mixture.Mixture method*), 124
 HINTCOND() (*Solvers.TwoFluid.Liquid method*), 162
 HINTCOND() (*Solvers.TwoFluid.Vapor method*), 151
 HINTEVAP() (*Solvers.Mixture.Mixture method*), 124
 HINTEVAP() (*Solvers.TwoFluid.Liquid method*), 163
 HINTEVAP() (*Solvers.TwoFluid.Vapor method*), 151
 HM() (*Solvers.Mixture.Mixture method*), 124
 HTOT() (*Solvers.Mixture.Mixture method*), 124
 HTOT() (*Solvers.TwoFluid.Liquid method*), 163
 HTOT() (*Solvers.TwoFluid.Vapor method*), 151
 HVTOT() (*Solvers.Mixture.Mixture method*), 124
 HWALEVAP() (*Solvers.Mixture.Mixture method*), 125
 HWALEVAP() (*Solvers.TwoFluid.Liquid method*), 163
 HWALEVAP() (*Solvers.TwoFluid.Vapor method*), 152
 HWALHEAT() (*Solvers.Mixture.Mixture method*), 125
 HWALHEAT() (*Solvers.TwoFluid.Liquid method*), 163
 HWALHEAT() (*Solvers.TwoFluid.Vapor method*), 152
 HWALL() (*Solvers.Mixture.Liquid method*), 133
 HWALL() (*Solvers.Mixture.Mixture method*), 125
 HWALL() (*Solvers.Mixture.Vapor method*), 117
 HWALLBOIL() (*Solvers.Mixture.Mixture method*), 125
 HWALLPOSTCBT() (*Solvers.Mixture.Mixture method*), 125
 Hydrodynamic equilibrium, 25
 Hydrodynamic non-equilibrium, 25
 |
 ID (*Inputs.Geometry attribute*), 85
 ID (*Inputs.Model attribute*), 77
 ID (*Inputs.Options attribute*), 88
 initializeBaseAndWave() (*Solvers.FourField.Film method*), 100
 initializeSolver() (*Solvers.FourField.FourFieldSolver method*), 96
 initializeSolver() (*Solvers.Mixture.MixtureSolver method*), 114
 initializeSolver() (*Solvers.ThreeField.ThreeFieldSolver method*), 136
 initializeSolver() (*Solvers.TwoFluid.TwoFluidSolver method*), 145
 Input (*class in Inputs*), 90
 Input() (*Inputs.Input method*), 91
 InputEnums (*module*), 65
 Inputs (*module*), 75
 InputSet (*class in Inputs*), 81
 inputSet (*Solvers.AbstractFilm attribute*), 168
 inputSet (*Solvers.AbstractSolver attribute*), 172
 inputSet (*Solvers.Mixture.Mixture attribute*), 119
 inputSet (*Solvers.Mixture.MixtureSolver attribute*), 113
 inputSet (*Solvers.ThreeField.ThreeFieldSolver attribute*), 136
 inputSet (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
 InputSet() (*Inputs.InputSet method*), 81
 INTAREA (*class in InputEnums*), 65
 INTAREA (*Inputs.Model attribute*), 77
 INTAREA() (*Solvers.TwoFluid.Liquid method*), 163
 INTAREA() (*Solvers.TwoFluid.Vapor method*), 152
 Interfacial area, 25
 Interfacial transfer, 25
 interpBoundaryConditions() (*Solvers.Mixture.MixtureSolver method*), 114
 INTHFLUX() (*Solvers.TwoFluid.Liquid method*), 164

INTHFLUX() (*Solvers.TwoFluid.Vapor method*), 152
INTLENGTH (*class in InputEnums*), 69
INTLENGTH (*Inputs.Model attribute*), 77
INTLENGTHLCST (*Inputs.Model attribute*), 77
INTLENGTHVCST (*Inputs.Model attribute*), 77
INTNU (*class in InputEnums*), 74
INTNU (*Inputs.Model attribute*), 77
INTNU() (*Solvers.TwoFluid.Liquid method*), 164
INTNU() (*Solvers.TwoFluid.Vapor method*), 152
INTNULCST (*Inputs.Model attribute*), 77
INTNUVCST (*Inputs.Model attribute*), 77
INTTRANSH (*class in InputEnums*), 66
INTTRANSH (*Inputs.Model attribute*), 77
isLegalPath() (*in module Session*), 92
ITR (*Solvers.AbstractField attribute*), 173
ITR (*Solvers.FourField.Base attribute*), 107
ITR (*Solvers.FourField.Film attribute*), 99
ITR (*Solvers.FourField.Wave attribute*), 101
ITR (*Solvers.Mixture.Mixture attribute*), 119
ITR (*Solvers.ThreeField.Drop attribute*), 139
ITR (*Solvers.ThreeField.Film attribute*), 143
ITR (*Solvers.TwoFluid.Liquid attribute*), 160
ITR (*Solvers.TwoFluid.Vapor attribute*), 148

J

J() (*Solvers.TwoFluid.Liquid method*), 164
J() (*Solvers.TwoFluid.Vapor method*), 153
JG() (*Solvers.Mixture.Mixture method*), 126
JL() (*Solvers.Mixture.Mixture method*), 126

K

KBLOCKRATIO (*Inputs.Model attribute*), 77
KDIST() (*Solvers.Mixture.Mixture method*), 126
KENH() (*Solvers.ThreeField.Drop method*), 141
KF (*Inputs.FluidProperties attribute*), 82
KG (*Inputs.FluidProperties attribute*), 83
KIDX() (*Solvers.Mixture.Mixture method*), 126
KL() (*Inputs.FluidProperties method*), 84
KLOC (*Inputs.Model attribute*), 77
KLOCZ() (*Solvers.Mixture.Mixture method*), 126
KLOSS (*Inputs.Model attribute*), 77
KLOSS() (*Solvers.Mixture.Mixture method*), 126
KTRELAX (*Inputs.Model attribute*), 77
KTRELAX() (*Solvers.Mixture.Mixture method*), 127
KTUNING (*Inputs.Model attribute*), 77
KV() (*Inputs.FluidProperties method*), 84

L

L() (*Solvers.TwoFluid.Liquid method*), 164
L() (*Solvers.TwoFluid.Vapor method*), 153
legend() (*Solvers.SolverPlotter method*), 171
LENGTH (*Inputs.Geometry attribute*), 85
LHGR() (*Solvers.Mixture.Mixture method*), 127
LHGR() (*Solvers.TwoFluid.Mixture method*), 157

Liquid (class in *Solvers.Mixture*), 132
 Liquid (class in *Solvers.TwoFluid*), 159
 liquid (*Solvers.Mixture.Mixture* attribute), 120
 liquid (*Solvers.TwoFluid.TwoFluidSolver* attribute), 145
 Liquid film, 25
 Liquid() (*Solvers.Mixture.Liquid* method), 133
 Liquid() (*Solvers.TwoFluid.Liquid* method), 159
 liquidInit (*Solvers.TwoFluid.TwoFluidSolver* attribute), 145
 listInputProperties() (*Inputs.Input* method), 91
 LOCLEVEL (class in *InputEnums*), 73
 LOCLEVEL (*Inputs.Model* attribute), 77
 Log (class in *Session*), 93
 log (*Session.Session* attribute), 92
 Log() (*Session.Log* method), 93
 log() (*Session.Log* method), 94
 log() (*Solvers.AbstractSolver* method), 172
 LOGFID (*Session.Log* attribute), 93
 logFileName (*Session.Log* attribute), 93
 logFilePath (*Session.Log* attribute), 94
 LogMode (class in *Session*), 94
 LOGMODE (*Session.Log* attribute), 93

M

makeSessionDirectory() (*Session.Session* method), 93
 MAXITER (*Inputs.Options* attribute), 88
 MBASE() (*Solvers.FourField.Wave* method), 104
 MCT() (*Solvers.Mixture.Mixture* method), 127
 MDEP() (*Solvers.FourField.Base* method), 110
 MDEP() (*Solvers.FourField.Wave* method), 105
 MDEP() (*Solvers.ThreeField.Drop* method), 141
 MDER (*Solvers.Mixture.Mixture* attribute), 119
 memoizeFunction() (*Solvers.AbstractField* method), 173
 MENT() (*Solvers.AbstractFilm* method), 169
 MENT() (*Solvers.FourField.Base* method), 110
 MENT() (*Solvers.FourField.Wave* method), 105
 MEVAP (*Solvers.FourField.Film* attribute), 99
 MEVAP (*Solvers.ThreeField.Film* attribute), 143
 MEVAP() (*Solvers.FourField.Base* method), 110
 MEVAP() (*Solvers.FourField.Wave* method), 105
 MFBT (class in *InputEnums*), 66
 MFBT (*Inputs.Model* attribute), 77
 mfbt (*Solvers.Mixture.Mixture* attribute), 120
 MFBT() (*Solvers.Mixture.Mixture* method), 127
 MFLOW (*Inputs.BoundaryConditions* attribute), 86
 MFLUX() (*Solvers.Mixture.Liquid* method), 133
 MFLUX() (*Solvers.Mixture.Mixture* method), 127
 MFLUX() (*Solvers.Mixture.Vapor* method), 117
 MFLUX() (*Solvers.TwoFluid.Mixture* method), 158
 MINT() (*Solvers.Mixture.Mixture* method), 127
 MINT() (*Solvers.TwoFluid.Liquid* method), 164
 MINT() (*Solvers.TwoFluid.Vapor* method), 153
 MINTCOND() (*Solvers.Mixture.Mixture* method), 128
 MINTCOND() (*Solvers.TwoFluid.Liquid* method), 165
 MINTCOND() (*Solvers.TwoFluid.Vapor* method), 153

`MINTEVAP()` (*Solvers.Mixture.Mixture method*), 128
`MINTEVAP()` (*Solvers.TwoFluid.Liquid method*), 165
`MINTEVAP()` (*Solvers.TwoFluid.Vapor method*), 153
`mix` (*Solvers.AbstractFilm attribute*), 168
`mix` (*Solvers.ThreeField.Drop attribute*), 139
`mix` (*Solvers.TwoFluid.Liquid attribute*), 160
`mix` (*Solvers.TwoFluid.Vapor attribute*), 149
`mixSolver` (*Solvers.ThreeField.ThreeFieldSolver attribute*), 136
`mixSolver` (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
`Mixture` (*class in Solvers.Mixture*), 118
`Mixture` (*class in Solvers.TwoFluid*), 156
`mixture` (*Solvers.Mixture.MixtureSolver attribute*), 113
`Mixture model`, 25
`Mixture()` (*Solvers.Mixture.Mixture method*), 118
`Mixture()` (*Solvers.TwoFluid.Mixture method*), 157
`mixtureInit` (*Solvers.Mixture.MixtureSolver attribute*), 113
`MixtureSolver` (*class in Solvers.Mixture*), 112
`MixtureSolver()` (*Solvers.Mixture.MixtureSolver method*), 112
`Model` (*class in Inputs*), 75
`model` (*Inputs.InputSet attribute*), 81
`Model()` (*Inputs.Model method*), 75
`MOECKCOEF` (*Inputs.Model attribute*), 77
`MOMENTBASE` (*class in InputEnums*), 69
`MOMENTBASE` (*Inputs.Model attribute*), 77
`MOMENTDROP` (*class in InputEnums*), 69
`MOMENTDROP` (*Inputs.Model attribute*), 78
`MOMENTFILM` (*class in InputEnums*), 72
`MOMENTFILM` (*Inputs.Model attribute*), 78
`MOMENTGAS` (*class in InputEnums*), 74
`MOMENTGAS` (*Inputs.Model attribute*), 78
`MOMENTLIQUID` (*class in InputEnums*), 74
`MOMENTLIQUID` (*Inputs.Model attribute*), 78
`MOMENTWAVE` (*class in InputEnums*), 68
`MOMENTWAVE` (*Inputs.Model attribute*), 78
`MRM` (*Solvers.Mixture.Mixture attribute*), 119
`MRMTIMEINT` (*class in InputEnums*), 72
`MRMTIMEINT` (*Inputs.Options attribute*), 88
`MTOT()` (*Solvers.AbstractFilm method*), 169
`MTOT()` (*Solvers.FourField.Base method*), 110
`MTOT()` (*Solvers.FourField.Wave method*), 105
`MTOT()` (*Solvers.Mixture.Mixture method*), 128
`MTOT()` (*Solvers.TwoFluid.Liquid method*), 165
`MTOT()` (*Solvers.TwoFluid.Vapor method*), 154
`MTRANSV()` (*Solvers.Mixture.Mixture method*), 128
`MTURB()` (*Solvers.FourField.Base method*), 110
`MTURB()` (*Solvers.FourField.Wave method*), 105
`MU()` (*Solvers.Mixture.Mixture method*), 128
`MUF` (*Inputs.FluidProperties attribute*), 83
`MUG` (*Inputs.FluidProperties attribute*), 83
`MUL()` (*Inputs.FluidProperties method*), 84
`Multi-field model`, 25
`Multi-wall channel`, 25
`MUV()` (*Inputs.FluidProperties method*), 84
`MWALEVAP()` (*Solvers.Mixture.Mixture method*), 128

MWALEVAP() (*Solvers.TwoFluid.Liquid method*), 165
 MWALEVAP() (*Solvers.TwoFluid.Vapor method*), 154
 MWAVE() (*Solvers.FourField.Base method*), 110

N

N() (*Solvers.FourField.Wave method*), 105
 name (*Session.Session attribute*), 92
 NEARWALL (*Solvers.Mixture.Mixture attribute*), 119
 NEARWALLEQOAF (*Inputs.Model attribute*), 78
 NEARWALLH (*class in InputEnums*), 66
 NEARWALLH (*Inputs.Model attribute*), 78
 NEARWALLHRATIO (*Inputs.Model attribute*), 78
 NEARWALLRATIO (*Inputs.Model attribute*), 78
 NEARWALLRELAX (*class in InputEnums*), 67
 NEARWALLRELAX (*Inputs.Model attribute*), 78
 NEARWALLRELAXCOEF (*Inputs.Model attribute*), 78
 NEARWALLRELAXT (*Inputs.Model attribute*), 78
 NEARWALLRELAXX (*Inputs.Model attribute*), 78
 newTile() (*Solvers.SolverPlotter method*), 171
 NNODES (*Inputs.Model attribute*), 78
 Non-equilibrium, 25
 NTIME (*Solvers.AbstractField attribute*), 173
 NTIME (*Solvers.FourField.Base attribute*), 107
 NTIME (*Solvers.FourField.Film attribute*), 99
 NTIME (*Solvers.FourField.Wave attribute*), 101
 NTIME (*Solvers.Mixture.Mixture attribute*), 119
 NTIME (*Solvers.Mixture.MixtureSolver attribute*), 113
 NTIME (*Solvers.ThreeField.Drop attribute*), 139
 NTIME (*Solvers.ThreeField.Film attribute*), 143
 NTIME (*Solvers.ThreeField.ThreeFieldSolver attribute*), 135
 NTIME (*Solvers.TwoFluid.Liquid attribute*), 160
 NTIME (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
 NTIME (*Solvers.TwoFluid.Vapor attribute*), 148
 NU() (*Solvers.Mixture.Liquid method*), 133
 NU() (*Solvers.Mixture.Vapor method*), 117
 NWALL() (*Inputs.Geometry method*), 85
 NZ (*Solvers.AbstractField attribute*), 173
 NZ (*Solvers.FourField.Base attribute*), 107
 NZ (*Solvers.FourField.Film attribute*), 99
 NZ (*Solvers.FourField.Wave attribute*), 101
 NZ (*Solvers.Mixture.Mixture attribute*), 119
 NZ (*Solvers.Mixture.MixtureSolver attribute*), 113
 NZ (*Solvers.ThreeField.Drop attribute*), 139
 NZ (*Solvers.ThreeField.Film attribute*), 143
 NZ (*Solvers.ThreeField.ThreeFieldSolver attribute*), 135
 NZ (*Solvers.TwoFluid.Liquid attribute*), 160
 NZ (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
 NZ (*Solvers.TwoFluid.Vapor attribute*), 148

O

OAF (*class in InputEnums*), 72
 OAF (*Inputs.Model attribute*), 78
 OAFBASERATIO (*Inputs.Model attribute*), 78
 OAFDROPRATIO (*Inputs.Model attribute*), 78

OAFENTRAINED (*class in InputEnums*), 67
OAFENTRAINED (*Inputs.Model attribute*), 78
OAFEQUILMAXITER (*Inputs.Options attribute*), 88
OAFEQUILTOL (*Inputs.Options attribute*), 88
OAFFILMSPLIT (*class in InputEnums*), 72
OAFFILMSPLIT (*Inputs.Model attribute*), 78
OAFIDX() (*Solvers.Mixture.Mixture method*), 129
OAFTRANSITION (*Inputs.Model attribute*), 79
OAFWL() (*Solvers.Mixture.Mixture method*), 129
OAFX() (*Solvers.Mixture.Mixture method*), 129
OAFX() (*Solvers.TwoFluid.Mixture method*), 158
OAFZ() (*Solvers.Mixture.Mixture method*), 129
OKAWACOEFS (*Inputs.Model attribute*), 79
OKAWACOEFS() (*Solvers.AbstractFilm method*), 169
OKAWAMENT() (*Solvers.AbstractFilm method*), 169
One-dimensional model, 25
openLog() (*Session.Log method*), 94
Options (*class in Inputs*), 87
options (*Inputs.InputSet attribute*), 81
Options() (*Inputs.Options method*), 87
overwriteFiles (*Session.Session attribute*), 92

P

P (*Solvers.Mixture.Mixture attribute*), 119
parentDir (*Session.Session attribute*), 93
PCRIT (*Inputs.FluidProperties attribute*), 83
PERIM (*Inputs.Geometry attribute*), 85
PERIOD() (*Solvers.FourField.Wave method*), 106
Phase, 25
Phase change, 25
PHI2F() (*Solvers.Mixture.Mixture method*), 129
PHI2K() (*Solvers.Mixture.Mixture method*), 129
plot() (*Inputs.BoundaryConditions method*), 87
plot() (*Inputs.FluidProperties method*), 84
plotK() (*Solvers.SolverPlotter method*), 171
plotOAF() (*Solvers.SolverPlotter method*), 171
plott() (*Solvers.FourField.FourFieldSolver method*), 97
plott() (*Solvers.Mixture.MixtureSolver method*), 114
plott() (*Solvers.ThreeField.ThreeFieldSolver method*), 137
plott() (*Solvers.TwoFluid.TwoFluidSolver method*), 146
plotz() (*Solvers.FourField.FourFieldSolver method*), 97
plotz() (*Solvers.Mixture.MixtureSolver method*), 115
plotz() (*Solvers.SolverPlotter method*), 171
plotz() (*Solvers.ThreeField.ThreeFieldSolver method*), 137
plotz() (*Solvers.TwoFluid.TwoFluidSolver method*), 146
plotzt() (*Solvers.AbstractField method*), 173
plotzt() (*Solvers.FourField.FourFieldSolver method*), 97
plotzt() (*Solvers.Mixture.MixtureSolver method*), 115
plotzt() (*Solvers.ThreeField.ThreeFieldSolver method*), 137
plotzt() (*Solvers.TwoFluid.TwoFluidSolver method*), 147
POSFILM (*Inputs.Model attribute*), 79
POWER (*Inputs.BoundaryConditions attribute*), 86
PRANDTLF (*Inputs.FluidProperties attribute*), 83
PRANDTLG (*Inputs.FluidProperties attribute*), 83

PRANDTLTLL() (*Inputs.FluidProperties method*), 84
 PRANDTLV() (*Inputs.FluidProperties method*), 84
 PRESSURE (*Inputs.BoundaryConditions attribute*), 86
 PRESSURE (*Inputs.FluidProperties attribute*), 83
 PROPERTIES (*Inputs.FluidProperties attribute*), 83
 PROPERTIES (*Inputs.Model attribute*), 79
 PTRIPLE (*Inputs.FluidProperties attribute*), 83

Q

Q() (*Solvers.Mixture.Liquid method*), 133
 Q() (*Solvers.Mixture.Vapor method*), 118
 QK (*Inputs.FluidProperties attribute*), 83

R

RANZMARSHALLCST (*Inputs.Model attribute*), 79
 RANZMARSHALLVCST (*Inputs.Model attribute*), 79
 RE() (*Solvers.AbstractFilm method*), 169
 RE() (*Solvers.Mixture.Liquid method*), 134
 RE() (*Solvers.Mixture.Mixture method*), 130
 RE() (*Solvers.Mixture.Vapor method*), 118
 RE() (*Solvers.ThreeField.Drop method*), 141
 RE() (*Solvers.TwoFluid.Liquid method*), 165
 RE() (*Solvers.TwoFluid.Vapor method*), 154
 REGIMES (*class in Solvers.TwoFluid*), 147
 REL() (*Solvers.Mixture.Mixture method*), 130
 REL() (*Solvers.TwoFluid.Vapor method*), 154
 Relaxation model, 25
 Relaxation time, 26
 RELAXCONDCOEFF (*Inputs.Model attribute*), 79
 RELAXCONDFO (*Inputs.Model attribute*), 79
 RELAXCONDFOCOEFF (*Inputs.Model attribute*), 79
 RELAXCONDT (*Inputs.Model attribute*), 79
 RELAXEVAPCOEFF (*Inputs.Model attribute*), 79
 RELAXEVAPFO (*Inputs.Model attribute*), 79
 RELAXEVAPFOCOEFF (*Inputs.Model attribute*), 79
 RELAXEVAPT (*Inputs.Model attribute*), 79
 RELAXFW (*Inputs.Options attribute*), 88
 RELAXHL (*Inputs.Options attribute*), 89
 RELAXHM (*Inputs.Options attribute*), 89
 RELAXHV (*Inputs.Options attribute*), 89
 RELAXPM (*Inputs.Options attribute*), 89
 RELAXT() (*Solvers.Mixture.Mixture method*), 130
 RELAXTB (*Inputs.Model attribute*), 79
 RELAXTCOND() (*Solvers.Mixture.Mixture method*), 130
 RELAXTCOND() (*Solvers.TwoFluid.Mixture method*), 158
 RELAXTEVAP() (*Solvers.Mixture.Mixture method*), 130
 RELAXTEVAP() (*Solvers.TwoFluid.Mixture method*), 158
 RELAXTW (*Inputs.Model attribute*), 79
 RELAXUB (*Inputs.Options attribute*), 89
 RELAXUD (*Inputs.Options attribute*), 89
 RELAXUF (*Inputs.Options attribute*), 89
 RELAXUL (*Inputs.Options attribute*), 89
 RELAXUV (*Inputs.Options attribute*), 89
 RELAXUW (*Inputs.Options attribute*), 89

RELAXWB (*Inputs.Options attribute*), 89
RELAXWF (*Inputs.Options attribute*), 89
RELAXWL (*Inputs.Options attribute*), 89
RELAXWM (*Inputs.Options attribute*), 89
RELAXWV (*Inputs.Options attribute*), 89
RELAXWW (*Inputs.Options attribute*), 89
RELAXX (*Inputs.Model attribute*), 79
RELVELCST (*Inputs.Model attribute*), 79
resizeFigure() (*Solvers.SolverPlotter method*), 171
REV() (*Solvers.FourField.Wave method*), 106
REV() (*Solvers.ThreeField.Drop method*), 141
REV() (*Solvers.TwoFluid.Liquid method*), 165
RHO() (*Solvers.Mixture.Mixture method*), 131
RHO() (*Solvers.TwoFluid.Mixture method*), 158
RHOF (*Inputs.FluidProperties attribute*), 83
RHOG (*Inputs.FluidProperties attribute*), 83
RHOL() (*Inputs.FluidProperties method*), 84
RHOV() (*Inputs.FluidProperties method*), 84
RWALL() (*Inputs.Geometry method*), 85

S

S() (*Solvers.TwoFluid.Liquid method*), 166
S() (*Solvers.TwoFluid.Vapor method*), 154
save() (*Solvers.AbstractSolver method*), 172
Session (*class in Session*), 92
session (*Inputs.InputSet attribute*), 81
Session (*module*), 92
Session() (*Session.Session method*), 92
setupLog() (*Session.Session method*), 93
setZs() (*Solvers.SolverPlotter method*), 171
SHAPEFACTOR() (*Solvers.FourField.Wave method*), 106
SHAPEFACTORCOEF (*Inputs.Model attribute*), 79
showWarnings (*Session.Log attribute*), 94
showWarnings (*Session.Session attribute*), 93
SIGMA (*Inputs.FluidProperties attribute*), 83
SLIP (*Inputs.Model attribute*), 80
solve() (*Solvers.FourField.FourFieldSolver method*), 98
solve() (*Solvers.Mixture.MixtureSolver method*), 116
solve() (*Solvers.ThreeField.ThreeFieldSolver method*), 138
solve() (*Solvers.TwoFluid.TwoFluidSolver method*), 147
Solver, 26
SOLVERDEPENDENTPROPS (*Inputs.Model attribute*), 80
solverName() (*Solvers.AbstractSolver method*), 172
SolverPlotter (*class in Solvers*), 170
SolverPlotter() (*Solvers.SolverPlotter method*), 170
Solvers (*module*), 168
Solvers.FourField (*module*), 98
Solvers.Mixture (*module*), 116
Solvers.ThreeField (*module*), 138
Solvers.TwoFluid (*module*), 147
SolverState (*class in Solvers*), 174
Source term, 26
SPACING() (*Solvers.FourField.Wave method*), 106
SPHTM (*class in InputEnums*), 70

SPHTM (*Inputs.Model* attribute), 80
 SPMTM (*class in InputEnums*), 66
 SPMTM (*Inputs.Model* attribute), 80
 SSCONVFW (*Inputs.Options* attribute), 89
 SSCONVH (*Inputs.Options* attribute), 89
 SSCONVP (*Inputs.Options* attribute), 89
 SSCONVU (*Inputs.Options* attribute), 90
 SSCONVUD (*Inputs.Options* attribute), 90
 SSCONVUF (*Inputs.Options* attribute), 90
 SSCONVW (*Inputs.Options* attribute), 90
 SSCONVWF (*Inputs.Options* attribute), 90
 SSMAXITER (*Inputs.Options* attribute), 90
 SSTSTEP (*Inputs.Options* attribute), 90
 STATE (*Solvers.AbstractSolver* attribute), 172
 Straight channel, 26
 struct() (*Solvers.AbstractField* method), 173

T

T() (*Inputs.FluidProperties* method), 84
 T() (*Solvers.Mixture.Liquid* method), 134
 T() (*Solvers.Mixture.Mixture* method), 131
 T() (*Solvers.Mixture.Vapor* method), 118
 T() (*Solvers.TwoFluid.Liquid* method), 166
 T() (*Solvers.TwoFluid.Vapor* method), 154
 TAUW() (*Solvers.Mixture.Mixture* method), 131
 TAUW() (*Solvers.Mixture.Vapor* method), 118
 TAUW() (*Solvers.TwoFluid.Liquid* method), 166
 TAUW() (*Solvers.TwoFluid.Mixture* method), 158
 TAUW() (*Solvers.TwoFluid.Vapor* method), 155
 TBASE() (*Solvers.FourField.Base* method), 111
 TDRY() (*Solvers.FourField.Base* method), 111
 Thermal equilibrium, 26
 Thermal non-equilibrium, 26
 THERMALNONEQ (*class in InputEnums*), 74
 THERMALNONEQ (*Inputs.Model* attribute), 80
 THERMALRELAX (*class in InputEnums*), 73
 THERMALRELAX (*Inputs.Model* attribute), 80
 THICK() (*Solvers.AbstractFilm* method), 169
 THICK() (*Solvers.FourField.Wave* method), 106
 THICKMIN() (*Solvers.FourField.Base* method), 111
 THINFILMFRIC (*class in InputEnums*), 68
 THINFILMFRIC (*Inputs.Model* attribute), 80
 THINFILMTHICK (*Inputs.Model* attribute), 80
 THINWAVETHICK (*Inputs.Model* attribute), 80
 Three-field model, 26
 ThreeFieldSolver (*class in Solvers.ThreeField*), 134
 ThreeFieldSolver() (*Solvers.ThreeField.ThreeFieldSolver* method), 135
 TIDX (*Solvers.AbstractField* attribute), 173
 TIDX (*Solvers.FourField.Base* attribute), 107
 TIDX (*Solvers.FourField.Film* attribute), 99
 TIDX (*Solvers.FourField.Wave* attribute), 101
 TIDX (*Solvers.Mixture.Mixture* attribute), 119
 TIDX (*Solvers.ThreeField.Drop* attribute), 139
 TIDX (*Solvers.ThreeField.Film* attribute), 143

TIDX (*Solvers.TwoFluid.Liquid* attribute), 160
TIDX (*Solvers.TwoFluid.Vapor* attribute), 148
TIME (*Inputs.BoundaryConditions* attribute), 86
TIME (*Solvers.AbstractField* attribute), 173
TIME (*Solvers.FourField.Base* attribute), 107
TIME (*Solvers.FourField.Film* attribute), 99
TIME (*Solvers.FourField.Wave* attribute), 101
TIME (*Solvers.Mixture.Mixture* attribute), 119
TIME (*Solvers.Mixture.MixtureSolver* attribute), 113
TIME (*Solvers.ThreeField.Drop* attribute), 139
TIME (*Solvers.ThreeField.Film* attribute), 143
TIME (*Solvers.ThreeField.ThreeFieldSolver* attribute), 135
TIME (*Solvers.TwoFluid.Liquid* attribute), 160
TIME (*Solvers.TwoFluid.TwoFluidSolver* attribute), 145
TIME (*Solvers.TwoFluid.Vapor* attribute), 148
TIME() (*Solvers.Mixture.Liquid* method), 134
TIME() (*Solvers.Mixture.Vapor* method), 118
TIME() (*Solvers.TwoFluid.Mixture* method), 158
TIMEINTERP (*Inputs.Options* attribute), 90
timeInterpolate() (*Solvers.Mixture.MixtureSolver* method), 116
TIN() (*Inputs.BoundaryConditions* method), 87
Title (*Solvers.SolverPlotter* attribute), 170
TPFM (class in *InputEnums*), 68
TPFM (*Inputs.Model* attribute), 80
TPHTM (class in *InputEnums*), 69
TPHTM (*Inputs.Model* attribute), 80
TPKM (class in *InputEnums*), 71
TPKM (*Inputs.Model* attribute), 80
transient() (*Inputs.FluidProperties* method), 84
transient() (*Solvers.AbstractField* method), 173
Ts (*Solvers.SolverPlotter* attribute), 170
TSAT (*Inputs.FluidProperties* attribute), 83
TSAT() (*Inputs.BoundaryConditions* method), 87
TSTEP (*Inputs.Options* attribute), 90
TWALL() (*Solvers.Mixture.Mixture* method), 131
Two-fluid model, [26](#)
Two-phase flow, [26](#)
TwoFluidSolver (class in *Solvers.TwoFluid*), [144](#)
TwoFluidSolver() (*Solvers.TwoFluid.TwoFluidSolver* method), [144](#)

U

U (*Solvers.FourField.Base* attribute), 107
U (*Solvers.FourField.Film* attribute), 99
U (*Solvers.FourField.Wave* attribute), 101
U (*Solvers.ThreeField.Drop* attribute), 139
U (*Solvers.ThreeField.Film* attribute), 143
U (*Solvers.TwoFluid.Liquid* attribute), 160
U (*Solvers.TwoFluid.Vapor* attribute), 148
U() (*Solvers.Mixture.Liquid* method), 134
U() (*Solvers.Mixture.Mixture* method), 131
U() (*Solvers.Mixture.Vapor* method), 118
U() (*Solvers.TwoFluid.Mixture* method), 158
UALGEBR() (*Solvers.AbstractFilm* method), 170
UALGEBR() (*Solvers.ThreeField.Drop* method), 142

UC (*Inputs.FluidProperties attribute*), 83
 UDEQUILMAXITER (*Inputs.Options attribute*), 90
 UDEQUILTOL (*Inputs.Options attribute*), 90
 UEQUIL() (*Solvers.AbstractFilm method*), 170
 UEQUIL() (*Solvers.ThreeField.Drop method*), 142
 UEQUILS() (*Solvers.AbstractFilm method*), 170
 UFEQUILMAXITER (*Inputs.Options attribute*), 90
 UFEQUILTOL (*Inputs.Options attribute*), 90
 UP() (*Solvers.Mixture.Mixture method*), 131
 USLIP() (*Solvers.ThreeField.Drop method*), 142
 USLIP() (*Solvers.TwoFluid.Liquid method*), 166
 USLIP() (*Solvers.TwoFluid.Vapor method*), 155
 UWALL() (*Solvers.AbstractFilm method*), 170

V

validateFunctionHandleInput() (*Inputs.Input static method*), 91
 validateInputEntry() (*Inputs.Input method*), 91
 Vapor (*class in Solvers.Mixture*), 116
 Vapor (*class in Solvers.TwoFluid*), 148
 vapor (*Solvers.Mixture.Mixture attribute*), 120
 vapor (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
 Vapor() (*Solvers.Mixture.Vapor method*), 117
 Vapor() (*Solvers.TwoFluid.Vapor method*), 148
 VAPORFRIC (*class in InputEnums*), 73
 VAPORFRIC (*Inputs.Model attribute*), 80
 VAPORFRICCST (*Inputs.Model attribute*), 80
 vaporInit (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145
 VF() (*Solvers.Mixture.Liquid method*), 134
 VF() (*Solvers.Mixture.Mixture method*), 132
 VF() (*Solvers.Mixture.Vapor method*), 118
 VF() (*Solvers.TwoFluid.Liquid method*), 166
 VF() (*Solvers.TwoFluid.Mixture method*), 158
 VF() (*Solvers.TwoFluid.Vapor method*), 155
 VISC() (*Solvers.TwoFluid.Liquid method*), 166
 VISC() (*Solvers.TwoFluid.Vapor method*), 155
 VOID (*class in InputEnums*), 72
 VOID (*Inputs.Model attribute*), 80
 VOLUME() (*Solvers.ThreeField.Drop method*), 142
 VR() (*Solvers.ThreeField.Drop method*), 142
 VR() (*Solvers.TwoFluid.Liquid method*), 167

W

W (*Solvers.FourField.Base attribute*), 107
 W (*Solvers.FourField.Film attribute*), 100
 W (*Solvers.FourField.Wave attribute*), 101
 W (*Solvers.Mixture.Mixture attribute*), 119
 W (*Solvers.ThreeField.Drop attribute*), 139
 W (*Solvers.ThreeField.Film attribute*), 143
 W (*Solvers.TwoFluid.Liquid attribute*), 160
 W (*Solvers.TwoFluid.Vapor attribute*), 149
 W() (*Solvers.Mixture.Liquid method*), 134
 W() (*Solvers.Mixture.Vapor method*), 118
 W() (*Solvers.TwoFluid.Mixture method*), 159
 WALEVAPRATIO() (*Solvers.Mixture.Mixture method*), 132

WALEVAPRATIO() (*Solvers.TwoFluid.Liquid method*), 167
WALEVAPRATIO() (*Solvers.TwoFluid.Mixture method*), 159
WALEVAPRATIO() (*Solvers.TwoFluid.Vapor method*), 155
Wall boiling, 26
Wall friction, 26
Wall heat flux, 26
WallIdx (*Solvers.SolverPlotter attribute*), 170
warning() (*Session.Log method*), 94
warning() (*Solvers.AbstractSolver method*), 172
warnings (*Inputs.Input attribute*), 91
Wave (*class in Solvers.FourField*), 101
wave (*Solvers.FourField.Film attribute*), 100
Wave() (*Solvers.FourField.Wave method*), 101
WAVEBASEINT (*class in InputEnums*), 66
WAVEBASEINT (*Inputs.Model attribute*), 80
WAVEDRAGCOEF (*Inputs.Model attribute*), 80
WAVEFREQUENCY (*class in InputEnums*), 68
WAVEFREQUENCY (*Inputs.Model attribute*), 80
WAVEMIXCOEF (*Inputs.Model attribute*), 80
WBOILINGN (*Inputs.Model attribute*), 81
WBOILINGXSAT (*Inputs.Model attribute*), 81
WBOILINGXSUB (*Inputs.Model attribute*), 81
WIDTH() (*Solvers.FourField.Wave method*), 106
WL() (*Solvers.AbstractFilm method*), 170
WL() (*Solvers.FourField.Wave method*), 106
WMESH (*Inputs.BoundaryConditions attribute*), 86
WMIN() (*Solvers.FourField.Base method*), 111
WMINL() (*Solvers.FourField.Base method*), 111
WPOWER (*Inputs.BoundaryConditions attribute*), 86
writeInputFile() (*Inputs.BoundaryConditions static method*), 87
writeInputFile() (*Inputs.Geometry static method*), 85
writeInputFile() (*Inputs.Input static method*), 91
writeInputFile() (*Inputs.Model static method*), 81
writeInputFile() (*Inputs.Options static method*), 90
WWALL() (*Solvers.Mixture.Mixture method*), 132

X

X() (*Solvers.AbstractFilm method*), 170
X() (*Solvers.Mixture.Liquid method*), 134
X() (*Solvers.Mixture.Mixture method*), 132
X() (*Solvers.Mixture.Vapor method*), 118
X() (*Solvers.ThreeField.Drop method*), 142
X() (*Solvers.TwoFluid.Liquid method*), 167
X() (*Solvers.TwoFluid.Vapor method*), 155
XEQ() (*Solvers.Mixture.Mixture method*), 132
XEQ() (*Solvers.TwoFluid.Mixture method*), 159
XIN() (*Inputs.BoundaryConditions method*), 87
xlim() (*Solvers.SolverPlotter method*), 171
XTR_ANN() (*Solvers.TwoFluid.Liquid method*), 167
XTR_ANN() (*Solvers.TwoFluid.Vapor method*), 156
XTR_CBT() (*Solvers.TwoFluid.Vapor method*), 156
XTR_ITM() (*Solvers.TwoFluid.Liquid method*), 167
XTR_ITM() (*Solvers.TwoFluid.Vapor method*), 156
XTR_SAT() (*Solvers.TwoFluid.Liquid method*), 167

XTR_SAT() (*Solvers.TwoFluid.Vapor method*), 156

XTR_SUB() (*Solvers.TwoFluid.Liquid method*), 167

XTR_SUB() (*Solvers.TwoFluid.Vapor method*), 156

Y

ylabels() (*Solvers.SolverPlotter method*), 171

ylim() (*Solvers.SolverPlotter method*), 171

YPLUS2THICK() (*Solvers.AbstractFilm method*), 170

Z

Z (*Solvers.AbstractField attribute*), 173

Z (*Solvers.FourField.Base attribute*), 108

Z (*Solvers.FourField.Film attribute*), 100

Z (*Solvers.FourField.Wave attribute*), 102

Z (*Solvers.Mixture.Mixture attribute*), 119

Z (*Solvers.Mixture.MixtureSolver attribute*), 113

Z (*Solvers.ThreeField.Drop attribute*), 139

Z (*Solvers.ThreeField.Film attribute*), 143

Z (*Solvers.ThreeField.ThreeFieldSolver attribute*), 135

Z (*Solvers.TwoFluid.Liquid attribute*), 160

Z (*Solvers.TwoFluid.TwoFluidSolver attribute*), 145

Z (*Solvers.TwoFluid.Vapor attribute*), 149

Z() (*Solvers.Mixture.Liquid method*), 134

Z() (*Solvers.Mixture.Vapor method*), 118

Z() (*Solvers.TwoFluid.Mixture method*), 159

Zs (*Solvers.SolverPlotter attribute*), 171